TESIS DOCTORAL POR COMPENDIO DE PUBLICACIONES

Predicción bioinformática de vías moleculares compensatorias en enfermedades raras y cáncer para su tratamiento.

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Málaga, 22 de septiembre de 2025

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Que D. Federico García Criado, Graduado en Bioquímica por la Universidad de Málaga, ha realizado bajo su dirección conjunta en el Departamento de Biología Molecular y Bioquímica de la Universidad de Málaga el trabajo de investigación correspondiente a su Tesis Doctoral que lleva por título "Predicción bioinformática de vías moleculares compensatorias en enfermedades raras y cáncer para su tratamiento". Que las publicaciones en coautoría en revistas científicas relevantes con revision por pares que avalan de forma idonea la presente Tesis Doctoral por compendio son: : "I*ntegrative transcriptomic and network-based analysis for the analysis of neuromuscular diseases*", "*Integrating Language Models and Network Embeddings to Uncover Hidden Relationships in Neuromuscular Diseases*" y "*Advancing edge-based clustering and graph embedding for biological network analysis: a case study in RASopathies*", las cuales no han sido utilizadas en tesis anteriores. Tras revisar y supervisar este trabajo, lo consideran apropiado para que se presente y defienda ante el tribunal de tesis correspondiente, por lo cual

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Producción científica

A continuación se listan los artículos publicados en revistas científicas. Los artículos que avalan esta tesis doctoral están señalados con el símbolo ‡.

- ‡ **García-Criado F**[†], Hurtado-García L[†], Rojano E[*], Esteban-Martos Á, Pérez-García J, Seoane P[†], Ranea JAG[†]. Integrative transcriptomic and network-based analysis for the analysis of neuromuscular diseases. *Int J Mol Sci*. 2025 Sep 18; accepted. [1][2]

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- ‡ **García-Criado F**, Pérez-García J, Rojano E, Ranea JAG, Seoane-Zonjic P[*]. Integrating Language Models and Network Embeddings to Uncover Hidden Relationships in Neuromuscular Diseases. *Lect Notes Bioinform*. Springer; 2025. accepted.[1]

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Acrónimos y abreviaciones

ALS: *Amyotrophic Lateral Sclerosis* (esclerosis lateral amiotrófica).
AUROC: *Area Under the Receiver Operating Characteristic curve* (Área bajo la curva ROC).
CFC: *Cardio-Facio-Cutaneous* (síndrome cardio-facio-cutáneo).
CDLIB: *Community Discovery Library* (librería para detección de comunidades).
DepMap: *Dependency Map Project*.
DMD: *Duchenne Muscular Dystrophy* (distrofia muscular de Duchenne).
EWS: *Exome-Wide Sequencing* (secuenciación de exoma a gran escala).
GCN: *Graph Convolutional Network* (red neuronal convolucional de grafos).
GNN: *Graph Neural Network*.
GO: *Gene Ontology*.
GWS: *Genome-Wide Sequencing* (secuenciación de genoma a gran escala).
HDBSCAN: *Hierarchical Density-Based Spatial Clustering of Applications with Noise* (Método de búsqueda de comunidades en datos vectoriales).
HGNC: *HUGO Gene Nomenclature Committee*.
HIPPIE: *Human Integrated Protein–Protein Interaction rEference*.
HLC: *Hierarchical Link Clustering* (detección de comunidades solapantes en redes).
HLCnp: Implementación en Python de *Hierarchical Link Clustering* optimizada para redes densas.
HPO: *Human Phenotype Ontology*.
LGMD: *Limb-Girdle Muscular Dystrophy* (distrofia muscular de cinturas).
LKCMsp: Implementación en R de *Link Communities* (basada en HLC) optimizada en memoria y tiempo.
lncRNAs: *long non-coding RNAs* (ARN largos no codificantes).
LoF: *Loss-of-Function* (variantes de pérdida de función génica).
MONDO: *Mondo Disease Ontology*.
NMDs: *Neuromuscular Disorders* (enfermedades neuromusculares).
node2vec: Algoritmo de embebido de nodos basado en *random walks*.
OMIM: *Online Mendelian Inheritance in Man*.
PETS: *Patients with Undiagnosed Genetic Diseases Using the Patient Exploration Tools Suite* (herramienta utilizada para actualizar/analizar el *diseasoma*).
PPI: *Protein–Protein Interaction* (interacción proteína–proteína).
PTC: *Premature stop codon* (codón de parada prematuro).
RNA: *ribonucleic acid* (Ácido ribonucleico).
RNA-seq: *RNA sequencing* (secuenciación de ARN).

ROC: *Receiver Operating Characteristic curve* (curva Característica Operativa del Receptor).
SAFE: *Spatial Analysis of Functional Enrichment.*
STRING: *Search Tool for the Retrieval of Interacting Genes/Proteins* (base de datos de interacciones proteína–proteína).
TA: *Transcriptional Adaptation.*
UMAP: *Uniform Manifold Approximation and Projection* (técnica de reducción de dimensionalidad no lineal).

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Parte I

Summary

Capítulo 1

Summary

1.1. Introduction

1.1.1. Genetic Diseases

Genetic diseases are caused by mutations in the human genome. Not all of them, however, have clinical consequences. Many genetic variants are neutral and exert little or no effect on gene or protein function, whereas pathogenic mutations disrupt gene activity and can give rise to disease. Such disorders may be present from birth or emerge later in life, with cancer representing one of the most frequent outcomes.

Notably, potentially pathogenic mutations are also found in healthy individuals where they do not produce clinical symptoms [1, 2]. This indicates the presence of biological mechanisms that buffer the effects of harmful mutations, a phenomenon known as compensation or robustness [3]. Understanding these mechanisms requires identifying the systems involved and analyzing how they influence disease onset or prevention.

This dissertation examines molecular mechanisms of genetic compensation, whereby dysfunction in one gene can be counterbalanced by the activity of another. Rare diseases offer a particularly relevant framework for this analysis. Most are caused by a single mutation, yet individuals with the same variant often show heterogeneous clinical symptoms [4]. Although each rare disease is uncommon, together they affect 3.5–5.9 % of the global population [5]. They are highly diverse both genetically and clinically [6], making them of particular scientific and clinical interest. Analyzing groups of rare diseases can therefore help reveal compensatory mechanisms.

In this thesis, two main groups of rare diseases are considered: neuromuscular disorders, notable for their heterogeneity and relevance to gene therapies, and RASopathies, which have potential links to cancer.

Neuromuscular diseases Neuromuscular diseases comprise a diverse set of disorders caused by mutations in genes involved in neuromuscular development and function [7]. They range from defects in neuromuscular transmission to abnormalities in muscle contraction. Despite some shared phenotypic features, their molecular characterization remains incomplete: of nearly 600 described neuromuscular diseases, only about 30 have been fully characterized at the genetic level [7].

This group is of particular interest because it represents one of the rare disease categories where gene therapies have been successfully applied [7]. Notably, therapeutic strategies based on genetic compensation have been developed, such as Nusinersen for spinal muscular atrophy [8, 9] and Ezutromid for Duchenne muscular dystrophy [10, 11], both of which leverage the overexpression of functionally similar genes to restore lost functions.

These cases highlight compensation as a potentially relevant phenomenon in neuromuscular diseases. Progress in this field requires improved classification and molecular characterization of these disorders, followed by the identification of compensatory systems and mechanisms that may serve as therapeutic targets.

RASopathies RASopathies are a group of rare diseases caused by germline mutations in genes encoding components of the RAS/MAPK signaling pathway [12, 13]. This pathway is essential for regulating fundamental cellular processes, including proliferation, differentiation and apoptosis [14]. Mutations affecting these genes result in abnormal activation of signaling, leading to a broad spectrum of clinical manifestations such as developmental defects [13], congenital heart anomalies [15], characteristic facial dysmorphisms [12], and an increased predisposition to cancer [16].

The principal RASopathies comprise Noonan syndrome, cardio-facio-cutaneous syndrome, Costello syndrome, and LEOPARD syndrome. Although these disorders exhibit considerable phenotypic variability, they share a common molecular basis in the dysregulation of the RAS/MAPK signaling pathway [12]. The combination of broad phenotypic diversity and low individual prevalence makes RASopathies a particularly informative model for investigating compensatory mechanisms.

The relation with cancer is particularly significant. The same RAS/MAPK pathway, frequently altered by somatic mutations in a wide range of solid and hematologic tumors, is disrupted in RASopathies through inherited germline variants [12, 16]. This parallelism suggests that insights from RASopathy research may not only shed light on compensatory mechanisms in rare diseases, but also identify therapeutic targets relevant to oncology.

Thus, RASopathies represent a paradigmatic example of how mutations in a key signaling pathway can produce both rare developmental disorders and tumorigenesis when occurring in somatic cells. Comparative analyses of these conditions

could therefore be fundamental to unraveling the relationship between genetic compensation, rare disease and cancer predisposition.

1.1.2. Graph theory applied to the study of biological and clinical networks

Graphs, also referred to as networks, have proven highly useful in biomedicine due to their ability to naturally model complex relationships among heterogeneous entities. A graph enables the representation, within a single mathematical structure, of both molecular interactions and clinical or epidemiological associations. This makes it a versatile tool for integrating information from different levels of biomedical analysis.

Illustrative examples include protein–protein interaction networks, which facilitate the identification of functional complexes [17]; gene co-expression networks, useful for discovering regulatory modules [18, 19]; biomedical knowledge graphs, integrating scientific literature, databases, and clinical terminologies [20, 21]; and patient networks, where nodes represent individuals and edges capture phenotypic similarities or shared risk factors [22]. These representations support critical tasks such as predicting disease-associated genes, identifying biomarkers, drug repurposing or stratifying patient subgroups.

Graph types and definitions A graph $G = (V, E)$ is defined by a set of nodes V and a set of edges E representing relationships between pairs of nodes. In biomedicine, nodes may correspond to genes, proteins or clinical entities, while edges capture interactions, co-expression, or functional similarity. Depending on the nature of these relationships, graphs can be directed (when edges (u, v) indicate asymmetric influences such as those in gene regulatory networks) or undirected (when edges represent symmetric associations like gene co-expression, where correlations are inherently mutual). Edges may also be weighted, with a function $w : E \to \mathbb{R}$ quantifying interaction strength, binding affinity, or confidence scores, as in protein–protein interaction networks; or unweighted, where only the presence or absence of a connection is considered. In addition, bipartite or multipartite graphs partition the node set into disjoint subsets, with edges restricted to nodes in different partitions, an approach especially valuable for modeling heterogeneous biomedical data, for instance, linking drugs and diseases (bipartite) or simultaneously integrating genes, diseases, and chemical compounds (multipartite). Across these graph types, a common set of topological metrics is applied at both global and local levels: the former describe properties of the entire network, while the latter capture the roles of individual nodes or substructures. This dual perspective is crucial in biomedicine, enabling both the analysis of emergent system-wide

properties and the identification of key molecular or clinical elements.

Global properties Global network properties provide insights into the overall structure of a graph and enable comparisons across networks of different nature or under distinct experimental conditions. Among the most relevant metrics in biomedicine are the number of nodes, which reflects the scale of the system (e.g., genes, proteins, patients or diseases); the edge density, which quantifies the proportion of observed connections relative to the maximum possible and indicates whether a network is highly interconnected or more modular; the global clustering coefficient, which measures the tendency of nodes to form tightly connected groups or communities through triangular relationships; and assortativity, which evaluates the correlation between the degrees of connected nodes, distinguishing networks where nodes preferentially connect to others of similar degree from those where high- and low-degree nodes are more likely to be linked. Together, these measures are particularly relevant for comparing molecular networks under physiological versus pathological conditions [23] or for detecting structural changes associated with pharmacological treatments [24].

Local properties Local properties describe the relevance and functional role of individual nodes within a network. Key measures include node degree, which reflects the number of direct connections and highlights hubs whose alteration may strongly impact system dynamics; centrality metrics, which assess the importance of a node according to its position, with variants such as betweenness (nodes controlling information flow), closeness (nodes efficiently spreading information), and eigenvector centrality (nodes connected to other influential nodes); connected components, defined as subsets of nodes internally connected but isolated from the rest of the network, useful for identifying independent functional modules; and communities, groups of nodes more densely connected to each other than to the rest of the network, often corresponding to protein complexes, metabolic pathways, gene modules, or patient subgroups. Together, these measures provide a robust framework for analyzing both the role of specific elements and their contribution to the overall architecture of biomedical systems, supporting hypothesis generation and knowledge discovery.

Matrix representation of a graph A graph $G = (V, E)$ can be represented in different ways, with matrix-based approaches being among the most widely used in computational analysis. These representations provide an algebraic framework that is well suited for linear algebra methods and machine learning applications. The most common is the adjacency matrix $A \in \mathbb{R}^{n \times n}$, where $n = |V|$ and A_{ij} equals 1 if an edge exists between nodes i and j (or the corresponding weight in

weighted graphs), with A being symmetric in the undirected case. From A, the diagonal degree matrix D is defined with entries $D_{ii} = \sum_{j=1}^{n} A_{ij}$, representing the degree of each node. Combining both, the Laplacian matrix $L = D - A$ captures global connectivity properties and constitutes a key operator in spectral graph theory, with the number of zero eigenvalues indicating the number of connected components [25]. The Laplacian also underlies the construction of graph kernels, diffusion algorithms, and spectral embedding methods. Altogether, these matrix formulations not only compactly describe network structure but also provide the basis for factorization algorithms, random walks, and deep learning architectures for the analysis of complex networks.

Gene–gene similarity graphs In the genomic context, it is possible to build weighted undirected graphs in which nodes represent genes and edges encode measures of similarity. Similarity can be defined in multiple ways, such as expression similarity (correlation of transcriptomic profiles [18]), functional similarity (based on Gene Ontology terms [26] or Reactome pathways [27]), genetic interaction similarity [28], clinical similarity (shared phenotypes from the Human Phenotype Ontology [29] or disease associations in MONDO [30]), or protein–protein interaction context from the STRING database [31]. These different notions of similarity can be exploited to search for disease-related functional mechanisms (by proximity to causal genes), to detect potential compensatory systems, or to identify communities corresponding to biologically relevant functions characterized at the phenotypic level.

1.1.3. Embedding in vector spaces

While graphs provide a natural representation for complex biomedical interactions, their direct manipulation has important limitations. Graph structural measures (degree, centrality, communities, etc.) are informative but often insufficient to capture higher-order relationships. Moreover, comparing nodes or substructures is computationally expensive, as it requires operating on the entire topology. In addition, many standard machine learning algorithms are defined in vector spaces and cannot be directly applied to graph-structured data [32].

To address these issues, node embedding techniques project each node into a lower-dimensional vector space while preserving, as much as possible, the original network structure and relationships. This representation facilitates the application of classification, prediction or clustering methods, and enables more efficient integration of heterogeneous biomedical data.

The distance problem Defining distance or proximity between two nodes in a graph is not trivial. Unlike Euclidean spaces, where distance metrics such as Euclidean or cosine are well established, graph distances depend on topology and can be defined in multiple ways. A common approach is to consider the length of the shortest path between two nodes. However, as shown in Figure 1.1this may fail to capture their true proximity. Alternative measures include the number of shared neighbors or the probability of reaching one node from another through random walks, which form the basis of many network propagation algorithms [33]. Each of these definitions emphasizes a different aspect of network structure, underscoring the complexity of working directly with raw graphs. In this context, *node kernels* emerged as an initial formal approach to quantify node similarity without the need to explicitly define a vector space [34].

Kernels Node kernels were developed as an initial solution to the distance problem, relying on matrix representations of the graph (mainly the adjacency and Laplacian matrices) to define similarity functions through spectral operations. Examples include the Exponential Laplacian Kernel, which models diffusion processes and depends critically on a scaling parameter [35]; the Commute Time Kernel, based on Markov chain theory and the pseudoinverse of the Laplacian, reflecting the expected number of steps for a random walker between nodes [36, 37]; and the Random Forest Kernel, defined in terms of random spanning trees and widely used in tools such as GeneMANIA for label propagation [38, 39, 40]. Collectively, these kernels exemplify the diversity of approaches for quantifying node similarity directly from network structure. They represented a milestone in formally defining distances on graph data and laid the groundwork for more advanced techniques, node embeddings, that project graphs into explicit vector spaces where standard similarity metrics naturally apply.

Node embedding The development of embedding techniques overcame kernel limitations by projecting nodes into a low-dimensional vector space that preserves network structure and relationships. Once in this space, standard distance or similarity measures (e.g., Euclidean or cosine) can be applied, enabling the use of well-optimized machine learning algorithms at lower computational cost than graph-based methods. In other words, embeddings act as a bridge, translating complex network topology into a vector space where analysis is more efficient and comparable.

Node embedding consists in mapping nodes into a vector space while preserving network structure as much as possible [32]. Early examples include the SAFE algorithm [41, 42], illustrated in Figure 1.2.

In this approach, the graph is represented using a force-directed layout and the

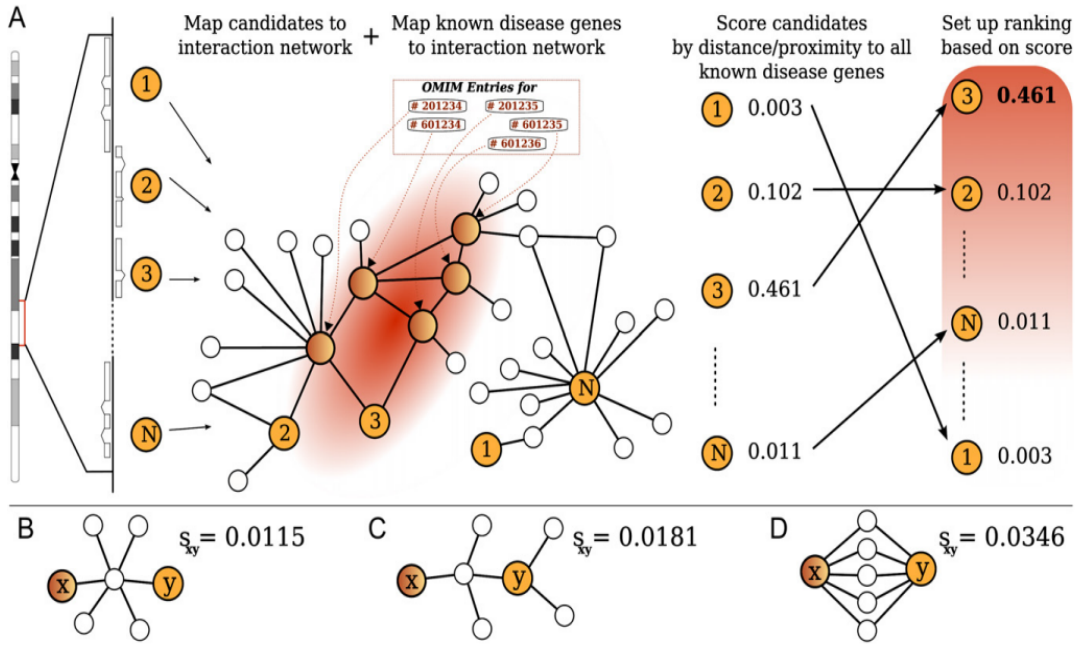

Figura 1.1: Prioritization of disease-associated genes. (A) All candidate genes located at the genome-wide level, together with previously known disease genes under study, are mapped onto the interaction network. The method assigns a score to each candidate gene by evaluating its relative position with respect to the known disease genes, using global distance measures in the network. Candidate genes are ranked according to their score to generate a prioritized list for experimental investigation. (B–D) Each of the three subgraphs shows a different configuration with the same number of nodes. The global distance between a hypothetical disease gene (x) and a candidate gene (y) differs in each case. In (B), proteins x and y are connected through a central hub node with many connections, resulting in a lower global similarity (s_{xy}) than in (C), where x and y are connected through a protein with fewer interactions than the hub. In contrast, nodes connected by multiple paths (D) receive a higher similarity than those connected by a single path. It is noteworthy that the shortest path length between x and y is identical in all three cases (B–D), so distance measures based solely on this local information cannot distinguish between these types of connections. In particular, an approach that considers only direct interactions with gene x would not identify gene y as a candidate in any of the three scenarios. Image extracted from [34].

resulting two-dimensional map is interpreted as a vector space. Similarity between nodes is then estimated using Euclidean distance in this representation, enabling functional enrichment analysis over node neighborhoods. However, the method is limited to two dimensions and may fail to capture key relationships requiring higher-dimensional representations.

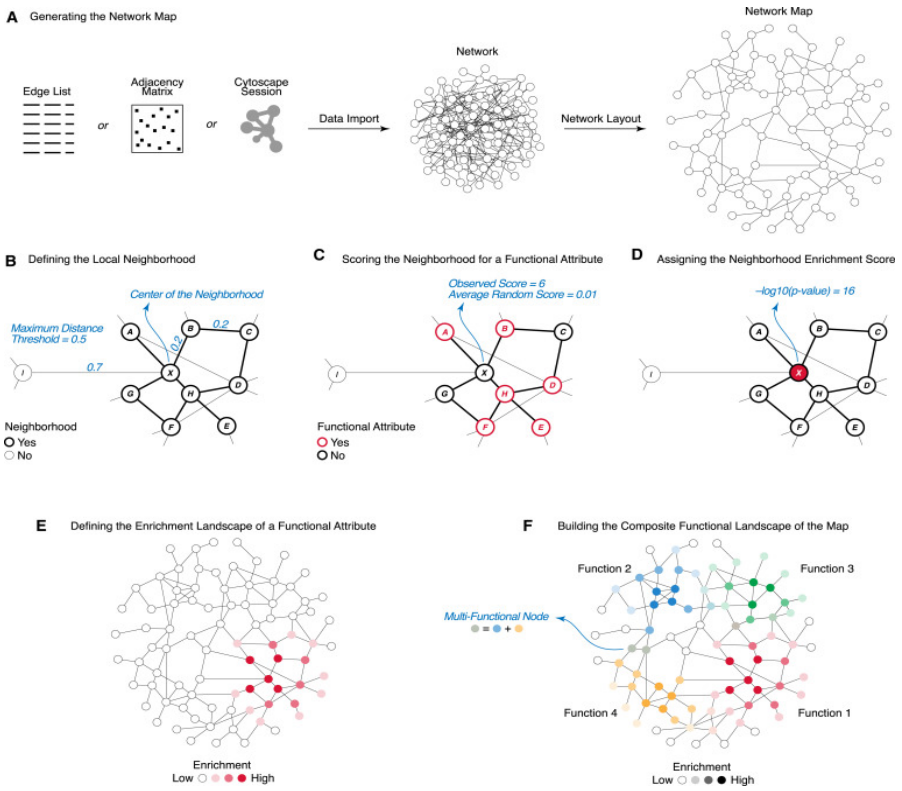

Figura 1.2: Schematic representation of SAFE operation. (A) Starting from a biological network, represented as an edge list or adjacency matrix, SAFE generates a two-dimensional map by applying a force-directed layout algorithm; alternatively, the map can be imported from Cytoscape [43]. (B) For each node X in the map, SAFE defines its local neighborhood by identifying all nodes (A–H) located within a distance smaller than a maximum threshold. (C) In each neighborhood, SAFE sums the values of the neighbors for a functional attribute of interest, compares the result to random expectation, and computes a p-value of significance. (D) The p-value, corrected for multiple testing across all attributes, is logarithmically transformed into a neighborhood enrichment score and assigned to node X. (E) The enrichment scores of all nodes for a given attribute define the enrichment landscape of that attribute; color intensity represents the variation in scores. (F) Each attribute is assigned a unique color, and the signals are combined proportionally to their relative enrichment in each neighborhood. Pure colors indicate regions with a predominant function, whereas hybrid colors reflect multifunctional regions. SAFE outputs text files containing enrichment scores, color assignments, and node–attribute associations. Image extracted from [41].

Types of node embeddings Current approaches to node embedding can be grouped into three main families: matrix factorization methods, random walk–based methods, and graph convolutional neural networks (GCNs). Matrix factorization

techniques decompose graph-associated matrices, such as the adjacency or similarity matrices, into lower-dimensional products that yield dense vector representations of nodes, with graph kernels representing a particular case of this strategy [44, 45]. Random walk–based methods, such as *DeepWalk* [46] and *node2vec* [47], simulate stochastic walks to generate node sequences and train distributed embeddings inspired by natural language processing models like *word2vec* [48], capturing local and structural context. More recently, GCNs [49] have emerged as powerful alternatives, jointly integrating graph topology and node attributes through iterative neighborhood aggregation, and proving highly effective in tasks such as node classification, link prediction, and multimodal biomedical integration [50, 51]. Together, these approaches enable the projection of nodes into vector spaces where standard machine learning algorithms can be applied efficiently. In this thesis, we initially worked with kernel-based methods and later incorporated random walk–based embeddings, extending the group's previous experience with graph kernels.

Applications to Biomedicine Graph node embeddings have proven useful across multiple biomedical applications, including candidate gene identification via proximity in the embedding space [52, 53], prediction of novel protein–protein interactions [54, 17], disease subtype characterization and clustering [55], and drug repurposing strategies [56, 21].

Accordingly, for studying the genetics of compensation, these techniques provide a robust framework to model gene–gene relationships and to explore mechanisms that might mitigate the impact of pathogenic mutations.

1.1.4. Application to the identification of compensatory genes

Identifying compensatory genes is essential for understanding the resilience of biological systems to pathogenic mutations. Studying these mechanisms not only elucidates the fundamentals of genetic compensation but also opens opportunities for innovative therapies, particularly in rare diseases and cancer.

Characterized natural mechanisms of compensation and systematic approaches for their identification Despite the limited characterization of many compensatory pathways with pathological relevance, several theoretical studies in the field of biological robustness support their existence [57, 58, 59] and, in fact, specific compensatory mechanisms have already been described.. A prominent example is overexpression-based compensation, characterized by El-Brolosy [60] and Ma [61], who showed how frameshift and nonsense mutations can induce

a compensatory response via overexpression of a gene with high sequence similarity [62]. This process, known as *transcriptional adaptation* (TA), is mediated by small RNAs and regulated through UPF1, and has been confirmed in zebrafish, mouse cells, *Caenorhabditis elegans*, and cultured human cells. Figure 1.3 provides a schematic representation of the process.

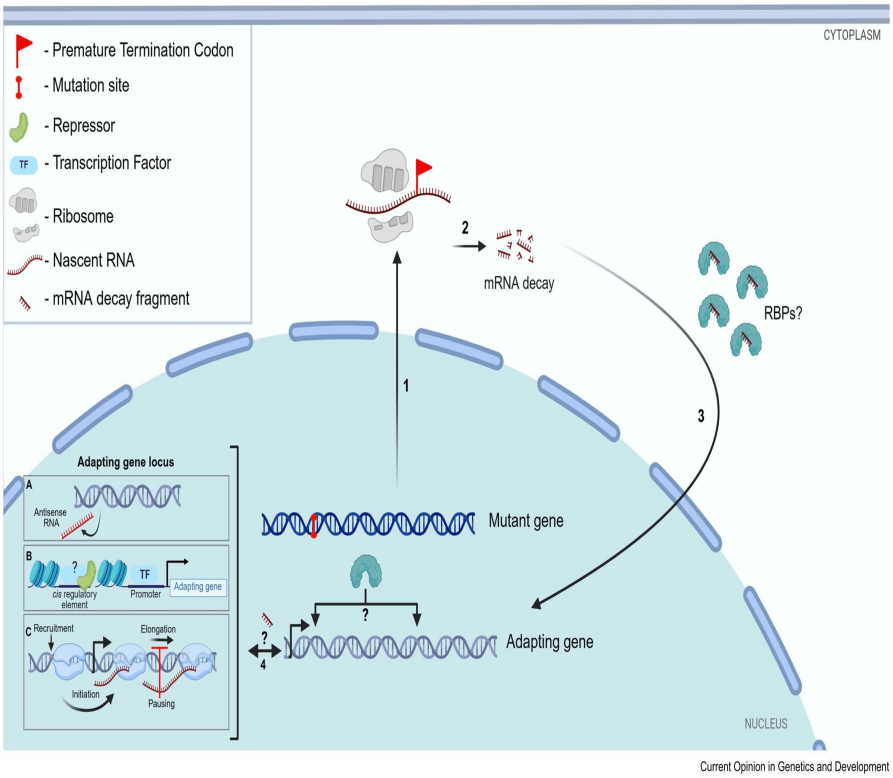

Figura 1.3: Proposed model of transcriptional adaptation (TA). (1) In the nucleus, mutations in a gene (mutation site) introduce a premature termination codon (PTC) in the mRNA. (2) The mRNA containing the PTC is degraded by the nonsense-mediated decay machinery. (3) The fragments of degraded mutant mRNA are presumably transported back to the nucleus, possibly guided by RNA-binding proteins (RBPs). (4) These fragments could induce the overexpression of adaptation genes through, for example: (a) binding to antisense transcripts derived from the adapter gene locus, (b) preventing repressors from binding to cis-regulatory elements, or (c) releasing transcriptional pausing. Other mechanisms of gene overexpression are also possible. Image extracted from [63]

Additional compensatory mechanisms have been proposed at different levels, including molecular buffering networks [64], gene duplication [65, 66], and redundancy-based compensation [67, 68]. Collectively, these findings suggest that similarity between genes, at the sequence, functional or interaction level, can serve as a useful indicator for predicting compensatory gene pairs. There are already therapeutic

examples in rare diseases that leverage these mechanisms, particularly in neuromuscular disorders [8, 11]. Nevertheless, much remains to be done in both basic characterization and clinical translation.

Causal gene prioritization systems and their application to genetic compensation Delving deeper into similarity networks, introduced earlier as a relevant case, we found that, over the past decade, numerous computational tools have been developed to identify causal disease genes from genomic, transcriptomic and clinical data. These approaches typically integrate gene–gene and protein–protein interaction networks [69], functional ontologies (GO, Reactome) [70], expression data [71], and measures of phenotypic and clinical proximity [72]. Several efforts also combine these sources [35, 73, 74, 75], employing integration schemes based on kernels [35], random walks [73], node embeddings [74], or GNNs [75].

These systems have proven highly effective for predicting disease-associated genes. Although their initial goal is to detect the causal gene of a pathology, the same tools can be adapted to propose functionally similar genes that may play compensatory roles. In this context, node embedding methods allow the identification of candidates based on topological proximity and functional similarity, providing a systematic approach to discovering potential disease modulators.

Compensation and cancer In oncology, genetic compensation plays a critical role in two main scenarios: treatment resistance [76] and the experimental identification of compensatory pairs [28]. Regarding drug resistance, studies have documented compensatory responses in cells resistant to MAPK pathway inhibitors [77], to nilotinib in chronic myeloid leukemia [78] and in lung adenocarcinoma [79], where resistance to *ARHGAP35* suppression is compensated through *SRC* and *STAT3*. For experimental identification, functional genomic screens (CRISPR screens) have revealed gene pairs whose combined loss leads to synthetic lethality [28], uncovering compensatory relationships that may be exploited therapeutically. In this regard, it is worth highlighting the case of PARP inhibitors in breast cancer, which are considered the first clinically approved drugs designed on the basis of the principle of synthetic lethality [80, 81].

Analyzing these functional dependencies not only advances our understanding of cancer biology but also points to new combinatorial therapeutic strategies, where the simultaneous inhibition of compensatory genes can enhance treatment efficacy and reduce the likelihood of resistance [82].Although many of these pathways have not yet been fully exploited in clinical practice due to the limited characterization of their molecular mechanisms, they represent an area of considerable interest as they increase the likelihood of identifying tractable therapeutic targets. Consequently, multiple drugs targeting pairs of compensatory genes are being developed

and are currently at different stages of clinical evaluation [80, 81]. These include crizotinib, used to inhibit the gene *ROS1* in the context of its synthetic lethality with *CDH1* mutations [83]; ceralasertib, which targets the gene *ATR* due to its synthetic lethality with *ARID1A* [84]; and onvansertib, directed against *PLK1* [85]. These cases underscore the growing clinical interest in exploiting the phenomenon of genetic compensation.

1.2. Hypotheses

The present work is structured according to the following hypotheses:

Hypothesis I Node embedding techniques allow the exploitation of the generated vector space to study the relationships among genes within a disease.

Hypothesis II Community detection is important to increase the accuracy of node embedding–based techniques. In particular, the application of overlapping communities detected in network embeddings further improves their performance.

Hypothesis III Node embedding techniques can be useful to build knowledge corpora customized to a specific group of diseases, such as the case of neuromuscular disorders.

Hypothesis IV The different known neuromuscular disorders may share common regulatory mechanisms.

Hypothesis V Additional genes modulate the Ras/MAPK pathway and may contribute to the pathophysiology of RASopathies, representing potential therapeutic targets.

Hypothesis VI The knowledge of gene–gene relationships generated in the study of genetic diseases and their causes can be applied to identify genes that may alleviate or compensate for the disease.

Hypothesis VII The compensatory pathways that can be identified in genetic diseases are also applicable to the identification of drug resistance mechanisms in cancer.

1.3. Objectives

In accordance with these hypotheses, the main objective of this doctoral dissertation is to characterize molecular mechanisms involved both in various genetic diseases and in cancer, and to apply this knowledge to identify distinct compensatory mechanisms. Specifically, this main objective is broken down as follows:

Objective I To design and develop a node embedding system that integrates overlapping community information and to implement a prioritization algorithm based on these developed systems.

Objective II To generate a specialized knowledge corpus on neuromuscular disorders by combining textual similarity networks with node embedding–based representations.

Objective III To integrate and analyze transcriptomic information from different neuromuscular disorders together with protein–protein interaction data in order to identify common regulatory mechanisms.

Objective IV To identify potential genes involved in modulating the Ras/MAPK pathway and to characterize their impact on the pathophysiology of RASopathies.

Objective V To generate a *diseasome*, understood as a network of diseases, updated for the validation of the causal gene prioritization system.

Objective VI To identify and characterize a set of disease–causal gene / compensatory gene pairs in the context of rare diseases and cancer.

1.4. Methodological procedure

The methodological framework of this thesis was structured around the design, development, and application of graph-based computational techniques to study genetic compensation mechanisms in rare diseases and cancer. The procedure was organized in relation with the research objectives, as follows:

Development of a community-constrained node embedding system (Objective I) To overcome the limitations of traditional overlapping community detection methods, a new node embedding approach was designed. This method,

based on constrained random walks, integrates overlapping community information to generate more faithful vector representations of nodes. The algorithm was implemented in Python within the *NetAnalyzer* library. Its performance was validated using human interactome data from STRING and biological pathways from Reactome. A Bayesian prioritization strategy was then developed to rank candidate genes in the embedding space, with a particular focus on RASopathies.

Knowledge integration for neuromuscular disorders (Objectives II–III)
Two complementary analyses were performed for neuromuscular diseases. First, a specialized corpus was built by combining ontological and textual similarity networks derived from PubMed articles, MONDO disease ontology, and the Human Phenotype Ontology (HPO). Node embedding methods (*node2vec*, *ggvec*, *prone*, *glove*) were compared and the best-performing representation was applied to cluster diseases using HDBSCAN and visualize the resulting groupings with UMAP. Functional enrichment analyses of associated genes and phenotypes were conducted to uncover shared molecular mechanisms.

Second, transcriptomic data from Duchenne muscular dystrophy (DMD), limb-girdle muscular dystrophy (LGMD), and amyotrophic lateral sclerosis (ALS) were integrated with protein–protein interaction data to identify common molecular signatures. For this purpose, the *NetAnalyzer* library was extended with modules for automated community detection, node embedding, and prioritization of genes using Fisher's combined probability test.

Identification of compensatory genes in RASopathies (Objective IV)
The newly developed embedding and prioritization framework was applied to study genes potentially involved in modulating the Ras/MAPK pathway. Networks constructed from STRING and Reactome data were embedded, and candidate genes were ranked according to their proximity to known RASopathy-related genes. This analysis revealed potential modulators of the pathway, providing novel insights into the molecular mechanisms underlying RASopathies.

Construction of a *diseasome* for causal gene prioritization (Objective V)
A comprehensive disease network (*diseasome*) was generated by integrating information from multiple biomedical databases, including STRING, HIPPIE, HPO, MONDO, GO, DepMap, Reactome and HGNC. This network served as a reference for validating the prioritization system. The integration was implemented in *NetAnalyzer* through two newly developed modules: (i) an *integration* module for combining similarity pairs, and (ii) a *topological analysis* module for computing global and local graph properties.

Systematic identification of compensatory gene pairs (Objective VI) To validate the proposed methodology in the context of rare diseases and cancer, a systematic search was conducted to identify disease–causal gene / compensatory gene pairs. Literature mining combined with manual curation was employed to build a gold standard of compensatory relationships. The prioritization system was then evaluated against this set, demonstrating that compensatory genes can be systematically identified using network-based embeddings. Additionally, case studies were performed on neuromuscular disorders, RASopathies and cancer, with results highlighting compensatory mechanisms relevant to disease pathophysiology and therapeutic resistance.

In summary, the methodological procedure followed a pipeline that combined (i) data collection and integration from heterogeneous biomedical sources, (ii) graph construction and embedding, (iii) algorithmic development for community-aware embeddings and prioritization, and (iv) application to rare disease and cancer case studies for validation and knowledge discovery.

1.5. Conclusions

From the present thesis, the following conclusions can be drawn:

Conclusion I The embedding space generated in various studies proved valuable for exploring the relationships between genes for multiple diseases.

Conclusion II By integrating overlapping clustering with node embedding, our approach enhanced the representation of pathways in the human interactome.

Conclusion III The use of language models and network embeddings to explore associations between genetic diseases allows clusterize of neuromuscular disorders and generate specific document corpora for each subgroup.

Conclusion IV The lncRNA gene *PAX8-AS1* was detected in both Duchenne muscular dystrophy and limb-girdle muscular dystrophy. Its previously reported association with Duchenne muscular dystrophy highlights it as a potential point of interest for further characterization in limb-girdle muscular dystrophy.

Conclusion V In the case of RASopathies, we identified new putative treatment gene targets such as *ALK* and *PIK3CD* with known associated drugs and *RAP1B*, *RASGRF1* genes as promising targets but without associated drug information.

Conclusion VI The causal gene prioritization system proved effective in identifying compensatory gene pairs. Several are associated with neuromuscular diseases, such as *DMD* in Duchenne muscular dystrophy and *SMN1* in spinal muscular atrophy, while others are linked to ciliopathies, including *CEP290*, or to hematological disorders such as α-thalassemia-2 caused by mutations in the *HBA2* gene.

Conclusion VII The prioritization system applied to compensatory genes was able to detect cases of compensation in cancer. In fact, some of these compensations occurred in the context of drug resistance.

Parte II

Introducción

Capítulo 2

Introducción

2.1. Enfermedades genéticas

Las enfermedades genéticas son ocasionadas por mutaciones en el genoma de un individuo. Se pueden clasificar dependiendo de si afectan a un gen exclusivo, a un conjunto de genes o a cromosomas completos. Otras también se producen por una combinación de factores de origen genético y ambientales.

A pesar del impacto de las mutaciones en el organismo, se debe puntualizar que no todas tienen consecuencias clínicas o un efecto patológico asociado. Existen mutaciones que no afectan a la función, expresión ni regulación de los genes. Sin embargo, aquellas que sí producen alteraciones en la estructura o función génica, afectando negativamente a la homeostasis y fisiología celular para derivar en el desarrollo de enfermedades, se consideran patogénicas.

Las enfermedades genéticas pueden manifestar sus síntomas desde el nacimiento, como los defectos congénitos entre los que se incluyen el síndrome de Down, o desarrollarse a lo largo de la vida del individuo como el resultado de las consecuencias del efecto de las mutaciones en el organismo, como sucede con el cáncer. Paradójicamente, se han registrado casos donde mutaciones potencialmente patogénicas no llegan a expresarse en individuos sanos [1, 2]. Todo ello apunta a la existencia de mecanismos biológicos capaces de contrarrestar los efectos de mutaciones potencialmente patogénicas.

Este hecho da lugar al denominado fenómeno de la compensación [3] o robustez [57, 86], entendida como la capacidad de los organismos para mantener su funcionamiento frente a perturbaciones internas o externas. Uno de los aspectos fundamentales en su estudio es identificar qué sistemas pueden ser responsables de paliar los efectos de una mutación, así como analizar los mecanismos que podrían estar involucrados en la aparición o prevención de distintas enfermedades. Como se puede inferir de lo descrito previamente, la presencia de mutaciones génicas se

suele relacionar con la manifestación de los efectos patológicos de las enfermedades genéticas. Sin embargo, hay casos en lo que esto no sucede. Esto se conoce gracias a los últimos avances en las tecnologías de secuenciación y genotipado a gran escala del exoma (EWS) y genoma (GWS), que han revelado un número sorprendentemente elevado de variantes en el genoma de individuos aparentemente sanos o que no presentan alteraciones clínicas reconocibles. Se estima que cada genoma contiene al menos un centenar de estas variantes, incluidas más de treinta variantes de pérdida de función (del inglés *loss-of-function*, LoF) presentes en estado homocigótico [1, 2]. Este hallazgo pone de relieve que la presencia de mutaciones, incluso aquellas que eliminan la función de un gen, no siempre son suficientes para el desarrollo de las manifestaciones clínicas de una enfermedad determinada, lo que enfatiza la importancia de analizar con detalle la interacción entre los mecanismos genéticos, moleculares y ambientales en la determinación del fenotipo. Este comportamiento, además, es consistente con modelos teóricos previos en el campo de la robustez biológica, en los que se ha demostrado la existencia de fenotipos capaces de permanecer inalterados frente a perturbaciones genómicas [58], y se ha sugerido que tales mecanismos pueden resultar fundamentales para la capacidad evolutiva (*evolvability*) de los organismos [59].

En la presente tesis doctoral se aborda el estudio de los mecanismos moleculares de la compensación de funciones biológicas, en el que la disfunción de un gen causada por una mutación puede ser compensada por la de otro. Para el estudio en humanos de los sistemas de compensación, uno de los casos de estudio más relevantes lo constituyen las enfermedades raras. La mayoría de estas patologías están originadas por una única mutación en el genoma, y se ha observado una amplia variabilidad en la sintomatología entre distintos individuos portadores de la misma alteración [4]. Además, aunque cada enfermedad rara presenta una prevalencia muy baja a nivel poblacional, en conjunto afectan aproximadamente entre el 3.5 % y el 5.9 % de la población mundial [5]. Por otro lado, este grupo de trastornos se caracteriza por una notable diversidad, tanto a nivel génico como en sus manifestaciones clínicas [6]. Esta heterogeneidad, junto con su impacto global y el hecho de que en la mayoría de los casos tienen una causa genética desencadenante, confiere a las enfermedades raras un especial interés científico y clínico. Asimismo, y a pesar de que existe un gran número de enfermedades raras y subtipos de las mismas, muchas de ellas están asociadas a mutaciones en genes que participan en los mismos procesos biológicos o muy similares. En este contexto, el análisis de determinadas agrupaciones de enfermedades raras adquiere una relevancia particular, ya que permite identificar y comprender los posibles mecanismos de compensación subyacentes.

A continuación, se presentan dos principales agrupaciones de enfermedades raras. Las enfermedades neuromusculares, como caso de interés por su heteroge-

neidad y aplicación sobre terapias génicas, y las RASopatías, como caso de enfermedades raras con potenciales asociaciones a cáncer.

2.1.1. Enfermedades neuromusculares

Las enfermedades neuromusculares constituyen un conjunto diverso de patologías causadas por mutaciones en genes implicados en los procesos de desarrollo y funcionamiento tanto del sistema nervioso como de los músculos [7]. Estas abarcan desde alteraciones en la transmisión de señales que comunican con las partes motoras, a nivel de los nervios, hasta defectos en la propia contracción muscular. A pesar de compartir similitudes fenotípicas, aún persiste una importante falta de caracterización a nivel molecular sobre las consecuencias de estas patologías. Es así que, de las casi seiscientas enfermedades neuromusculares descritas, aproximadamente treinta cuentan con una caracterización detallada a nivel genético, mientras que del resto aún se desconocen [7].

Resulta especialmente interesante destacar que este es uno de los principales grupos de enfermedades raras en los que se han aplicado terapias génicas de manera efectiva [7]. En particular, se han desarrollado estrategias terapéuticas basadas en la compensación genética, como es el caso del medicamento Nusinersen para la atrofia muscular espinal [8, 9] o el caso de Ezutromid para la distrofia muscular de Duchenne [10, 11], donde se aprovecha la sobreexpresión de genes funcionalmente similares para suplir la función perdida.

Este hecho es de gran interés para el objeto de estudio de la presente tesis doctoral, ya que la compensación aparece como un fenómeno potencialmente aplicable a las enfermedades neuromusculares. Para avanzar en este campo, resulta imprescindible mejorar la clasificación y la caracterización de estas patologías, con el fin de identificar los sistemas y mecanismos involucrados en los procesos compensatorios para así explorarlos como posibles dianas terapéuticas.

2.1.2. RASopatías

Las RASopatías constituyen un grupo de enfermedades raras causadas por mutaciones germinales en genes que codifican componentes de la vía de señalización RAS/MAPK [12, 13]. Esta ruta es esencial para la regulación de procesos celulares fundamentales, como la proliferación, la diferenciación y la apoptosis [14]. Las mutaciones que afectan a estos genes producen una activación anómala de la señalización, lo que se traduce en un espectro amplio de manifestaciones clínicas que incluyen alteraciones en el desarrollo [13], defectos cardíacos congénitos [15], anomalías faciales características [12] y una predisposición aumentada al cáncer [16].

Entre las principales RASopatías se encuentran el síndrome de Noonan, el síndrome cardio-facio-cutáneo, el síndrome de Costello y el síndrome de Leopard, todas ellas con fenotipos variables pero con un origen molecular común en la desregulación de la vía RAS/MAPK [12]. Esta diversidad fenotípica, junto con su baja prevalencia individual, convierte a las RASopatías en un modelo de especial interés para el estudio de los mecanismos compensatorios.

El vínculo con el cáncer resulta particularmente relevante. La misma vía RAS/MAPK, cuya disfunción en células somáticas está implicada en numerosos tumores sólidos y hematológicos, se ve alterada en las RASopatías por mutaciones germinales heredadas [12, 16]. Este paralelismo abre la posibilidad de que el conocimiento derivado del estudio de las RASopatías pueda aportar claves no solo para comprender los mecanismos de compensación presentes en enfermedades raras, sino también para identificar dianas terapéuticas aplicables en oncología.

En este sentido, las RASopatías representan un ejemplo paradigmático de cómo una mutación en una vía clave puede dar lugar tanto a una enfermedad rara con manifestaciones del desarrollo como a procesos tumorales cuando ocurre en células somáticas. Por ello, el análisis comparativo de estas condiciones podría resultar fundamental para desentrañar la relación entre compensación genética, enfermedad rara y predisposición oncológica.

2.2. Teoría de grafos aplicada al estudio de redes biológicas y clínicas

Los grafos, también denominados redes, han demostrado ser de gran utilidad en biomedicina debido a su capacidad para modelar de manera natural relaciones complejas entre entidades heterogéneas. Un grafo permite representar, en una misma estructura matemática, tanto interacciones moleculares como asociaciones clínicas o epidemiológicas. Esto lo convierte en una herramienta versátil para integrar información proveniente de distintos niveles de análisis biomédico.

Algunos ejemplos ilustrativos de su aplicación incluyen las redes de interacción proteína–proteína, que facilitan la identificación de complejos funcionales [17]; las redes de coexpresión génica, útiles para descubrir módulos reguladores [18, 19]; los grafos de conocimiento biomédico, que integran literatura científica, bases de datos y terminologías clínicas [20, 21]; y las redes de pacientes, donde cada nodo corresponde a un individuo y las aristas reflejan similitudes fenotípicas o compartición de factores de riesgo [22]. Estas representaciones permiten abordar tareas de gran importancia, como la predicción de genes asociados a enfermedades, la identificación de biomarcadores, el reposicionamiento de fármacos o la caracterización de subgrupos de pacientes.

2.2.1. Definiciones en tipos de grafos

Un grafo $G = (V, E)$ está definido por un conjunto de nodos V y un conjunto de aristas E que representan relaciones entre pares de nodos. Como se ha mencionado, en el contexto biomédico, los nodos pueden corresponder a genes, proteínas o entidades clínicas, mientras que las aristas reflejan relaciones de interacción, coexpresión o similitud funcional. Entre algunos de los distintos tipos de grafos con los que se trabajan en esta tesis, nos centramos especialmente en trabajar con los siguientes grafos.

En primer lugar, los *grafos dirigidos* se definen como $G = (V, E)$ con $E \subseteq V \times V$, de modo que cada arista es un par ordenado (u, v) que indica la existencia de una relación que parte de u y llega a v. Este formalismo permite modelar relaciones asimétricas como las redes de regulación génica, donde la acción de un gen regulador sobre otro no implica reciprocidad.

Por otro lado, los *grafos no dirigidos* se expresan como $G = (V, E)$ con $E \subseteq \{\{u, v\} \mid u, v \in V,\ u \neq v\}$, es decir, cada arista es un conjunto no ordenado de una pareja de nodos que refleja una relación simétrica entre ellos. Esta definición resulta adecuada para modelar redes de coexpresión de genes, en las que la correlación entre la actividad de dos genes es siempre mutua.

En cuanto a la ponderación de las aristas, los *grafos ponderados* incorporan una función $w : E \to R$ que asigna a cada arista un valor numérico que cuantifica la intensidad, fuerza o confianza de la relación. Un caso típico se da en redes de interacción proteína-proteína, donde $w(u, v)$ puede corresponder a la afinidad de unión entre proteínas u y v. En contraste, en los *grafos no ponderados* no existe tal función de pesos y se considera únicamente la existencia o ausencia de una relación, es decir, todas las aristas tienen peso unitario: $w(u, v) = 1$ si $(u, v) \in E$ y $w(u, v) = 0$ en caso contrario.

Finalmente, los *grafos bipartitos o multipartitos* se caracterizan por una partición del conjunto de nodos. En un grafo bipartito, V puede dividirse en dos subconjuntos disjuntos V_1 y V_2 tales que $V = V_1 \cup V_2$ y $E \subseteq V_1 \times V_2$. De forma más general, en un grafo k-partito, el conjunto de nodos se descompone como $V = V_1 \cup V_2 \cup \cdots \cup V_k$ con $V_i \cap V_j = \emptyset$ para todo $i \neq j$, y las aristas únicamente conectan nodos que pertenecen a distintos subconjuntos. Esta representación es especialmente útil en biomedicina, por ejemplo, para modelar asociaciones entre fármacos y enfermedades en un grafo bipartito, o para integrar simultáneamente genes, enfermedades y compuestos químicos en un grafo tripartito o multipartito, permitiendo estudiar interacciones complejas entre múltiples niveles de información.

Dentro de esta clasificación de los distintos tipos de grafos, es importante destacar que todos ellos comparten un conjunto de métricas diseñadas para caracterizar su topología, las cuales pueden dividirse en dos niveles de análisis: global y local.

A nivel global se consideran las medidas que describen propiedades estructurales de la red en su conjunto, mientras que a nivel local se estudian aquellas métricas que caracterizan propiedades particulares de cada nodo o subestructura dentro del grafo. Esta distinción resulta fundamental, ya que en biomedicina interesa tanto comprender el comportamiento emergente de un sistema biológico en su totalidad como identificar elementos clave que desempeñan funciones críticas en procesos moleculares o clínicos.

2.2.2. Propiedades globales

Las propiedades a nivel de red permiten analizar la estructura global de un grafo y facilitan la comparación entre redes de distinta naturaleza o bajo diferentes condiciones experimentales. Entre las métricas globales más relevantes que permiten caracterizar una red biomédica se encuentran las siguientes:

Número de nodos El tamaño de la red viene dado por el número de nodos, que corresponde al cardinal del conjunto V, es decir $|V| = n$. Este valor refleja la magnitud del sistema representado, pudiendo equivaler, por ejemplo, al número de genes, proteínas, pacientes o enfermedades incluidas en el análisis.

Densidad de aristas La densidad de aristas cuantifica la proporción de conexiones observadas respecto al número máximo de conexiones posibles. Para un grafo no dirigido sin bucles, se define como $D = \frac{2|E|}{|V|(|V|-1)}$, donde $|E|$ es el número de aristas. Valores altos indican sistemas altamente interconectados, mientras que valores bajos sugieren una estructura más dispersa o modular.

Coeficiente de agrupamiento global El coeficiente de agrupamiento global mide la tendencia de los nodos a formar triángulos, reflejando la cohesión de la red. Se define como el promedio de los coeficientes de agrupamiento locales: $C = \frac{1}{|V|}\sum_{v\in V} C(v)$, donde $C(v) = \frac{2T(v)}{k_v(k_v-1)}$, k_v representa el grado del nodo v y $T(v)$ el número de triángulos en los que participa dicho nodo. Un valor alto de C sugiere la presencia de módulos o comunidades densamente interconectadas.

Asortatividad La asortatividad mide la correlación entre los grados de los nodos conectados. Su expresión general es $r = \frac{\sum_{uv\in E}(k_u-\mu)(k_v-\mu)}{\sum_{uv\in E}(k_u-\mu)^2}$, donde k_u y k_v son los grados de los nodos u y v, respectivamente, y μ es el grado medio de la red. El coeficiente r toma valores entre -1 y 1. Redes con $r > 0$ presentan una tendencia de los nodos a conectarse con otros de grado similar, mientras que en redes con $r < 0$ los nodos de alto grado se conectan preferentemente con nodos de bajo grado.

Estas métricas son especialmente relevantes en biomedicina, ya que permiten, por ejemplo, comparar redes moleculares en condiciones fisiológicas con aquellas alteradas en distintos estados patológicos [23], así como detectar modificaciones estructurales asociadas a la acción de tratamientos farmacológicos [24].

2.2.3. Propiedades locales

En contraposición, las propiedades locales describen la relevancia y el papel funcional de nodos individuales dentro de la red. Algunas de las más utilizadas son:

Grado de un nodo El grado de un nodo v, denotado como $deg(v)$, corresponde al número de conexiones directas que mantiene con otros nodos. Este valor constituye un indicador primario de la relevancia estructural de un nodo: en redes biológicas, un nodo con alto grado puede representar una proteína altamente interactuante (*hub*) cuya alteración afecte significativamente la dinámica del sistema.

Medidas de centralidad La centralidad busca cuantificar la importancia relativa de un nodo en función de su posición en la red. Existen múltiples variantes: la *centralidad de intermediación* (*betweenness*) mide cuántas veces un nodo se encuentra en los caminos más cortos entre otros nodos, identificando puntos críticos para el flujo de información; la *centralidad de cercanía* (*closeness*) evalúa la distancia media de un nodo hacia todos los demás, reflejando su capacidad para difundir información; y la *centralidad de vector propio* (*eigenvector centrality*) asigna un valor de importancia a cada nodo de manera recursiva, de forma que un nodo recibe una puntuación alta si está conectado con otros nodos que también presentan una alta centralidad. Esta métrica permite identificar nodos relevantes dentro de estructuras jerárquicas o altamente interconectadas.

Componentes conexas Un componente conexo se define como un subconjunto dentro de la red en que existe al menos un camino entre cualquier par de sus nodos, sin que ninguno de ellos se encuentre conectado con nodos externos a dicho componente. El análisis de componentes conexas permite identificar grupos aislados dentro de una red, lo cual es útil, por ejemplo, para detectar subconjuntos de genes o proteínas que interactúan únicamente entre sí y que podrían corresponder a módulos funcionales independientes.

Comunidades Las comunidades corresponden a agrupaciones de nodos que presentan una densidad de conexiones interna significativamente mayor que con el resto de la red. La detección de comunidades resulta particularmente valiosa en

biomedicina, ya que permite identificar complejos proteicos, rutas metabólicas, módulos génicos o grupos de pacientes con perfiles fenotípicos similares.

En conjunto, estas métricas proporcionan un marco analítico robusto que permite comprender tanto la arquitectura global de los sistemas biomédicos como el papel estratégico de nodos individuales, facilitando la generación de hipótesis biológicas y el descubrimiento de nuevos conocimientos.

2.2.4. Representación matricial de un grafo

Un grafo $G = (V, E)$ puede representarse de diversas formas, siendo una de las más utilizadas en el análisis computacional las representaciones matriciales. Estas ofrecen una descripción algebraica de la red que resulta especialmente conveniente para la aplicación de métodos de álgebra lineal y de aprendizaje automático.

Matriz de adyacencia Definida como $A \in \mathbb{R}^{n \times n}$, donde $n = |V|$ corresponde al número de nodos. Cada entrada A_{ij} se define como $A_{ij} = 1$ si existe una arista entre los nodos i y j, y $A_{ij} = 0$ en caso contrario. En el caso de grafos ponderados, el valor A_{ij} toma el peso de la arista entre i y j. La matriz A es simétrica cuando el grafo es no dirigido.

Matriz diagonal de grados En este caso, la matriz $D \in \mathbb{R}^{n \times n}$, en la cual cada elemento diagonal D_{ii} corresponde al grado del nodo i, es decir, al número de aristas incidentes en dicho nodo: $D_{ii} = \sum_{j=1}^{n} A_{ij}$.

Matriz Laplaciana A partir de A y D se define la matriz laplaciana como $L = D - A$. El laplaciano captura propiedades de conectividad global y constituye un operador fundamental en teoría espectral de grafos. Entre sus propiedades destaca que el número de valores propios nulos de L coincide con el número de componentes conexas del grafo [25]. Asimismo, L se emplea en la definición de múltiples *graph kernels*, en algoritmos de difusión y en técnicas de embebido espectral de nodos, que se presentará más adelante.

Estas representaciones matriciales no solo proporcionan un marco algebraico compacto para describir la estructura de un grafo, sino que también sirven como base para la aplicación de algoritmos de factorización, *random walks* y arquitecturas de *deep learning* en redes complejas.

2.2.5. Grafos de similitud gen a gen

En el contexto genómico, es posible construir grafos ponderados no dirigidos en los que los nodos son genes y las aristas representan una medida de su similitud. Este puede definirse en función a distintos parámetros, como son la similitud de expresión (correlación entre perfiles transcriptómicos [18]), similitud funcional (por términos en *la Gene Ontology* [26] o rutas metabólicas en *Reactome* [27]), similitud en base a interacción génica [28], similitud clínica (relación con fenotipos comunes en base a la *Human Phenotype Ontology* [29] o enfermedades asociadas por MONDO [30]) o incluso similitud en contexto de interacción proteína-proteína almacenados en la base de datos STRING [31]. Con todas estas distintas maneras de definir la similitud, podemos emplearlo para la búsqueda de mecanismos funcionales asociados a la enfermedad (por proximidad o similitud a genes causales), detectar potenciales sistemas de compensación o encontrar comunidades con funciones biológicas claves previamente caracterizadas a nivel fenotípico.

2.3. Embebido en espacios vectoriales

Si bien los grafos constituyen una representación natural para describir interacciones biomédicas complejas, su manipulación directa presenta limitaciones importantes. En primer lugar, las medidas estructurales en grafos (grado, centralidad, comunidades, etc.) resultan útiles pero no siempre suficientes para capturar la totalidad de las relaciones de alto orden presentes en la red. Además, la comparación entre nodos o subestructuras es costosa desde el punto de vista computacional, ya que requiere operar sobre la topología completa. Por otra parte, muchos algoritmos de aprendizaje automático estándar están definidos en espacios vectoriales y no pueden aplicarse de manera directa sobre datos en forma de grafo [32].

Para superar estas restricciones, se recurre al embebido de nodos, es decir, a la proyección de cada nodo en un espacio vectorial de menor dimensión que preserve, en la medida de lo posible, la estructura y las relaciones originales de la red. Esta representación facilita la aplicación de técnicas de clasificación, predicción o agrupamiento, y permite integrar de forma más eficiente la información proveniente de diferentes niveles de datos biomédicos.

2.3.1. El problema de la distancia

Definir la distancia o proximidad entre dos nodos en un grafo no es una tarea trivial. A diferencia de los datos definidos en espacios euclídeos, donde la noción de distancia y proximidad suele estar bien establecida (por ejemplo, la distancia euclídea o la similitud por coseno), en un grafo la métrica de distancia depende de la topología y puede adoptar múltiples formas. Una posibilidad es considerar

la longitud del camino más corto entre dos nodos. Sin embargo, como muestra la Figura 2.1, esto puede conllevar serias distorsiones a la hora de mostrar la proximidad real entre dos nodos [34]. Otro modo de definir medidas de similitud es por el número de vecinos comunes o en la probabilidad de alcanzar un nodo desde otro mediante recorridos aleatorios, proceso muy utilizado para algoritmos de propagación en red [33]. Cada una de estas definiciones captura un aspecto distinto de la estructura de la red, lo que pone de manifiesto la dificultad de trabajar directamente con el grafo crudo. En este contexto surgieron los *kernels de nodos* como primera aproximación formal para cuantificar similitudes entre nodos sin necesidad de definir un espacio vectorial explícito.

Kernels Los kernels de nodos fueron desarrollados como una primera aproximación para abordar esta problemática. La estrategia se fundamenta en el uso de representaciones matriciales del grafo, principalmente la matriz de adyacencia y la matriz laplaciana, a partir de las cuales es posible definir funciones que capturan la similitud entre nodos. Dichas funciones se construyen aplicando operaciones espectrales sobre estas matrices, aprovechando sus propiedades algebraicas para codificar relaciones estructurales dentro de la red. A continuación, se presentan algunos ejemplos de este tipo de *kernels*.

Un primer ejemplo es el **Exponential Laplacian Kernel** o kernel de difusión exponencial, definido como $K_{\mathrm{EL}} = \exp(-\beta L)$ [35], donde el parámetro β controla la magnitud de la difusión en la red. Este kernel modela procesos de propagación de información o calor sobre el grafo, y su rendimiento depende críticamente de la calibración del parámetro β: valores excesivamente altos pueden inducir un comportamiento lineal, mientras que valores demasiado bajos pueden provocar una pérdida de la no linealidad característica del proceso de difusión.

Otro caso relevante es el **Commute Time Kernel** (CT), que se basa en la teoría de cadenas de Markov y modela la probabilidad de transición entre estados, es decir, de moverse de un nodo a otro en la red. Formalmente, el kernel se define como $K_{\mathrm{CT}} = L^{+}$, donde L^{+} es la pseudoinversa de Moore–Penrose del laplaciano [36, 37]. Este kernel surge del cálculo del número promedio de pasos que un caminante aleatorio en un grafo necesita para ir de un nodo a otro y volver.

Finalmente, el **Random Forest Kernel** (RF) [38] constituye otra alternativa basada en el laplaciano, y se define como $K_{\mathrm{RF}} = (I + L)^{-1}$. Este kernel tiene una interpretación en términos del número de árboles generados al azar desde el grafo donde los dos nodos se encuentran [39]. Este kernel es utilizado por el algoritmo GeneMANIA en el contexto de la propagación de etiquetas mediante campos aleatorios gaussianos [40].

En conjunto, estos kernels ejemplifican la diversidad de aproximaciones existentes para cuantificar similitudes entre nodos a partir de la estructura de un grafo.

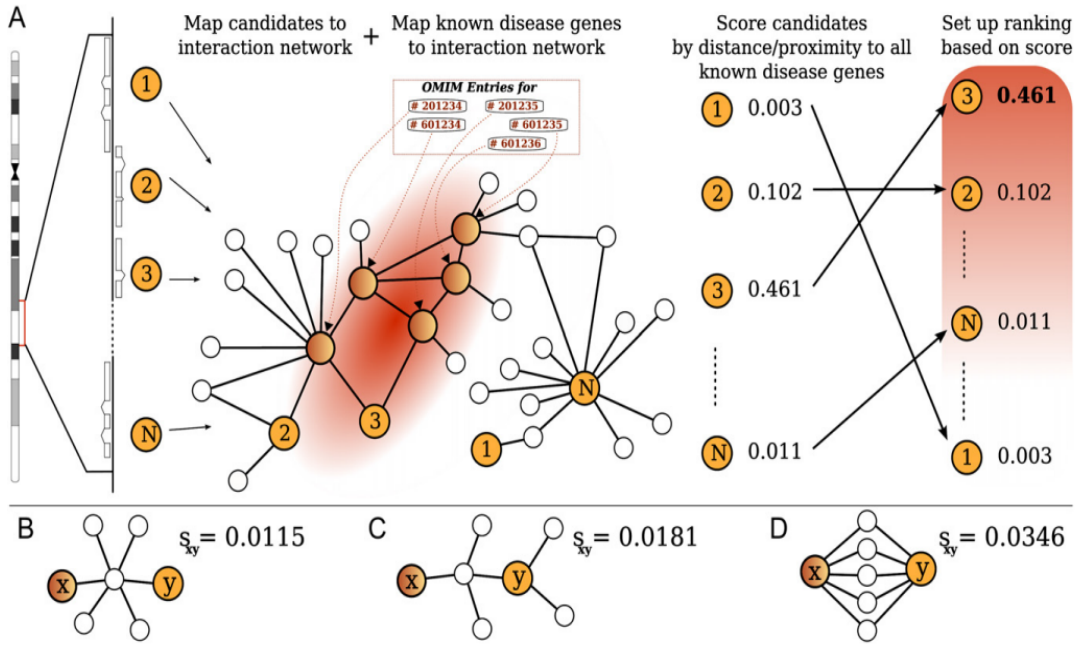

Figura 2.1: Priorización de genes asociados a enfermedad. (A) Todos los genes candidatos contenidos próximos a nivel de genoma, junto con los genes previamente conocidos de la enfermedad en estudio, se mapean a la red de interacciones. El método asigna una puntuación a cada gen candidato evaluando su localización relativa respecto a los genes de la enfermedad conocidos, utilizando medidas de distancia global en la red. Los genes candidatos se ordenan según su puntuación para generar una lista priorizada de candidatos a investigar experimentalmente. (B–D) Cada uno de los tres subgrafos muestra una configuración distinta con el mismo número de nodos. La distancia global entre un gen hipotético de enfermedad (x) y un gen candidato (y) difiere en cada caso. En (B), las proteínas x e y están conectadas a través de un nodo central (hub) con muchas conexiones, lo que resulta en una similitud global (s_{xy}) menor que en (C), donde x e y se conectan mediante una proteína con menos interacciones que las del hub. En cambio, nodos conectados por múltiples caminos (D) reciben una mayor similitud que aquellos conectados por un único camino. Cabe destacar que la longitud del camino más corto entre x e y es idéntica en los tres casos (B–D), por lo que las medidas de distancia basadas solo en esta información local no pueden distinguir entre estos tipos de conexión. En particular, un enfoque que solo considere interacciones directas con el gen x no identificaría al gen y como candidato en ninguno de los tres escenarios. Imagen extraída de [34].

Su desarrollo marcó un hito al posibilitar los primeros intentos de definir la distancia sobre datos en forma de red, y sentó las bases para la evolución posterior hacia métodos más sofisticados, el embebido de nodos. En esta nueva estrategia,

pasaremos a definir directamente un espacio vectorial. Es decir, en lugar de definir una proximidad dentro del grafo, pasaremos a mover el grafo a un espacio donde dicha métrica ya exista.

2.3.2. Embebido de nodos

El desarrollo posterior de técnicas de embebido permitió superar las limitaciones de los kernels, al proyectar los nodos de un grafo en un espacio vectorial de baja dimensión que preserva, en la medida de lo posible, la estructura y las relaciones originales de la red. La motivación central de este enfoque es que, una vez en dicho espacio, es posible aplicar métricas estándar de distancia o similitud (como la euclídea o la coseno) y, sobre ellas, emplear de forma directa algoritmos clásicos de aprendizaje automático, ampliamente optimizados y con menor coste computacional que los métodos definidos directamente sobre grafos. En otras palabras, el embebido actúa como un puente que traslada la complejidad topológica de la red a un espacio vectorial donde el análisis resulta más eficiente y comparable.

El embebido de los nodos de un grafo consiste en proyectar estos en un espacio vectorial de manera que se preserve, en la medida de lo posible, la estructura y las relaciones originales de la red [32]. Existen diversas aproximaciones para llevar a cabo esta estrategia. Como ejemplo ilustrativo, puede mencionarse una de las primeras propuestas en este ámbito: el algoritmo SAFE [41, 42].

En este sistema, como se muestra en la Figura 2.2, el grafo es representado mediante un algoritmo de visualización denominado *spring-embedded*, y dicha representación gráfica es posteriormente tratada considerando el espacio vectorial bidimensional generado durante la proyección. De este modo, los autores pudieron estimar la similitud o la distancia entre nodos utilizando la distancia euclídea en esa representación. Esto les valió para poder realizar enriquecimientos funcionales sobre la vecindad de cada nodo. Sin embargo, se trata de una aproximación inicial que, con el tiempo, ha sido ampliamente superada. Una de las limitaciones más evidentes es que el sistema está restringido a dos dimensiones, cuando en ocasiones estas representaciones pueden omitir relaciones clave al considerar espacios de mayor dimensionalidad.

2.3.3. Tipos de embebido de nodos

En la actualidad, los principales enfoques para la obtención de embebido de nodos se agrupan en tres grandes familias: métodos basados en factorización matricial, en recorridos aleatorios (*random walks*) y en redes neuronales convolucionales de grafos (*Graph Convolutional Networks*, GCN). A continuación, se describen cada una de ellas.

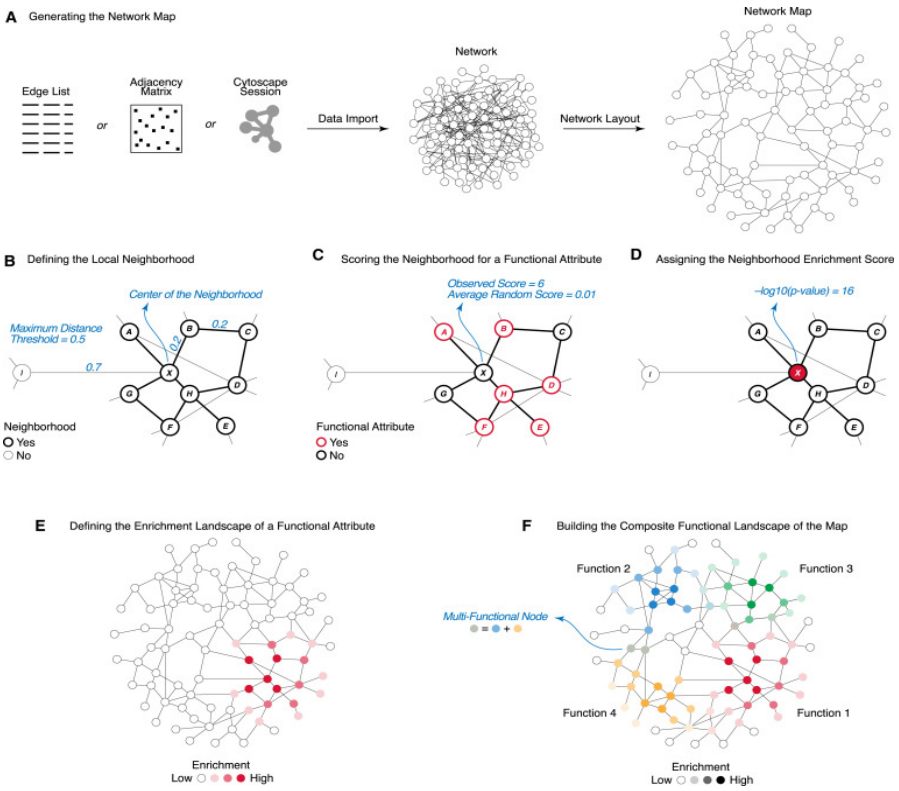

Figura 2.2: Esquema del funcionamiento de SAFE. (A) A partir de una red biológica, representada como lista de aristas o matriz de adyacencia, SAFE genera un mapa bidimensional aplicando un algoritmo de diseño de fuerza dirigida; alternativamente, puede importar el mapa desde Cytoscape [43]. (B) Para cada nodo X del mapa, SAFE define su vecindario local identificando todos los nodos (A–H) situados a una distancia menor que un umbral máximo. (C) En cada vecindario, SAFE suma los valores de los vecinos para un atributo funcional de interés, compara el resultado con la expectativa aleatoria y calcula un valor p de significancia. (D) El valor p, corregido para las pruebas múltiples entre todos los atributos, se transforma logarítmicamente en una puntuación de enriquecimiento de vecindario y se asigna al nodo X. (E) Las puntuaciones de enriquecimiento de todos los nodos para un atributo definen el paisaje de enriquecimiento de dicho atributo; la intensidad del color representa la variación en las puntuaciones. (F) A cada atributo se le asigna un color único y las señales se combinan proporcionalmente a su enriquecimiento relativo en cada vecindario. Colores puros indican regiones con una función predominante, mientras que colores híbridos reflejan regiones multifuncionales. SAFE genera como salida archivos de texto con las puntuaciones de enriquecimiento, asignaciones de color y asociaciones nodo–atributo. Imagen extraída de [41].

Factorización matricial Este tipo de métodos se fundamenta en la descomposición de matrices asociadas al grafo, tales como la matriz de adyacencia A o

matrices de similitud derivadas de ella. El objetivo es aproximar A mediante el producto de dos matrices de menor dimensión, de manera que se obtengan representaciones vectoriales densas para los nodos [44, 45]. En términos generales, se busca la factorización $A \approx UV^\top$, donde $U \in \mathbb{R}^{n\times d}$ y $V \in \mathbb{R}^{n\times d}$, con $d \ll n$ siendo la dimensión del espacio embebido. Cada fila de U o V representa el vector embebido de un nodo, y la similitud entre dos nodos i y j puede calcularse como el producto interno $\langle U_i, V_j\rangle$. Los *graph kernels*, descritos anteriormente, pueden considerarse un caso particular de este enfoque, ya que definen matrices de similitud que pueden ser posteriormente factorizadas para obtener representaciones latentes.

Métodos basados en random walks Los enfoques de este grupo utilizan trayectorias aleatorias sobre el grafo para capturar el contexto local y estructural de cada nodo. La idea consiste en simular recorridos estocásticos a partir de cada nodo y, a partir de las secuencias obtenidas, entrenar un modelo de representación distribuida inspirado en técnicas de procesamiento de lenguaje natural, como *word2vec* [48]. En este marco, cada nodo $v \in V$ se representa mediante un vector $z_v \in \mathbb{R}^d$, aprendido de manera que la probabilidad condicional de observar un nodo u en el contexto de v se maximice. Formalmente, el objetivo es optimizar: $\max_z \sum_{v\in V} \sum_{u\in\mathcal{N}(v)} \log \Pr(u|z_v)$, donde $\mathcal{N}(v)$ corresponde al conjunto de nodos que aparecen en el contexto de v durante los recorridos aleatorios. Métodos como *DeepWalk* [46] y *node2vec* [47] se basan en esta estrategia, diferenciándose en la forma en que generan los recorridos para balancear exploración y explotación de la red.

Redes neuronales convolucionales de grafos (GCN) Las GCN representan la aproximación más reciente y potente, al integrar de manera conjunta la información topológica del grafo y los atributos asociados a los nodos [49]. En este paradigma, el vector de características de cada nodo se actualiza a través de un proceso de agregación de la información de sus vecinos, siguiendo una regla de propagación del tipo: $H^{(l+1)} = \sigma\Big(\tilde{D}^{-1/2}\tilde{A}\,\tilde{D}^{-1/2}H^{(l)}W^{(l)}\Big)$, donde $\tilde{A} = A + I$ es la matriz de adyacencia con autoconexiones, $\tilde{D}$ es su matriz diagonal de grados, $H^{(l)}$ representa las características de los nodos en la capa l, $W^{(l)}$ son los parámetros de la capa y σ una función de activación no lineal. Tras varias capas, se obtiene una representación latente $H^{(L)}$ en la que cada nodo codifica tanto sus atributos originales como la información estructural de su vecindario. Este tipo de métodos ha demostrado gran eficacia en tareas de clasificación de nodos, predicción de enlaces e integración multimodal en biomedicina [50, 51].

Estas técnicas permiten aplicar algoritmos estándar de *machine learning* sobre representaciones vectoriales y comparables de entidades biomédicas. Asimismo, al

proyectar los nodos de un grafo en un espacio vectorial, se obtiene una organización coherente en términos de proximidad dentro de la red. En el marco de esta tesis, se inició el trabajo con métodos de kernels y, posteriormente, se incorporaron enfoques basados en *random walk*. Todo ello, siguiendo y expandiendo los trabajos previamente desarrollados por el grupo en materia de *kernels*.

2.3.4. Aplicación a Biomedicina

El embebido de nodos de un grafo ha demostrado ser útil en múltiples aplicaciones biomédicas. Entre ellas, la identificación de genes candidatos mediante la proximidad en el espacio de embebido [52, 53], la predicción de nuevas interacciones proteína-proteína [54, 17], la caracterización de subtipos de enfermedades y potenciales agrupaciones [55], así como propuestas para estrategias de reposicionamiento de fármacos [56, 21].

De tal modo, para el estudio de la genética de la compensación, estas técnicas ofrecen un marco robusto para modelar relaciones entre genes y explorar mecanismos que podrían mitigar el impacto de mutaciones patogénicas.

2.4. Aplicación a identificación de genes compensatorios

La identificación de genes compensatorios constituye un paso esencial para comprender la resiliencia de los sistemas biológicos frente a mutaciones patogénicas. El estudio de estos mecanismos no solo permite desentrañar la biología fundamental de la compensación genética, sino que además abre vías terapéuticas innovadoras, en particular en el contexto de enfermedades raras y cáncer.

2.4.1. Mecanismos naturales caracterizados para compensación y aproximaciones sistemáticas para su identificación

A pesar de la limitada caracterización de muchas vías compensatorias con relevancia en procesos patológicos, diversos estudios teóricos en el campo de la robustez biológica respaldan su existencia [57, 58, 59] y, de hecho, ya se han descrito mecanismos concretos de compensación. Un ejemplo destacado es la compensación por sobreexpresión, caracterizada por El-Brolosy [60] y Ma [61], quienes demostraron por qué mutaciones de cambio de marco (*frameshift*) o mutaciones sin sentido (*nonsense*) son capaces de inducir una respuesta de compensación génica de la sobreexpresión de un gen con alta similitud a nivel de secuencia [62]. Este proceso,

conocido como *transcriptional adaptation* (TA), está mediado por pequeños RNA y regulado a través de la proteína UPF1 y ha sido confirmado en pez cebra, células de ratón, *Caenorhabditis elegans*, y células humanas en cultivo. Un esquema del proceso puede observarse en la Figura 2.3.

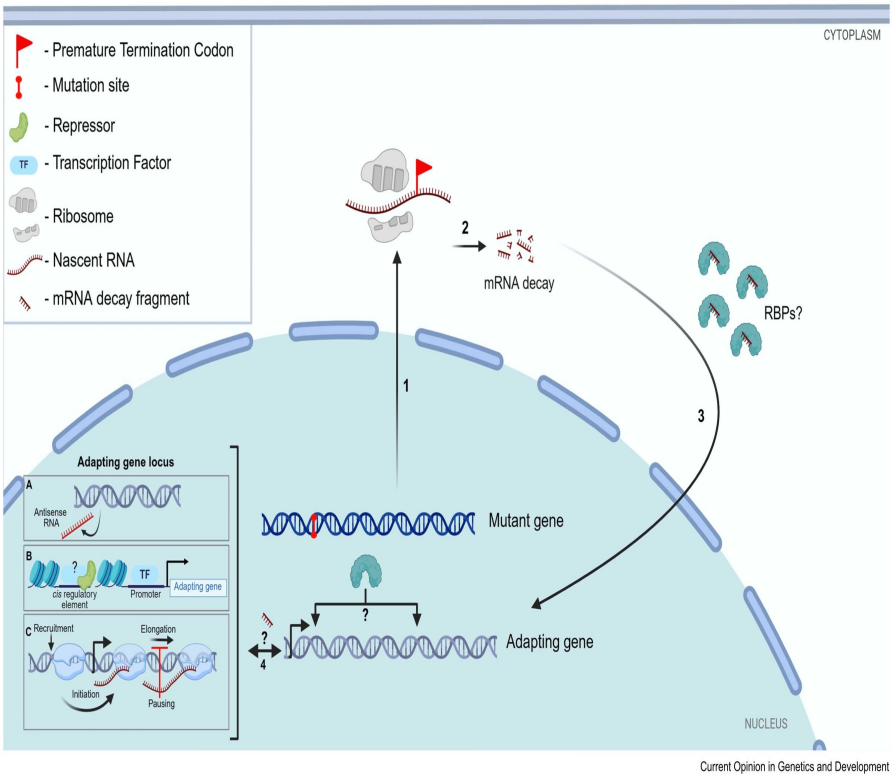

Figura 2.3: Modelo propuesto de adaptación transcripcional (TA). (1) En el núcleo, las mutaciones en un gen (sitio de mutación) introducen un codón de terminación prematuro (PTC) en el ARNm. (2) El ARNm que contiene el PTC es degradado por la maquinaria de *nonsense-mediated decay*. (3) Los fragmentos de ARNm mutante degradado son presumiblemente transportados de nuevo al núcleo, posiblemente guiados por proteínas de unión a ARN (RBPs). (4) Dichos fragmentos podrían inducir la sobreexpresión de los genes de adaptación mediante, por ejemplo: (a) unión a transcritos antisentido derivados del locus del gen adaptador, (b) prevención de la unión de represores a elementos cis-regulatorios, o (c) liberación de pausas transcripcionales. Otros mecanismos de sobreexpresión génica también son posibles. Imagen extraída de [63]

Asimismo, se han propuesto otros mecanismos compensatorios en diferentes niveles, como las redes de *buffering* molecular [64], o la duplicación génica [65, 66] y la compensación por redundancia funcional [67, 68]. En conjunto, estos hallazgos sugieren que la similitud entre genes, ya sea a nivel de secuencia, función o interacción molecular, puede constituir un indicador útil para la predicción de pares

de compensación génica. Actualmente, ya existen casos de fármacos aplicados a enfermedades raras que aprovechan estos mecanismos, especialmente en el caso de enfermedades neuromusculares [8, 11]. Sin embargo, mucho quedar por hacer tanto a nivel de su caracterización como de aplicación clínica.

2.4.2. Sistemas de priorización de genes causales y aplicación a casos de compensación génica.

Observando la aplicación de redes de similitud, que constituyen un caso de interés como se ha señalado previamente, en la última década se han desarrollado múltiples herramientas computacionales orientadas a la identificación de genes causales de enfermedad a partir de datos genómicos, transcriptómicos y clínicos. Estas herramientas suelen emplear información procedente de redes de interacción gen-gen y proteína-proteína [69], ontologías funcionales (GO, Reactome) [70], datos de expresión [71], así como medidas de proximidad fenotípica y clínica [72]. Por supuesto, también existen distintos intentos de integrar este tipo de información [35, 73, 74, 75]. Todos ellos utilizando distintos sistemas de integración basados en Kernels [35], random walks [73], embebido de nodos [74] o GNNs [75].

La implementación de este tipo de sistemas ha demostrado ser de gran utilidad para la predicción de genes asociados a enfermedades. Si bien su objetivo inicial es la detección del gen causal de una patología, estas mismas herramientas pueden adaptarse para proponer genes con funciones similares que potencialmente ejerzan un papel compensatorio. En este contexto, los métodos de embebido de nodos permiten identificar candidatos basados en la proximidad topológica y la similitud funcional, aportando así un enfoque sistemático para la búsqueda de posibles genes moduladores de la enfermedad.

2.4.3. Compensación y cáncer.

En oncología, la compensación genética adquiere un rol crítico en dos escenarios principales, la resistencia a tratamientos [76] y la identificación experimental de parejas compensatorias [28]. En relación con la resistencia a fármacos, se ha documentado en múltiples contextos,como en células resistentes a inhibidores de la vía MAPK [77], en el uso de nilotinib en leucemia mieloide crónica [78], y en adenocarcinoma de pulmón [79], donde la resistencia a la supresión de *ARHGAP35* es compensada mediante la intervención de *SRC* y *STAT3*.

Por otra parte,en la idenfiticación experimental, estudios de cribado funcional (CRISPR screens) han permitido detectar genes cuya pérdida conjunta conduce a letalidad sintética [28], revelando relaciones de compensación que podrían explotarse terapéuticamente. En este sentido, cabe destacar el caso de los inhibidores

de PARP, considerados los primeros fármacos en cáncer clínicamente aprobados diseñados a partir del principio de letalidad sintética [80, 81], al dirigirse contra mecanismos de reparación del DNA en tumores con mutaciones en BRCA1/2. Tal es el grado de compensación entre PARP y BRCA1/2, que se ha descrito resistencia al tratamiento causada por mutaciones secundarias que restauran parcialmente la función de BRCA [87].

El análisis de estas dependencias funcionales no solo contribuye a comprender la biología del cáncer, sino que también sugiere nuevas estrategias terapéuticas combinatorias [80, 81], en las que la inhibición simultánea de genes compensatorios puede aumentar la eficacia del tratamiento y reducir la probabilidad de resistencia [82].Aunque muchas de estas vías aún no han podido ser explotadas plenamente en la práctica clínica debido a la limitada caracterización de sus mecanismos moleculares, representan un área de gran interés al ofrecer una mayor probabilidad de identificar dianas tratables. En consecuencia, se están desarrollando múltiples fármacos dirigidos a parejas de genes compensatorios que se encuentran en distintas fases de evaluación clínica [80, 81]. Entre ellos se incluyen el crizotinib, empleado para inhibir el gen *ROS1* en el contexto de su letalidad sintética con mutaciones en gen *CDH1* [83]; el ceralasertib, que actúa sobre el gen *ATR* por su letalidad sintética con *ARID1A* [84]; y el onvansertib, dirigido contra *PLK1* [85]. Estos casos ponen de manifiesto el creciente interés clínico en explotar el fenómeno de la compensación genética.

Parte III

Hipótesis y Objetivos

Capítulo 3

Hipótesis y objetivos

El presente trabajo se estructura en función de las siguientes hipótesis:

Hipótesis I Las técnicas de embebido de nodos permiten aprovechar el espacio vectorial generado para estudiar las relaciones entre genes dentro de una enfermedad.

Hipótesis II La detección de comunidades es importante para aumentar la precisión de las técnicas basadas en embebido de nodos.

Hipótesis III Las técnicas de embebido de nodos pueden ser útiles para construir corpus de conocimiento específicos de un conjunto concreto de enfermedades como es el caso de las enfermedades neuromusculares.

Hipótesis IV Las distintas enfermedades neuromusculares conocidas posiblemente tengan mecanismos reguladores comunes.

Hipótesis V Existen genes adicionales que modulan la vía Ras/MAPK y que podrían contribuir a la fisiopatología de las RASopatías y representar potenciales blancos terapéuticos.

Hipótesis VI El conocimiento sobre las relaciones entre genes generado en el estudio de enfermedades genéticas y sus causas se puede aplicar para identificar genes que puedan paliar o compensar la enfermedad.

Hipótesis VII Las vías de compensación que se puedan identificar en enfermedades genéticas también son aplicables para la identificación de mecanismos de resistencia a fármacos en cáncer.

Conforme a estas hipótesis, el objetivo principal de esta tesis doctoral consiste en caracterizar mecanismos moleculares implicados tanto en diversas enfermedades génicas como en cáncer y aplicar este conocimiento para identificar distintos mecanismos de compensación. Concretamente, este objetivo principal se desglosa de la siguiente manera:

Objetivo I Diseñar y desarrollar un sistema de embebido de nodos que integre la información de comunidades solapantes e implementar un algoritmo de priorización basado en estos sistemas desarrollados.

Objetivo II Generar un corpus de conocimiento especializado en enfermedades neuromusculares mediante la combinación de redes de similitud textual y representaciones basadas en embebido de nodos.

Objetivo III Integrar y analizar información transcriptómica de distintas enfermedades neuromusculares junto con datos de interacción proteína–proteína, con el fin de identificar mecanismos reguladores comunes.

Objetivo IV Identificar potenciales genes implicados en modular la vía Ras/MAPK y caracterizar su impacto en la fisiopatología de las RASopatías.

Objetivo V Generar un *diseasoma*, entendido como una red de enfermedades, actualizado para la validación del sistema de priorización de genes causales.

Objetivo VI Identificar y caracterizar un conjunto de pares gen causal de la enfermedad – gen compensador, en el contexto de enfermedades raras y cáncer.

Parte IV

Desarrollo de nuevas técnicas de análisis de redes basadas en detección de comunidades

Capítulo 4

Modelado e implementación de embebido de nodos en base a *random walk* constreñido por comunidades

El análisis de redes biológicas, como aquellas que representan interacciones proteína-proteína, asociaciones gen-enfermedad o relaciones sintomatológicas, genera estructuras de gran complejidad, caracterizadas por su alta dimensionalidad y heterogeneidad. Si bien estas redes constituyen una fuente valiosa de conocimiento, su tamaño y complejidad dificultan la extracción directa de patrones interpretables. En este sentido, los métodos de embebido de nodos resultan fundamentales, ya que permiten transformar la información estructural de la red en representaciones vectoriales de menor dimensión, preservando al mismo tiempo las propiedades topológicas y semánticas más relevantes. Esta transformación no sólo facilita la aplicación de técnicas de posteriores aprendizaje automático, sino que también habilita la integración de información procedente de diferentes dominios (molecular, clínica y textual). De este modo, el desarrollo de métodos robustos de embebido de nodos se convierte en una necesidad, ya que permite optimizar el análisis y la interpretación de redes complejas. Esto facilita la identificación de patrones ocultos, la integración de información heterogénea y la generación de nuevo conocimiento a partir de estructuras altamente interconectadas.

En este capítulo se presenta el desarrollo de un nuevo método de embebido aplicado a grafos. El enfoque propuesto, basado en estrategias de tipo *random walk*, tiene como objetivo obtener una representación más fiel de las comunidades solapantes presentes en la red. Este planteamiento permite una mejor caracterización de las distintas redes biológicas de interés, mostrando además un caso de aplicación concreto en el estudio de las RASopatías. El trabajo posee un doble

interés: por un lado, técnico, al superar las limitaciones computacionales de métodos tradicionales de detección de comunidades solapantes; y por otro, biológico, al posibilitar un análisis integrador de las distintas RASopatías consideradas.

4.1. Optimización de detección de comunidades solapantes y aplicación sobre embebido de nodos.

Mi contribución:

A continuación, se presenta una publicación en la revista *Briefings in Bioinformatics*, en la cual soy primer autor. He llevado a cabo la validación de las implementaciones del algoritmo *Hierarchical Link Clustering* desarrolladas en el laboratorio. Una vez comparado el rendimiento computacional y la precisión de las distintas estrategias, pasé a diseñar y desarrollar un nuevo algoritmo de embebido de nodos, capaz de integrar la información de comunidades solapantes en el proceso. Esto implicó tanto la formulación de un nuevo tipo de *random walk* como su implementación a nivel de código en la librería de Python *NetAnalyzer*.

Tras implementar el sistema de embebido, comprobé que era capaz de representar las rutas biológicas con mayor coherencia en el contexto del interactoma humano. he trabajado con datos de interacción de la base de datos *STRING*, mientras que las rutas biológicas fueron extraídas de *Reactome*. Finalmente, el sistema validado lo utilicé para analizar nuevos genes y rutas biológicas potencialmente involucrados en diversas RASopatías. Para ello, diseñé una estrategia bayesiana que permitiera priorizar los nodos (genes) en función del espacio vectorial generado por el embebido.

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Case Study

Advancing edge-based clustering and graph embedding for biological network analysis: a case study in RASopathies

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Abstract

Understanding and predicting biological processes from protein–protein interaction (PPI) networks requires accurate and efficient representations of their structure. However, many existing methods fail to capture the complex, overlapping modular structure of biological systems. To address this, we propose a network embedding strategy that improves both biological interpretability and predictive power. By transforming networks into a low-dimensional space while preserving key topological properties, embedding enables the discovery of novel functional relationships. Pre-clustering a network before embedding enhances representation quality, i.e. the ability to preserve meaningful structural and functional properties in the embedding space. However, traditional non-overlapping clustering methods can introduce bias by ignoring the overlapping nature of biological communities. We overcome this limitation by integrating the Hierarchical Link Clustering (HLC) algorithm into an embedding workflow tailored for large, weighted, undirected networks. First, we introduce two optimized HLC implementations for Python and R, both outperforming existing methods in clustering accuracy and scalability. Then, by restricting random walks to HLC-defined communities, we improve the representation of biological pathways, as shown using Reactome on the human PPI network. We also apply our full cluster embedding workflow to analyze RASopathies, a group of interrelated disorders with a diverse range of phenotypes, caused by mutations in genes from the RAS/MAPK pathway. This approach was used not only to represent known pathways, but also to identify potential novel gene candidates associated with RASopathies, including Noonan and Costello syndrome. HLC implementations are available in the CDLIB library (https://github.com/GiulioRossetti/cdlib), and at https://github.com/jimrperkins/linkcomm for Python and R, respectively.

Keywords: RASopathies; network embedding; overlap community; HLC; protein-protein interaction

Introduction

Network analysis is fundamental to bioinformatics and systems biology, where these techniques have been applied to molecular interactions (e.g. protein networks [1], metabolites [2]) and phenotypic traits such as comorbidities [3, 4]. In recent years, network embedding has emerged as a powerful addition to these approaches, allowing the transformation of nodes into low-dimensional vector spaces that preserve topological and functional relationships [5]. These embeddings facilitate tasks such as module detection, gene prioritization, and disease classification by revealing hidden patterns and improving the integration with machine learning models [6].

Methods for embedding include matrix factorization methods such as Laplacian eigenmap [7], Singular Value Decomposition [8], and network kernels [9, 10]. In addition, powerful and pioneering methods for network embedding based on deep-learning have emerged in recent years. These include random walk-based methods like DeepWalk [11] and Node2vec [5], which have been applied in diverse tasks, including link prediction [12, 13], node classification [6, 14], community detection [15–17] and single-cell analysis [18, 19].

In these random walk-based approaches, custom random walks are integrated into a word2vec framework to generate embeddings [5, 11, 20]. Various algorithms have refined this scheme, such as node2vec, which balances breadth-first and depth-first sampling to enhance representation quality [5]. More recently, these methods have been improved by incorporating classical clustering techniques, making them community-aware and showing promising results in embedding biological networks [21–24].

Choosing an appropriate clustering method is crucial when integrating classical network techniques with network

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embeddings and other deep learning approaches. Most methods use non-overlapping clustering, assigning each node to a single community. However, in biological networks, genes and proteins often belong to multiple pathways or functional modules [25], making overlapping clustering methods essential [26]. To address these challenges, we propose a novel embedding pipeline that integrates overlapping edge-based clustering with random walk–based node embedding. Specifically, we incorporate Hierarchical Link Clustering (HLC) [25, 27] to identify overlapping communities, and we constrain random walks to remain within these communities. This strategy enhances the functional coherence of the embeddings and improves biological pathway representation.

Recent work has explored alternative strategies to improve network representations, preserving properties such as structural roles [28], high-order node proximity [29], and integrating global structural information [30], each capturing complementary aspects of network complexity. In contrast, our approach prioritizes the preservation of overlapping community structure, a defining characteristic of biological networks, by incorporating an edge similarity–based clustering step i.e. computationally efficient and scalable to large, weighted networks. This allows the integration of biologically meaningful community information into the embedding process without incurring substantial computational overhead.

Despite these advantages, current HLC implementations face limitations. The widely used R package linkcomm [31] supports weighted networks but struggles with large datasets, while the Python package CDLIB [32] offers only an unweighted implementation, limiting its applicability to weighted biological networks.

Network embedding can be applied to analyze genes related to diseases, such as RASopathies. These developmental disorders are caused by germline mutations in genes of the RAS/MAPK signaling pathway [33], involved in multiple processes including angiogenesis [34] and cancer [35]. Despite significant advances in genetic testing, the full mechanisms driving these conditions and the relationships among the various disorders remain unclear. Further investigation into how these genes interact and regulate the pathway may provide crucial insights into their overlapping yet distinct clinical features, as well as providing potential novel drug targets.

Applying graph embedding and overlapping clustering to study RASopathies highlights its potential to uncover crucial insights into gene interactions and pathway regulation. This not only helps explain their overlapping yet distinct clinical features but also suggests novel drug targets. Given the molecular links between RASopathies and cancer, our approach may further support the repurposing of cancer therapies for these disorders, offering new avenues for targeted treatment.

The main contributions of this work are as follows:

- We present a new node embedding framework which integrates community information obtained from HLC.
- We optimize the HLC algorithm for large-scale, weighted biological networks, providing efficient Python and R implementations.
- We demonstrate that the new embedding enhances the representation of known biological pathways in the human interactome, outperforming multiple baseline node embedding methods.
- We apply the method to RASopathies, providing novel representation of known genes and revealing novel candidate genes associated with the disorders.

Materials and methods

Implementation of HLC

We developed new implementations for analyzing weighted undirected networks for both R and Python. They extend original methods, which also apply to unweighted networks.

In R, the original linkcomm clustering method (LKCM) has significant memory limitations when analyzing large networks. We addressed this issue by utilizing sparse matrices, resulting in the implementation referred to as linkcomm sparse (LKCMsp).

We modified the linkcomm implementation to calculate Tanimoto coefficients following formulations proposed by Ahn et al. [25]. In our approach, the Tanimoto coefficient, $S(e_{ik}, e_{jk})$, is computed as shown in Equation (1).

$$S(e_{ik}, e_{jk}) = \frac{a_i \cdot a_j}{a_i \cdot a_i + a_j \cdot a_j - a_i \cdot a_j}. \quad (1)$$

where e_{ik} represents the edge between nodes i and k. a_i or a_j define the i and j rows of the adjacency matrix, known as adjacency vectors, respectively. In the original implementation [25], each ith position of a_i assigns the average degree of the ith node, whereas in linkcomm this value is set to 1. We refer to these formulations as the linkcomm and the Ahn formulation, respectively.

In Python, CDLIB includes a weighted HLC implementation derived from a repository related to the Ahn formulation; however, the authors have not fully adapted it to their library interface. In this study, we addressed this gap. The HLC algorithm is optimized for low-density networks and represents adjacency using a dictionary data structure.

We propose an alternative, HLC NumPy (HLCnp), that uses NumPy array representation and SciPy operations to vectorize the Tanimoto coefficient calculation, which is developed within the CDLIB library. This strategy is particularly efficient for dense networks in terms of execution time, although it may not optimize memory usage.

Moreover, using SciPy operations in hierarchical clustering allows for linkage methods beyond single linkage. An intermediate filtering step can also be added to eliminate spurious edge similarities, i.e. those with values below a given threshold.

All different symbols and acronyms used in this study are presented in Table 1.

Benchmarking of HLC: measuring technical performance and accuracy

We used eight different weighted networks, ranging from 100 to 12 000 nodes, with the number of edges varying from 2000 to 472 000 (Supplementary Fig. S1). Most of these networks are biological in nature and are also commonly used in bioinformatics, including protein–protein interaction (PPI) and semantic similarity networks, as described in the Supplementary Section.

For accuracy, we applied the metrics proposed in the Ahn study [25], specifically community and overlap coverage. Community coverage quantifies the total number of nodes in the network that belong to at least one non-trivial cluster (at least three nodes) while overlap coverage is the average number of non-trivial communities to which a node belongs.

For technical performance, we measured the percentage of CPU usage, total runtime, and maximum memory usage using the GNU/Linux time command. Each run was conducted on a Lenovo SR645 node equipped with 156 cores and 512 GB of physical memory.

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Table 1. Glossary of terms and definitions used in the study

Term	Definition
Formulation	
Ahn	Tanimoto formula proposed by Ahn and collaborators [25] to compute the edge similarity.
Linkcomm	Version of computation by Kalinka and collaborators [31] to compute edge similarity.
HLC implementations	
HLC	Original python CDLIB implementation
HLCnp	Python CDLIB implementation based in numpy array operations
LKCM	Original linkcomm HLC implementation
LKCMsp	Linkcomm HLC implementation with sparse matrices operations
Networks	
huST700	Interactome network of proteins in *Homo sapiens* with a cutoff of 700
huST900	Interactome network of proteins in *Homo sapiens* with a cutoff of 900
ecST700	Interactome network of proteins in *Escherichia coli* with a cutoff of 700
ecST900	Interactome network of proteins in *Escherichia coli* with a cutoff of 900
coORPHA	Comorbidity network with data extracted from ORPHANET
coOMIM	Comorbidity network with data extracted from OMIM
coMERGED	Comorbidity network with data merged from ORPHANET and OMIM
Wikipedia	Interaction network from Wikipedia

For the execution of the different algorithms, the following parameters were used: HLC with default parameter values; HLCnp with *hcmethod* = *single* and *min_edges* = 2; LKCMsp with *hcmethod* = *single*, *diagnorm* = *TRUE*, *minedges* = 3; and LKCM with *edglim* = 10^8 and *removetrivial* = *TRUE*. All algorithms were configured to remove clusters with fewer than two edges, except for HLC, which required the removal of trivial clusters in subsequent steps to achieve comparable results. For LKCM, the *edglim* was set to a higher value than the number of edges in any network used in the study (10^8) to minimize the number of disk write operations.

Network embedding using implemented HLC

Our network embedding system builds upon previous approaches, such as node2vec [5], which consists of an initial generation of random walks followed by the application of a skip-gram model to generate node embeddings. For the first step, we build on the work of Zhang and collaborators with the CRARE algorithm [21], modifying their approach to develop a custom random walk model that incorporates information about overlapping communities. In CRARE, walks are generated using a random walk process that allows jumps not only between neighboring nodes but also between nodes belonging to the same community or sharing the same structural properties, such as nodes with the same degree.

This implementation has been incorporated into and made available through NetAnalyzer [10], a Python library developed by our group for advanced network analysis and embedding.

In our implementation, we made the following changes: (i) we removed the jumps in the random walk between nodes with the same structural property, (ii) the method was generalized to use any clustering algorithm and adapted for overlapping clustering, and (iii) the transition probability from one node to a node in another community is weighted by the inverse of the community size. Our random walk model can then be formulated as follows:

Given n_0 as the source node and n_t as the visited node at step t, we defined a flexible random walk strategy using the following transition probability:

$$P(n_t = y \mid n_{t-1} = x) = \begin{cases} \Pi_{y,x} & \text{if } (x,y) \in E \text{ or } y \in C_x, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

where E is the edge set, C_x represents the union of all communities to which x belongs, and $\Pi_{y,x}$ is the combined probability defined in Equation (3).

$$\begin{aligned} \Pi_{y,x} &= \frac{1}{2}\left(\frac{w_{neigh}}{w_{neigh}+w_{comm}}\right) p_{neigh}(y \mid x) \\ &\quad + \frac{1}{2}\left(\frac{w_{comm}}{w_{neigh}+w_{comm}}\right) p_{comm}(y \mid x) \\ &= (1-q)p_{neigh}(y \mid x) + q p_{comm}(y \mid x), \quad q \in [0,1]. \end{aligned} \quad (3)$$

Here, w_{neigh} and w_{comm} are parameters that define the relevance of being part of the same neighbor or the same community, respectively. Unlike classical methods such as DeepWalk [11] and node2vec [5], which rely on uniform or biased random walks over direct neighbors, our model introduces a dual mechanism that considers both structural proximity (via neighbors) and functional context (via community membership). This formulation enables the embedding to capture overlapping modular structures, which are common in biological networks, while maintaining local connectivity. Additionally, by adjusting the parameter q, the walker can be tuned to favor intra-community transitions, something that traditional approaches cannot directly control. The probabilities p_{neigh} and p_{comm} are simply defined as uniform distributions (as defined in Equation (4) and Equation (5)).

$$p_{neigh}(y \mid x) = \begin{cases} \frac{1}{\text{degree}(x)} & \text{if } (x,y) \in E, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

$$p_{comm}(y \mid x) = \begin{cases} \frac{1}{|C_x|} & \text{if } y \in C_x, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

For each node, ten random walks with a length of 100 are generated to be used as corpus for Word2vec (vector_size=128, window=10, epochs=5, sg=1, hs=1, min_count=0), a popular skip-gram model [36]. Finally, the parameters of the model are optimized by the stochastic gradient descent [37].

Network embedding benchmark

The network embedding implementation was applied to the human PPI network from human STRING (huST) using two different score cutoffs, 700 and 900, producing the networks huST700 and huST900, respectively [1]. We used the combined score from STRING, which integrate the information from different channels in this database.

During the network embedding process, we used the communities identified by HLC in each PPI network. As a reference, we also included Louvain clustering, since it is used by the CRARE algorithm. We explored different parameter settings: baseline (random selection of edges, with parameters: $w_{neigh} = 0$, $w_{comm} = 0$); neigh (random walk between connected nodes, with

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parameters: $w_{neigh} = 1, w_{comm} = 0$); HLC_comm and Louvain_comm (random selection of nodes inside the same community, with parameters: $w_{neigh} = 0, w_{comm} = 1$); and neigh-HLC_comm and neigh-Louvain_comm (random walk combining both neigh and comm strategies, with parameters: $w_{neigh} = 1, w_{comm} = 1$).

In addition to our custom embedding strategy, we incorporated a diverse set of existing network embedding methods for comparative analysis. These included *node2vec* [5], *DeepWalk* [11], *Role2Vec* [28], *NodeSketch* [29], *GloVe* [38], *GraRep* [30], *GGVec*, *NetMF* [39], *HOPE* [40] and *NMFADMM* [41]. For each of these methods, we used the default parameter settings provided by the karateclub [42] and nodevectors Python libraries.

To assess the improvement in embedding, we retrieved gene annotations from the Reactome database (https://reactome.org/download/current/Ensembl2Reactome.txt) [43]. We then evaluated the normalized distance between genes within the same Reactome pathways in the embedding space, comparing results across the different parameter settings. The normalized distances were computed starting from the euclidean distances in the embedding space is extracted. Then, all pair or nodes are z-normalized and the median of the normalized distance inside every group of nodes is obtained. Additionally, we computed the size and internal edge density for each pathway. Statistical comparisons between distributions across methods were conducted using t-tests with *P*-values adjusted using the Benjamini–Hochberg (BH) procedure.

Finally, we used the Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP) learning technique to visually represent the embedding spaces generated for each parameter setting. UMAP was computed using the Python umap library (https://pypi.org/project/umap-learn) with the following parameters: $n_neighbors = 15, min_dist = 0.1, n_components = 2, metric =' euclidean'$.

Using the network embedding to improve our understanding of RASopathy related genes

We applied our methodology to RASopathy related genes. To obtain a list of RASopathies and associated genes, we used a semi-automated approach using the MONDO (Human Disease Ontology) and Monarch databases. The methodology was as follows:

1. Identification of Rasopathy Terms in MONDO: We began by capturing the child terms related to the term "RASopathy" (MONDO:0021060) in the MONDO ontology.
2. Manual Grouping of Disorders: After retrieving the relevant terms, we manually grouped disorders that are considered to represent the same underlying condition, based on expert curation, e.g. Noonan syndrome 1 and Noonan syndrome 2 were both considered Noonan Syndrome. Full details of the grouping are given in Supplementary Table S1.
3. Gene Association Using Monarch: For each identified RASopathy child term, we used the Monarch database (https://monarchinitiative.org/) to associate the relevant genes with the respective disorders.

We used the network embedding obtained for the human PPI networks, huST700 and huST900 to further investigate the RASopathy related genes. This was done firstly by visualizing the RASopathy genes using the UMAP representation of the network and showing each of the RASopathy related genes in different colors, corresponding to the RASopathy grouping.

Then, to perform more formal analysis, we performed prioritization analysis, to find potential new RASopathy related genes.

To prioritize genes for each RASopathy, we applied a Bayesian integration approach. Rather than ranking genes solely based on their average distance in the embedding space, we estimated the probability that each gene is associated with a causal gene of the disease. Genes with higher probabilities were given higher priority in the ranking, as described in the next Equation (6).

$$p_{\text{disease}}(gene_i) = 1 - \prod_{gene \in \text{disease}} (1 - p_{gene}(gene_i)) \quad (6)$$

where $p_{\text{disease}}(gene_i)$ represents the probability of gene i being associated with the set of genes related to the disease, and $p_{gene}(gene_i)$ denotes the probability of association between gene i and a particular gene.

To compute the probabilities of association, we trained a logistic regression model using the cosine similarity between gene embeddings as scores, pair of nodes from edges in the PPI network were considered positive labels, and the remaining cases were randomly selected as negatives labels. All the edges and the equivalent number of negative where used during the training step.

Results

Comparison of community detection across HLC implementations

To illustrate these differences, we analyzed results from smaller graphs. In Fig. 1A, we compared the linkcomm formulation (implemented in LKCM) with the Ahn formulation (used in LKCMsp, HLCnp, and HLC) across four weighted graphs. The Ahn formulation identified highly connected edge communities that were not detected by linkcomm. Consequently, in these cases, Ahn's approach outperformed linkcomm in terms of overlapping coherence and the identification of additional clusters across different network structures.

This trend is further supported by the overlap coverage metrics shown in Fig. 1B, where the linkcomm formulation consistently led to slightly lower values in terms of overlap and community coverage across the studied networks. Notably, ecST700 exhibited greater overlap coverage than ecST900, as the higher cutoff value of 900 reduced the number of connections between proteins, limiting their potential membership in multiple communities. In all three cases, overlap coverage remained above one, highlighting the necessity of clustering methods that account for overlapping communities.

Additionally, no major differences were observed in community coverage metrics between the chosen cutoff values, indicating that the proportion of nodes assigned to at least one community remained stable. However, it is worth noting that the Ahn formulation consistently achieved higher community coverage than linkcomm, despite never reaching full coverage (i.e. 1).

To assess the consistency among the three implementations of the Ahn formulation, we evaluated the coherence of the detected clusters in two widely used benchmark networks: The Karate Club and Les Misérables (Supplementary Fig. S2). In both cases, HLC, HLCnp, and LKCMsp achieved a maximum normalized mutual information value of 1 (Supplementary Fig. S2C), indicating that they identified identical clusters.

In contrast, LKCM, which deviates from the Ahn formulation, produced notably different clustering results. This divergence is

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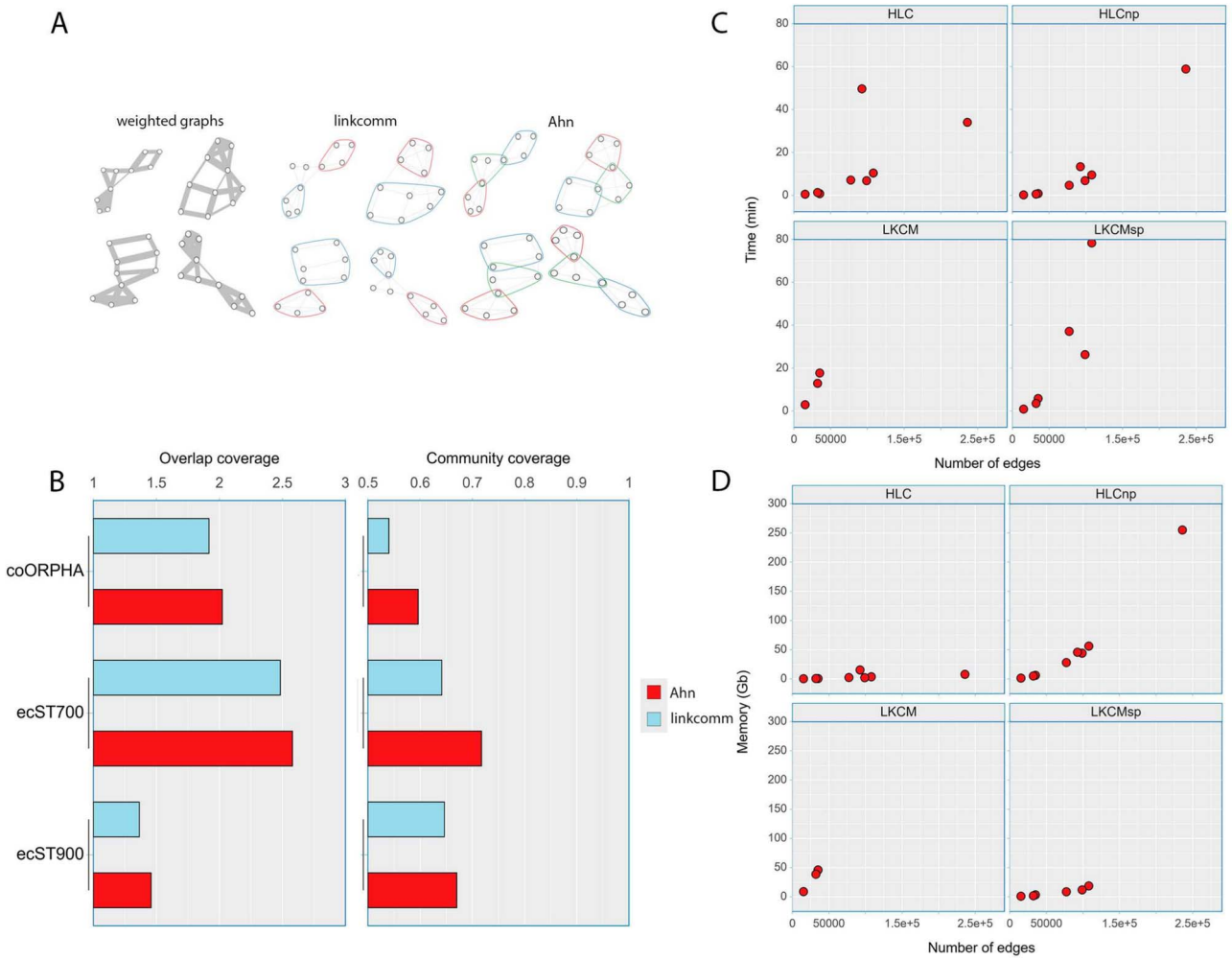

Figure 1. A) Graph representation of differences between the Ahn formulation (HLC, HLCnp and LKCMsp) and the linkcomm formulation (LKCM) using four different small networks. (B) Overlap and community coverage metrics measured for three real biological networks, calculated using the cases where LKCM could be executed. (C and D) Time and memory usage, respectively, across the different implementations.

reflected in its lower mutual information score and is visually evident in the graph plots (Supplementary Fig. S2A and B).

Technical performance and scalability of HLC implementations

With regard to technical performance, Fig. 1C and D show that both HLCnp and LKCMsp outperformed LKCM in terms of execution time (reductions of ×19.4 and ×3.3, respectively; Table 2) and memory usage (reductions of ×7.18 and ×15.21, respectively; Table 2). Notably, results could be generated for all eight networks in the cases of HLC and HLCnp, whereas LKCM only executed successfully for three networks and LKCMsp for six. Failures in execution were attributed to memory limitations (using more than 450 GB) or excessive execution time (lasting more than three days).

Although the original CDLIB implementation (HLC) demonstrated greater memory efficiency compared to our Python implementation, HLCnp (×53.4 and ×7.2, respectively; Table 2), its execution time increases significantly when handling dense networks (Supplementary Fig. S4). In contrast, our NumPy-based matrix approach exhibits superior scalability, as it uses vectorized operations to optimize performance. This is particularly evident for the Wikipedia network, where HLC requires considerably more time due to its high edge density

Table 2. Mean ratio for time and memory used in all four HLC implementations with respect to LKCM, expressed as fold-change, i.e. how much faster it take to run, or how much less memory is used

Implementation	Time speedup	Memory reduction
LKCM	1×	1×
LKCMsp	3.312×	15.214×
HLCnp	19.404×	7.185×
HLC	12.77×	53.39×

Ratio is computed against LKCM time or memory values.

(Fig. 1C and Supplementary Fig. S2A), as further supported by Supplementary Fig. S1.

HLCnp introduces new functionalities that enhance performance by utilizing NumPy and SciPy for customizable linkage procedures (Supplementary Fig. S3) and Tanimoto matrix distance filtering (Fig. 2). The latter significantly reduces memory usage and execution time while preserving overlap and community coverage, particularly for low cutoffs such as 0.2 (Fig. 2, panels A and B).

Additionally, HLCnp supports multiple linkage methods, expanding beyond single linkage to include complete linkage, which can further improve coverage and overlap. This is

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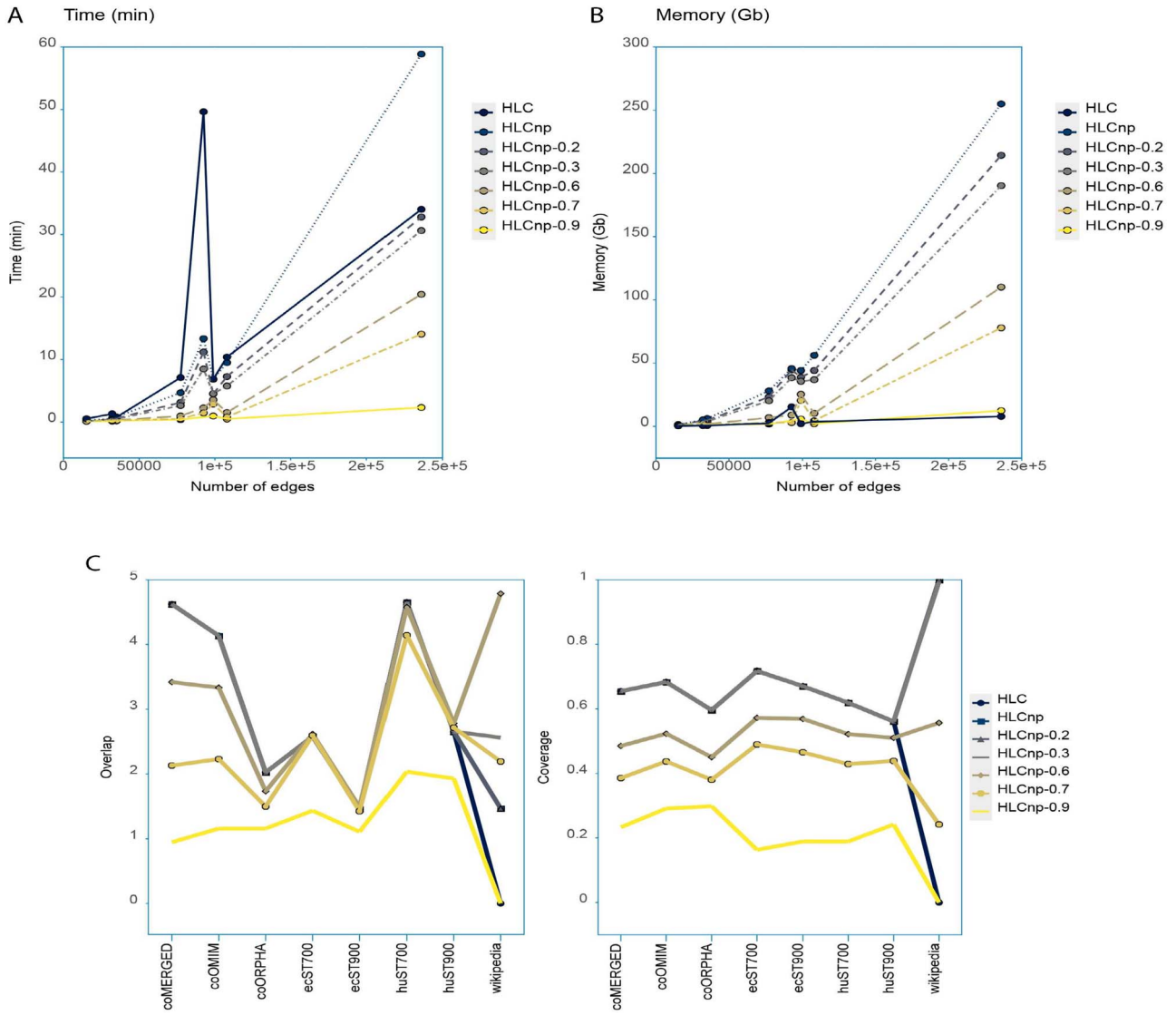

Figure 2. Application of filters to the Tanimoto matrix. Time (A), memory (B), coverage and overlap (C) are measured with respect to different cutoff filters. HLC and HLCnp were added to the analysis as controls for every dataset. The cutoff values applied to the filters are specified in the format "HLCnp-*cutoff value*".

demonstrated in Supplementary Fig. S3, where complete linkage consistently outperforms single linkage across nearly all tested networks.

Embedding analysis and evaluation of pathway cohesion in the human interactome

To evaluate the potential of different HLC algorithms in enhancing network representation learning in PPI networks, we first assessed whether pathways with high internal coherence (i.e. pathways whose members are highly connected) are better represented when HLC communities were integrated into the graph embedding procedure. As shown in Fig. 3, we observed that the distances in embedding space between genes within the same pathway decrease as the internal edge density increases. This trend is further enhanced when HLC clustering is incorporated into the graph embedding, suggesting that HLC improves the capture of pathway cohesion.

Building on these results, we evaluated the proximity of genes within coherent pathways (internal edge density > 0.75) in the embeddings generated from two human PPI networks (huST700 and huST900). We normalized the embedding distances between pathway genes as z-scores based on the overall distribution of distances in the embedding; negative z-scores indicate that pathway genes are closer together than average, whereas positive z-scores indicate larger distances.

As shown in Fig. 4A and B, integrating both node neighborhood and HLC communities produced a significant improvement, leading to the shortest average distance across the parameter settings tested. We further compared the HLC-based method against 11 baseline approaches (Supplementary Fig. S5), and in all cases, our method outperformed the alternatives. A t-test with BH correction confirmed the statistical significance of these results, with all adjusted *P*-values falling below.01. In contrast, when this combination was performed with Louvain clustering, the resulting embedding exhibited the poorest distance representation. This is likely because the Louvain algorithm, which produces a small number of large, non-overlapping communities (see Supplementary Fig. S6) can negatively impact the embedding process by distorting the natural structure of the network. This can occur when clustering is too coarse to reflect the finer-grained relationships,

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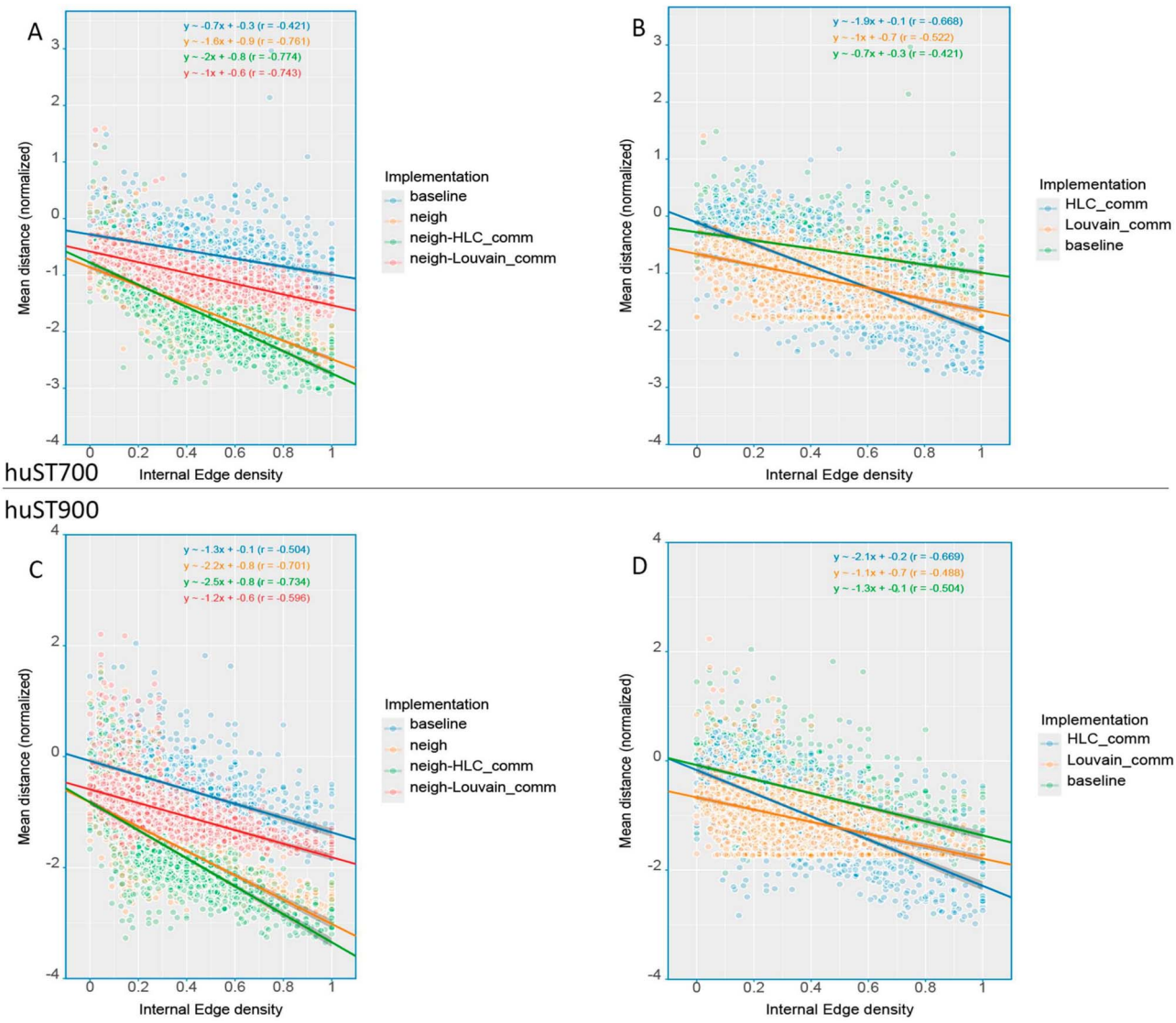

Figure 3. Relationship between embedding distance and internal edge density for huST700 (A and B) and huST900 (C and D) using various embedding strategies: baseline (random edge selection), neigh (random walk between connected nodes), HLC_comm (random selection within the same HLC community), Louvain_comm (random selection within the same Louvain community), and neigh-HLC_comm or neigh-Louvain_comm (random walk combining both neighborhood and community strategies). Linear regression is shown with the associated confidence region.

reducing the ability of embedding to capture meaningful local distinctions in the network. Additionally, since random walks play a key role in learning node representations, the rigid community structure imposed by Louvain may bias these walks by forcing them to remain within clusters longer than they naturally would.

We further explored the impact of the clustering methods on the embedding space of the huST networks (Supplementary Figs S7 and S8). Analysis of the top 20 clusters from each method revealed that Louvain clusters cover almost the entire network, while HLC clusters represent a smaller portion (Supplementary Fig. S7B vs. A for huST700 and Supplementary Fig. S8B vs. A for huST900). This difference greatly impacts the embedding: as illustrated in Supplementary Figs 7D and 8D, Louvain clustering groups nodes by community membership in a way that disrupts the original network structure (see Supplementary Figs S7B and S8B), whereas the HLC method better preserves this structure (see Supplementary Figs S7C and S8C).

Finally, we examined the top ten largest Reactome pathways (internal edge density > 0.75) to illustrate how our analysis represents biological pathways. Figures 4C and D show that using HLC clusters results in pathway members being closer together in the embedding space compared to those generated using Louvain clusters. Further UMAP analyses of the top five and ten largest pathways are shown in Figs 5 and 6, respectively. We see clear differences between the HLC and Louvain methods, with the former tending to show much clearer separation between pathways.

Case study: RASopathies

To further explore the biological relevance of candidate genes identified through our analysis, we conducted a focused case study examining their potential involvement in RASopathies and RASopathy-associated signaling. We selected a subset of genes for manual curation based on their presence in the prioritization lists obtained through embedding. For each gene, we assessed the strength of the evidence linking it to RASopathy phenotypes by integrating information from published literature, known signaling pathways, and documented pathogenic variants. This analysis

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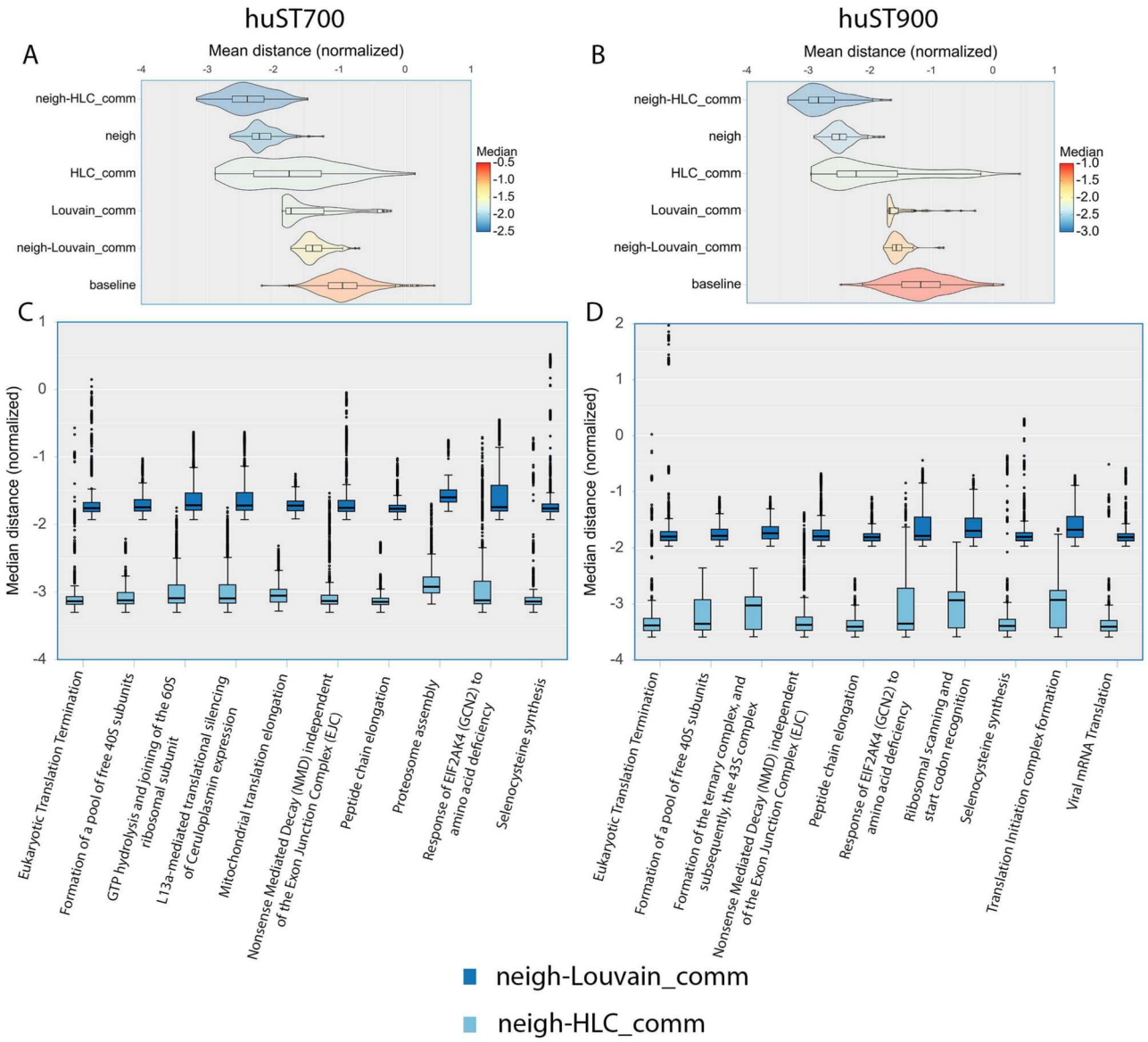

Figure 4. Embedding results using different node selection strategies: baseline (random selection of edges), neigh (random walk between connected nodes), HLC_comm (random selection of nodes within the same HLC community), Louvain_comm (random selection of nodes within the same Louvain community), and neigh-HLC_comm or neigh-Louvain_comm (random walk combining both neighborhood and community strategies). (A and B) Distributions of mean z-scores for distances between nodes within the same high-cohesion pathway (edge density > 0.75) in huST700 and huST900, respectively; (C and D) Distributions of mean z-scores for distances within the ten largest high-cohesion pathways in huST700 and huST900, highlighting neigh-HLC_comm (light blue) and neigh-Louvain_comm (dark blue).

allowed us to distinguish between genes with strong functional or genetic evidence of involvement in RASopathies, those with more tentative or indirect associations, and those with no clear evidence of relevance to the RASopathy disease spectrum.

In total, 21 RASopathy related genes, belonging to 9 different RASopathy groups were compiled through analysis of MONDO and Monarch, as shown in Table 3. The positions of these genes in the UMAP representation of the graph-embedding of the human PPI network are shown in Fig. 7A. We see that the HLC informed graph embedding approach (neigh-HLC_comm) leads to clear groupings of genes in close, but distinct areas of the UMAP representation.

For full details of the top 20 prioritized results for each of the RASopathies investigated, please see Supplementary Tables S2 and S3, for the huST700 and huST900 networks respectively.

We can also see, in Fig. 7B and C, that the top prioritized non-RASopathy related genes tend to lie close to the RASopathy related genes. Interestingly, we see a small group of prioritized genes far from the main groupings, these are all prioritized genes for Noonan syndrome-like disorder with loose anagen hair, and are close in the UMAP to the gene *PPP1CB* (Fig. 7D).

Our prioritization algorithm using all known RASopathy related genes together for the huST700 network identified several candidate genes with potential relevance to RASopathies among the top 20 genes based on the prioritization score. For instance, SHC domain proteins like SHC1 are involved in RAS-mediated signal activation [44]. In addition, MAPK7 (a lesser-known MAP kinase) and MAP3K2 also emerged among the top candidates; kinases from this family have key roles in the RAS/MAPK signaling pathway, particularly MAPK3 and MAPK1 [45].

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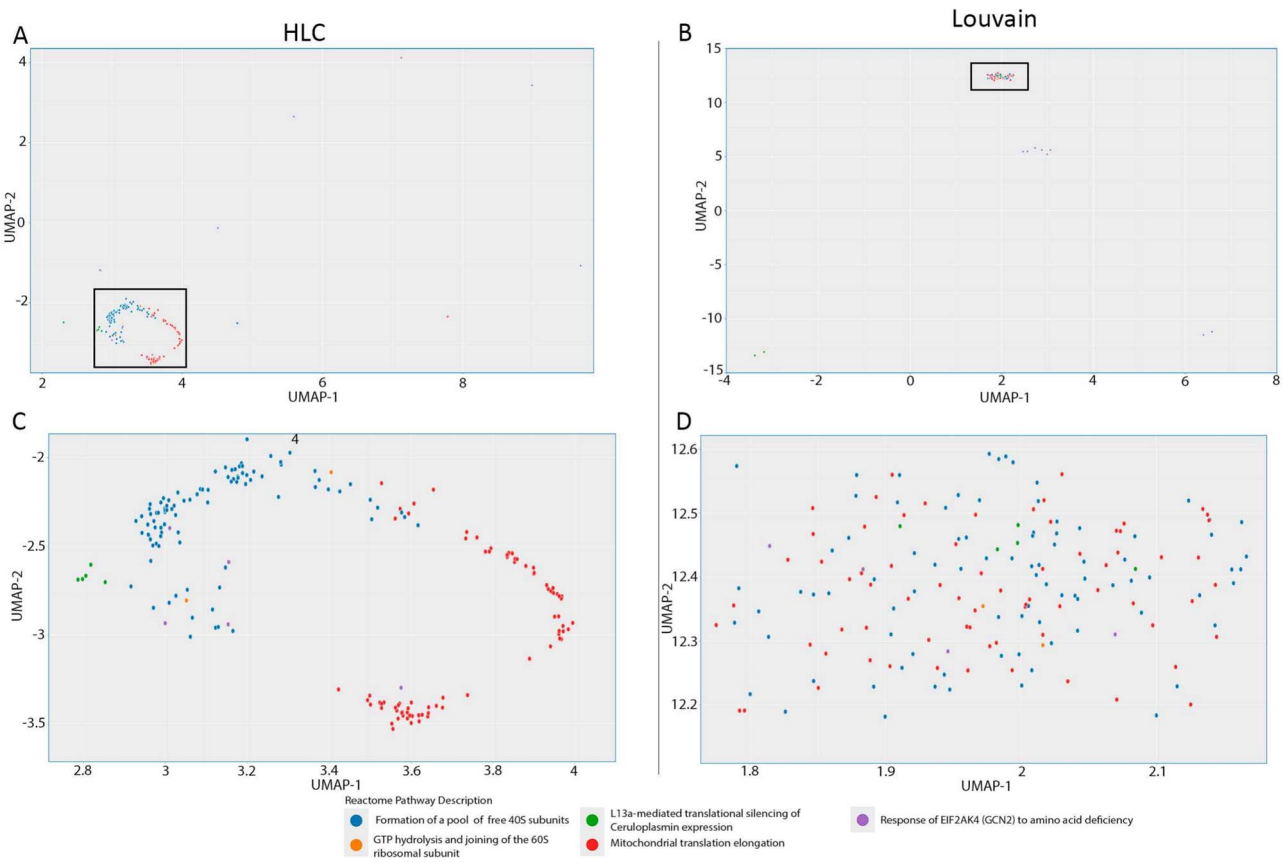

Figure 5. UMAP representation of the embeddings generated for huST700. The top five largest Reactome biological pathways (with an internal edge density > 0.75) are shown in different colors. (A and C) Embedding representation using HLC; (B and D) the representation obtained using Louvain clustering.

Table 3. RASopathies under study and their associated genes

RASopathy	Genes
CBL-related disorder	*CBL*
Costello syndrome	*HRAS*
Legius syndrome	*SPRED1*
Noonan syndrome	*BRAF*, *KRAS*, *LZTR1*, *MAP2K1*, *MAPK1*, *MRAS*, *NRAS*, *PTPN11*, *RAF1*, *RIT1*, *RRAS2*, *SOS1*, *SOS2*, *SPRED2*
Noonan syndrome with multiple lentigines	*BRAF*, *PTPN11*, *RAF1*
Noonan syndrome-like disorder with loose anagen hair	*PPP1CB*, *SHOC2*
RASopathy	*BRAF*, *CBL*, *HRAS*, *KRAS*, *LZTR1*, *MAP2K1*, *MAP2K2*, *MAPK1*, *MRAS*, *NF1*, *NRAS*, *PPP1CB*, *PTPN11*, *RAF1*, *RIT1*, *RRAS2*, *SHOC2*, *SOS1*, *SOS2*, *SPRED1*, *SPRED2*
Watson syndrome	*NF1*
cardiofaciocutaneous syndrome	*BRAF*, *KRAS*, *MAP2K1*, *MAP2K2*
neurofibromatosis type 1	*NF1*

Notably, *RAP1A* and *RAP1B*, members of the Ras subfamily implicated in cellular adhesion and morphogenesis, were also prioritized and have been shown to influence cell proliferation [46]. Rap1 has also shown possible association to Kabuki syndrome [47], a disorder increasingly recognized as part of the RASopathy spectrum due to its overlapping features and involvement of genes in the RAS/MAPK signaling pathway [48]. Another prioritized gene, *NKS1A* (also known as Odin) is a PTB-domain adaptor that regulates ER export of receptor tyrosine kinases (RTKs). It binds EphA2/ErbB2 in the ER and loads them into COPII vesicles.

The gene *ANKS1A* also acts downstream of EGFR and Eph receptors via its ankyrin and PTB domains [49]. By controlling the trafficking of RTKs that signal through RAS, ANKS1A can modulate Ras-MAPK activity. SKAP1 (Src kinase–associated phosphoprotein 1, or SKAP-55) is an adaptor in T cell signaling. It mediates RapL (NORE1B/RASSF3) membrane localization and LFA-1 activation in a process dependent on its PH domain and PI3K activity [50]. Although SKAP1 has not been studied in RASopathies, this role in PI3K-driven Rap1 signaling suggests a possible crosstalk between RAS and PI3K networks. Another highly ranked gene *RAPGEF5* (MR-GEF) is a Rap guanine exchange factor expressed during development. Mutations have been reported associated with congenital heart defects and neurological disorders, reflecting its developmental importance [51]. Because RapGEF5 activates Rap (a Ras-family GTPase) and influences Wnt/cell polarity pathways, it may interface with Ras/MAPK signaling in embryogenesis.

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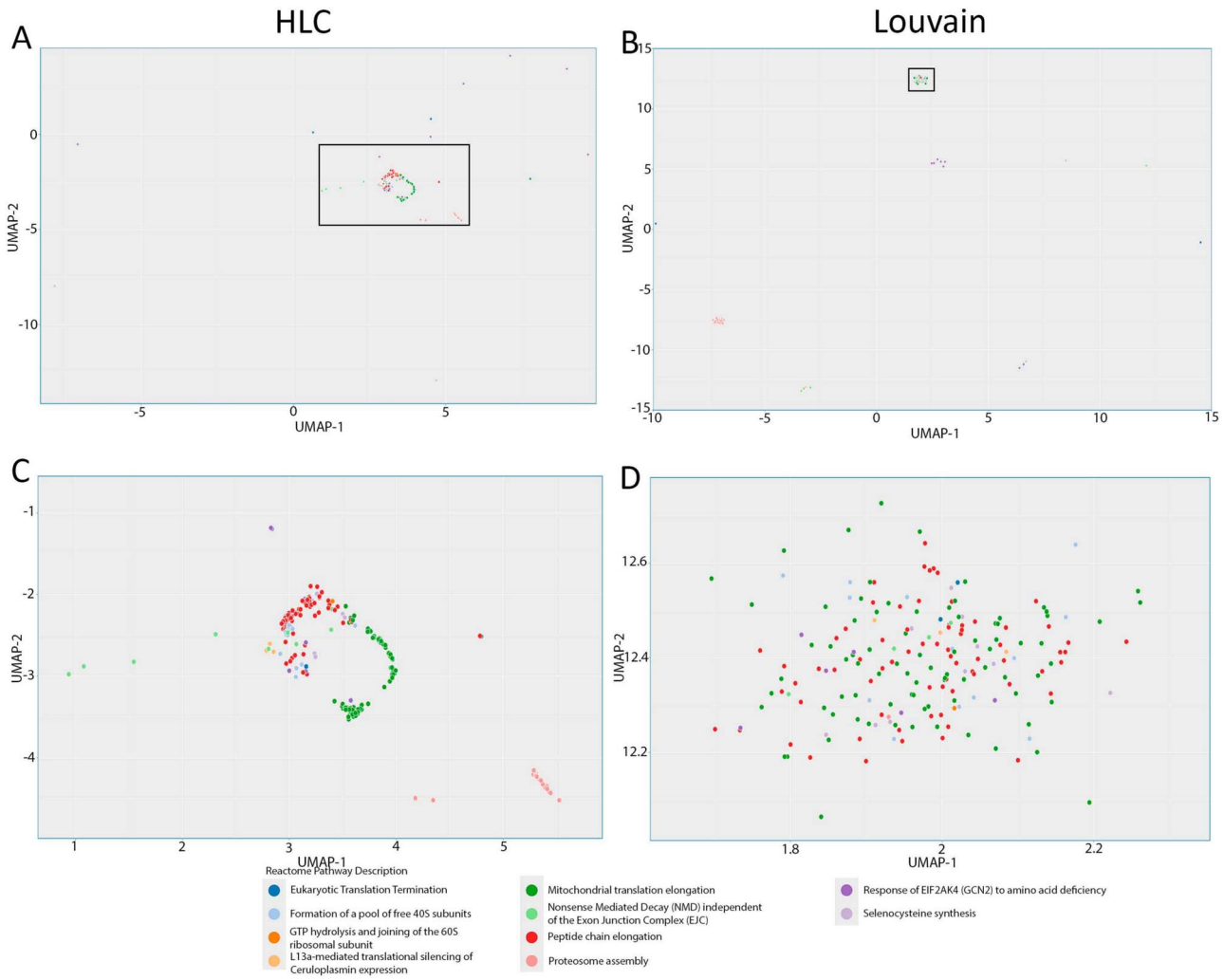

Figure 6. UMAP representation of the embeddings generated for huST700. The top ten largest Reactome biological pathways (with an internal edge density > 0.75) are depicted in different colors. (A and C) Embedding representation using HLC; (B and D) the representation obtained using Louvain clustering.

There were also a number of PI3K/AKT pathway members among the prioritized genes. These include *PIK3R1*, which encodes the p85α regulatory subunit of PI3K. Mutations in the C-terminal SH2 domain of p85α impair binding with p110α in PI3K complexes, which may indirectly affect Ras-pathway output [52]. These findings suggest cross talk between PI3K and Ras/MAPK signaling, which is to be expected given that PI3K is one of the main effectors of Ras [52]. Interestingly PI3K/AKT/mTOR pathway often cause developmental syndromes that share some similarity to RASopathies at the phenotypic level [53]. These findings show the potential use of our methodology to identify and investigate related pathways in the context of disease.

Furthermore, our analysis identified several prioritized genes that represent potential drug targets. For instance, ALK, a RTK known to be implicated in multiple cancers, has been successfully targeted using inhibitors such as crizotinib, which has shown efficacy in non-small cell lung cancer [54]. In addition, the identification of PRKCA (protein kinase C alpha) suggests that pharmacological modulators of PKC activity might have therapeutic implications in disorders characterized by altered RAS/MAPK signaling [45]. These findings demonstrate that our method not only prioritizes known RASopathy-related genes but also uncovers additional candidates with potential therapeutic relevance.

When the methodology was applied to the genes associated with specific RASopathies, we found additional genes of interest among the priotization results. For example, when using the Noonan syndrome associated genes, *RASA2* emerged as a high-ranking gene; it encodes a Ras GTPase-activating protein whose loss-of-function mutations have been associated with Noonan syndrome-like phenotypes [55, 56]. We also found *HRAS* and *NF1*, important genes associated with different RASopathies (neurofibromatosis type 1 and Costello syndrome respectively) in the top 20. In addition, important RAS pathway related genes were also found, including *BRA* and various Guanine nucleotide exchange factors involved in Ras signaling. Interestingly, the results using the Costello syndrome associated genes were broadly similar, with the top twenty including multiple RAS genes involved in other RASopathies. For CFC syndrome, multiple additional RAS related genes are also found. Together, these findings highlight the molecular overlap between distinct RASopathies and support the idea that shared dysregulation within the RAS/MAPK pathway underlies their related phenotypic features, enabling the identification of novel common associations across the spectrum.

We also inspected the results of the huST900 network analysis. For prioritization involving all genes associated with RASopathies,

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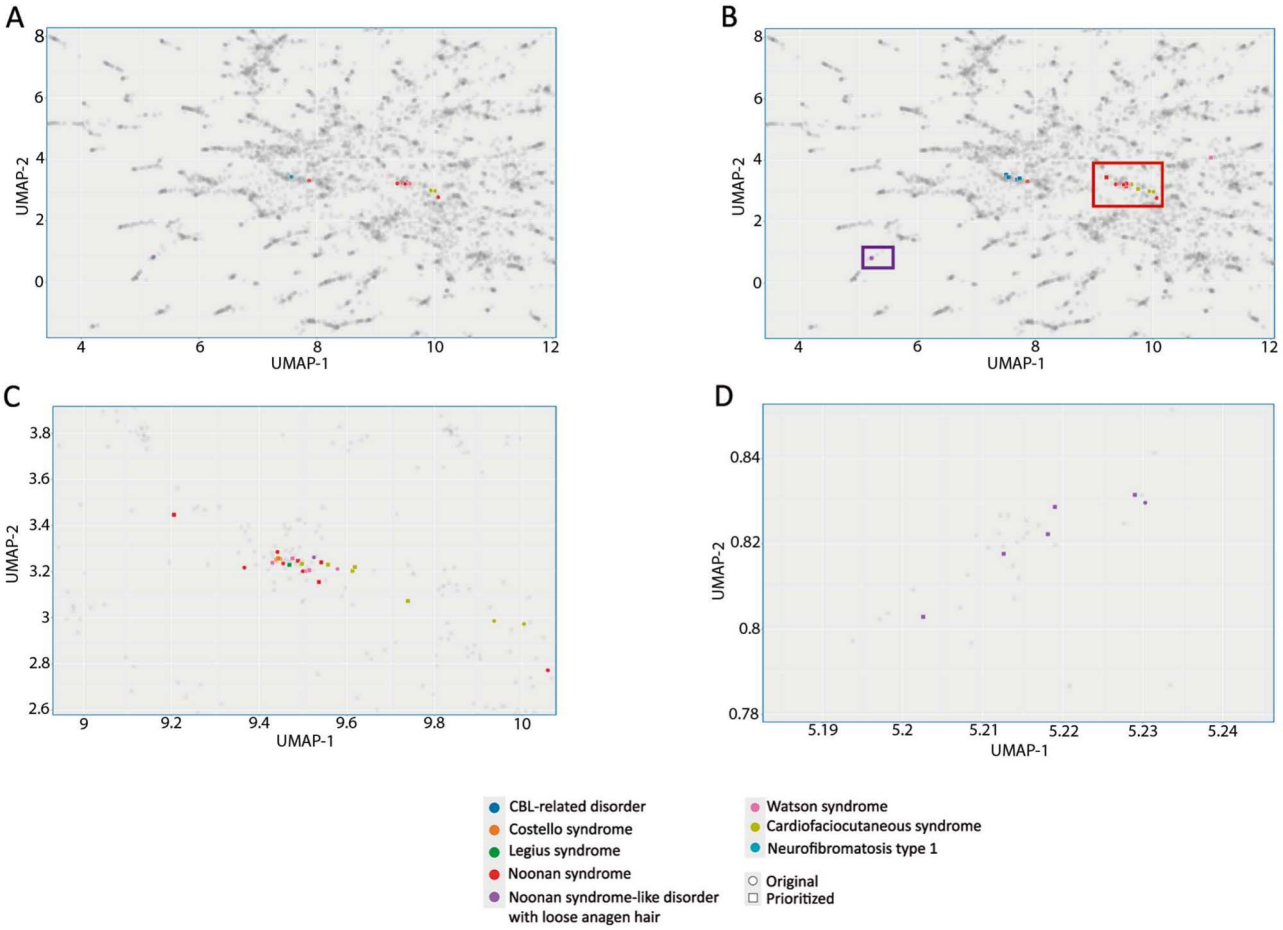

Figure 7. UMAP representation of the embeddings generated for huST700. Note that we have zoomed into a small part of the entire UMAP to better visualize the distances between these genes and other, non RASopathy-related genes. (A) All RASopathy-related genes. (B) All RASopathy-related genes, plus the top 5 prioritized genes for each RASopathy. (C) Zoomed in on the main cluster of RASopathy related genes (red box in B). (D) Zoomed in on the single Noonan gene found far from the other RASopaty genes, along with the other priotized Noonan-related genes that lie close in the UMAP representation (violet box in B).

results were broadly simiar to the huST700 analysis, with some differences. For example, RASA2, the Noonan-associated target was found. More generally, our analysis highlighted several other genes, including *RAP1B*, *RASGRF1*, *OSBPL3*, *RASGRF2*, *ARHGEF12*, *IL3RA*, *CNKSR1*, *CSF2RB*, *KIAA1549*, *GAB2*, and *RAPGEF5* that, although not classically linked to RASopathies, play integral roles in modulating Ras signaling. Given their involvement in the regulation of the Ras-MAPK pathway, these genes may represent novel risk factors or mechanistic contributors to developmental disorders associated with dysregulation of Ras signaling [33].

In addition to examining RASopathy-associated genes, we investigated the relative positions of the pathways to which these genes belong using UMAP visualization (Fig. 8). Notably, the genes involved in MAPK1 activation can be found in close proximity to those participating in the CSF3 signaling pathway and signaling by phosphorylated juxtamembrane, extracellular and kinase domain KIT mutants pathway genes. KIT signaling have been shown to activate the RAS/MAPK cascade, and that GM-CSF and c-Kit together promote proliferation [57, 58]. Given that MAPK1 (also known as ERK2) is a key effector in the RAS pathway, such spatial proximity suggests that cross-talk or co-regulation between these pathways may contribute to the abnormal signal transduction observed in RASopathies. However, it should be noted that the actual RASopathy-associated genes from the MAPK1 activation pathway (MAPK1 and MAP2K2) are found relatively far from the other members of this pathway in the UMAP visualization.

Moreover, the UMAP visualization also revealed that genes related to FGFR4 signaling, PDGFRA, and insulin signaling are positioned near a MAPK1 activation gene and, again, in the vicinity of CSF3 pathway genes. This arrangement is significant because FGFR4 is an established upstream activator of RAS/MAPK signaling [59], PDGFRA has been linked to RAS in a proliferation model, and the insulin signaling cascade is known active the Ras-MAPK pathway following the binding of Grb2 and SOS to IRS proteins [60]. Again, proximity of these pathways implies a potential convergence of signals that could amplify or modulate RAS activity. Further analysis could involved examining and investigating the genes that do not belong to these pathways, but fall between them in the UMAP representation.

Discussion

In this study, we developed and evaluated two enhanced implementations of the HLC algorithm to address challenges in embedding PPI networks. Our results show that clustering the network before embedding improves pathway representation by increasing coherence—meaning that functionally related genes are grouped more closely together in the embedding space, making it easier to identify biologically meaningful relationships. These results

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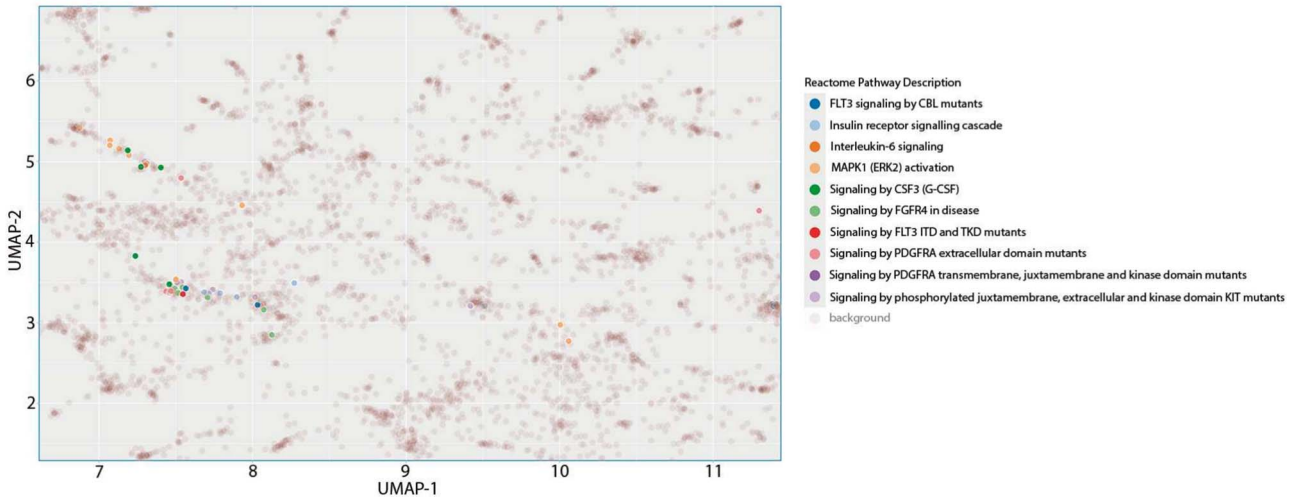

Figure 8. UMAP representation of the embeddings generated for huST700. Various Reactome biological pathways (with an internal edge density > 0.75) to which RASopathy associated genes belong are shown in different colors.

emphasize the value of incorporating community-aware strategies into network representation learning, as they enhance the ability to capture biologically relevant structures.

To address the computational limitations of HLC when applied to large biological networks, we introduced two novel implementations: HLCnp in Python and LKCMsp in R. LKCMsp adheres more closely to the original Ahn formulation while improving execution time and memory efficiency. HLCnp, integrated with CDLIB, extends functionality by optimizing dense network processing, providing greater flexibility in clustering, and enabling efficient filtering of the Tanimoto distance matrix. Benchmarking these tools on diverse weighted networks, including large biological datasets and dense graphs like Wikipedia, demonstrated their scalability and effectiveness.

We used our HLC implementation to enhance the embedding of the human proteome network. Among the tested approaches, integrating HLC clusters yielded the best performance (Fig. 4A and B). In contrast, the Louvain algorithm performed poorly, likely due to distortions in its embedding space (Supplementary Fig. S7). (Supplementary Fig. S7). More specifically, Louvain tends to identify a few large clusters, which can artificially separate functionally related genes if a Reactome pathway is spread across multiple clusters. Since the embedding process preserves local structures within clusters but not necessarily between them, nodes assigned to different Louvain clusters are distributed in isolated regions of the embedding space. As a result, the distances between pathway members increase, disrupting biological coherence and making it harder to capture meaningful functional relationships. In addition, within the clusters identified by Louvain, the algorithm was unable to reflect any meaningful internal structure, instead displaying a homogeneous distribution pattern.

For example, in UMAP analyses of the top five largest pathways (Fig. 5), the HLC method clearly differentiates pathways such as "Formation of a pool of free 40S subunits," "Mitochondrial translation elongation," and "L13a-mediated translational silencing of Ceruloplasmin expression," whereas the Louvain method groups them into a single cluster. This pattern persists when representing the full set of the top ten largest pathways (Fig. 6), where additional related pathways appear connected in the HLC embedding but remain indistinguishable in the Louvain embedding. Notably, in the "Proteome assembly" pathway, although members cluster together with both methods, the HLC algorithm positions them closer to the previously described pathways, indicating that Louvain clustering distorts the embedding space and fails to accurately capture the relationships between biological pathways.

Furthermore, while both methods grouped protein degradation pathways (e.g. proteasome-related) in close proximity, only HLC positioned them near protein synthesis processes, accurately reflecting their functional relationship. In contrast, Louvain left them isolated (Fig. 6). These findings illustrate how biological insights can be extracted from the UMAP representation of network embeddings.

Interestingly, Louvain clustering appears to perform worse than no clustering at all, which is in contrast with previous findings, such as those reported in CRARE [21]. To better understand this discrepancy, future work should explore different clustering parameters for embedding across a range of networks. Reducing the weight assigned to clustering integration might help mitigate distortions in the embedding space. In fact, a similar pattern was observed in CRARE, where performance deteriorated when a high weight was given to clustering. Furthermore, testing alternative clustering methods could provide additional insights into whether the issue arises from the Louvain algorithm itself or from the specific way it interacts with the embedding process.

These findings demonstrate that HLC produces a biologically meaningful organization of pathways, highlighting the value of overlapping cluster structures in complex biological networks. Consequently, this improved embedding could enhance deep learning applications in genomics and proteomics, where embedding serves as an essential preprocessing step [6, 12–14]. Moreover, the embedding representation itself holds significant potential, as demonstrated in this study. Researchers can map genes of interest onto the embedding space, explore their connectivity to biological pathways, and interpret their functional roles within this context.

In benchmark comparisons with other node embedding methods, our strategy consistently outperformed the alternatives, with node2vec [5] and DeepWalk [11] ranking as the next best performers. Interestingly, Role2Vec [28] exhibited substantial variability in its results. This may be due to the heterogeneous nature of biological pathways, with some of them having common structural properties that are more favorable to the Role2Vec methodology.

The results of the application of our method to RASopathy related genes found multiple novel candidate genes that may contribute to the pathogenesis of RASopathies. Several of these

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genes were found across multiple RASopathies, highlighting a significant overlap in the underlying biological processes shared by these syndromes.. The identification of RASA2, a gene already implicated in Noonan syndrome-like phenotypes, supports the algorithm's validity, while the discovery of additional modulators of Ras signaling such as RAP1B, RASGRF1, and others opens new avenues for research into the molecular mechanisms underlying these disorders. Moreover, the recognition of druggable targets such as ALK and PIK3CD shows the potential for identifying existing targeted therapies for repurposing in RASopathy contexts, although efforts to validate these candidate genes through functional assays and clinical studies are essential to translate these findings into effective therapeutic strategies.

While our approach demonstrates strong performance across diverse networks, several limitations should be acknowledged. First, the accuracy of biological interpretation depends on the quality and completeness of the input PPI data, which may carry biases due to underrepresented or poorly annotated proteins [61, 62]. Second, although our method performs well on the human proteome network, its generalizability to other species or to highly sparse or noisy biological graphs remains to be evaluated.

Future work will focus on extending our approach to multi-omics integration, enabling embeddings that simultaneously capture transcriptomic, epigenomic, and metabolomic relationships. Moreover, applying these methods across a broader range of diseases and biological systems will help to assess the robustness and transferability of our framework. Finally, deeper validation of candidate gene predictions through experimental and clinical collaborations will be essential to fully realize the translational potential of our approach.

In conclusion, our work underscores the importance of overlapping clustering in biological networks and introduces computationally efficient tools to enhance graph embeddings. By leveraging community-aware embeddings, we establish a robust framework for uncovering biologically meaningful insights, particularly in pathway and disease analysis within human PPI networks.

Key Points

- Optimized Edge-Based Clustering for Large Networks.We introduce HLCnp (Python/CDLIB) and LKCMsp (R/linkcomm), optimized implementations of the Hierarchical Link Clustering (HLC) algorithm. These tools enhance efficiency and accuracy in clustering large, weighted, undirected networks, incorporating complete linkage and Tanimoto filtering for improved performance.
- Enhanced Graph Embedding via Community-Constrained Random Walks. By integrating overlapping clustering with network embedding, our approach improves pathway representation in the human interactome. Constraining random walks to pre-computed communities improve embedding space with respect to biological patwhay.
- Application to RASopathies for Gene Prioritization. We apply our method to multiple RASopathies, including Noonan and Costello syndrome, improving the visualization of gene associations and prioritizing novel causal genes and additional pathway-related genes.
- Accessible Implementations and Benchmarking We provide open-source tools and benchmarking workflows, enabling researchers to apply and extend our methods in diverse biological and biomedical network studies.

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Author contributions

J.A.G.R., F.G.C., J.R.P., and P.S. conceived the methodology. F.G.C., P.S., and J.R.P. developed the software that implements the protocol. J.A.G.R., F.G.C., J.R.P., P.S., and E.R. analyzed the results and provided interpretation. P.S.Z., F.G.C., J.R.P., and E.R. wrote the manuscript. J.A.G.R., J.R.P., and P.S. were involved in planning of the study, contributed to the acquisition of research funding and headed the project. All authors read and approved the final version of the manuscript.

Supplementary data

Supplementary data is available at *Briefings in Bioinformatics* online.

Conflict of interest: No competing interest is declared.

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Data availability

HLCnp is available in the CDLIB library (https://github.com/GiulioRossetti/cdlib), and LKCMsp can be accessed at https://github.com/jimrperkins/linkcomm. Workflow with the benchmarking process and datasets corresponding to HLC implementations and network embedding are available at https://github.com/seoanezonjic/network_hlc_benchmark.git and https://github.com/federedef/embedding_hlc_benchmark.

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Case Study

Supplementary tables

Table 1. Classification of RASopathy-related disorders based on MONDO ontology, showing the relationships between general disorder categories and their specific subtypes.

General Disorder Category	Specific Subtypes
Costello syndrome	Costello syndrome
Legius syndrome	Legius syndrome
Noonan syndrome with multiple lentigines	LEOPARD syndrome 2
Noonan syndrome with multiple lentigines	LEOPARD syndrome 3
Noonan syndrome with multiple lentigines	LEOPARD syndrome 1
RASopathy	RASopathy
CBL-related disorder	CBL-related disorder
cardiofaciocutaneous syndrome	cardiofaciocutaneous syndrome 1
cardiofaciocutaneous syndrome	cardiofaciocutaneous syndrome 2
cardiofaciocutaneous syndrome	cardiofaciocutaneous syndrome 3
cardiofaciocutaneous syndrome	cardiofaciocutaneous syndrome 4
neurofibromatosis type 1	neurofibromatosis, familial spinal
neurofibromatosis type 1	chromosome 17q11.2 deletion syndrome, 1.4Mb
neurofibromatosis type 1	neurofibromatosis type 1 due to NF1 mutation or intragenic deletion
Noonan syndrome	Noonan syndrome 1
Noonan syndrome	Noonan syndrome 2
Noonan syndrome	Noonan syndrome 3
Noonan syndrome	Noonan syndrome 4
Noonan syndrome	Noonan syndrome 5
Noonan syndrome	Noonan syndrome 6
Noonan syndrome	Noonan syndrome 7
Noonan syndrome	Noonan syndrome 8
Noonan syndrome	Noonan syndrome 9
Noonan syndrome	Noonan syndrome 10
Noonan syndrome	Noonan syndrome 14
Noonan syndrome	Noonan syndrome 11
Noonan syndrome	noonan syndrome 12
Noonan syndrome	Noonan syndrome 13
Watson syndrome	Watson syndrome
Noonan syndrome-like disorder with loose anagen hair	Noonan syndrome-like disorder with loose anagen hair 2
Noonan syndrome-like disorder with loose anagen hair	Noonan syndrome-like disorder with loose anagen hair 1

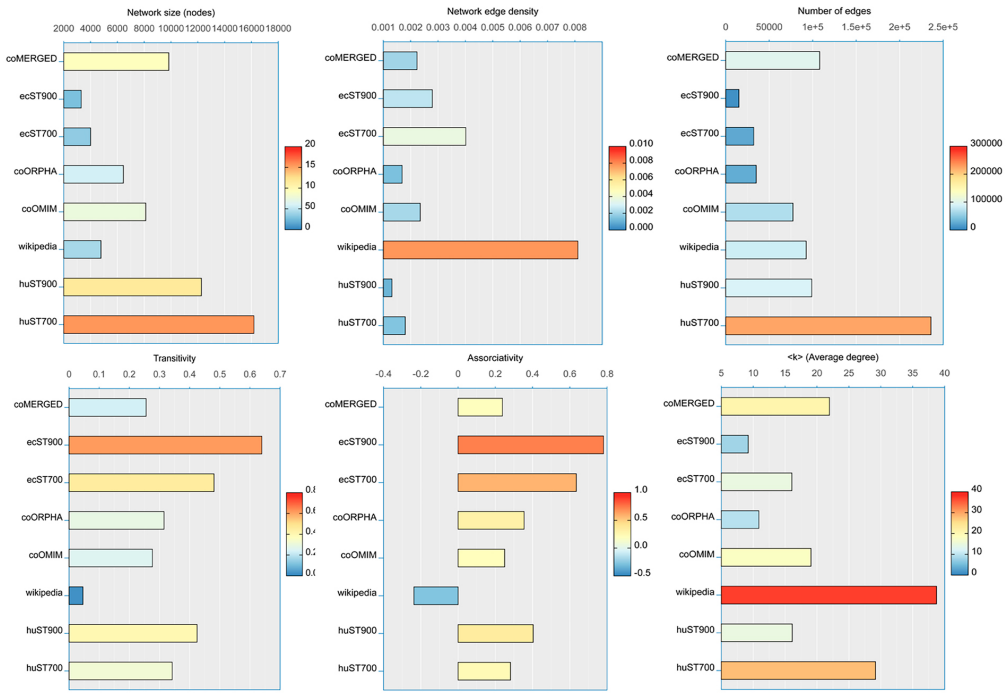
Table 2. Top 20 prioritized genes for every RASopathy disorder in huST700.

Disorder	Associated Genes
CBL-related disorder	GAB1, SHC2, EGFR, INPPL1, AXL, UBASH3A, SH3GL3, SHC4, CRKL, GAB2, SH3GLB2, SHC3, GRB2, SHC1, SH2B2, SPRY1
Costello syndrome	KRAS, RRAS2, RASA4, RASA2, MRAS, SPRED1, RGL1, RASGRP1, RASA3, RASGRP2, RAPGEF5, RASAL2, NRAS, RALGDS, RGL3, KSR1, RASAL1, RASGRP4
Legius syndrome	KRAS, RRAS2, RASA4, RASA2, RGL1, RASGRP1, RASA3, RASGRP2, RAPGEF5, SPRED2, RASAL2, NRAS, EGFL7, RGL3, KSR1, HRAS, RASAL1, YES1, RASGRP4
Noonan syndrome	RRAS, DAB2IP, RASGRF2, RALB, RASA2, RGL4, SPRED1, RCE1, RASA3, NF1, SIPA1L2, CNKSR2, KSR2, BRAP, RASGRF1, HRAS, RGL2
Noonan syndrome-like disorder with loose anagen hair	PPP1R13B, C20orf27, PPP1R37, CSRNP2, SH2D4A, PPP1R1C, PPP1R16B, PPP1R14A, PPP1CA, PPP1CC, TP53BP2, PPP1R15B, PPP1R10, C6orf62, SAMD14, RASSF5, TOX4, PPP1R9B
RASopathy	PIK3R2, RAP1B, RAP1GAP2, PIK3R3, MAPK7, SKAP1, ICMT, PDPK1, ANKS1A, RAP1A, VAV2, ALK, MAP3K2, SHC1, PRKCA, THUMPD2, PIK3R1, YES1
Watson syndrome	EVI2A, KRAS, RRAS2, RASA4, RASA2, MRAS, SPRED1, RGL1, EVI2B, RASA3, RASSF1, RASAL2, NRAS, KSR1, BRAP, HRAS, RASAL1, RASGRP4
Cardiofaciocutaneous syndrome	ARHGEF6, RAP1GAP2, RRAS2, MRAS, ARAF, SPRED1, RGL1, RASA3, NRAS, RAP1A, RALGDS, CNKSR2, KSR1, MAP3K1, MAP3K2, KSR2, HRAS, YES1
Neurofibromatosis type 1	EVI2A, KRAS, RRAS2, RASA4, RASA2, MRAS, SPRED1, RGL1, EVI2B, RASA3, RASSF1, RASAL2, NRAS, KSR1, BRAP, HRAS, RASAL1, RASGRP4

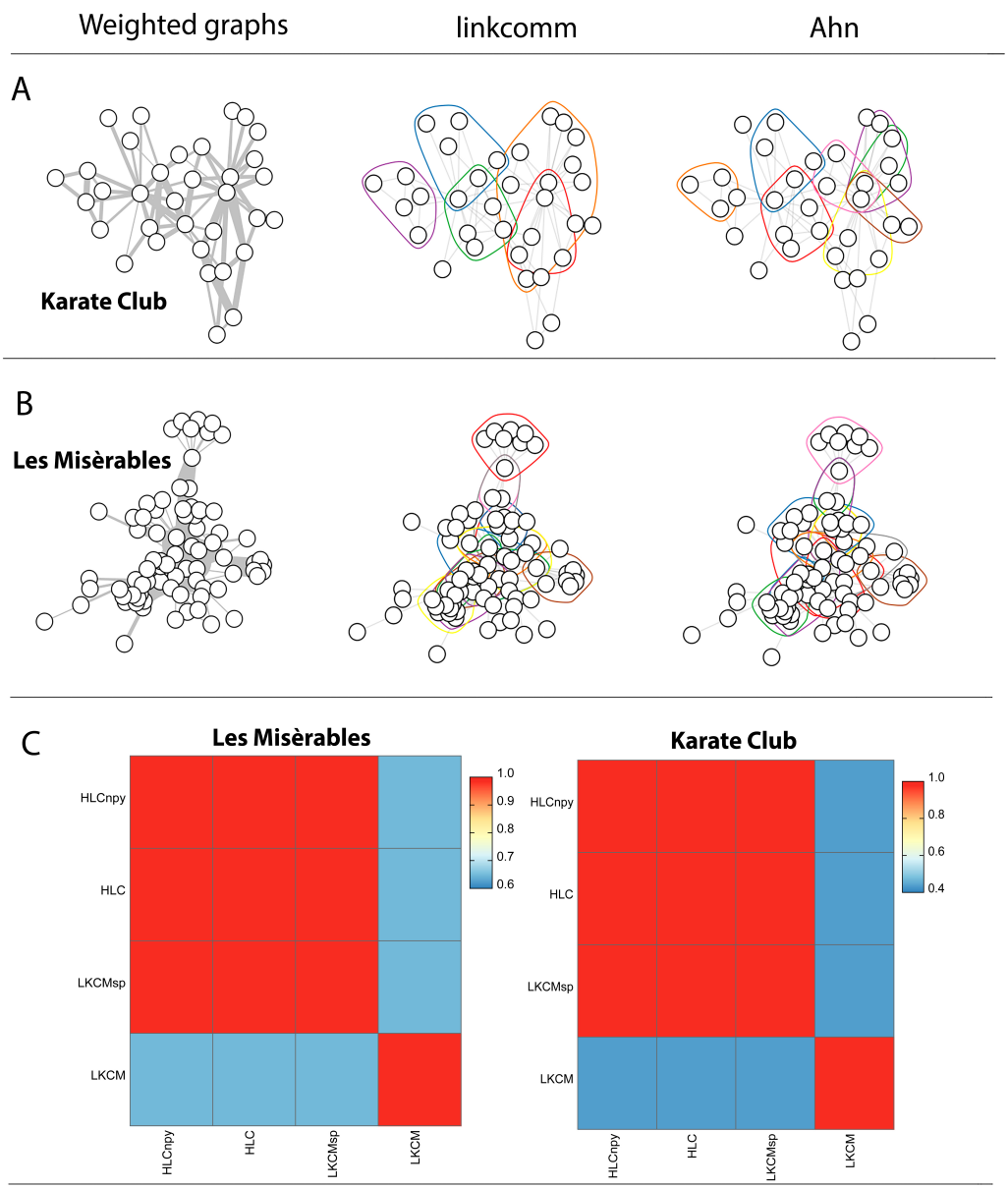
Table 3. Top 20 prioritized genes for every RASopathy disorder in huST900.

Disorder	Associated Genes
CBL-related disorder	BCR, BLNK, CD320, FYN, GAB2, GRAP2, GRB2, LCP2, SH3KBP1, SHC2, SHC4, SORBS2, SPRY1, SPRY2, TCN1, TCN2, UBASH3A, UBASH3B
Costello syndrome	BRAP, KRAS, MRAS, NF1, NRAS, RALGDS, RAPGEF2, RAPGEF5, RASA2, RASGRP1, RASGRP2, RASGRP4, RGL1, RGL2, RIN1, RRAS, RRAS2, SOS2
Legius syndrome	ARAF, BRAP, HRAS, KRAS, MRAS, NF1, NRAS, OSBPL3, RALGDS, RAPGEF2, RAPGEF5, RASA2, RASGRP4, RASSF1, RGL2, RRAS, RRAS2
Noonan syndrome	BRAP, HRAS, MAP3K1, NF1, OSBPL3, RALGDS, RAPGEF2, RAPGEF5, RASA2, RASGRP1, RASGRP2, RASGRP4, RASSF1, RGL1, RGL2, RIN1
Noonan syndrome-like disorder with loose anagen hair	PASK, PPP1CA, PPP1CC, PPP1R10, PPP1R11, PPP1R12C, PPP1R14A, PPP1R15B, PPP1R16A, PPP1R16B, PPP1R2, PPP1R3F, PPP1R7, PPP1R8, PPP1R9B, SAMD14, TOX4, TP53BP2
RASopathy	ALK, ARHGEF12, BRAP, CNKSR1, CSF2RB, EML4, GAB2, IL3RA, KIAA1549, OSBPL3, PIK3CD, RAP1B, RAPGEF5, RASA2, RASGRF1, RASGRF2, SPRY2
Watson syndrome	BRAP, HRAS, KRAS, MRAS, NRAS, OSBPL3, RALGDS, RAPGEF2, RAPGEF5, RASA2, RASGRP4, RASSF1, RGL1, RGL2, RIN1, RRAS, RRAS2, SPRED1
Cardiofaciocutaneous syndrome	ARAF, ARHGEF6, BRAP, HRAS, KSR1, KSR2, MAP3K1, MRAS, NF1, NRAS, RALGDS, RAPGEF5, RASA2, RASGRP2, RASGRP4, RRAS, RRAS2
Neurofibromatosis type 1	BRAP, HRAS, KRAS, MRAS, NRAS, OSBPL3, RALGDS, RAPGEF2, RAPGEF5, RASA2, RASGRP4, RASSF1, RGL1, RGL2, RIN1, RRAS, RRAS2, SPRED1

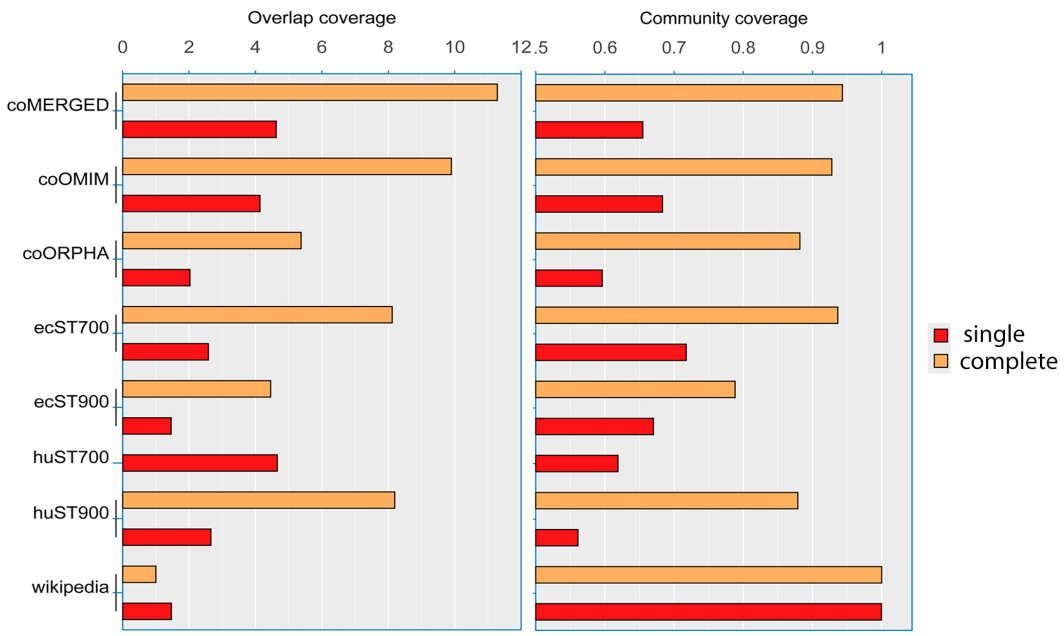
Supplementary figures



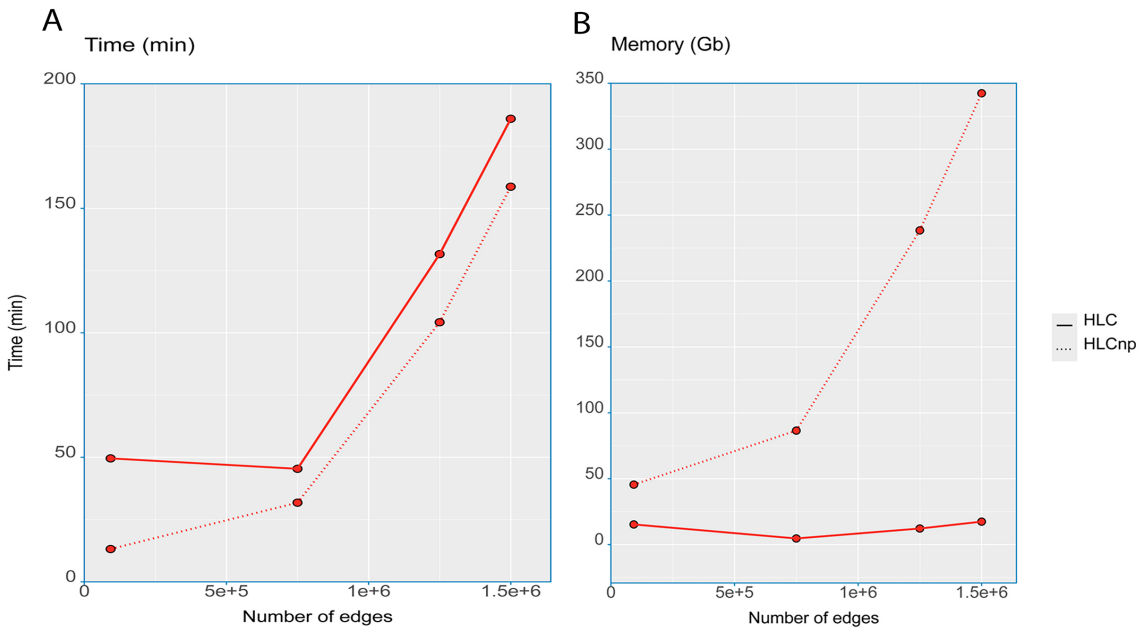
Supplementary Figure 1. Main network characteristics for all considered networks for technical performance analysis: number of nodes, edge density, number of edges, transitivity (global clustering), assortativity and mean degree of nodes.



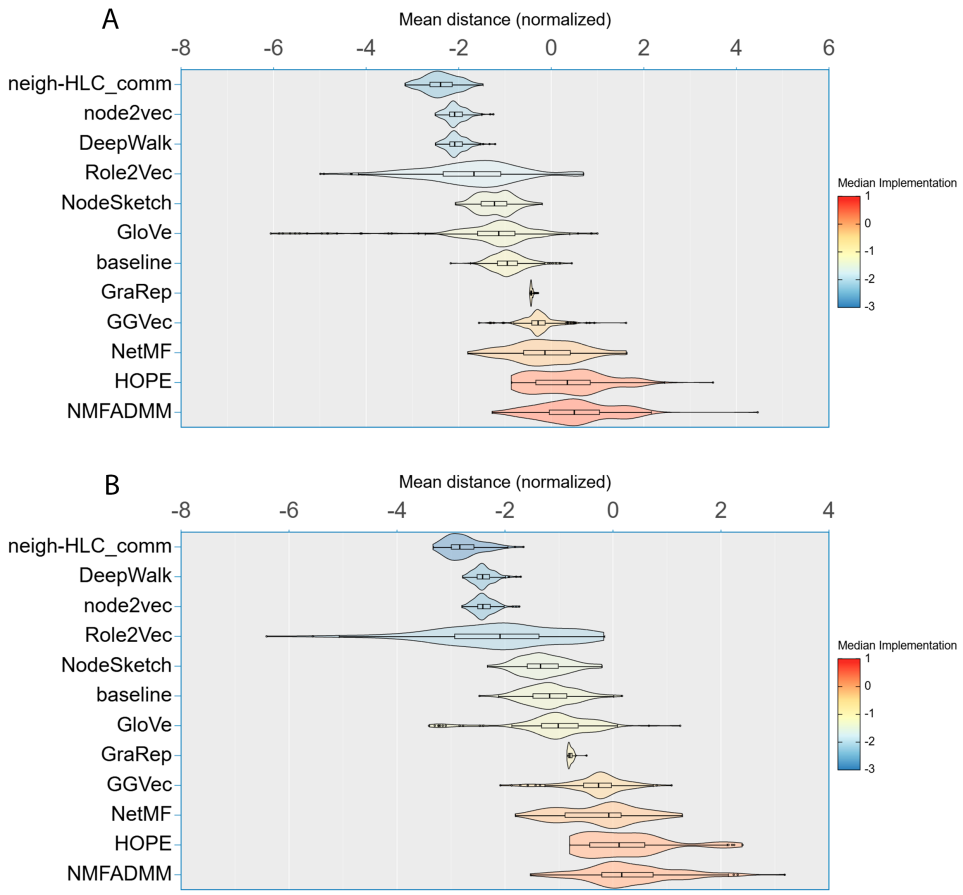
Supplementary Figure 2. Visual quantitative comparative between the Ahn (LKCMsp, HLC and HLCnp) and linkcomm formulations (LKCM). In A) and B), communities obtained are presented for Karate Club and Les Miserables datasets, respectively. Finally, an extension of the Normalized Mutual Information (NMI) score for overlapping clustering is used to compute similarity between implementations. Graph plotting was performed using the iGraph library and mutual information with CDLIB evaluation module.



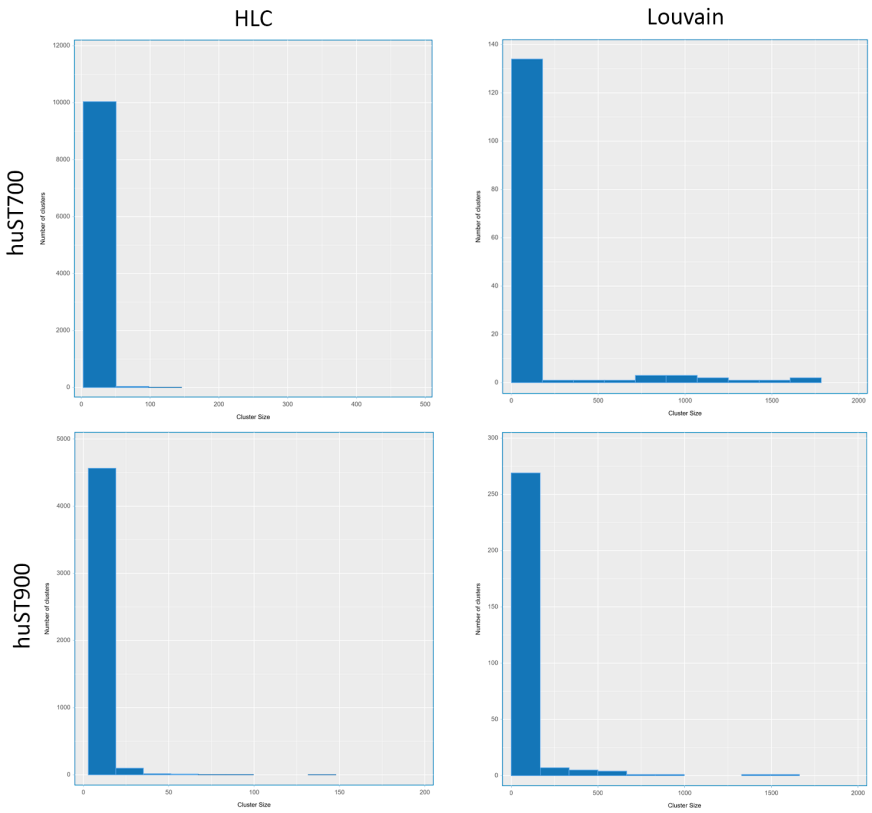
Supplementary Figure 3. Community coverage and overlap coverage difference between HLCnp with single and complete linkage.



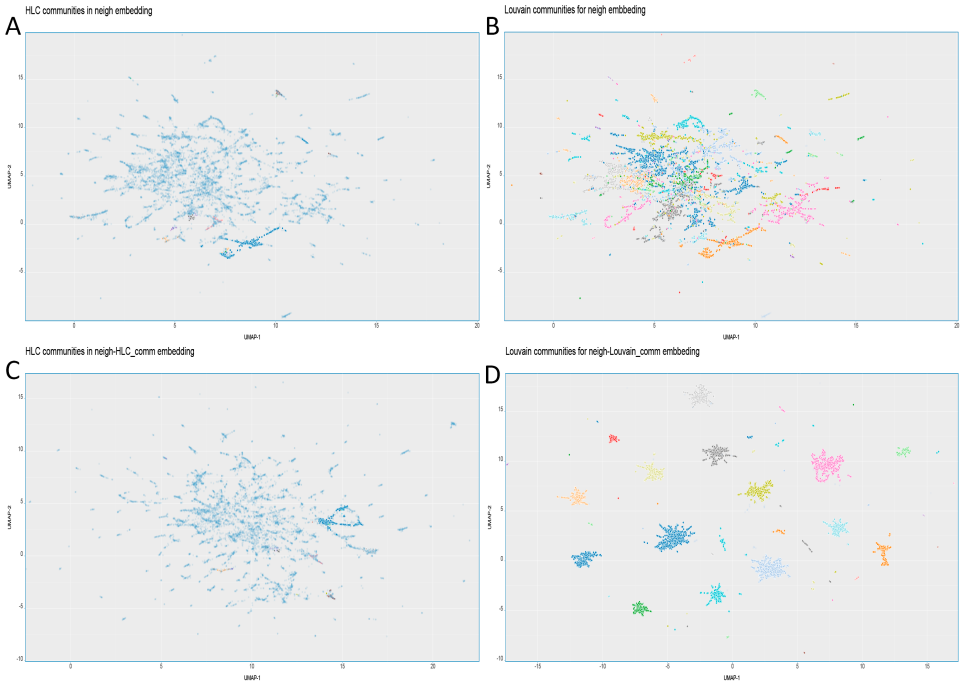
Supplementary Figure 4. Execution time (A) and memory usage (B) on dense networks. Graphs used for the analysis were the Wikipedia graph and random graphs with a density value of 0.06, 0.1, 0.12. Single linkage was applied during the hierarchical clustering. Random graphs were obtained by node size and density constraint with NetworkX and weights were added following a normal distribution.



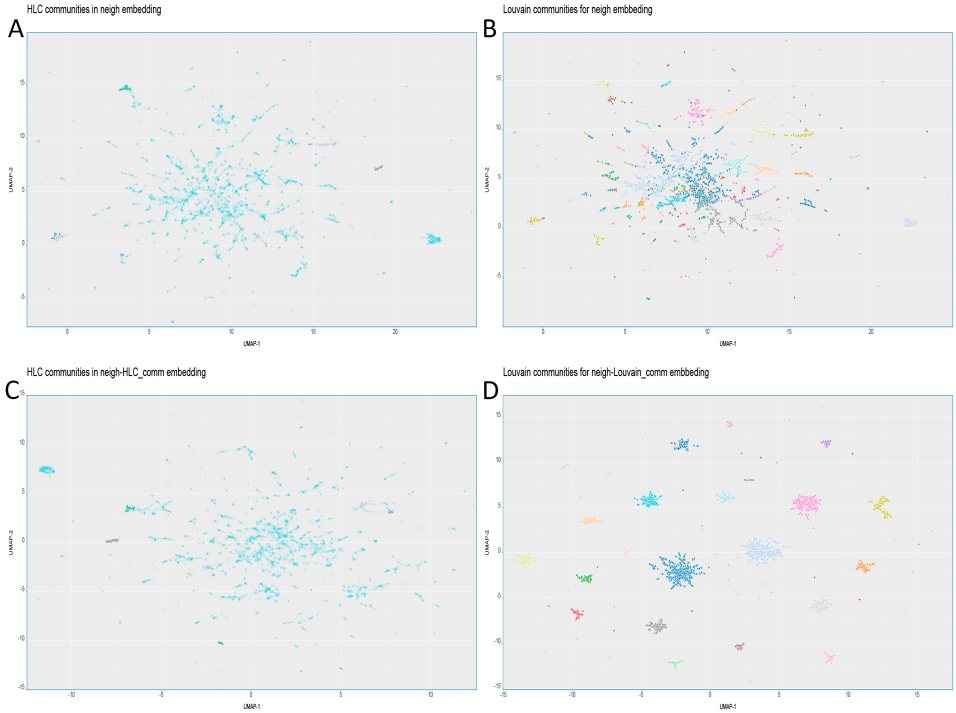
Supplementary Figure 5. Embedding results using the neigh-HLC_comm approach, baseline (random selection of edges) and different predefined node embeddings. (A, B) Distributions of mean z-scores for distances between nodes within the same high-cohesion pathway (edge density > 0.75) in huST700 and huST900, respectively.



Supplementary Figure 6. Cluster size distribution for each clustering method used (HLC or Louvain) and each dataset analyzed (huST700 and huST900).



Supplementary Figure 7. UMAP representation of the embeddings generated for huST700 is presented with the top 20 largest communities from HLC and Louvain. In panels A and B, representation of communities from HLC and Louvain are represented in the neigh embedding space. In panels C and D, representation of HLC and Louvain communities are shown in neigh-HLC_comm and neigh-Louvain_comm embeddings, respectively.



Supplementary Figure 8. UMAP representation of the embeddings generated for huST900 is presented with the top 20 largest communities from HLC and Louvain. In panels A and B, representation of communities from HLC and Louvain are represented in the neigh embedding space. In panels C and D, representation of HLC and Louvain communities are shown in neigh-HLC_comm and neigh-Louvain_comm embeddings, respectively.

Parte V

Exploración de enfermedades raras mediante técnicas de análisis de redes.

Capítulo 5

Análisis de factores comunes entre las distintas enfermedades neuromusculares

Las enfermedades neuromusculares constituyen un grupo heterogéneo de trastornos poco frecuentes que afectan de manera significativa a la función neuromotora. Su amplia variabilidad clínica y genética dificulta el estudio sistemático, lo que plantea un desafío relevante en el ámbito biomédico a la hora de integrar el conocimiento disponible. En este contexto, resulta de gran interés el empleo de sistemas capaces de integrar distintos tipos de información, incluyendo tanto información de tipo relacional, como similitud o interacción entre las enfermedades, como datos textuales o de expresión molecular.

Con este propósito, en el presente capítulo se desarrolla un análisis desde dos perspectivas complementarias. En primer lugar, se examina la similitud entre enfermedades neuromusculares a partir de enfoques ontológicos y textuales, con el fin de establecer relaciones globales dentro de este grupo de patologías. En segundo lugar, se aborda un estudio a nivel molecular centrado en tres enfermedades neuromusculares con características similares: distrofia muscular de Duchenne (DMD,por sus siglas en inglés), distrofia muscular de cinturas (LGMD, por sus siglas en inglés) y esclerosis lateral amiotrófica (ALS, por sus siglas en inglés). Estas patologías se caracterizan por mecanismos moleculares complejos. Por ello, se combina el análisis de expresión génica con el estudio de redes de interacción proteína-proteína, lo que permite profundizar en los mecanismos subyacentes.

5.1. Integración de modelos de lenguaje y representaciones de redes para descubrir relaciones ocultas en enfermedades neuromusculares.

Mi contribución:

En el marco del I Congreso de la Sociedad Española de Inteligencia Artificial en Biomedicina (CIABiomed), publiqué como primer autor un trabajo en la revista *Lecture Notes in Bioinformatics* (LNBI). Mi contribución en este estudio se centró en el análisis de 368 enfermedades neuromusculares (NMDs) previamente caracterizadas.

Para ello, desarrollé un enfoque basado en medidas de similitud textual entre enfermedades, utilizando el conjunto de artículos científicos disponibles en PubMed. Dichas similitudes se calcularon a partir de modelos de lenguaje implementados en el laboratorio y aplicados específicamente a las enfermedades neuromusculares registradas en la ontología MONDO. El mismo procedimiento lo extendí a los fenotipos, codificados según la *Human Phenotype Ontology* (HPO).

A partir de esta información, construí diferentes redes de similitud enfermedad–enfermedad, definidas en función de genes, artículos o fenotipos compartidos, así como de la similitud ontológica dentro de MONDO. Sobre estas redes evalué distintos sistemas de embebido de nodos, concretamente *node2vec*, *ggvec*, *prone* y *glove*. Cada uno de ellos fue comparado en términos de la correlación de *Spearman* entre la similitud de coseno en el espacio de embebido y la similitud original dentro de la ontología MONDO. Tras seleccionar el embebido con mejor rendimiento, lo apliqué tanto a la detección de comunidades de enfermedades neuromusculares mediante el algoritmo *HDBSCAN* como a su visualización en un espacio bidimensional utilizando *UMAP*.

Finalmente, sobre las comunidades resultantes, realicé un análisis de enriquecimiento funcional de los genes asociados y un estudio de los fenotipos comunes. Esta aproximación me permitió identificar posibles mecanismos compartidos en la etiología de las enfermedades, así como patrones de sintomatologías recurrentes entre ellas.

Integrating Language Models and Network Embeddings to Uncover Hidden Relationships in Neuromuscular Diseases

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Abstract. Neuromuscular diseases (NMDs) are a heterogeneous group of rare disorders that significantly impair motor function and quality of life. Their clinical and genetic variability makes systematic study and knowledge integration particularly challenging. We present a scalable, automated framework that combines natural language processing and network analysis to uncover hidden relationships among NMDs. Using Sentence Transformer models, we identified NMDs and their phenotypes in open-access documents. Then, we use this data to build NMD networks to infer new relations using embedding techniques and clustering approaches structuring a document corpus for 328 NMDs basis on disease type and associated to phenotypes and genes. These findings show the value of combining language models and network embeddings for large-scale rare disease analysis.

Keywords: Neuromuscular diseases · language models · network embeddings · translational bioinformatics · rare diseases

1 Introduction

Neuromuscular diseases (NMDs) are a diverse group of disorders that impair the function of muscles and the peripheral nerves controlling them, leading to progressive muscle weakness, motor dysfunction, and substantial disability [21, 12]. Uncovering the molecular and genetic relationships shared among different NMDs offers a valuable opportunity to advance diagnostic and therapeutic strategies [21]. To extract meaningful information about NMDs, it is essential to

recognize that the biomedical field has accumulated a large body of knowledge over decades of research [6]. However, a large portion of this information resides in unstructured formats, primarily within free-text scientific publications, making it inaccessible to programmatic querying and large-scale computational analysis [9]. To address this challenge, the use of standardized vocabularies and biomedical ontologies has become increasingly important. Ontologies are formal, structured representations of domain knowledge that define entities and their relationships, enabling logical inference and semantic interoperability across datasets [5, 1]. Despite their utility, much of the knowledge contained in the literature is not explicitly expressed using ontology-aligned terminology. Therefore, it is necessary to develop methods that map free-text expressions to standardized ontological concepts, bridging the gap between unstructured information and structured, machine-readable representations.

Recent advances in natural language processing (NLP), particularly the emergence of large language models (LLMs), have dramatically improved the ability to extract structured knowledge from unstructured text [19]. Models based on the Bidirectional Encoder Representations from Transformers (BERT) architecture [4] have demonstrated remarkable performance in tasks such as named entity recognition (NER), enabling automated identification and classification of entities within unstructured text, including names of individuals, locations, biological entities and ontological terms [3]. Applying such models to literature related to NMDs allows exploit all available text data.

Beyond entity extraction, representing these concepts and their relationships in the form of networks offers powerful systems-level insights. Recent developments in network embedding techniques, which transform high-dimensional network structures into compact vector representations, have facilitated the discovery of latent topological patterns and biologically meaningful clusters. These embeddings enable more accurate clustering, gene prioritization, and pathway inference by capturing complex topological relationships, thereby enhancing the identification of functionally relevant modules and improving our ability to detect potential biomarkers and therapeutic targets within a systems-level framework [2, 13].

In this study, we focus on the set of NMDs defined by the MONDO ontology, developed by the Monarch Initiative [20]. For each disease term, we retrieve related scientific literature using semantic search powered by Sentence Transformers. We then apply the same models to extract phenotypic descriptions from the identified documents, mapping them to concepts in the Human Phenotype Ontology (HPO) [7]. The resulting phenotype-disease associations are used to construct disease similarity networks, which are subsequently embedded into low-dimensional spaces using algorithms such as node2vec [8]. These embeddings allow us to uncover hidden relationships between NMDs and reveal potential shared molecular and genetic underpinnings.

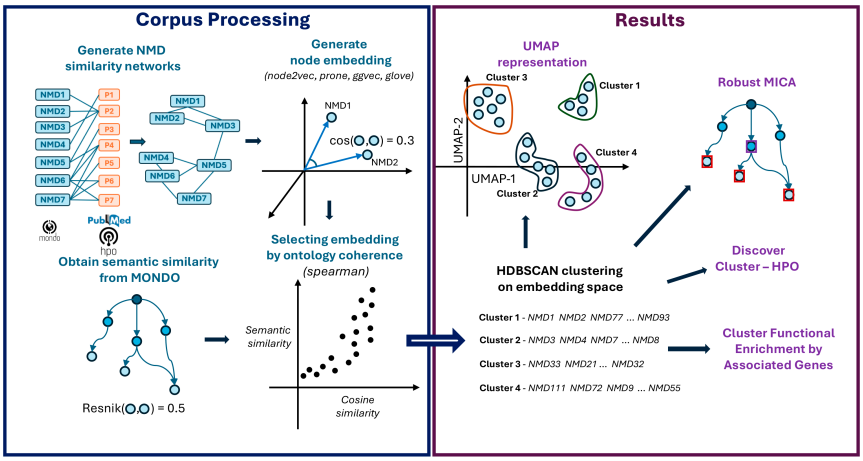

Fig. 1. Diagram of the corpus generation process based on neuromuscular diseases (NMDs) and the generated results from the communities identified within the selected embedding space.

2 Materials and Methods

2.1 Identify neuromuscular diseases in full-text articles

The text corpus chosen for this work was composed of all full-text articles available in PubMed Central. Articles were downloaded via the National Center for Biotechnology Information (NCBI) FTP service (https://ftp.ncbi.nlm.nih.gov), from the "/pub/pmc/oa bulk" directory, including documents from the "oa comm", "oa noncomm" and "oa other" folders, encompassing both baseline and incremental files. This yielded 6,132,543 articles, which were further processed by hierarchically splitting them into paragraphs and then into sentences, using full stops and commas as delimiters. The text queries consisted of disease terms from the MONDO ontology [20] and phenotype terms from the HPO ontology [7]. For diseases, we extracted all child terms of "neuromuscular disease" (MONDO:0019056) and retained only the leaf terms (those without further descendants) resulting in 755 terms. And for phenotypes, all descendants of "Phenotypic abnormality" (HP:0000118) were retrieved, totaling 17,957 terms. For both diseases and phenotypes, their names and available synonyms were collected for each term.

Both the query and corpus texts were processed using the embedding model "all-mpnet-base-v2" from the Sentence Transformers library (v3.2) [18] (https://huggingface.co/sentence-transformers/all-mpnet-base-v2), a diagram of this process is shown in Figure 1, This model generates high-dimensional vector embeddings that capture the semantic content of text, enabling the recognition

and quantification of semantic similarity between text fragments. After embedding, all query–document fragment pairs were compared using cosine similarity, retaining only those with a score ≥ 0.7. Articles with fewer than two links to a given disease were excluded from further analysis. Finally, the ontology distribution of the annotated terms was evaluated using the py_semtools package (Pérez-García *et al.*, in press) to assess the quality of the text mining process.

2.2 Disease-disease network building and embedding

We built two bipartite networks linking diseases to phenotypes and to articles. Network analyses were conducted using our py_netanalyzer Python library [11]. The disease–article network was generated from the previously described connections. To build the disease–phenotype network, we first created a tripartite network combining disease–article and article–phenotype associations. Diseases were represented by MONDO terms, phenotypes by HPO terms, and articles by their PubMed IDs (PMIDs). HPO terms identified in NMD articles were assigned to the corresponding MONDO terms annotated in the same articles to weight disease-phenotype (or MONDO-HPO) associations. For this, we applied a Simpson projection to the tripartite disease–article–phenotype network, retaining only MONDO–HPO relations with a Simpson similarity 0.9. In this way, we obtain a directly inferred and a weighted MONDO-HPO networks.

Next, since the MONDO ontology is a directed acyclic graph (DAG), we transformed it into a disease–disease network to serve as a reference for evaluating information preservation in subsequent analyses. This network was filtered to include only "neuromuscular disease" (MONDO:0019056) and its descendants. The resulting network, termed the "dis_ont" network, was used as both a control and reference throughout our experiments.

All the bipartite networks described previously were converted in disease-disease networks using different algorithms. The relations MONDO-article are projected using Jaccard but all associations not supported by at least two MONDO terms are removed, we name it as the "dis_text" network. The same process was applied for the weighted MONDO-HPO, generating the network "dis_phen_jacc". Since phenotypes are represented by HPO terms, we applied ontology-based semantic similarity to the disease–phenotype network to calculate phenotypic similarity between all NMDs based on their HPO annotations. We used a normalized version of the ERIC metric, which accounts for potential noise in HPO annotations [14]. The similarity matrix was then filtered with a threshold of 0.2 to generate the "dis_phen_sim" network.

The disease-disease networks were embedded using several state-of-the-art algorithms, including node2vec [8], Prone [22], GGVec, and GloVe [16]. For node2vec, we applied the following parameters: `window=5`, `num_walks=30`, `q=1`, `random_seed=69`, `walk_length=50`, `dimensions=128`. The Prone and GGVec algorithms were executed using their recommended default hyperparameters, as specified in their original publications. GloVe was run with a vector dimension of 128 to ensure comparability across embedding methods.

2.3 Clustering disease identification and analysis

Based on the coordinates of the selected embedding, clustering was performed using HDBSCAN [15]. This method is particularly suitable for detecting non-linear patterns in high-dimensional embedding spaces. The computation was carried out with the *scikit-learn* library, using the following parameters: `min_cluster_size=10, min_samples=10, metric='cosine', cluster_selection_method='leaf'`. Then, for each MONDO-based cluster, we computed the average shortest path based on the DAG structure of MONDO. This metric was used to assess the internal coherence of each cluster in relation to the ontology. To further evaluate whether the clusters aligned with specific NMD classifications, we defined a *robust* Maximum Informative Common Ancestor (MICA). This was determined by identifying all ancestral terms shared by at least 75% of the diseases within the cluster, ensuring resistance to small subsets of outlier terms with divergent ancestry. Among these shared terms, the one with the highest Resnik information content was selected as the *robust* MICA. A diagram of all the process is presented in Figure 1.

Two types of enrichment analyses were performed on the identified clusters. First, a phenotype-based enrichment was carried out using the MONDO–HPO associations previously obtained. A hypergeometric test was applied to determine which HPO terms were significantly over-represented within each set of MONDO terms. Second, a functional enrichment analysis was performed using the Biological Process category of the Gene Ontology (GO). Each MONDO term was linked to its associated genes, as reported by the Monarch Initiative [17]. For each cluster, enrichment analysis was conducted using the *clusters_to_enrichment.R* script from the ExpHunter Suite tool developed by our group [10], applying the Benjamini–Hochberg false discovery rate (FDR-BH) procedure with an adjusted p-value threshold of 0.05.

3 Results and discussion

3.1 Assessing the neuromuscular diseases domain and its phenotypes

As an initial step, we evaluated the coherence and coverage of the Sentence Transformer–based annotations by analyzing the distribution of terms within the MONDO and Human Phenotype Ontology (HPO). This assessment aimed to validate the alignment between the literature-derived annotations and established biomedical ontologies. In the case of NMDs inferred from the articles (Figure 2A), the annotation frequency was notably low. This reduction is explained by the wide distribution of disease annotations across a large number of articles (382,519 in total) relative to the 743 MONDO terms considered. This can be explained by the focus of each article toward a specific NMD, in this way, few studies describe several diseases. In addition, there are NMD domains such as “akinetopsia" and “atrophic muscular disease" with very low annotations. Furthermore, the two domain diseases with the most representation are “hereditary

neuromuscular disease" and “peripheral neuropathy". However, when these papers are annotated to phenotypes (HPO, Figure 2B) we see that all phenotypes have noticeable representation. This is due to the fact that neuromuscular diseases are complex systemic diseases that affects several biological systems and causes very different pathological phenotypes. However, we can see that “abnormality of the musculoskeletal system", “abnomarlity of limbs", “abnormality of the nervous system" and “abnormality of the metabolism/homeostasis" have a large presentation with the first three categories clearly related to NMDs. Both phenomena, the focus at the article level and the overlap at the phenotype level, reflect the nature of current research practices, where diseases are often studied in isolation, one by one, even though they may share similar phenotypes. Importantly, the low annotation frequency also raises questions about the completeness and potential bias of the extracted data, as certain diseases may be systematically underrepresented or overlooked.

From the several networks that we have build, we measure which network and which embedding is the most appropriate to capture the relations between the NMDs. We use as reference the ontological semantic similarity between all NMD terms listed in the MONDO ontology and its Spearman correlation against their cosine similarity in the network embedding (Figure 2C, dis_ont row). We can see that this reference network has the second high correlation values (maximum in 0.32) whereas the dis_text network has the highest correlation values (maximum in 0.35). The phenotype networks have correlation values near to zero and for this reason we focus in the dis_text network. The limited performance of the phenotype networks may be attributed to the need for more precise associations between diseases and their corresponding phenotypes, as the current approach links all identified phenotypes to all diseases within a document. A potential strategy to address this limitation would be to filter out phenotypes with low information content or those that occur with high frequency across the entire NMD corpus, or alternatively to weight the associations between MONDO terms according to the information content of their phenotype intersections. With respect to the embedding methods tested, node2vec consistently achieved the highest performance values across the networks, and was therefore selected for subsequent analyses.

Finally, we take the dis_text network embedding and applied a UMAP representation and colored the diseases using the domain disease schema of Figure 2A. In Figure 3A we can see that diseases of each disease domain are grouped together in their own UMAP space and that the domains converge in a whole group. When a HDBSCAN clustering is applied to embedding coordinates of the diseases (Figure 3B) emerges 12 clusters and 402 diseases are not clustered (pink transparent points). As we can see, the unclustered diseases are placed in the center of the main group where in Figure 3A are placed in the convergence area of disease domains. The identified clusters range in size from 12 to 63 diseases (Table 1). When mapped onto the MONDO ontology, the diseases within each cluster are closely related, with an average shortest path length between

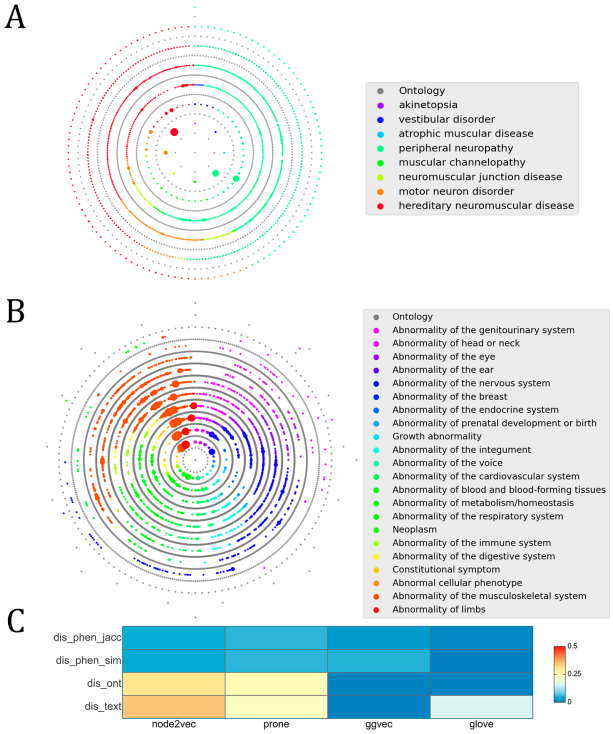

Fig. 2. Circular plots showing the distribution of MONDO term annotations. Inner rings correspond to higher-level ontology categories, while outer rings represent more specific terms. Point size indicates the frequency of each term within the dataset. Color represents a child of the selected root terms, “neuromuscular disease" to highlight disease domains or “phenotypical abnormality" in the case of phenotypes. The annotation counts are transferred from child to parent. Illustrated sets are articles associated to A) disease terms and B) phenotypes terms. C) Heatmap with correlation values obtained when ontological similarity and embedding cosine similarity are compared. Spearman values are show for each combination of network (rows) and embedding algorithm (columns).

2.48 and 4.85 hops. For reference, two disease variants typically have a distance of two hops: one to their common parent and another to reach the other variant.

Regarding the cluster distribution, we observe that “hereditary neuromuscular disease" is split into clusters 10 and 11, whose MICAs are “hereditary spastic paraplegia" and “muscular dystrophy", respectively, corresponding to subdivisions of the “hereditary neuromuscular disease" term in the MONDO ontology. For the “motor neuron disorder" domain, clusters 2 and 8 correspond to the “spinal muscular atrophy" and “familial amyotrophic lateral sclerosis" subdivisions. A special case is the heterogeneous domain “peripheral neuropathy", which

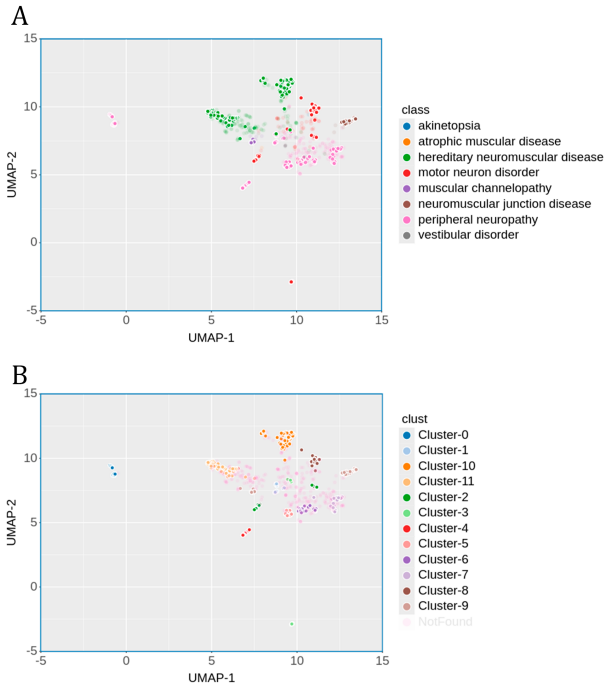

Fig. 3. UMAP representation based on the coordinates of the node2vec embedding over the dis_paper network. Each dot represents a MONDO of Neuromuscular Diseases. In A), color represent the classification based on the main 8 subdivision of NMD terms. In B), each color represent the clusters obtained with the HDBSCAN algorithm. Nodes without a clusters assigned were presented in a transparent tone. For UMAP dimensionality reduction we used the following parameters: `n_neighbors=30`, `min_dist=0.1`, `n_components=2`, `metric='euclidean'`, and `random_seed=123`

includes all diseases related to the peripheral nervous system and therefore encompasses clusters 0, 1, 4, 5, 6, and 7.

The cluster 1 has the general MICA “hereditary peripheral neuropathy" and clusters 5-6 have “peripheral neuropathy" and this is due to the heterogenous diseases grouped in these clusters. However, there are very specialized clusters like cluster 0 with “Zellweger spectrum disorders" that is related with peroxisome diseases, the cluster 4 with the MICA “sphingolipidosis" and the cluster 7 with the “Charcot-Marie-Tooth disease" MICA. This splitting explain the reason of the sparse distribution of the peripheral neuropathy and even explain why we have identified with high frequency phenotypes related with metabolism terms in figure 2B. Additionally, cluster 3 was associated with the parental category “coenzyme Q10 deficiency", further supporting the metabolic component underlying several of the identified phenotypes.

Table 1. Attributes for clusters obtained from dis_text network. Size is the number of diseases in the clusters. Avg_sht_path is the average shortest path calculated for each cluster in the MONDO ontology. MONDO parental name is the calculated MICA for diseases of the cluster.

Cluster	Size	Avg_sht_path	MONDO parental name
Cluster-0	14	4.15	Zellweger spectrum disorders
Cluster-1	17	3.36	hereditary peripheral neuropathy
Cluster-2	50	3.96	spinal muscular atrophy
Cluster-3	12	3.27	coenzyme Q10 deficiency
Cluster-4	12	4.28	sphingolipidosis
Cluster-5	25	4.08	peripheral neuropathy
Cluster-6	23	4.85	peripheral neuropathy
Cluster-7	63	3.51	Charcot-Marie-Tooth disease
Cluster-8	31	2.48	familial amyotrophic lateral sclerosis
Cluster-9	24	4.13	hereditary neuromuscular disease
Cluster-10	39	2.97	hereditary spastic paraplegia
Cluster-11	18	4.34	muscular dystrophy

In fact, we performed an enrichment analysis in each cluster using the identified phenotypes with the language model. Specifically, we employed the disease–phenotype pairs obtained during the bipartite network generation. A hypergeometric test was applied to assess enrichment within each cluster, and for illustration, we report the most significant HPO term for each cluster that is below a FDR cut-off of 0.05 (Table 2). We then highlight and discuss the most notable cases. We can se that cluster 8 has the phenotype “Amyotrophic lateral sclerosis" that corresponds with eh MONDO term “familial amyotrophic lateral sclerosis". In the case of cluster 0 that is related with peroxisome diseases, is connected to know phenotype “Aplasia/hypoplasia affecting bones of the axial skeleton". Cluster 2 corresponds with spinal muscular atrophy and has associated the “Delayed ability to roll over" phenotype. Cluster 10 is hereditary spastic paraplegia that has “pulverulent cataract", a feature observed in certain syndromic presentations. Finally, cluster 9 that corresponds with hereditary neuromuscular disease is connected with the phenotype “Periodic hyperkalemic paralysis".

As a final application, we performed gene enrichment analysis on the identified clusters and compared the results, as shown in Figure 4. Statistically significant categories are connected based on shared genes. Our primary goal was to demonstrate the functional coherence within each disease cluster. The enrichment results reveal distinct gene specializations and shared biological functions among diseases, providing strong evidence of the internal consistency and biological relevance of the identified clusters.

4 Conclusion

Our work demonstrates the use of language models and network embeddings to explore associations between genetic diseases. In a case study on NMDs, we

Table 2. HPO enrichment analysis with phenotype terms obtained from the text annotation procedure. A FDR cut-off of 0.05 was applied

Cluster_id	Phenotype	FDR
Cluster-8	Amyotrophic lateral sclerosis	3.55E-14
Cluster-4	Bullet-shaped hallux phalanx	7.49E-12
Cluster-0	Aplasia/hypoplasia affecting bones of the axial skeleton	9.92E-10
Cluster-1	Elevated circulating C6 acylcarnitine concentration	1.29E-08
Cluster-2	Delayed ability to roll over	7.39E-08
Cluster-11	Triangular epiphyses of the 3rd finger	8.87E-07
Cluster-3	Aplasia/Hypoplasia of the 4th finger	3.66E-06
Cluster-5	Triangular shaped proximal phalanx of the 5th finger	0.00028
Cluster-7	Elevated circulating C5 acylcarnitine concentration	0,00079
Cluster-10	Pulverulent cataract	0,001
Cluster-6	Abnormal line of Schwalbe morphology	0,0075
Cluster-9	Periodic hyperkalemic paralysis	0,0081

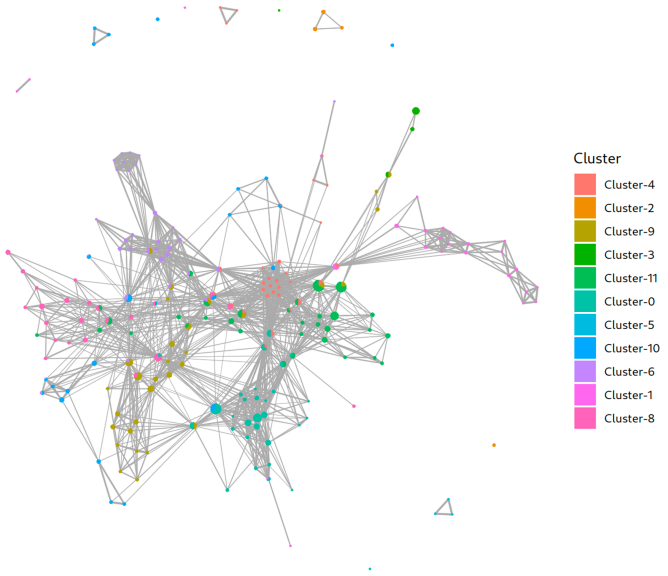

Fig. 4. Network representation of functional enrichment results. Each node represents Gene Ontology (GO) Biological Process (BP) terms, with colors indicating the cluster (HDBSCAN-derived) assignment of the contributing genes. Edges connect BP terms that share common genes, with edge thickness corresponding to the number of shared genes.

show that analyzing only full open-access articles can reconstruct known disease relationships from ontology structures. A key advantage of our approach is the ability to generate specific document corpora for each disease cluster, linking

them to phenotypes and related genes to facilitate the discovery of shared disease mechanisms.

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Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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5.2. Análisis integrativo transcriptómico y basado en redes para el estudio de las enfermedades neuromusculares.

Mi contribución:

En el siguiente trabajo, publicado en la revista *International Journal of Molecular Sciences* y en el que figuro como primer autor, se presenta el estudio de mecanismos comunes en enfermedades neuromusculares a partir de la integración de datos de expresión génica y redes de interacción proteína-proteína, aplicado a la distrofia muscular de Duchenne (DMD), la distrofia muscular de cinturas (LGMD) y la esclerosis lateral amiotrófica (ALS).

Mi contribución se centró en la implementación de los métodos necesarios a nivel de análisis de redes para poder llevar a cabo el estudio presentado en este trabajo en lo que respecta a la interacción proteína–proteína. Para ello, amplié la herramienta *NetAnalyzer*, de modo que fuera posible realizar automáticamente la detección de comunidades, el embebido de nodos y el análisis de la proximidad de genes dentro del espacio de embebido. En esta ocasión, para el sistema de priorización en función de la proximidad, diseñé e implementé una aproximación basada en el test combinado de probabilidades de Fisher. Además, incorporé un nuevo módulo en la librería *NetAnalyzer*, denominado *net_explorer*, con el que es posible obtener de manera automatizada los informes de todos estos procesos.

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Integrative Transcriptomic and Network-Based Analysis of Neuromuscular Diseases

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Abstract

Neuromuscular diseases (NMDs) like Duchenne muscular dystrophy (DMD), limb–girdle muscular dystrophy (LGMD), and amyotrophic lateral sclerosis (ALS) are rare, progressive disorders with complex molecular mechanisms. Traditional transcriptomic analyses often struggle to capture systems-level dysregulation, especially given the small sample sizes typical of rare disease studies. Our differential expression analysis of eight public RNA-seq datasets from various cell types in DMD, LGMD, and ALS revealed not only disease-relevant pathways but also unexpected enrichments, such as renal development, suggesting systemic impacts beyond muscle tissue. To address limitations in capturing broader molecular mechanisms, we applied an integrative systems biology approach combining differential expression data, protein–protein interaction (PPI) networks, and network embedding techniques. Comparative functional enrichment revealed shared pathways, including glycosaminoglycan binding in both DMD and *FUS*-related ALS, implicating extracellular matrix–protein interactions in *FUS* mutation effects. Mapping DEGs onto the human PPI network and assessing their proximity to causal genes uncovered dysregulated non-coding RNAs, such as *PAX8-AS1*, *SBF2-AS1*, and *NEAT1*, potentially indicating common regulatory roles. We also found candidate genes within disease-proximal clusters, like *HS3ST3A1*, which may contribute to pathogenesis. Overall, this integrative approach reveals shared transcriptional programs and novel targets, advancing our understanding and potential treatment strategies for NMDs.

Keywords: neuromuscular diseases; integrative analysis; differential gene expression; network-based analysis

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1. Introduction

Neuromuscular diseases (NMDs) encompass a diverse group of disorders that affect the function of muscles and the nerves that control them, often leading to progressive

weakness, loss of motor function, and significant disability [1–4]. Among these, Duchenne muscular dystrophy (DMD) and amyotrophic lateral sclerosis (ALS) are two of the most extensively studied NMDs [5,6]. Although less prevalent than DMD or ALS, limb–girdle muscular dystrophy (LGMD) represents an important subgroup within the muscular dystrophies due to its phenotypic similarities with DMD, particularly in terms of progressive weakness affecting the shoulder and pelvic girdle muscles. While DMD is primarily characterized by progressive muscle degeneration due to mutations in the *DMD* gene [7], ALS involves the degeneration of motor neurons, with a multifactorial etiology involving both genetic and environmental factors [8]. Regarding LGMD, it encompasses a genetically heterogeneous set of disorders caused by mutations in various genes (including *CAPN3*, *TNPO3* and *LMNA*, among others) involved in muscle fiber integrity, membrane repair, or sarcomere organization [9,10]. Even the age of onset and rate of progression may vary; LGMD often mirrors the muscle degeneration patterns seen in DMD, which can be useful in studies exploring shared molecular mechanisms among diseases [11]. Despite the clinical and pathological differences between DMD and LGMD and ALS, these diseases have overlapping clinical features, including progressive muscle weakness and dysphagia [12], suggesting they may share common molecular pathways that remain poorly understood, contributing to delays in diagnosis and limited therapeutic options [13].

One of the significant challenges in the study of NMDs is the heterogeneity of their pathological phenotypes and the associated genetic profiles [14]. For instance, while DMD is typically linked to a single gene mutation [7], its clinical progression can vary due to modifier genes and environmental influences [15]. In contrast, ALS exhibits remarkable genetic heterogeneity [16], with mutations in over 40 genes implicated in its familial and sporadic forms [17]. Similar to ALS, LGMD includes multiple subtypes of the disease depending on the gene affected [9]. This complexity complicates the establishment of universal diagnostic biomarkers and therapeutic targets [18], resulting in prolonged diagnostic timelines and suboptimal treatment strategies. Moreover, the lack of comprehensive insights into the molecular mechanisms underlying these diseases hinders the development of effective interventions [1].

Identifying common molecular mechanisms and genetic factors shared by NMDs could provide critical insights into their pathogenesis and open new avenues for diagnostic and therapeutic advancements [1]. Transcriptomic studies, which analyze gene expression changes at the RNA level, offer a powerful approach to unravel the molecular underpinnings of these diseases [19,20]. Differential gene expression analysis, in particular, has been instrumental in identifying genes and pathways dysregulated in NMDs [20,21]. However, interpreting these data in isolation often falls short of capturing the complex interactions among genes and their regulatory networks.

To address this limitation, network-based approaches have emerged as complementary tools to integrate transcriptomic data and elucidate the relationships between differentially expressed genes (DEGs) [22]. By mapping DEGs onto biological networks, such as protein–protein interaction (PPI) networks, researchers can identify clusters of DEGs that are close to the disease-causal gene and determine pathways that could play central roles in disease pathogenesis [23]. More recently, the application of network embedding techniques has further advanced this field by enabling the transformation of complex network topologies into low-dimensional vector spaces that preserve neighborhood and connectivity information. In biological terms, these techniques allow us to convert a large, intricate interactome, comprising thousands of proteins and their interactions, into mathematical representations (vectors) where proteins with similar interaction patterns are placed close together in the new space. Unlike traditional clustering methods that rely uniquely on topological features, like node degree (number of direct connections between nodes) or

direct neighbors (nodes directly connected to a node of interest), embeddings can capture higher-order structures, such as indirect connections or community contexts, which are often biologically meaningful. For example, two genes that are not directly connected but share similar interaction environments may be projected nearby in the embedded space, reflecting potential co-regulation or shared functional roles. These embeddings facilitate more accurate clustering, gene prioritization, and pathway inference, thereby improving the detection of functionally relevant modules and enhancing our capacity to pinpoint potential biomarkers and therapeutic targets in a systems-level context [24–26]. With these considerations, this integrative framework not only enhances the understanding of molecular mechanisms but also facilitates the prioritization of potential biomarkers and therapeutic targets.

In this study, we aim to leverage differential expression and network-based analyses to identify shared genetic factors and molecular mechanisms in DMD, LGMD, and ALS. By integrating RNA-seq data with PPI networks, we seek to uncover critical genes and pathways implicated in these diseases. Our findings have the potential to enhance the understanding of NMD pathogenesis, improve diagnostic accuracy, and inform the development of targeted therapies that address common aspects of these debilitating conditions.

2. Results

The results of the read pre-processing indicated good overall quality across all samples. In nearly all samples, the percentage of reads uniquely mapped to the reference genome was very high, with the exception of one control and one patient from the ALS_iN_C9ORF72 dataset (Supplementary Figure S1G). No signs of adapter contamination or major sequencing artifacts were found (Supplementary Figure S2). In addition, for each dataset, Supplementary Figure S3 illustrates the total number of reads remaining after pre-processing, the number of reads that aligned uniquely to the reference genome, and, among these, the subset that mapped to annotated genes.

2.1. Quality Assessment and Differential Expression Analysis Across Datasets

Differential expression analysis was performed with ExpHunter Suite independently for each neuromuscular disease (NMD) dataset. It includes a data quality control module that performs principal component analysis (PCA) to assess the distribution of samples and a Hierarchical Clustering on Principal Components (HCPC) that groups samples in clusters sharing similar characteristics. For the DMD datasets, in the case of the DMD_myot dataset (Figure 1a), it is noteworthy that the control iCtrl_R2_120 does not appear grouped with the rest of the controls in the principal component of the PCA, which accounts for 33.20% of the explained variability. In the HCPC of this comparison, we can see three well-differentiated clusters of samples: two with controls and patients perfectly grouped in their categories and another one including iCtrl_R2_120 along with the DMD_R3_120 patient sample (Supplementary Figure S5A). However, in the case of DMD_pCard (Supplementary Figure S4A), a clear separation of the samples between the two groups is observed: control (“ctrl”) and patient (“treat”). Specifically, the sample iCtrl_R1_RF is the one that is furthest from the control group, as can be seen in the HCPC (Supplementary Figure S5B). Regarding the DMD_cfib dataset, the PCA revealed certain separation between control and DMD patient samples (Supplementary Figure S4B); however, it is observed that Ctrl_1 and Ctrl_2 samples are somehow closer to DMD samples than to the rest of controls. This is also clearly observed in the HCPC (Supplementary Figure S5C), in which both controls are grouped with DMD samples in two different clusters, and Ctrl_3 and Ctrl_4 are included in another cluster. And in the case of the DMD_myob dataset, a clear separation between control and DMD samples is observed

(Supplementary Figure S4C). Additionally, three distinguishable groups of patient samples are also observed. This is confirmed by the HCPC (Supplementary Figure S5D). These results are consistent with the original study, since three replicates were extracted from each individual.

Regarding the results obtained for the analysis of the datasets corresponding to LGMD, in the case of the LGMD_myob dataset, a clear separation is observed between the two groups of samples in the PCA (Supplementary Figure S4D), in which the principal component accounts for 96.38% of the explained variability, as well as a very close clustering between the samples that are part of the sick group, while there is more distance between the samples of the control group, with Ctrl_1 being the most distant. In the HCPC, this separation of samples is reflected in three distinct clusters, one of them including the patient samples and two with control samples (Supplementary Figure S6A). The LGMD_pbmc dataset (Figure 1b) exhibits substantial heterogeneity among its samples. The first principal component accounts for 22.64% of the explained variability and reveals the presence of distinct sample groupings, such as the control samples Ctrl_1, Ctrl_2, and Ctrl_3, and the patient samples LGMD_2, LGMD_3, and LGMD_10. This pattern is further supported by the dendrogram generated through HCPC analysis (Supplementary Figure S6B), which identifies up to seven distinct clusters: two clusters composed exclusively of control samples, three containing only patient samples, and two including a mix of both sample types.

Regarding the PCA results for ALS datasets, in the case of ALS_iN_C9ORF72 we carried out the comparison of controls vs. ALS patients for cytoplasmic samples. As can be seen in Figure 1c, there is a clear separation between control and patient groups, with the exception of the samples labeled R3. These do not cluster with their expected groups: the control R3 sample clusters with ALS patients, while the ALS R3 sample appears separated from both groups (Supplementary Figure S6C). In the case of the ALS_fib_FUS dataset (Figure 1d), PCA reveals a separation between both groups of samples (controls and FUS-mutated ALS samples) not very well defined between the control and patient groups on the basis of the second component, which accounts for 9.79% of the explained variability. Samples ALS_1 and ALS_10 are the most distant in the PCA space compared to the others; however, we chose not to exclude them from the analysis to maintain sample balance in the comparison. As observed in the HCPC for this dataset (Supplementary Figure S6D), the samples cluster into different groups. Samples ALS_1, ALS_3, ALS_6, and ALS_10 form clusters on their own. There are two clusters that include only control samples and one with four ALS patient samples and a final cluster that combines both sample types.

For each dataset, differential expression analysis was performed by comparing samples as specified in Section 4. Table 1 summarizes the number of differentially expressed genes (DEGs) identified in each dataset, specifying how many were over-expressed and how many were down-regulated based on the $\log_2$ fold change ($\log_2$FC). Supplementary Tables S1 and S2 present, for DMD and for LGMD and ALS, respectively, the top five over-expressed and down-regulated genes identified in each dataset, ranked by their $\log_2$FC values.

Overall, the average number of expressed genes across the eight datasets is 13,429.1. Among these, a higher number of expressed genes is observed in samples taken from different tissues, like iNeurons, cardiac fibroblasts, and myoblasts, than from PBMC and fibroblasts (Table 1). Likewise, a higher number of over-expressed genes is observed than down-regulated genes, except in the case of DMD_pcard and LGMD_myob, where 91.3% and 63.6% of DEGs, respectively, have negative $\log_2$FC values.

Table 1. Summary of expressed and differentially expressed genes (DEGs) across datasets, including counts of over-expressed and down-regulated genes. DMD: Duchenne muscular dystrophy; LGMD: limb–girdle muscular dystrophy; ALS: amyotrophic lateral sclerosis.

Dataset	Expressed Genes	DEGs	Over-Expressed	Down-Regulated
DMD_myot	13,085	176	137	39
DMD_pCard	12,832	437	38	399
DMD_cfib	13,157	80	30	50
DMD_myob	14,466	580	356	224
LGMD_myob	13,838	548	199	349
LGMD_pbmc	13,598	160	108	52
ALS_iN_C9ORF72	13,453	325	284	41
ALS_fib_FUS	13,004	171	96	75

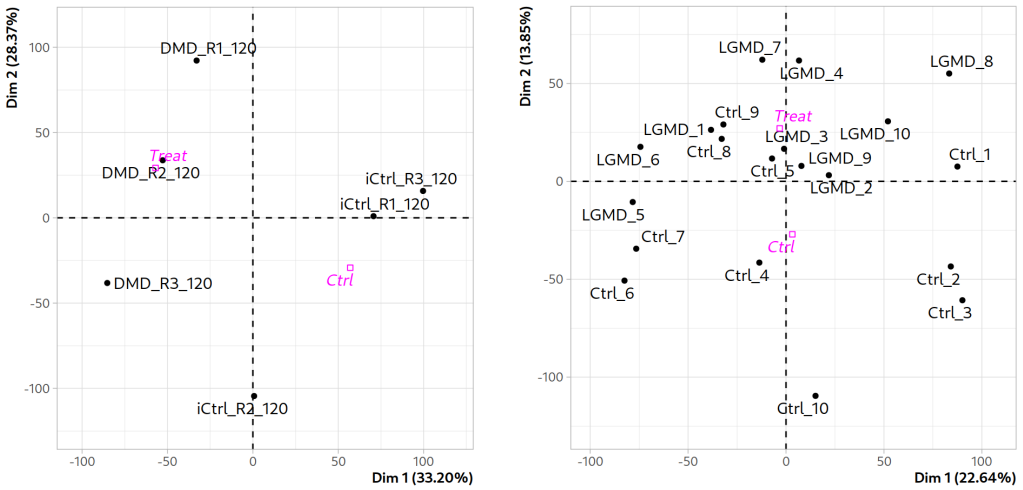

Figure 1. Principal component analysis (PCA) performed for the following datasets (each one representing a different causal gene): (**a**) DMD_myot (*DMD*), (**b**) LGMD_pbmc (*TNPO3*), (**c**) ALS_iN_C9ORF72 (*C9ORF72*), and (**d**) ALS_fib_FUS (*FUS*). The centroid of each group of samples compared (Ctrl and Treat) is displayed. DMD: Duchenne muscular dystrophy; LGMD: limb–girdle muscular dystrophy; ALS: amyotrophic lateral sclerosis; iCtrl: isogenic control. The centroid of the control and treatment groups is represented in magenta.

Differential expression analysis across the DMD and LGMD datasets revealed both dataset-specific and, in some cases, shared transcriptional alterations. Notably, a subset of the most significantly expressed genes is common to different datasets, suggesting potentially conserved molecular signatures. For instance, *MEG3* and ENSG00000225746 are over-expressed in both DMD_myot and DMD_myob; *PAX8-AS1* appears among the top over-expressed genes in DMD_pCard and DMD_cfib; *GSTM1* and *TRPV2* are shared between DMD_myob and LGMD_myob; and *SFRP4* is down-regulated in both DMD_cfib and ALS_fib_FUS. The results of the functional enrichment analysis for each dataset, presented as over-representation analysis (ORA) graphs generated by ExpHunter Suite, are shown in Supplementary Figure S7. In DMD_myot and DMD_myob, DEGs were enriched in biological processes related to renal system development, synaptic organization, and immune responses, including type I interferon signaling and hypoxia response. In contrast, DMD_cfib (the dataset with the fewest DEGs) had more limited enrichment, mainly involving extracellular matrix organization and nutrient response. In DMD_pCard, over-expressed genes, such as *PAX8-AS1* and *GJD2*, were linked to functions in connective tissue development, ossification, coagulation, and nervous system regulation. In LGMD_myob, strong over-expression of *ACSM5* and *GSTM1* reflected robust enrichment in muscle-specific processes, including muscle cell differentiation, sarcomere organization, and cardiac function. On the other hand, LGMD_pbmc showed a distinct immune-related signature, with DEGs involved in chemokine response, leukocyte migration, and immune cell activation, consistent with high expression of *MMP1*, *CXCL5*, and *SERPINB2*. The two ALS datasets also showed contrasting profiles. ALS_iN_C9ORF72 exhibited over-expression of *TUNAR* and *TMEM132E*, with enriched functions overlapping with those observed in DMD_myot, including renal morphogenesis, axon guidance, and glial differentiation. In contrast, ALS_fib_FUS displayed more moderate expression changes and lacked enrichment for neuronal processes. Instead, functional categories were related to cardiac muscle tissue growth, blood circulation, and mesenchyme morphogenesis, suggesting a fibroblast-specific response.

Comparative Functional Profiling Across Neuromuscular Disease Transcriptomes

To explore the functional relationships among DEGs across NMDs, we used ExpHunter Suite to perform a comparative functional enrichment analysis (Figure 2). This approach enabled us to identify enriched Gene Ontology (GO) biological processes associated with each dataset to determine which functional categories are shared among different NMDs and which are unique to specific conditions. We used a p-value of 0.05 to select significant annotations.

There are several functional categories shared among the eight datasets, as well as clusters specific to each of them, which share some connection with more general functions, establishing a highly interconnected network. Exceptionally, some isolated functions are observed, generally belonging to the same dataset. The functional categories extracellular matrix binding and glycosaminoglycan binding are shared by most of the datasets. Other categories, such as actin binding and structural constituent of muscle, are shared by both LGMD_myob and DMD_pCard, and both functions connect directly with other, more dataset-specific functions, such as filament actin binding and calmodulin binding, specific to the LGMD_myob dataset, and with cadherin binding and cytokine binding, specific to the DMD_pCard dataset. There are also shared functions among several datasets of different diseases taken from similar tissues, such as integrin binding for DMD_myob, LGMD_myob, and DMD_pCard and growth factor binding and collagen binding for the DMD_pCard, DMD_cfib, and LGMD_myob datasets. It is also worth noting that some of the functions associated with the DEGs of DMD_myob, LGMD_myob, and

LGMD_pbmc form clusters of functions characterized exclusively for those datasets. Also noteworthy is the peptidase regulator activity function, shared between the LGMD_pbmc and ALS_iN_C9ORF72 datasets, belonging, respectively, to two very different NMDs in phenotypic terms.

Figure 2. Comparative functional enrichment network. It represents the top enriched functional categories, specifically Gene Ontology (GO) biological process terms, identified across the different datasets (DMD_myot, DMD_pCard, DMD_cfib, DMD_myob, LGMD_myob, LGMD_pbmc, ALS_iN_C9ORF72, and ALS_fib_FUS). Nodes correspond to GO biological process terms (when both parent and child terms are present in the network, only the child terms are retained, and the parent terms are removed), and edges indicate functional relationships based on shared DEGs. Edge thickness reflects the number of shared DEGs connecting two GO terms, with thicker edges denoting stronger functional overlap.

2.2. Mapping Differential Expression Profiles onto the Human Interactome

Combining protein–protein interactions from all datasets allows the construction of a comprehensive network capturing interactions consistently present across multiple NMD models, providing a global perspective on how disease-causal genes relate to the observed differential expression profiles and enabling the identification of shared modules, conserved pathways, and core disease mechanisms that might not be apparent in individual datasets. We constructed a unified network composed of all expressed genes across the eight datasets (Figure 3). The network layout is generated using an attraction–repulsion algorithm that organizes nodes by connectivity: highly connected hubs cluster toward the center, while barely connected or isolated nodes are displaced to the periphery. We will refer to this network as the dataset view to differentiate it from other representations used in this study.

In this dataset view representation, DEGs from each dataset were mapped onto the human protein–protein interaction network. To improve visualization clarity, DEGs from the four DMD-related datasets (linked to the *DMD* gene) and the two LGMD-related datasets (linked to *TNPO3*) were merged into single DEG sets. Additionally, the four disease-causal genes corresponding to each NMD were used as seed nodes (starting reference points in the network) to explore and analyze the relationships between these causal genes and the DEGs. For further details and a complete network exploration, the interactive report is available in the integrative analysis results repository https://github.com/ElenaRojano/INTRINSED_datasets/blob/main/network_all_SAFE.html (accessed on 27 July 2025).

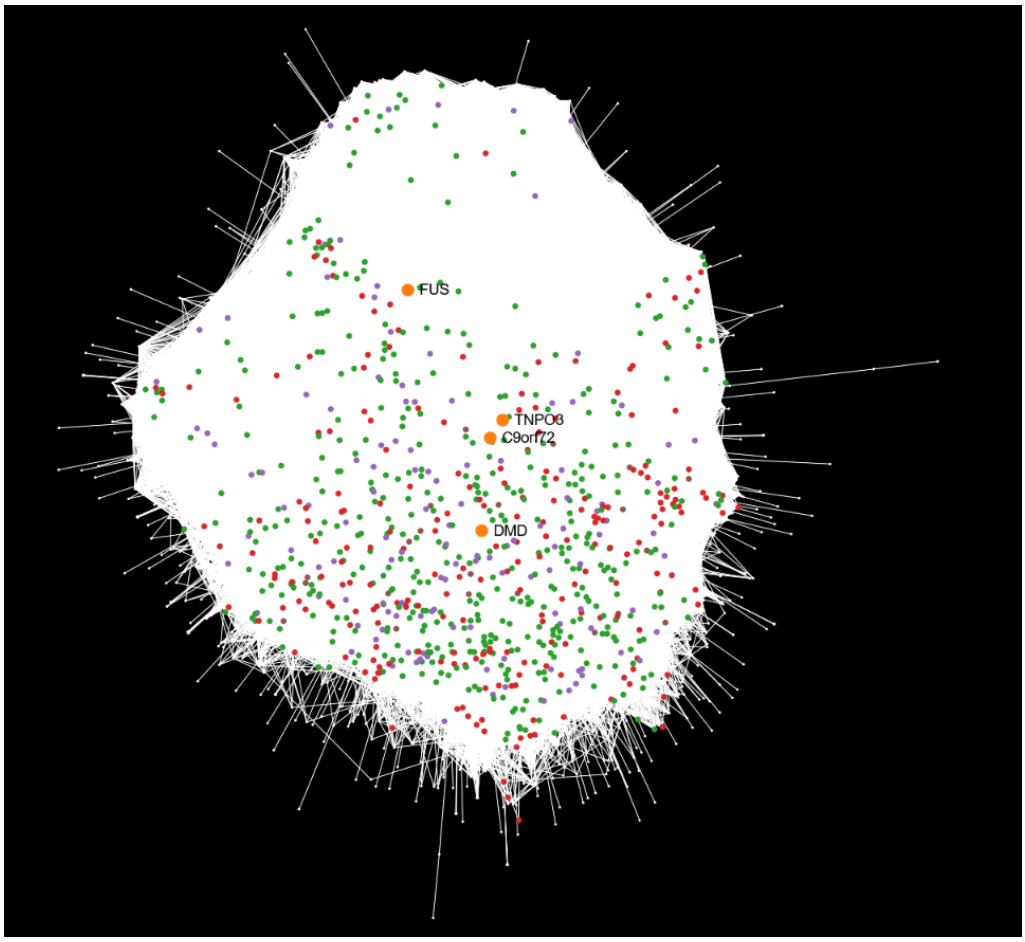

Figure 3. Protein–protein interaction network belonging to the dataset view, based on the union of all expressed genes across all the datasets (white body). Each differentially expressed gene list is represented in different colors, depending on the dataset to which they belonged. Those from the four DMD-related and the two LGMD-related datasets were collapsed into two aggregated DEG lists for visualization clarity. Green: DMD-related datasets; red: LGMD-related datasets; purple: ALS_iN_C9ORF72 dataset; and brown: ALS_fib_FUS dataset. Disease-causal genes (*DMD*, *TNPO3*, *C9ORF72*, and *FUS*) for all datasets are represented as orange nodes.

This integrated dataset view representation reveals notable patterns of proximity among disease-causal genes. Interestingly, despite being linked to distinct disorders, *C9ORF72* (ALS) and *TNPO3* (LGMD) are positioned closely within the network. The DEGs surrounding these two genes primarily originate from the DMD- and LGMD-related datasets (Figure 3, green and red nodes, respectively), rather than from ALS_iN_C9ORF72 (Figure 3, purple nodes). In the case of *TNPO3*, this is consistent with the phenotypic similarities between DMD and LGMD, both phenotypically similar NMDs [11]. Conversely, *DMD* is situated closer to *C9ORF72* than to *TNPO3*, despite the stronger phenotypic

resemblance between DMD and LGMD. In addition, *FUS* (also associated with ALS) appears distant from *C9ORF72* in the network, suggesting that they may be involved in distinct molecular pathways or operate within separate topological contexts. Interestingly, the disease-causal gene *DMD* shows a marked spatial separation from its nearest DEGs in the interaction network, in contrast to the other three disease-causal genes, which are surrounded by DEGs in close proximity. In addition, the distribution of DEGs in the ALS_fib_FUS dataset is particularly striking, as the disease-causal gene *FUS* is spatially distant from its associated DEGs (Figure 3, brown nodes), which instead appear closer in the network to the *DMD* gene.

To consolidate this information, Table 2 summarizes the integrative analysis across all datasets, including the number of DEGs successfully mapped to the human protein–protein interaction network. Notably, the number of mapped DEGs varies substantially between datasets. For example, DMD_pCard (247 mapped DEGs) and DMD_myob (239) showed the highest number of DEGs mapped onto the interactome. LGMD_myob and ALS_iN_C9ORF72 also showed substantial mapping, with 206 and 127 DEGs, respectively. In contrast, datasets such as DMD_myot (34), DMD_cfib (13), LGMD_pbmc (19), and ALS_fib_FUS (21) exhibited much lower numbers of mapped DEGs. It is worth noting that these datasets yielded fewer DEGs in the differential expression analysis compared to the other datasets (all with ≥300 DEGs, Table 1), which partly explains their lower number of mapped genes.

Table 2. Summary of the integrative analysis results across datasets. It includes the number of differentially expressed genes (DEGs) that mapped against the interaction network (Mapped DEGs), the number of DEGs that were left out of the network of interactions, referred to as unmapped differentially expressed genes (uDEGs), the number of coding uDEGs (Coding uDEGs), non-coding RNAs (ncRNAs), and pseudogenes (Pseudo), and the number of DEG clusters. DMD: Duchenne muscular dystrophy; LGMD: limb–girdle muscular dystrophy; ALS: amyotrophic lateral sclerosis.

Dataset	Mapped DEGs	uDEGs	Unmap. Coding	ncRNA	Pseudo	Clusters
DMD_myot	34	17	2	13	2	8
DMD_pCard	247	31	6	22	3	17
DMD_cfib	13	14	2	9	3	4
DMD_myob	239	164	14	114	30	13
LGMD_myob	206	120	11	85	20	15
LGMD_pbmc	19	53	12	19	19	4
ALS_iN_C9ORF72	127	50	3	40	2	10
ALS_fib_FUS	21	36	5	25	3	4

Analysis of Unmapped Differentially Expressed Genes

Looking closely at the information provided in Table 1 on the total DEGs for each dataset, there are a number that do not map against the human interactome network. In this work, we referred these DEGs as *unmapped differentially expressed genes* (uDEGs). The reasons why these DEGs are left out of the network may be either due to the absence of known interactions or annotation limitations. uDEGs are classified into three categories: coding uDEGs, which are protein-coding genes not represented in the interaction network; non-coding RNAs (ncRNAs), which mainly include long non-coding RNAs (lncRNAs); and pseudogenes.

In our study, ncRNAs were the most represented category among uDEGs in all datasets, followed by pseudogenes and unmapped coding genes. The datasets DMD_myob and LGMD_myob presented the highest numbers of uDEGs, with 164 and 120, respectively. In both cases, the majority of uDEGs were ncRNAs (114 and 85, respectively), followed by

pseudogenes and unmapped coding genes. Similarly, ALS_iN_C9ORF72 (50 uDEGs) and ALS_fib_FUS (36 uDEGs) also showed a predominance of ncRNAs (40 and 25, respectively), suggesting a potential regulatory contribution of these transcripts in ALS pathology. In contrast, datasets such as DMD_myot (17 uDEGs), DMD_cfib (14 uDEGs), and DMD_pCard (31 uDEGs) had a lower number of unmapped genes overall, with ncRNAs still representing the majority. LGMD_pbmc, despite its relatively low number of mapped DEGs (19), had a notable 53 uDEGs, including 19 ncRNAs and 19 pseudogenes, suggesting a potential enrichment of non-coding elements in this dataset. In Supplementary Table S4 we include all the uDEGs shared between datasets. Most of them are lncRNAs. For example, *PAX8-AS1* exhibits a striking expression pattern: down-regulated in DMD myoblasts ($\log_2$FC = −4.97) but consistently over-expressed in all other DMD datasets, as well as in the LGMD_myob dataset ($\log_2$FC = 4.69). A similar pattern is observed for *MEG3*, which is strongly over-expressed in DMD_myot ($\log_2$FC = 9.32) and DMD_myob ($\log_2$FC = 8.56) but down-regulated in DMD_pCard ($\log_2$FC = −2.16). Notably, *SBF2-AS1* and *NEAT1* emerged as differentially expressed lncRNAs in different diseases and tissues, but with opposing expression patterns: *SBF2-AS1* was over-expressed in LGMD_myob ($\log_2$FC = 3.71) yet down-regulated in ALS_iN_C9ORF72 ($\log_2$FC = −2.26), while *NEAT1* was over-expressed in DMD_myob ($\log_2$FC = 2.15) and down-regulated in ALS_fib_FUS ($\log_2$FC = −1.46). No differentially expressed ncRNAs were found in common between LGMD_pbmc and the rest of the datasets.

2.3. Mapping Disease-Associated Transcriptional Profiles onto the Human Interactome

To gain deeper insights into the molecular mechanisms underlying NMDs, we performed a network-based integrative analysis by projecting the DEGs from each dataset onto the human protein–protein interaction network. We refer to this representation as the disease view to distinguish it from the previously described dataset view. In this context, we define *functional groups* as clusters of DEGs that interact closely with one another in the network and are also connected to the disease-associated gene specific to each dataset: *DMD* for the DMD_myot, DMD_pCard, DMD_cfib, and DMD_myob datasets, *TNPO3* for both LGMD_myob and LGMD_pbmc datasets, *C9ORF72* for the ALS_iN_C9ORF72 dataset, and *FUS* for the ALS_fib_FUS dataset. For simplicity, we selected representative datasets for each disease: DMD_myot, LGMD_myob, ALS_iN_C9ORF72, and ALS_fib_FUS (Figure 4A–D, respectively). The rest of the datasets are included in Supplementary Figure S8. For each of the datasets, the report with the interaction network resulting from the integrative analysis and the functional analysis of the clusters that will be detailed in the following sections is available in the results repository at https://github.com/ElenaRojano/INTRINSED_datasets/blob/main/network_all_SAFE.html (accessed on 27 July 2025).

In the case of the DMD-related datasets, a pairwise pattern of similarity emerges. The DMD_myot dataset (Figure 4A) shows clusters with fewer DEGs that are more widely dispersed across the network. A similar distribution is observed in the DMD_cfib dataset (Supplementary Figure S8B), which also displays a reduced number of clusters. This is consistent with the low number of DEGs identified in this dataset (80; see Table 1), although some remain in close proximity to the disease-causal gene *DMD*. In both datasets, DEGs exhibit a dispersed pattern throughout the network, potentially suggesting a more heterogeneous or less coordinated transcriptional response. In the case of DMD_pCard and DMD_myob (Supplementary Figure S8A,C), both display comparable network organizations, with multiple DEG clusters positioned near the disease-causal gene *DMD*, suggesting a more coherent functional or topological response to disease perturbation in these tissues. A similar pattern to that observed in the DMD_myot and DMD_cfib datasets was found in LGMD_pbmc (Supplementary Figure S8D), which contains a relatively low number

of DEGs (160) compared to LGMD_myob (Figure 4B). This difference likely contributes to the reduced number of clusters detected in LGMD_pbmc. In LGMD_myob, DEGs are more heterogeneously distributed across the network, with several clusters forming around the disease-causal gene *TNPO3*, as well as additional clusters located in more peripheral regions. In contrast, LGMD_pbmc displays two smaller DEG clusters located at greater topological distances from *TNPO3*, although some genes within another cluster appear relatively close to the disease-associated gene in the network. Regarding the ALS-related datasets, ALS_iN_C9ORF72 (Figure 4C) exhibits a higher number of clusters with a denser distribution of DEGs compared to ALS_fib_FUS (Figure 4D). In this ALS_fib_FUS dataset, DEGs are strikingly distant from the disease-causal gene *FUS* within the interactome, reinforcing the observation from the dataset view that these genes occupy separate topological regions in the network.

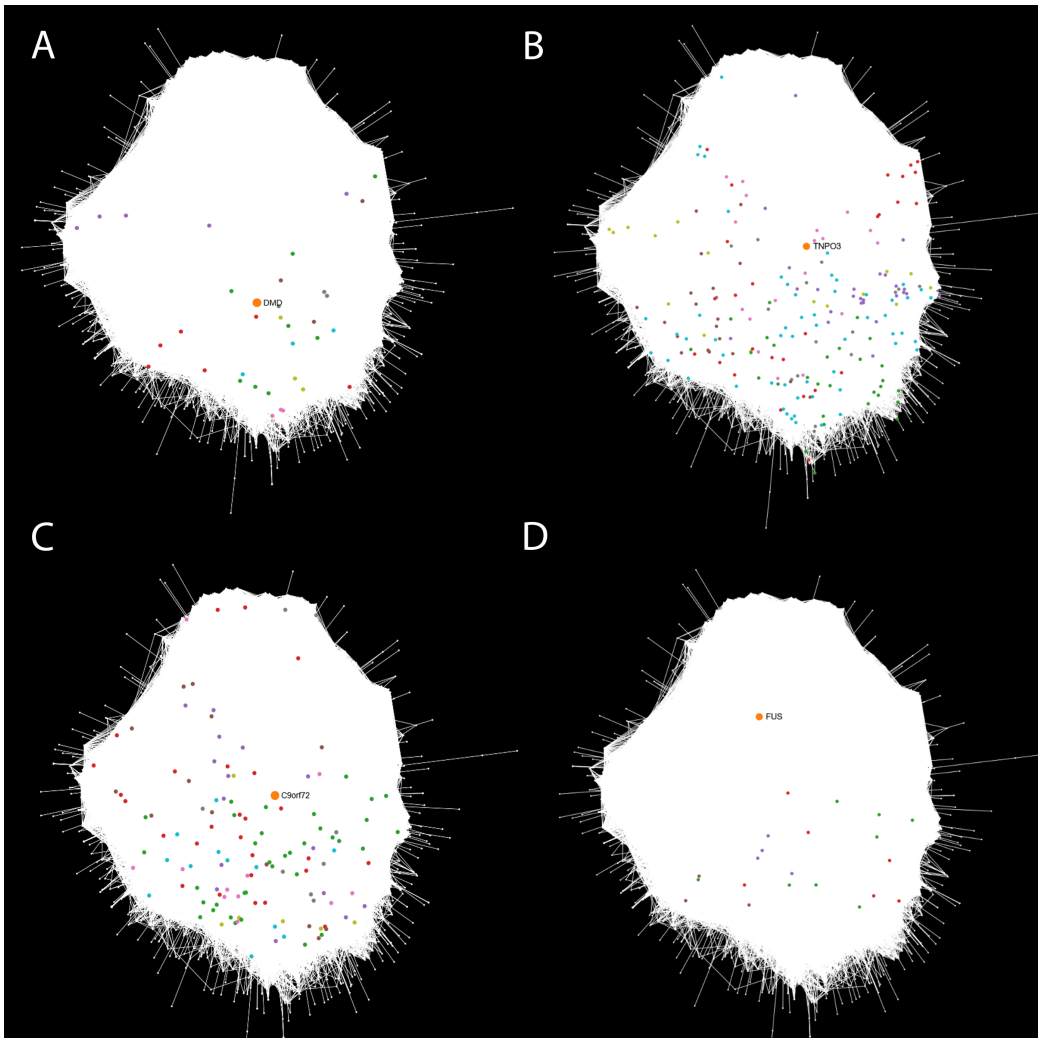

Figure 4. Protein–protein interaction networks from the disease view representation. Colored nodes correspond to clusters of differentially expressed genes (DEGs) for each dataset. Orange dots represent the disease-related gene used as a reference. (**A**) DMD_myot (*DMD*), (**B**) LGMD_myob (*TNPO3*), (**C**) ALS_iN_C9ORF72 (*C9ORF72*), and (**D**) ALS_fib_FUS (*FUS*).

These intra-group analyses reveal notable differences in the network organization of DEGs across datasets associated with the same disease gene. When comparing across disease groups, particularly between DMD, LGMD, and ALS datasets, distinct spatial patterns emerge: DMD-related datasets tend to have more compact DEG neighborhoods around *DMD*, while ALS datasets, especially ALS_fib_FUS, show more scattered DEG

distributions. Furthermore, analyzing the distribution of the number of clusters and the number of DEGs per cluster provides insight into the heterogeneity and connectivity patterns of disease-associated expression changes within the network. In Table 2, we observe the highest number of DEG clusters (17) for the DMD_pCard dataset, followed by LGMD_myob (15) and DMD_myob (13) datasets, which also exhibited a substantial number of clusters. The DMD_cfib, LGMD_pbmc, and ALS_fib_FUS datasets showed the lowest number of clusters (4). Certainly, it is observed that datasets derived from non-specific tissues, like fibroblasts (DMD_cfib and ALS_fib_FUS) and PBMCs (LGMD_pbmc), show a lower number of DEGs per cluster compared to those obtained from specific tissues. For example, as can be observed in Supplementary Figure S10B, the median number of genes per cluster is 3 for DMD_myot and DMD_cfib and 5 for ALS_fib_FUS, whereas in DMD_myob the median rises to 20 and in DMD_pCard to 15.

Regarding the distribution of DEGs per cluster, in the four DMD datasets (Supplementary Table S15) we observe a lower number of DEGs per cluster in the datasets with a lower number of clusters (DMD_myot and DMD_cfib), compared to the DMD_pCard and DMD_myob datasets. Interestingly, in these last two datasets, the first two clusters have the same number of DEGs (32 and 29, respectively, in clusters 0 and 1), but there are no shared genes between any of the clusters between the different datasets. In the case of the LGMD datasets (LGMD_myob and LGMD_pbmc, Supplementary Table S16), more clusters and a higher abundance of DEGs are observed in the LGMD_myob dataset. This dataset, along with DMD_myob, both from samples of the same cell type (myoblasts), are the ones that returned the most DEGs in the differential expression analysis. And, despite not being the cluster with the highest number of DEGs, the ALS_iN_C9ORF72 dataset has the cluster with the most DEGs per cluster (35).

2.4. Embedding-Based Prioritization of DEG Clusters Relative to the Disease-Causal Gene

Given the complexity of the network structure, considering the large number of DEG-derived clusters and the substantial set of isolated genes, traditional topological measures may be insufficient to capture the nuanced relationships required to identify disease-relevant genes [27]. Classical approaches often rely on direct connectivity and may overlook functionally important nodes that are not topologically central or well-connected but may still be biologically proximal to the disease-causal gene.

In this context, we adopt a network embedding strategy to project the high-dimensional interactome into a continuous latent space, which enables a more sensitive assessment of gene proximity and reflects functional similarity beyond direct edges. Importantly, embedding allows us to prioritize both clustered and non-clustered (or "isolated") DEGs according to their spatial proximity to the disease-causal gene under the hypothesis that genes closely located in the embedded space are more likely to participate in shared biological processes or pathways relevant to disease mechanisms. Thus, embedding provides a robust framework for integrative prioritization in complex disease networks where gene interactions may be indirect or context-dependent.

2.4.1. DEG Cluster Prioritizazion and Functional Analysis

Applying network embedding to each dataset reveals a higher number of clusters and mapped DEGs in datasets derived from differentiated tissues, such as LGMD_myob (Figure 5D), ALS_iN_C9ORF72 (Figure 5C), DMD_pCard (Supplementary Figure S9A), and DMD_myob (Supplementary Figure S9C). In contrast, a lower number of clusters is observed in the datasets that are of undifferentiated tissues: DMD_cfib (Supplementary Figure S9B), LGMD_pbmc (Supplementary Figure S9D), and ALS_fib_fus (Figure 5D). It is interesting to remark that, in the case of both DMD_myot and DMD_myob datasets

(Supplementary Figure S9B and D, respectively), a noticeable spatial separation between the disease-causal gene (*DMD*) and the surrounding DEGs in the closest clusters is observed. This trend can be also seen in the LGMD_pbmc (Supplementary Figure S9D) and ALS_fib_FUS (Figure 5D) datasets. The opposite trend is observed for the rest of datasets (all from differentiated tissues). For example, in the case of both DMD_pCard and ALS_iN_C9ORF72 datasets (Supplementary Figure S9A and Figure 5C, respectively), DEGs in the clusters nearest to the respective causal genes (*DMD* and *C9ORF72*) are placed in a similar region within the interaction network, suggesting closer functional relationships. Furthermore, the number of genes per cluster is clearly higher in these tissue-specific datasets compared to the others.

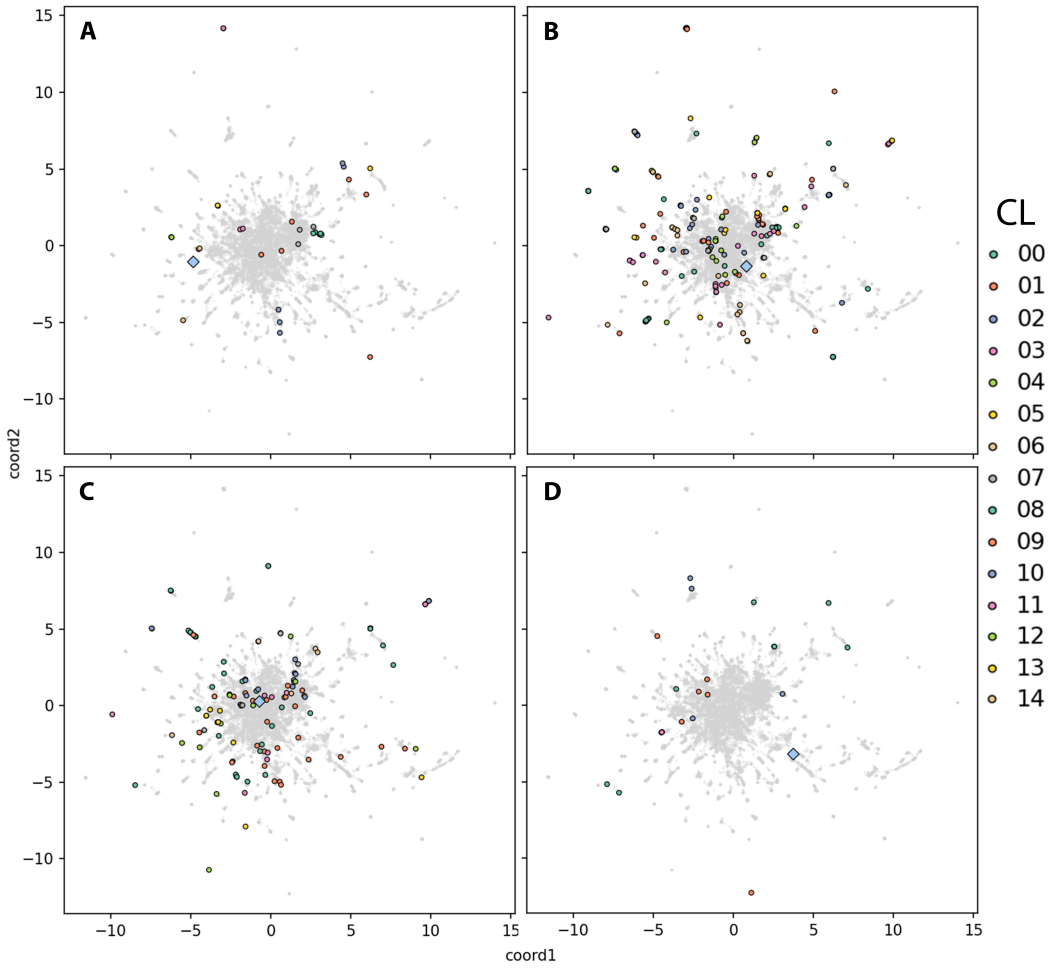

Figure 5. Two-dimensional embedding of gene interaction networks. Each panel represents the reduced vector space (coord1 vs. coord2) of the full gene interaction network based on STRING "Experimental" evidence. Background grey dots correspond to all expressed genes included in the network. The disease-causal gene (seed) for each condition (*DMD*, *TNPO3*, *C9ORF72*, and *FUS*) is represented as a blue rhombus. Panels correspond to the DMD_myot (**A**), LGMD_myob (**B**), ALS_iN_C9ORF72 (**C**), and ALS_fib_FUS (**D**) datasets. Colored dots represent differentially expressed genes (DEGs), grouped by cluster (CL).

To avoid possible biases in the visual representation of the data and the possibility of overlapping of the causal gene with any cluster, in Supplementary Tables S5 and S6 the ranked information, based on their similarity score, of the clusters closest and furthest to the causal gene is included. With this information, we can confirm that the closest cluster with respect to the *DMD* causal gene is observed for the DMD_cfib (cluster 0, score: 1.00) and DMD_pCard (cluster 8, score: 1.00) datasets, closely followed by DMD_myob (cluster 7,

score: 0.94). In the case of DMD_myot, cluster 3 is the one with the highest clustering score for this dataset (score: 0.77). The rest of the clusters, except for cluster 10 for the DMD_myob dataset (score: 0.11), present score values very close to 0. No clusters are observed that could be relevant for the LGMD-related datasets and their association with the disease causal gene (*TNPO3*), and in ALS, cluster 9 of the ALS_iN_C9ORF72 dataset has a score of 0.31, a substantially lower clustering score value compared to those calculated for the mentioned DMD datasets. In the fibroblast-derived ALS dataset, cluster 2 is the closest to the *FUS* causal gene, with a similarity score of 0.02, the lowest among the four datasets for the cluster nearest to its corresponding causal gene. The close proximity of these clusters to the causal gene suggests that the associated DEGs may interact directly with it and be involved in related biological processes. Based on this assumption, we performed a functional enrichment analysis of the DEGs within each cluster, prioritizing those closest to the causal gene.

The functional enrichment analysis of DEGs in cluster 0 (*DMD* and *ANLN*) for the DMD_cfib dataset revealed a strong association with molecular functions central to cytoskeletal organization and muscle integrity (Supplementary Table S8). Among the significantly enriched terms were dystroglycan binding, nitric-oxide synthase binding, vinculin binding, actin binding, and structural constituent of muscle. These functions are closely related to the biological role of dystrophin. It is interesting to mention that in the comparison performed for the differential expression analysis of this dataset, the *DMD* gene appears strongly down-regulated ($\log_2$FC = -2.12). In the case of the DMD_pCard dataset (Supplementary Table S9), cluster 8 included genes enriched in biological processes related to the negative regulation of coagulation and hemostasis, including negative regulation of blood coagulation, negative regulation of hemostasis, and innervation among them. These functions may suggest that dystrophin deficiency in cardiac tissue may also influence vascular homeostasis and neural regulation. Functional analysis of cluster 7 DEGs in the DMD_myob dataset included key molecular functions associated with extracellular matrix organization and protein processing, like dystroglycan binding, apolipoprotein binding, extracellular matrix structural constituent, serine-type endopeptidase activity, and serine-type peptidase activity (Supplementary Table S10). These functions suggests a possible role for this cluster of DEGs in maintaining tissue structure and regulating proteolytic activity, processes that are often dysregulated in DMD. For the DMD_myot dataset, cluster 3 included DEGs enriched in terms related to cardiac and muscle function, including the regulation of cardiac muscle cell contraction, specifically in ventricles, bundle of His cell-to-Purkinje myocyte communication, and actin-mediated cell contraction (Supplementary Table S7). These processes are critical for maintaining coordinated cardiac conduction and contractile function, suggesting that genes in this cluster may reflect the broader impact of dystrophin deficiency beyond skeletal muscle, potentially contributing to the cardiomyopathic features frequently observed in DMD. Regarding the ALS datasets, in the case of ALS_iN_C9ORF72 we found that genes in cluster 9 were enriched in biological functions related to synaptic structure modification and mitochondrial apoptotic processes, like modification of postsynaptic structure, positive regulation of release of cytochrome c from mitochondria, and apoptotic mitochondrial changes (Supplementary Table S11). These results suggest a disruption in both neuronal connectivity and mitochondrial integrity in *C9ORF72*-associated ALS. And in the case of the ALS_fib_FUS dataset (Supplementary Table S12), cluster 2 DEGs were enriched in biological process that are mainly related to connective and adipose tissue development, Wnt signaling regulation, and pattern specification, which may point to potential alterations in mesenchymal tissue remodeling and developmental signaling pathways and contribute to disease mechanisms in ALS linked to *FUS* mutations.

2.4.2. Prioritization of Isolated DEGs

In addition to clustered DEGs, we also identified a subset of network-mapped DEGs that do not form significant interactions with other DEGs and thus remain outside of defined functional clusters. We refer to these as "isolated DEGs". Despite their lack of clustering, these genes may still play important roles in disease pathogenesis, particularly if they are located in close proximity to the disease-causal gene within the human interactome, as this spatial closeness can suggest potential functional relevance through shared or convergent biological processes.

The top ten isolated DEGs for each dataset are ranked by their similarity score, from highest to lowest, and included in Supplementary Table S13 for the DMD-related datasets and in Supplementary Table S14 for LGMD and ALS datasets. In the case of the DMD_myot dataset, the *DMD* gene, which is the causal gene for DMD, is excluded from the identified gene clusters. This is an intriguing observation, as it suggests that despite being central to the disease, *DMD* may not directly interact with the rest of DEGs. It is interesting that the gene *CDH12* appears shared between two datasets of the same disease, DMD_pCard and DMD_cfib, but even more so is the case of *HS3ST3A1*, which appears shared in the datasets LGMD_myob and ALS_fib_FUS, both belonging to two different diseases caused by mutations in different genes. In the case of the DMD_myob dataset, it should also be noted that the *EPSTI1* gene appears with a very high score (0.99), and, despite not having clustered with other DEGs, this fact suggests some kind of functional relationship. In addition, it is interesting to mention *ABCB11* and *PTPRN* genes in DMD_pCard and DMD_cfib, respectively, as the isolated DEGs with the highest scores for these two DMD datasets. In the LGMD_myob dataset, the high-scoring isolated genes *ST6GAL1* (score: 0.99), *SRPK3* (score: 0.96), and *P2RX5* (score: 0.89) may reflect downstream effects of *TNPO3* dysfunction. In the case of the LGMD_pbmc dataset, the isolated genes *CHERP* (score: 0.84) and *EPHA1* (0.82) may also represent functionally relevant downstream targets of *TNPO3*-related dysregulation due to its proximity in the network to the causal gene of LGMD. And regarding the DEGs that did not cluster in the ALS datasets, the proximity of *NREP* (neuronal regeneration-related protein, score: 0.93) to *C9ORF72* in the network may reflect shared roles in neuronal plasticity, cytoskeletal remodeling, and cellular stress responses. In addition, the second closest DEG to *C9ORF72* is *MAP1LC3C*, which encodes a protein involved in the regulation of microtubule dynamics and autophagy. Finally, the isolated genes *JMJD6* and *VRK1*, in the ALS_fib_FUS dataset, showed high similarity scores with *FUS* in the interaction network, suggesting potential functional connections.

3. Discussion

Understanding neuromuscular diseases (NMDs), such as Duchenne muscular dystrophy (DMD), limb–girdle muscular dystrophy (LGMD), and amyotrophic lateral sclerosis (ALS), requires a comprehensive exploration of their genetic and molecular bases. They are severe NMDs with distinct genetic origins: mutations in the *DMD* gene for DMD [7], different genes depending on the LGMD subtype (like *TNPO3* [28]), and alterations in more than 40 genes in ALS [16], including *C9ORF72* [29,30] and *FUS* [31,32] among the most common ALS-associated genes, along with *SOD1* and *TARDBP* [33]. Despite their phenotypic differences, they converge in the disruption of fundamental biological processes, including cellular homeostasis, inflammation, and synaptic function [11,34]. Thus, comprehending the molecular basis of these NMDs is crucial for the development of targeted and effective therapeutic strategies [1].

From a molecular perspective, transcriptomic analysis has proven to be a powerful approach for identifying gene expression alterations associated with neuroinflammatory and neurodevelopmental diseases [35], offering valuable insights into dysregulated

pathways and potential biomarkers [36]. In the context of DMD, LGMD, and ALS, transcriptomic analysis through differential gene expression (RNA-seq) is crucial to elucidate the downstream molecular consequences of genetic alterations in key disease-associated genes, such as *DMD* [37,38], *TNPO3* [28], *C9ORF72* [29,39], and *FUS* [32], which may lead to altered expression profiles contributing to pathogenesis. First of all, to perform this study, we made sure to have at least two datasets for each of the diseases we analyzed, and we also performed sample selection from different tissue types to ensure biological relevance and enhance the interpretability of transcriptomic signatures. With this consideration, we selected samples from differentiated tissues, such as myoblasts (in the case of the DMD_myob and LGMD_myob datasets), cardiac fibroblasts (DMD_cfib dataset), iNeurons (ALS_iN_C9ORF72), and myotubes derived from reprogrammed skin fibroblasts (DMD_myot), and undifferentiated tissue samples, including fibroblasts, as in the DMD_cfib (cardiac fibroblasts) and ALS_fib_FUS datasets, and peripheral blood mononuclear cells (PBMCs) in the LGMD_pbmc dataset. Fibroblasts serve as a useful reference due to their accessibility and broad transcriptomic representation but may not fully capture tissue-specific pathological mechanisms. Therefore, we included cardiomyocytes for DMD, given the cardiac involvement commonly associated with dystrophin deficiency [40], and induced neurons (iNeurons) for ALS, as they more closely reflect the neurodegenerative processes central to the disease [41]. In addition, we incorporated datasets derived from myoblasts and myotubes to better characterize muscle-specific alterations in DMD and LGMD [42] and PBMCs to investigate systemic immune responses in LGMD [43], thus ensuring a broader representation of disease-relevant biological contexts. This strategy enabled a comprehensive view of both shared and tissue-specific alterations across NMDs, despite the challenge of limited high-quality data due to their rarity and our selection criteria.

3.1. Differential Expression Analysis Reveals Novel Insights from NMD Dataset-Specific Comparisons

Our differential expression results largely aligned with original studies, supporting the robustness of our pipeline and the biological relevance of the findings. Careful sample selection and dataset-specific comparisons enabled the identification of novel insights across NMDs. In all datasets except ALS_fib_FUS and DMD_cfib, we recovered expected processes such as morphogenesis, synaptic regulation, muscle integrity, inflammation, and extracellular matrix organization [43–48]. In DMD_cfib, unlike Soussi et al. [49], we did not detect glycolysis or mitochondrial dysfunction but found structural organization functions. Similarly, in ALS_fib_FUS, our findings contrasted with those of Kumbier et al. [31], who reported enrichment in metabolism, gene expression, and antigen processing. Instead, we detected functions such as cardiac tissue development, kidney morphogenesis, and blood circulation regulation, also present in other differentiated tissue datasets (Supplementary Figure S7). These differences likely reflect divergent methodological aims, as Kumbier et al. focused on subtle progression-related biomarkers through machine learning [31], while our approach emphasized broader transcriptional patterns.

3.2. Comparative Functional Enrichment Extends Original DEG Findings

The comparative functional enrichment analysis of DEGs across the different NMD datasets revealed a complex and interconnected landscape of biological functions (Figure 2). While each dataset reflects specific disease contexts, our results show an overlap in functional categories, indicating shared molecular mechanisms that transcend clinical and genetic heterogeneity. Among the most recurrently enriched functions were extracellular matrix binding and glycosaminoglycan binding, present across most datasets. These terms suggest a conserved dysregulation of extracellular matrix (ECM) interactions across NMDs,

which may contribute to impaired tissue integrity and aberrant cell signaling in both myogenic and neurogenic contexts [50–52]. A noticeable results is the frequent occurrence of the glycosaminoglycan binding term in most DMD-related datasets and in the ALS_fib_FUS dataset. Glycosaminoglycans are vital ECM components that regulate growth factor signaling, neuronal growth, synaptic plasticity, and tissue organization, dysregulation of which has been previously associated with muscular dystrophies like DMD [53,54]. In the context of ALS, and particularly *FUS*-associated ALS, recent studies have emphasized the role of ECM alterations and synaptic dysfunction mediated by RNA-binding proteins like *FUS* [55]. The enrichment of glycosaminoglycan binding shown in the ALS_fib_FUS dataset could suggest that ECM–protein interactions may be implicated in the downstream effects of *FUS* mutations, potentially influencing synaptic integrity and motor neuron survival. And the convergence of this function between DMD and ALS_fib_FUS datasets, despite their genetic heterogeneity, could reflect the potential involvement of ECM dysregulation as a shared pathological mechanism across these diseases. In addition, muscle-related functions, such as actin binding and structural constituent of muscle, were shared specifically between DMD and LGMD datasets derived from cardiomyocytes and myoblasts (DMD_pCard and LGMD_myob, respectively), underscoring common disruptions in cytoskeletal and contractile protein networks. These findings align with the known involvement of cytoskeletal integrity in muscle pathophysiology and suggest that even when different genes are causative, downstream effects on actin filament organization and contractility are preserved across muscular dystrophies [56]. Interestingly, functions such as integrin binding, growth factor binding, and collagen binding were found in DMD- and LGMD-related datasets of similar tissue origins, indicating that tissue context strongly influences which shared pathways are activated or disrupted. Moreover, the detection of peptidase regulator activity as a shared functional category between LGMD_pbmc and ALS_iN_C9ORF72, despite their distinct pathological phenotypes, raises the possibility of convergent regulatory mechanisms involving proteostasis or inflammatory modulation [34].

3.3. Insights from Mapping Differential Expression onto the Human Interactome

Despite differences in disease type and cell context across the datasets, the identification of shared genes and conserved biological functions reveals core mechanisms of neuromuscular pathology that are likely of broad significance. This principle drives our integrative systems-level methodology: by mapping differential expression onto the human interactome, we aim to uncover molecular signatures and pathways that consistently emerge across heterogeneous datasets, providing insights not only into disease mechanisms but also into potential therapeutic targets.

Mapping disease-causal genes and their associated DEGs onto the human interactome revealed unexpected patterns of proximity among NMDs, according to results obtained from the dataset view. Notably, despite belonging to clinically distinct diseases, the causal genes *C9ORF72* (ALS) and *TNPO3* (LGMD) are positioned closely in the network and share nearby DEGs primarily from DMD- and LGMD-related datasets. This suggests a potential molecular overlap involving *C9ORF72* and *TNPO3*, despite their association with distinct diseases, and reinforces the idea of a shared pathogenic basis between LGMD and ALS, which may help explain some overlapping molecular features as previously described in the comparative functional enrichment analysis of DEGs. Conversely, *FUS* and *C9ORF72*, both causally linked to ALS, are topologically distant in the interactome, suggesting involvement in distinct molecular mechanisms. The spatial separation observed between *FUS* and its associated DEGs, and similarly between *DMD* and its DEGs, may reflect complex regulatory relationships or tissue-specific expression patterns. These findings demonstrate

that, despite the datasets' limitations, our integrative approach offers new insights into shared pathways and potential therapeutic targets across NMDs.

3.4. Biological Relevance of Unmapped DEGs

A notable subset of DEGs across datasets remained unmapped to the human interactome and were classified as unmapped DEGs (uDEGs). Of special interest is *PAX8-AS1*, which is known to participate in transcriptional regulation and chromatin remodeling, with roles in modulating inflammatory and fibrotic responses characteristics of DMD cardiomyopathy [57]. Its over-expression in differentiated tissues and in LGMD_myob may reflect shared pathological mechanisms related to muscle remodeling or stress responses in muscular dystrophies. Given the shared cardiac and fibrotic manifestations in DMD and LGMD [40], this expression pattern raises the possibility that *PAX8-AS1* could serve as a novel biomarker for disease activity or progression in these NMDs. In the case of the *MEG3* gene, it regulates cell differentiation and apoptosis, with roles that vary depending on tissue context [58]. Its over-expression in skeletal muscle cells may reflect a compensatory response to dystrophin deficiency, promoting regenerative or stress-related pathways, while its down-regulation in cardiac cells could indicate a distinct regulatory landscape where *MEG3*-mediated mechanisms are suppressed or differently modulated, possibly contributing to cardiac-specific aspects of DMD [59]. *SBF2-AS1*, an lncRNA involved in regulating cell proliferation, migration, and stress responses [60], shows opposite expression patterns in LGMD myoblasts and ALS iNeurons. Its over-expression in LGMD may reflect a compensatory response to muscle degeneration and inflammation [61], while down-regulation in ALS suggests impaired stress responses linked to *C9ORF72*-related neuronal dysfunction. These tissue-specific differences underscore the role of *SBF2-AS1* as a context-dependent regulatory hub influencing disease progression, making it a promising target to explore shared molecular mechanisms across NMDs. And regarding *NEAT1*, a critical lncRNA for paraspeckle formation and RNA regulation [62], shows contrasting expression in NMDs: It is over-expressed in DMD myoblasts, potentially as an adaptive response to muscle stress and inflammation, helping to maintain transcriptional stability and protect muscle cells. Conversely, its down-regulation in ALS fibroblasts with *FUS* mutations aligns with paraspeckle dysfunction observed in ALS [63], which may impair RNA metabolism and stress responses in neurons. These opposing patterns emphasize tissue-specific paraspeckle roles in disease, and may position *NEAT1* as a promising molecular link and biomarker across muscular dystrophies and neurodegenerative disorders. These findings suggest that lncRNAs may contribute to disease- and tissue-specific regulatory programs in NMDs and represent a layer of molecular complexity not captured by current interactome-based analyses. Nonetheless, these findings should be validated through experimental approaches.

3.5. Integrating Differential Expression Profiles with the Human Interactome to Elucidate Disease Mechanisms

The spatial proximity of DEG clusters to causal genes in our integrative network analysis suggests functional associations relevant to disease pathogenesis. In the DMD_myot dataset, the proximity of a DEG cluster enriched in cardiac electrophysiology and muscle contraction functions to the causal gene *DMD* underscores the systemic impact of dystrophin deficiency on both skeletal and cardiac muscle, including disrupted Purkinje fiber signaling and actin-mediated contraction [40,64]. In DMD_pCard, cluster 8 was enriched in processes related to coagulation and hemostasis, supporting the involvement of vascular dysregulation in cardiac manifestations of DMD. Similarly, in DMD_cfib, although fewer DEGs were detected, cluster 0 showed enrichment in cytoskeletal organization and extracellular matrix anchoring, functions directly tied to the role of dystrophin in cardiac fibroblasts

and indicative of remodeling processes in the DMD heart. Cluster 7 in the DMD_myob dataset was enriched in genes involved in tissue architecture, extracellular proteolysis, and dystroglycan and apolipoprotein binding, reflecting the impact of dystrophin loss on membrane dynamics, lipid metabolism, and inflammatory remodeling during myogenesis. In the ALS_iN_C9ORF72 dataset, cluster 9 DEGs were enriched in functions related to neuronal homeostasis, including autophagy, synaptic regulation, and mitochondrial maintenance, processes central to C9ORF72-related ALS. Lastly, ALS_fib_FUS cluster 8 showed enrichment in connective tissue development and Wnt signaling regulation, indicating potential systemic and developmental effects of *FUS* mutations in non-neuronal cells [65]. The identified gene clusters and their associated biological processes represent plausible candidates for involvement in disease mechanisms; however, experimental validation is required to confirm their functional relevance.

3.6. Biological Relevance and Interpretation of Isolated DEGs

The analysis of isolated DEGs provides insights into specialized or context-specific roles not captured by co-regulated modules. These genes may function in parallel or non-canonical pathways, act in distinct tissues or disease stages, or reflect dynamic regulatory events not apparent in the current dataset. We identified *HS3ST3A1* as a high-scoring, isolated DEG in both LGMD_myob and ALS_fib_FUS, suggesting a previously unrecognized shared role in LGMD and ALS despite lacking direct interaction with *TNPO3* or *FUS*. Similarly, *CDH12* emerged as a consistently isolated DEG in DMD_pCard and DMD_cfib, supporting its involvement in fibroblast-mediated cardiac remodeling in DMD [66] and pointing it out as a potential therapeutic target. Several isolated DEGs in the DMD datasets are biologically relevant despite not clustering. In DMD_myob, *EPSTI1* is linked to interferon signaling and tissue remodeling, suggesting a role in inflammatory responses [67]. In DMD_pCard, *ABCB11* may affect cardiac energetics via lipid transport under stress [68], and in DMD_cfib, *PTPRN* may modulate intercellular communication. These DEGs, although isolated, are functionally proximal to *DMD* and may contribute to tissue-specific manifestations or disease progression. Notably, *DMD* itself was significantly down-regulated in DMD_myot, consistent with previous studies [48] and supporting the robustness of our approach. In the LGMD-related datasets, isolated DEGs found in LGMD_myob, such as *ST6GAL1*, *SRPK3*, and *P2RX5*, may reflect downstream effects of *TNPO3* dysfunction, impacting glycosylation, splicing, and membrane signaling in myoblasts. In LGMD_pbmc, *CHERP* and *EPHA1* suggest systemic effects involving splicing and immune signaling, reinforcing the broad functional impact of *TNPO3* [43]. For the ALS-related datasets, in ALS_iN_C9ORF72, isolated DEGs like *NREP* and *MAP1LC3C* are associated with neuronal regeneration, autophagy, and microtubule dynamics, processes disrupted in ALS [30,69]. Their proximity to *C9ORF72* supports their potential contribution to neurodegeneration [70]. In ALS_fib_FUS, *JMJD6* (involved in chromatin remodeling) and *VRK1* (linked to motor neuron degeneration) both show strong proximity to *FUS*, suggesting convergence on transcriptional and epigenetic dysregulation in ALS pathogenesis [71,72].

3.7. Study Limitations

The datasets analyzed in this study encompass different neuromuscular disease contexts (DMD, LGMD, and ALS) and diverse cell types, including myoblasts, myotubes, and fibroblasts, among others. While this heterogeneity introduces variability, it also enables the identification of molecular mechanisms that are consistently deregulated across distinct disease models. Using our integrative approach, we identified convergent alterations, such as shared DEGs and common functional categories, that show potential core disease mechanisms transcending individual cell types or experimental conditions. We note, however,

that cell-type-specific conclusions cannot be directly drawn from this analysis, and the findings should be interpreted as reflecting broad, conserved pathways underlying NMDs.

Another limitation consists of the small sample size per dataset, common in rare disease research, which limits statistical power and may hide subtle but biologically relevant expression changes. Access to patient-derived RNA-seq data is further constrained by ethical and legal restrictions, impeding broader data integration. Our network-based framework is also constrained by the incompleteness of current interaction databases, particularly regarding poorly annotated genes and ncRNAs. This gap can lead to the exclusion of potentially important regulators from clustering and prioritization analyses. Future efforts in expanding and curating interaction datasets, especially for lncRNAs, are essential to better capture the regulatory landscape of NMD. Additionally, variability in RNA-seq library preparation protocols across datasets introduces technical noise. PolyA-selected libraries enrich for mature protein-coding RNAs but miss non-polyadenylated transcripts, while ribodepletion includes a broader range of RNAs at the cost of sequencing depth per transcript. These protocol differences should be carefully considered in cross-study analyses and indicate the need for methodological harmonization. Expanding this approach to additional LGMD and ALS subtypes, as more datasets become available (e.g., via SRA), will help refine disease-specific versus shared molecular signatures. Moreover, future versions of our pipeline will incorporate regulatory gene–lncRNA relationships, enabling deeper insight into the non-coding architecture of NMDs. Altogether, our findings generate testable hypotheses on shared disease mechanisms and identify candidate genes for further functional validation.

4. Materials and Methods

We selected RNA-seq datasets of two neuromuscular disorders (NMDs) with clinically similar phenotypes [28], Duchenne muscular dystrophy (DMD), caused by mutations in *DMD* [73], and limb–girdle muscular dystrophy type D2 (also known as LGMD *TNPO3*-related or LGMDD2), a rare condition caused by mutations in the *TNPO3* gene [28], to explore shared molecular signatures. Additionally, we included two datasets from amyotrophic lateral sclerosis (ALS), a clinically distinct NMD, each involving different causal genes, to assess disease-specific and convergent mechanisms across diverse pathological contexts.

4.1. Dataset Description

We conducted three independent advanced searches in the NCBI Sequence Read Archive (SRA) online platform (https://www.ncbi.nlm.nih.gov/sra), accessed on 20 May 2025, to identify RNA sequencing (RNA-seq) datasets for DMD, LGMD, and ALS. Search queries included the terms “Duchenne Muscular Dystrophy”, “Limb-Girdle Muscular Dystrophy”, and “Amyotrophic Lateral Sclerosis”, restricting results to the *Homo sapiens* organism, “transcriptomic” (RNA) as the source, “Illumina” as the sequence platform, and “fastq” as the file type. Retrieved datasets were manually curated by selecting only those with at least three samples per condition, excluding studies that did not meet this minimum requirement. Additionally, we prioritized datasets generated using paired-end sequencing over single-end reads. Following this selection process, eight datasets were chosen for analysis: four corresponding to DMD with mutations in the *DMD* gene; two for LGMDD2 with mutations in *TNPO3*; and two for ALS, one involving mutations in *C9ORF72* and the other in *FUS* genes.

To facilitate dataset tracking, we assigned standardized names based on disease and cell type and, in the case of ALS datasets, causal gene studied: DMD_myot, DMD_pCard, DMD_cfib, DMD_myob, LGMD_myob, LGMD_pbmc, ALS_iN_C9ORF72,

and ALS_fib_FUS. These datasets encompass different Illumina sequencing platforms and sequencing layouts (single-end and paired-end). Additionally, we ensured that samples were obtained from different tissues, including, in the case of DMD, myotubes derived from reprogrammed skin fibroblasts (DMD_myot dataset), cardiomyocytes differentiated from iPSCs reprogrammed from peripheral blood mononuclear cells (PBMCs) in the DMD_pCard dataset, and human-induced pluripotent stem cell (hiPSC) lines derived from cardiac fibroblasts (DMD_cfib dataset) and myoblasts (DMD_myob dataset); in the case of LGMD, primary myoblasts (LGMD_myob dataset) and PBMCs (LGMD_pbmc dataset); and in the case of ALS, induced neurons (iNeurons) for the ALS_iN_C9ORF72 dataset and fibroblasts for the ALS_fib_FUS dataset. More information on these datasets is provided in Table 3.

Table 3. Summary of datasets used in the study, including the name of the dataset, number of samples per dataset, causal gene, average read size (ARS) per dataset, minimum read length (MRL) set to trim reads, ExpHunter Suite parameters, including minimum libraries selected (MLS) and $\log_2$ fold change ($\log_2$FC), and reference of the samples' original studies. ALS: amyotrophic lateral sclerosis, DMD: Duchenne muscular dystrophy, LGMD: limb–girdle muscular dystrophy.

Dataset	Samples	Causal Gene	ARS	MRL	MLS	$\log_2$FC	Ref.
DMD_myot	6 (3 isogenic controls + 3 DMD)	*DMD*	75	60	2	1.5	[48]
DMD_pCard	6 (3 CRISPR-Cas9 corrected + 3 DMD)	*DMD*	74	60	2	1.5	[44]
DMD_cfib	8 (4 healthy controls + 4 DMD)	*DMD*	100	85	2	1	[49]
DMD_myob	12 (3 controls + 9 patients)	*DMD*	150	135	2	1.5	[46]
LGMD_myob	6 (3 controls + 3 patients)	*TNPO3*	150	135	2	2.5	[47]
LGMD_pbmc	20 (10 controls and 10 patients)	*TNPO3*	76	61	2	1	[43]
ALS_iN_C9ORF72	12 (6 controls + 6 C9ORF72-ALS)	*C9ORF72*	150	135	2	0.6	[45]
ALS_fib_FUS	25 (13 controls + 12 FUS-ALS)	*FUS*	150	135	8	0.6	[31]

It is important to note that in our study, the differential expression analyses focused not on the specific location of the mutations within the genes analyzed, but rather on the broader impact of these genetic alterations on the expression of other genes. Furthermore, we carefully selected samples for each dataset, prioritizing controls from healthy individuals over isogenic controls, as they better reflect differential expression between healthy and diseased conditions. When isogenic controls were used, we chose those generated with the most effective gene-editing techniques to minimize potential biases in our results. It is also important to consider that our integrative analysis methodology is specifically designed with the limitations of rare disease research in mind, particularly the typically low number of samples available per dataset. Further details of the selected RNA-seq datasets are included in Supplementary Table S3.

For the DMD_myot dataset, Paredes-Redondo et al. [48] investigated neuromuscular junction defects and their potential impact on DMD pathogenesis. DMD patient fibroblasts carrying the c.10141C>T (p.R3381X) nonsense mutation, which affects all tissue-specific dystrophin isoforms, were reprogrammed into expanded potential stem cells, which were differentiated into embryoid bodies, and an isogenic control line was generated by correcting the mutation via CRISPR-Cas9. To determine whether the impaired differentiation and fusion observed in DMD-cells stemmed from abnormal myogenic gene expression, they conducted transcriptome sequencing and analysis of DMD patient-derived myotubes and the corrected DMD myogenic cultures at 0, 24, and 120 h during secondary differentiation. To capture more mature transcriptional profiles that better reflect the functional consequences of the DMD mutation, we selected samples belonging to 120 h of differentiation. A

total of three samples per condition (CRISPR/Cas9 isogenic-corrected controls and DMD patient samples) were analyzed.

Regarding the DMD_pCard dataset, Atmanli et al. [44] investigated the structural, functional, and transcriptional differences in cardiomyocytes derived from iPSCs reprogrammed from PBMS of a DMD patient. The original study comprised nine samples: three from the patient and six from CRISPR/Cas9 isogenic-corrected controls in which the *DMD* open reading frame was restored either by reframing (three samples) or exon skipping (three samples). For our comparison, we selected samples corrected by reframing, as this approach aims to preserve the near-full-length protein structure, potentially leading to a more physiologically accurate rescue of the dystrophin gene [74].

In the case of the DMD_cfib dataset, Soussi et al. [49] analyzed the gene expression profiles from hiPSC-derived cardiac fibroblasts obtained from DMD patients and healthy controls. DMD samples lack the full-length dystrophin isoform due to gene mutations, leading to impaired actin microfilament organization and a metabolic shift from oxidative phosphorylation to glycolysis. These cells also show disrupted mitochondrial networks, reduced mitochondrial respiration, and an enhanced myofibroblast phenotype in response to profibrotic stimuli. As cardiac fibrosis is a hallmark of DMD-related cardiomyopathy, this model provides insights into how dystrophin deficiency in non-cardiomyocyte cells contributes to disease progression. We used the eight samples available at the SRA, with four corresponding to control individuals and four corresponding to patients with mutations in the *DMD* gene.

Regarding the latest DMD dataset (DMD_myob), Lemoine et al. [46] carried out a study demonstrating the effectiveness of a single guide RNA CRISPR strategy to delete exon duplications in the DMD gene (exon 2, exons 2–9, and exons 8–9) in patient-derived myogenic cells, demonstrating that correction restored dystrophin expression and normalized related gene pathways, as shown by immunostaining and RNA-seq. In our study, we did not investigate the effects of CRISPR correction. Instead, we focused on performing differential expression analysis using samples from patients with different DMD mutations and a healthy control. Regarding this consideration, we analyzed the samples corresponding to myoblasts from patients carrying exon 2, exon 2–9, and exon 8–9 duplications in the DMD gene, along with samples from an immortalized myoblasts cell line (C25) belonging to a healthy individual. Each condition was represented by three replicates.

For the case of LGMD, we selected two datasets corresponding to LGMDD2, the *TNPO3*-related subtype of LGMD. For clarification, throughout the manuscript we refer to this disease subtype as LGMD. For the first dataset, LGMD_myob, published by Poyatos-Garcia et al. [47], they used a patient-derived immortalized myoblast model that recapitulates disease features, including *TNPO3* over-expression, impaired muscle differentiation, and autophagy dysregulation. CRISPR-Cas9 correction of the mutation reversed these phenotypes, eliminating the mutant protein and restoring 44% of transcriptomic alterations and 50% of dysregulated miRNAs. Muscle biopsies were obtained from the tibialis anterior of a 33-year-old male patient diagnosed with LGMD and from the quadriceps of a 38-year-old healthy male donor. Primary human myoblasts derived from these biopsies were immortalized through lentiviral transduction with hTERT and CDK4 vectors, followed by clonal selection. For each condition (LGMD and healthy control), three independent clonal lines were established and used as biological replicates. These immortalized myoblasts were cultured under standard growth conditions and subsequently differentiated to analyze disease-associated molecular alterations. In our study, we compared the three samples belonging to the healthy control against the LGMD patient samples.

In the second LGMD dataset (LGMD_pbmc) Diez-Fuertes et al. [43] explored the molecular basis of LGMD, performing a transcriptome analysis of PBMCs from LGMD

patients with the c.2771delA mutation in the *TNPO3* gene against healthy individuals, revealing differentially expressed genes (DEGs) with pro-inflammatory and antiviral process functions associated. This dataset includes 20 samples: 10 from LGMD patients with the deletion c.2771delA in the *TNPO3* gene and 10 samples from healthy individuals.

For the ALS_iN_C9ORF72 dataset, Castelli et al. [45] investigated the effects of neuroprotection in induced neurons (iNeurons) derived from ALS patients carrying expansion mutations in the *C9ORF72* gene. The study focused on a gene therapy approach aimed at inhibiting the SRSF1-dependent nuclear export of C9ORF72 repeat transcripts. For our analysis, we first selected 12 human samples from induced neurons (iNeurons) derived from three healthy individuals and three C9ORF72-ALS patients from whole-cell (WCT) and cytoplasmatic (CyT) transcriptomes. However, we realized that for the ALS_iN_C9ORF72 dataset, the samples belonging to the control (Ctrl_R2_WCT) and patient (ALS_R2_WCT) R2, in the whole-cell transcriptome (WCT), presented a high percentage of reads with incomplete alignments (Supplementary Figure S1G), and, consequently, both were discarded from the analysis. Therefore, due to the limited number of samples in each group, which would not lead to a robust differential expression analysis (even though this comparison would have been the most consistent with the rest of this study), we decided to analyze the comparison between control and ALS patient samples from the cytoplasmic transcriptome (CyT) instead.

And regarding the ALS_fib_FUS dataset, Kumbier et al. [31] investigated whether fibroblasts derived from ALS patients could be used to identify phenotypic heterogeneity in both sporadic ALS and FUS-mutated ALS cases. From the original dataset, we selected 25 samples, comprising 13 controls and 12 FUS-mutated ALS samples.

4.2. Dataset Processing for Differential Expression Analysis

All FASTQ files from the datasets were processed using the analysis methodology previously described in [75]. It performs a first quality analysis over raw sequence files using FastQC and then a pre-processing of the reads using SeqtrimBB, an in-house tool built on the BBmap framework [76]. The minimum Phred quality per nucleotide for all datasets was set to 26. The pre-processed read files were aligned against the human reference genome (version GRCh38.p13) using STAR [77], and the count tables obtained for each sample were aggregated into a single file of counts per gene and sample to analyze differential gene expression with the R/Bioconductor package ExpHunter Suite [78].

To ensure consistent and biologically meaningful detection of gene expression across datasets, we applied a filtering threshold of at least two counts per million (CPM) in a minimum number of libraries, adapted proportionally to the sample size of each comparison. For instance, in comparisons with smaller groups (e.g., three vs. three), genes were retained if they reached this expression level in at least two libraries per group. In larger datasets, the threshold was scaled accordingly, requiring expression in at least two-thirds of the libraries per group. This proportional filtering strategy helped minimize noise from lowly expressed or sporadic transcripts, ensuring a more reliable identification of DEGs while maintaining comparability across datasets of varying sizes. All differential expression analyses were conducted using DESeq2 [79], as implemented in the ExpHunter Suite [78], and $\log_2$FC and *p*-value information is available in Table 3. We selected different $\log_2$FC cutoffs depending on the number of DEGs obtained for the comparisons performed for each dataset, with the aim of balancing sensitivity and specificity to detect relevant genes and minimize noise, respectively.

Genes that passed the filtering criteria and were identified by DESeq2 were considered DEGs and selected for functional analysis. Functional enrichment was performed using the ExpHunter Suite module based on clusterProfiler [80], with annotations from Gene Ontol-

ogy (GO) [81]. The over-representation analysis (ORA) method was applied, evaluating both over-expressed and down-regulated DEGs through a hypergeometric test for each term in the selected functional categories. An adjusted *p*-value threshold of 0.05, using the Benjamini–Hochberg (BH) method, was used across all datasets.

4.3. Integrative Analysis

To perform the integrative analysis, we constructed a protein–protein interaction (PPI) network using data from the "Experimental" channel of the STRING database (v12.0 [82], https://string-db.org (accessed on 26 May 2025)), which compiles physical interactions supported by laboratory evidence. The distribution of the confidence score for this STRING channel is displayed in Supplementary Figure S10. To ensure reliability, we retained only interactions with a confidence score above 150. ENSEMBL protein identifiers were mapped to their corresponding gene identifiers, and we kept all the genes that were found to be expressed in at least one of the eight datasets. Additionally, a connected component filter was applied, removing all nodes that belonged to connected components (subsets of nodes in which every node is reachable from any other) with fewer than five nodes. Then, DEGs from each study were projected onto the interactome. The resulting DEG subgraphs were clustered, and for each study, we computed the proximity between each cluster and the corresponding causal gene (*DMD*, *TNPO3*, *C9ORF72*, and *FUS*) to identify biological pathways and potential therapeutic targets. All network analyses were conducted using NetAnalyzer [83], a Python library developed by our group (v1.0, https://pypi.org/project/NetAnalyzer (accessed on 27 May 2025)). The complete analysis workflow is available at https://github.com/lhurtadogarcia/degs2net (accessed on 10 July 2025), and the interactive reports for all datasets analyzed are available in the integrative analysis results repository at https://github.com/ElenaRojano/INTRINSED_datasets (accessed on 29 July 2025).

The PPI network derived from each dataset was clustered using the Louvain method [84], following previous studies [85], and implemented through the main module of NetAnalyzer [86]. To ensure structural coherence and biological relevance, we kept clusters containing three or more nodes. Then, the proximity between each resulting cluster and the corresponding causal gene was computed. To achieve this, a similarity matrix was constructed by applying the *node2vec* algorithm [87] with parameters `dimensions = 128, walk_length = 100, num_walks = 10, p = 1, q = 1, window = 10` to the degree-normalized adjacency matrix of the network. The resulting similarity matrix represents the dot product of node vectors embedded in a high-dimensional space. To account for differences in vector magnitudes, we normalized these values using the cosine similarity metric. With this matrix, the *ranker* module of NetAnalyzer was used to estimate the average proximity between all genes in each cluster and the causal gene.

Rather than relying solely on the arithmetic mean of proximities, we defined average proximity as the probability that a causal gene is functionally associated with the genes within a given cluster. To estimate this probability, we used a logistic regression model trained on the similarity scores to predict the likelihood that two genes were originally connected in the PPI network. For each gene within a cluster, the model provided a probability of association with the causal gene. Next, we aggregated these probabilities for all genes within the cluster using Fisher's combined probability test, which allows for robust prioritization of candidate genes. The resulting score value is then rank-normalized based on the distribution of proximities for all genes in the network. In this framework, clusters with a higher average proximity to the causal gene were those in which the causal gene had the highest rank in terms of proximity.

In addition to the quantitative network analyses, an exploratory visual analysis was conducted to facilitate the interpretation and navigation of the PPI network. For this purpose, we generated a representation of the identified clusters using *net_explorer*, a visualization module included in the NetAnalyzer library. This tool offers both an interactive network graph visualized using the Sigma.js v2 JavaScript library, with 200 layout iterations to optimize the positioning of nodes, and a UMAP-based dimensionality reduction plot, enabling intuitive exploration of gene–gene relationships in a two-dimensional space.

For the Uniform Manifold Approximation and Projection (UMAP) visualization, node embeddings were first computed using the *node2vec* algorithm (with parameters `dimensions = 64, walk_length = 30, num_walks = 200, p = 1, q = 1, window = 10`). These embeddings provided the coordinates for each node, which were then used as input for UMAP projection [88] (`n_neighbors = 15, min_dist = 0.1, n_components = 2, metric = 'euclidean'`). This method allowed us to create a simplified, yet informative, representation of the gene interaction network, making it easier to explore patterns within gene clusters and to generate hypotheses about which genes are functionally close to the disease-causing genes.

Finally, a functional analysis of the clusters was performed using *clusterProfiler* with GO biological process annotations. A threshold of 0.05 was applied to the *p*-values, which were adjusted for multiple testing using the BH method. Figure 6 illustrates a conceptual scheme of the methodology developed for this study.

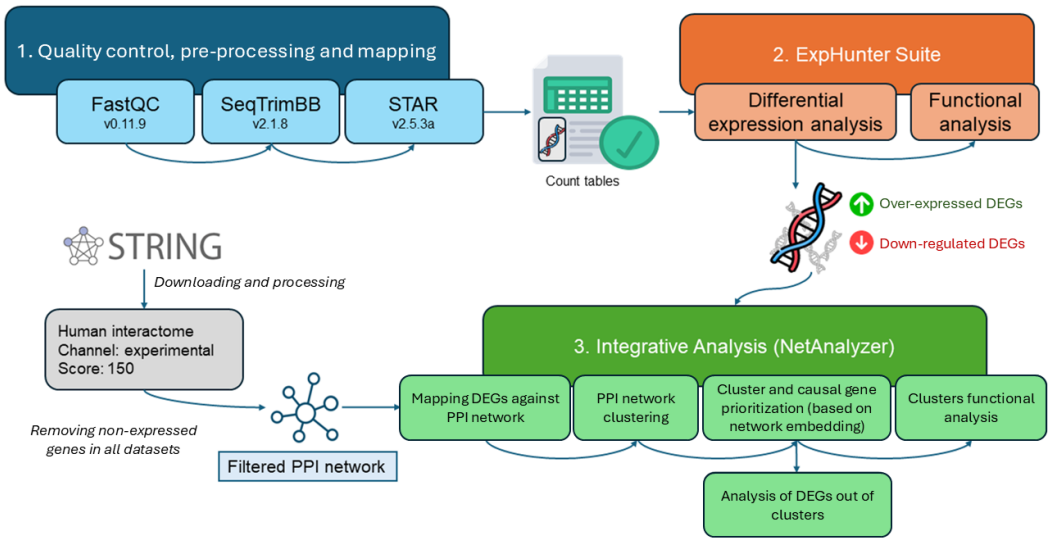

Figure 6. Workflow overview of the analysis pipeline. (1.) RNA-seq data undergo quality control, trimming, and alignment to the hg38 reference genome. (2.) ExpHunter Suite is used to identify differentially expressed genes (DEGs) and perform functional enrichment analysis. (3.) An integrative network analysis is conducted using a filtered STRING interactome built from the union of expressed genes across all datasets, allowing the exploration of DEG clustering around each disease-causal gene.

5. Conclusions

Our integrative approach uncovers biologically and clinically relevant patterns across diverse neuromuscular diseases (NMDs), revealing mechanisms not detectable by differential expression alone. Comparative functional enrichment analysis of DEGs across all datasets revealed shared functions among several diseases, including glycosaminoglycan binding enrichment in both DMD and *FUS*-related ALS, suggesting that extracellular matrix–protein interactions may contribute to the downstream effects of *FUS* mutations. By combining transcriptomic data with protein–protein interaction networks and clustering based on proximity to causal genes, we identified shared and specific modules enriched in

immune response, RNA metabolism, and cytoskeletal organization. Recurrently dysregulated lncRNAs, such as *PAX8-AS1*, *SBF2-AS1*, and *NEAT1*, indicate the regulatory roles of ncRNAs and their potential as cross-disease biomarkers. The prioritization of genes like *HS3ST3A1*, located near both *TNPO3* and *FUS*, illustrates how network context can reveal hidden connections between genetically distinct disorders. Our framework is applicable to other rare and common diseases, offering a scalable tool for uncovering molecular mechanisms and identifying novel therapeutic targets through systems-level analysis.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/ijms26199376/s1.

Author Contributions: J.A.G.R., E.R. and P.S. conceived the methodology. L.H.-G., F.G.-C., Á.E.-M., J.P.-G., P.S. and E.R. developed the software that implements the protocol. L.H.-G., F.G.-C., Á.E.-M., J.P.-G., P.S. and E.R. analyzed the results and provided interpretation. L.H.-G., F.G.-C., P.S. and E.R. wrote the manuscript. J.A.G.R., E.R. and P.S. were involved in planning of this study, contributed to the acquisition of research funding, and headed the project. All authors have read and agreed to the published version of the manuscript.

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Abbreviations

The following abbreviations are used in this manuscript:

ALS	Amyotrophic Lateral Sclerosis
ARS	Average Read Size
BH	Benjamini–Hochberg
CPM	Counts Per Million
CyT	Cytoplasmatic Transcriptome
DEG	Differentially Expressed Gene
DMD	Duchenne Muscular Dystrophy
ECM	Extracellular Matrix
GEO	Gene Expression Omnibus
GO	Gene Ontology
HCPC	Hierarchical Clustering on Principal Components
hiPSC	Human-Induced Pluripotent Stem Cell
iPSCs	Induced Pluripotent Stem Cells
LGMD	Limb–Girdle Muscular Dystrophy
LGMDD2	Limb–Girdle Muscular Dystrophy D2
lncRNAs	Long Non-Coding RNAs
MLS	Minimum Libraries Selected
MRL	Minimum Read Length
NCBI	National Center for Biotechnology Information
NMDs	Neuromuscular Diseases
ORA	Over-Representation Analysis
PBMCs	Peripheral Blood Mononuclear Cells
PCA	Principal Component Analysis
PPI	Protein–Protein Interaction
RNA-seq	RNA Sequencing
SRA	Sequence Read Archive
UMAP	Uniform Manifold Approximation and Projection
WCT	Whole-Cell Transcriptome

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Supplementary material

Table S1. Top ten differentially expressed genes (DEGs) ranked by their log_2FC (from highest to lowest) for each dataset DMD dataset (DMD_myot, DMD_pCard, DMD_cfib and DMD_myob).

ID	log_2FC	Gene Symbol	Dataset
ENSG00000214548	9.329897687	*MEG3*	DMD_myot
ENSG00000287080	8.680853606	*H3C3*	DMD_myot
ENSG00000155008	8.501931016	*APOOL*	DMD_myot
ENSG00000225746	8.46109789	*ENSG00000225746*	DMD_myot
ENSG00000162623	7.769389631	*TYW3*	DMD_myot
ENSG00000171094	-3.492998891	*ALK*	DMD_myot
ENSG00000114654	-3.042854954	*EFCC1*	DMD_myot
ENSG00000170290	-2.810693768	*SLN*	DMD_myot
ENSG00000133020	-2.565352591	*MYH8*	DMD_myot
ENSG00000120057	-2.432159111	*SFRP5*	DMD_myot
ENSG00000189223	5.88822191	*PAX8-AS1*	DMD_pcard
ENSG00000159248	5.434954133	*GJD2*	DMD_pcard
ENSG00000171864	3.421100698	*PRND*	DMD_pcard
ENSG00000110799	3.14559456	*VWF*	DMD_pcard
ENSG00000048540	2.849823199	*LMO3*	DMD_pcard
ENSG00000124107	-8.794610564	*SLPI*	DMD_pcard
ENSG00000077274	-8.58463897	*CAPN6*	DMD_pcard
ENSG00000124205	-8.001103953	*EDN3*	DMD_pcard
ENSG00000155011	-7.868336554	*DKK2*	DMD_pcard
ENSG00000196090	-7.819216639	*PTPRT*	DMD_pcard
ENSG00000169436	9.024092122	*COL22A1*	DMD_cfib
ENSG00000189223	6.566983729	*PAX8-AS1*	DMD_cfib
ENSG00000082482	5.883638995	*KCNK2*	DMD_cfib
ENSG00000144834	5.868433019	*TAGLN3*	DMD_cfib
ENSG00000171502	4.995137047	*COL24A1*	DMD_cfib
ENSG00000012504	-7.474132943	*NR1H4*	DMD_cfib
ENSG00000234444	-6.104334548	*ZNF736*	DMD_cfib
ENSG00000106483	-5.865551797	*SFRP4*	DMD_cfib
ENSG00000081842	-5.178089797	*PCDHA6*	DMD_cfib
ENSG00000224549	-5.022565823	*ENSG00000224549*	DMD_cfib
ENSG00000214548	8.564267813	*MEG3*	DMD_myob
ENSG00000134184	7.908815517	*GSTM1*	DMD_myob
ENSG00000225746	7.744089875	*ENSG00000225746*	DMD_myob
ENSG00000187688	7.686184497	*TRPV2*	DMD_myob
ENSG00000175161	7.531063627	*CADM2*	DMD_myob
ENSG00000174469	-7.7155905	*CNTNAP2*	DMD_myob
ENSG00000157601	-6.814552143	*MX1*	DMD_myob
ENSG00000228318	-6.809012077	*ENSG00000228318*	DMD_myob
ENSG00000184486	-6.625125528	*POU3F2*	DMD_myob
ENSG00000130303	-6.312210882	*BST2*	DMD_myob

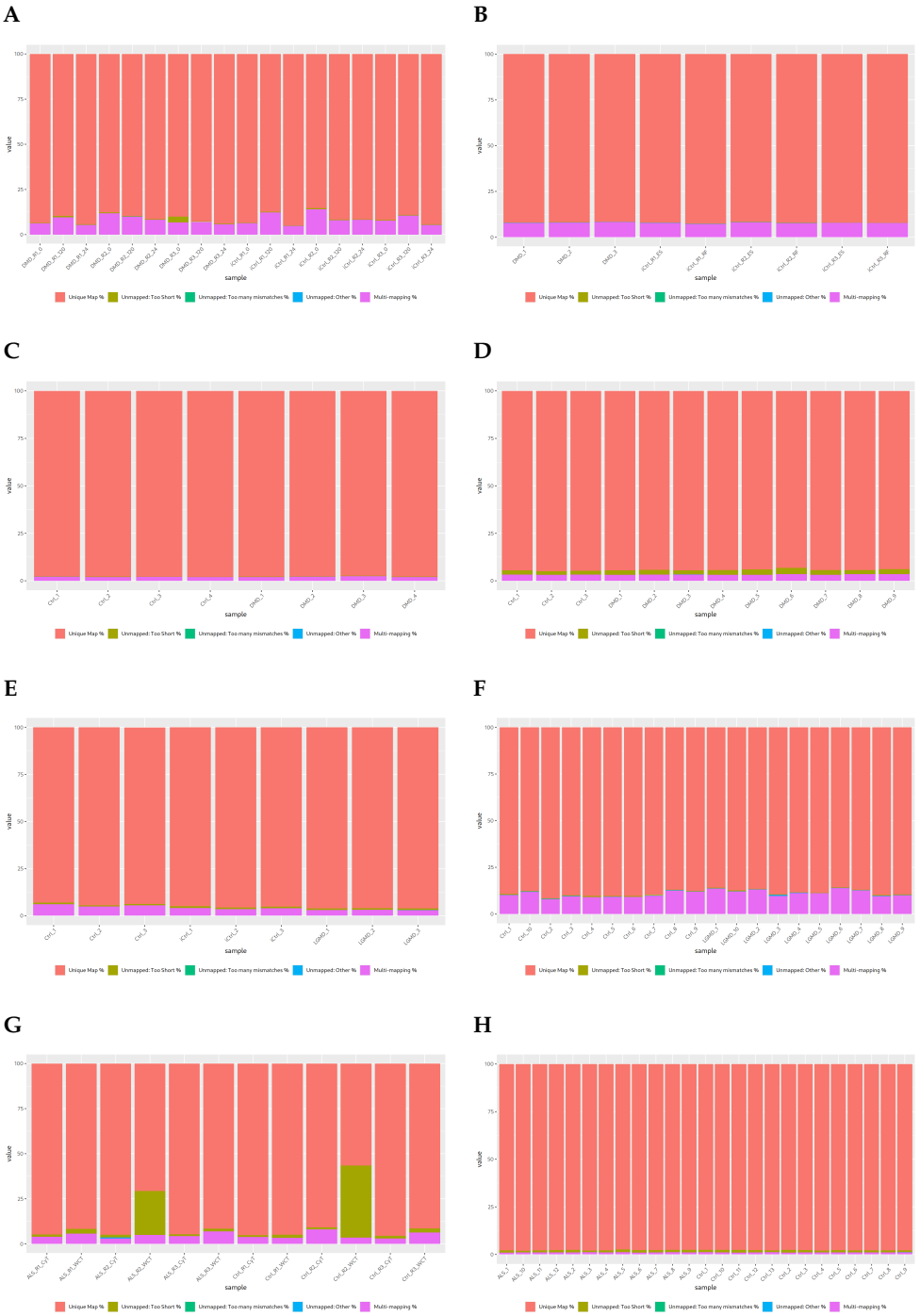

Figure S1. Distribution of the percentage of reads and how they mapped against the reference genome using STAR. Datasets: (A) DMD_myot, (B) DMD_pCard, (C) DMD_cfib, (D) DMD_myob, (E) LGMD_myob, (F) LGMD_pbmc, (G) ALS_iN_C9ORF72 and (H) ALS_fib_FUS. For each sample, the percentage of uniquely mapped reads (red bars), unmapped reads, including too short (olive green), many mismatches (green) and others (blue), along with multimapping reads (pink) is represented.

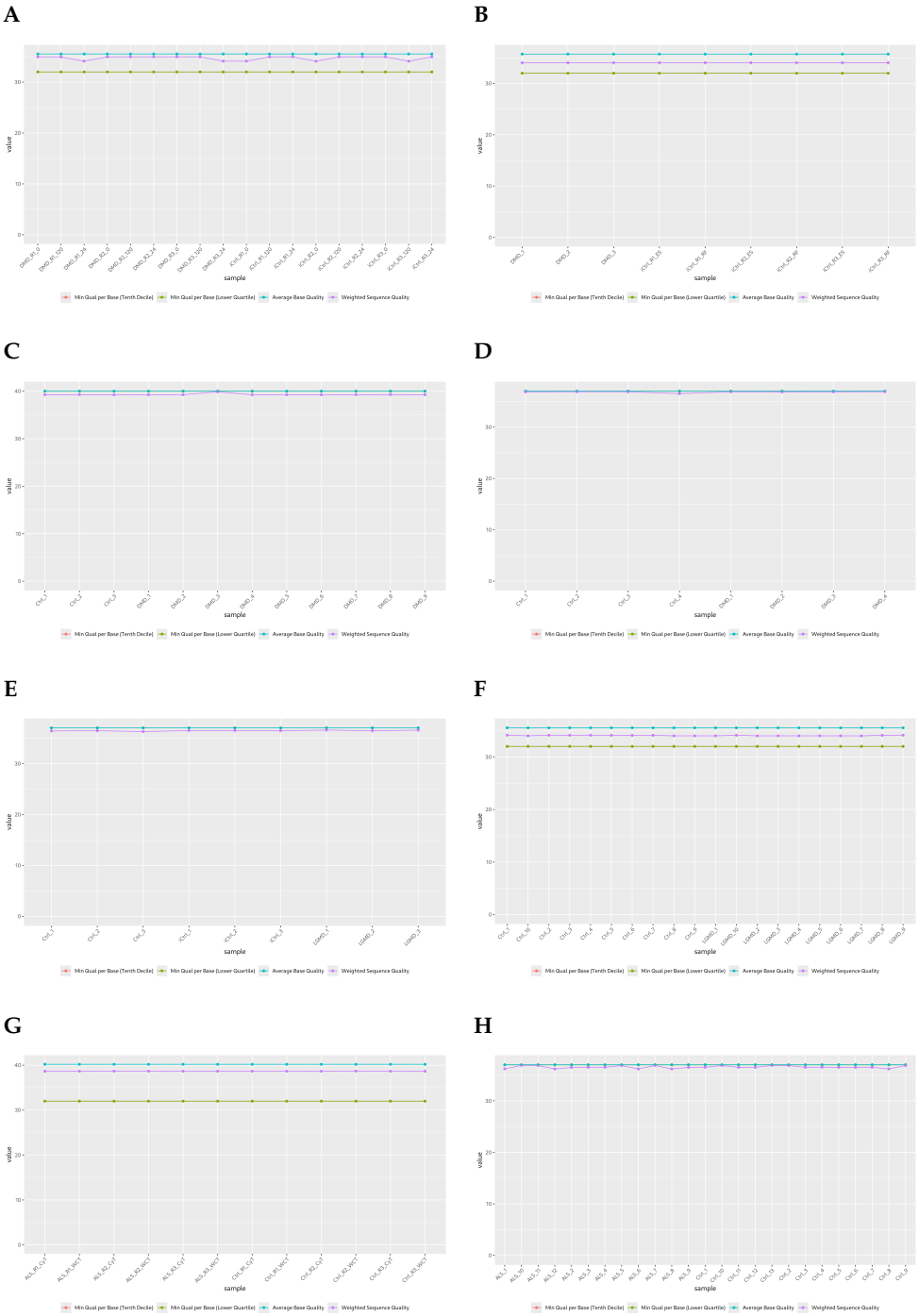

Figure S2. Quality scores after pre-processing samples with SeqTrimBB. Information is retrieved from FastQC. The minimum QUAL per base in the tenth decile (red), lower quartile (green), average base quality (light blue) and weighted sequence quality (mean Phred score across its bases) is included. Datasets: (A) DMD_myot, (B) DMD_pCard, (C) DMD_cfib, (D) DMD_myob, (E) LGMD_myob, (F) LGMD_pbmc, (G) ALS_iN_C9ORF72 and (H) ALS_fib_FUS.

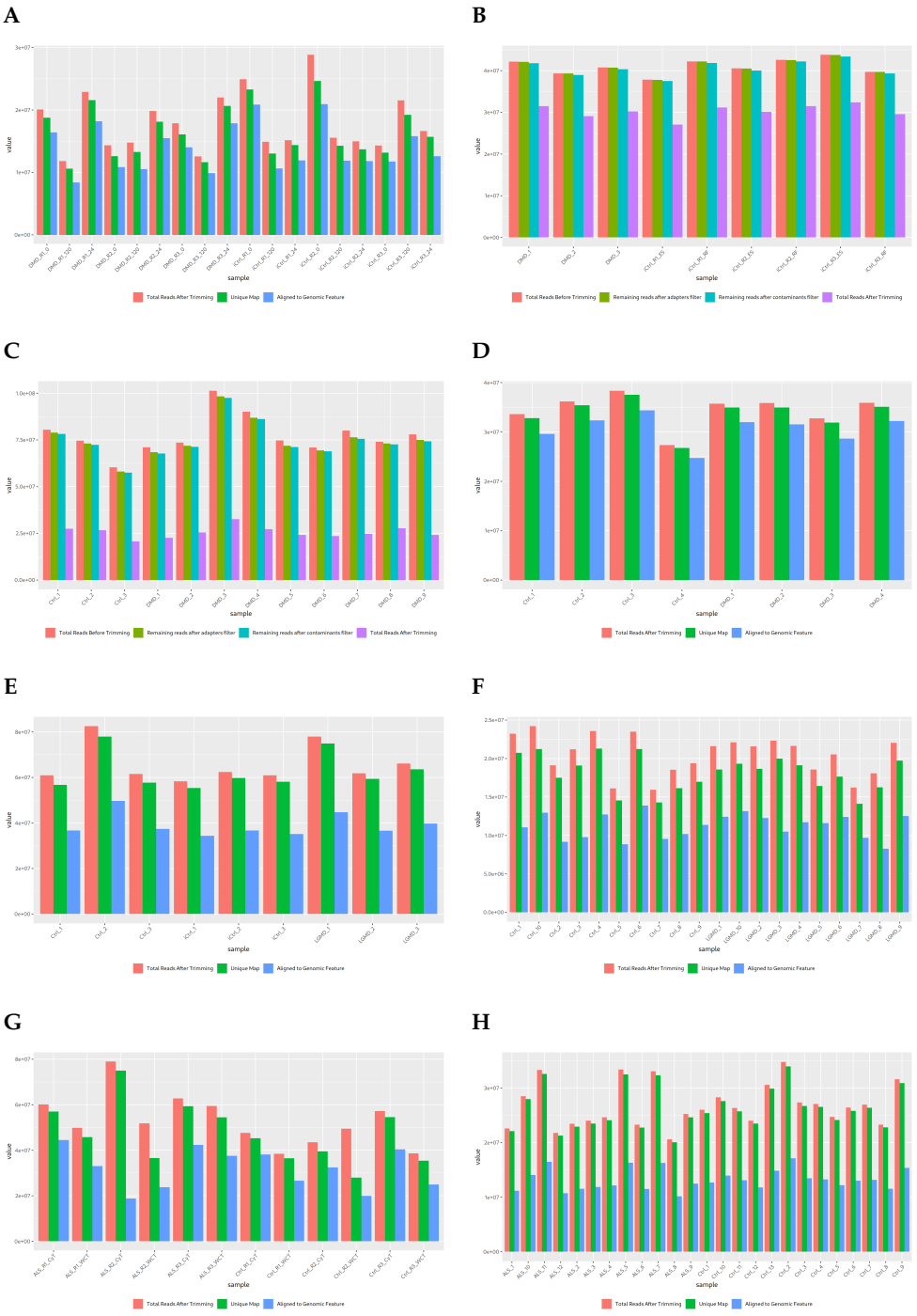

Figure S3. Summary of the mapping statistics for each dataset. For every dataset included in the study, the bar chart displays the total number of reads remaining after the pre-processing step (red bars), the number of reads that aligned uniquely to the reference genome (green bars), and, among these, the subset that mapped to annotated genes (blue bars). Datasets: (A) DMD_myot, (B) DMD_pCard,(C) DMD_cfib, (D) DMD_myob, (E) LGMD_myob, (F) LGMD_pbmc, (G) ALS_iN_C9ORF72 and (H) ALS_fib_FUS.

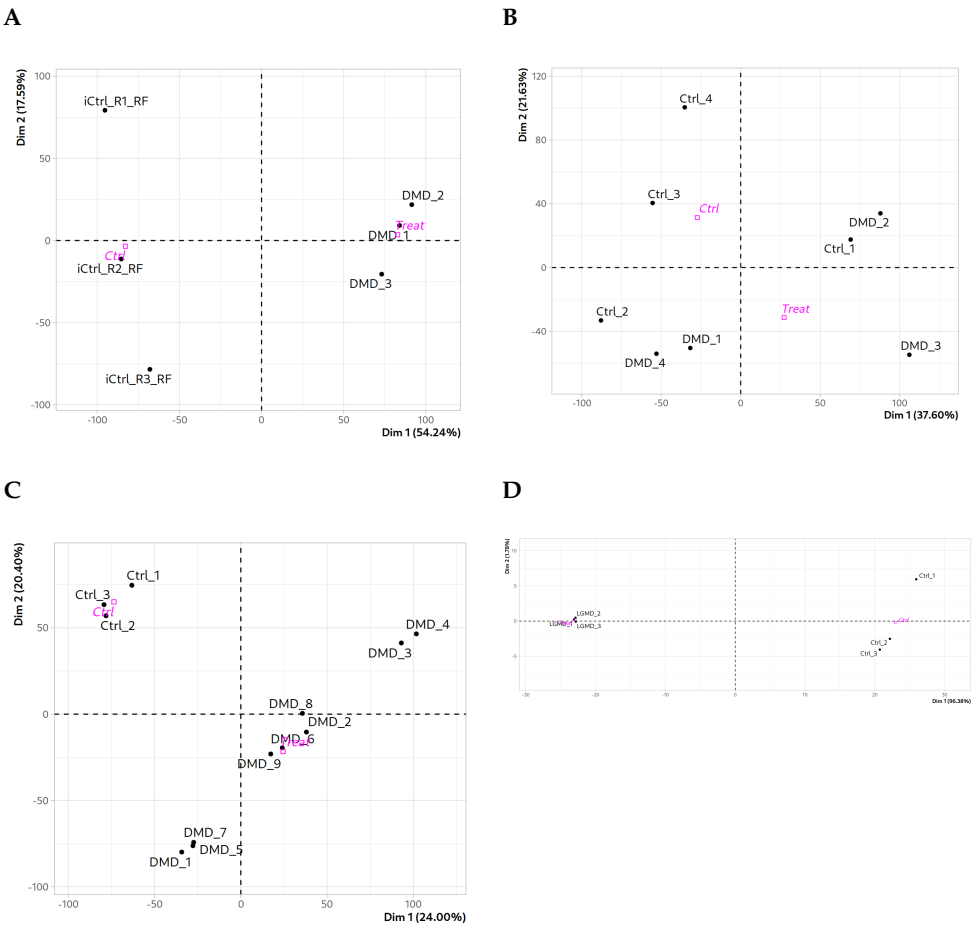

Figure S4. Principal component analysis (PCA) performed for the following datasets: A) DMD_pCard, B) DMD_cfib, C) DMD_myob and D) LGMD_myob. The centroid of each group of samples compared (Ctrl and Treat) is displayed. DMD: Duchenne muscular dystrophy, LGMD: Limb-girdle muscular dystrophy.

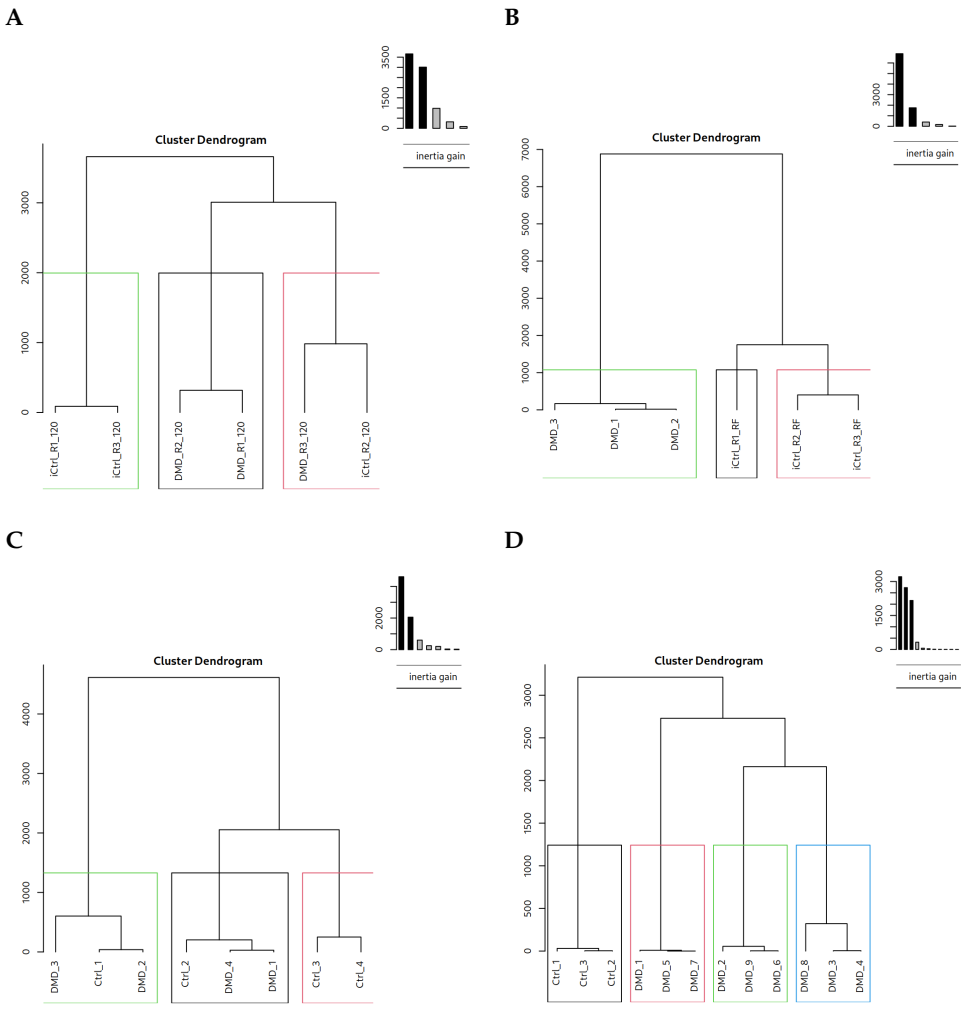

Figure S5. Hierarchical Clustering on Principal Components (HCPC) performed over the different datasets for DMD datasets: DMD_myot (A), DMD_pCard (B), DMD_cfib (C), DMD_myob (D). Distance is displayed in the Y-axis. Inertia gain is visualized as a bar graph where each bar represents the increase in inertia at a potential cut point in the dendrogram; individual branching events are explicitly shown, and the optimal cut is determined at the branching point with the highest inertia gain.

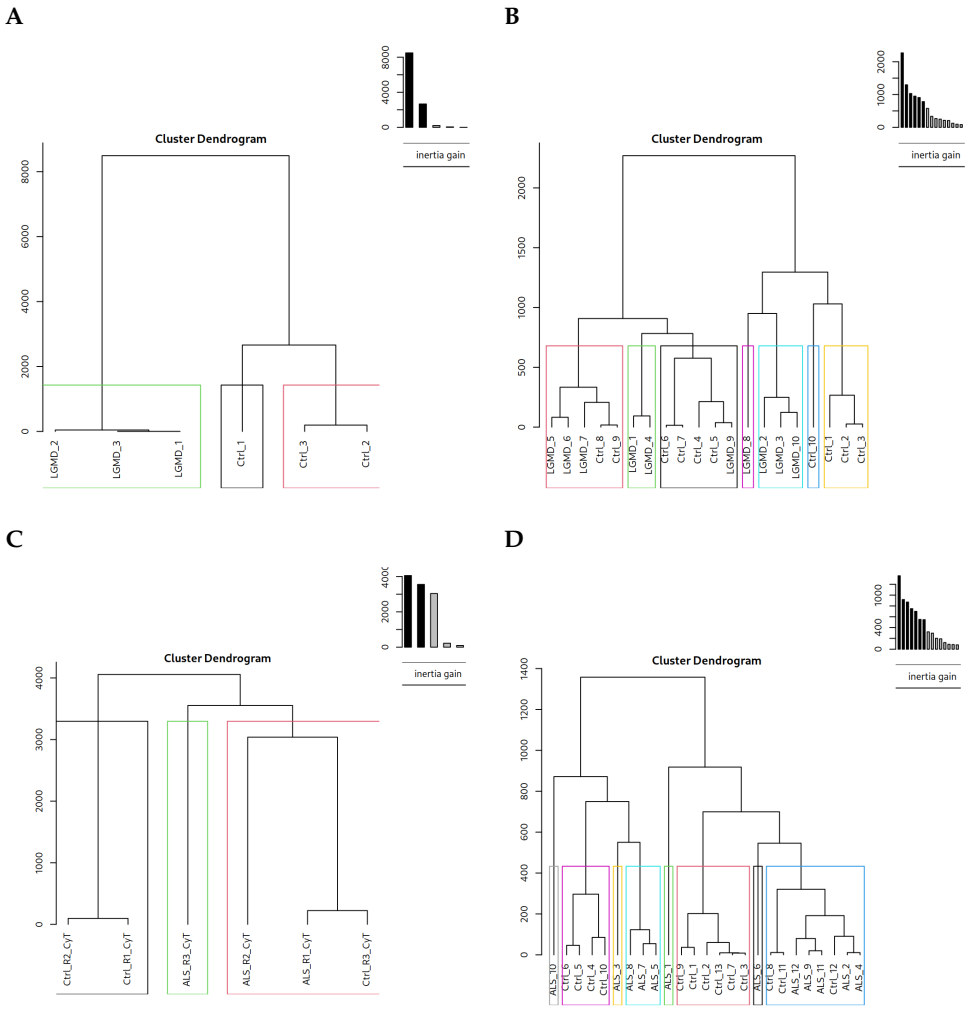

Figure S6. Hierarchical Clustering on Principal Components (HCPC) performed over the differ-ent datasets for LGMD datasets: LGMD_myob (A) and LGMD_pbmc (B), and for ALS datasets: ALS_iN_C9ORF72 (C), ALS_fib_FUS (D). Distance is displayed in the Y-axis. Inertia gain is visual-ized as a bar graph where each bar represents the increase in inertia at a potential cut point in the dendrogram; individual branching events are explicitly shown, and the optimal cut is determined at the branching point with the highest inertia gain.

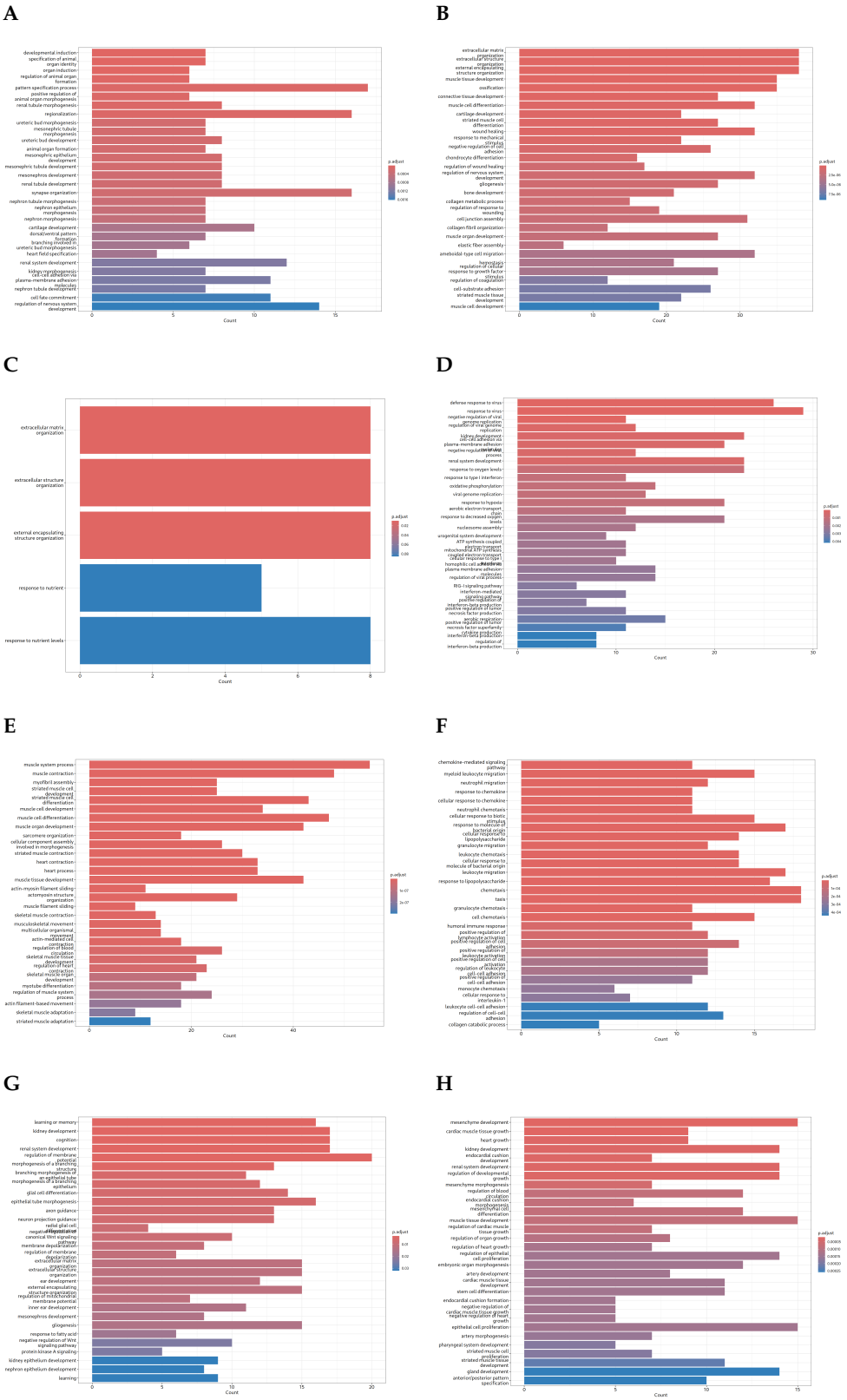

Figure S7. Barplots from the Over Representation Analysis (ORA) performed over differential ex-pressed genes (DEGs) from the different datasets: (A) DMD_myot, (B) DMD_pcard, (C) DMD_cfib,(D) DMD_myob, (E) LGMD_myob, (F) LGMD_pbmc, (G) ALS_iN_C9ORF72 and (H) ALS_fib_FUS. Categories shown belong to the Gene Ontology biological process.

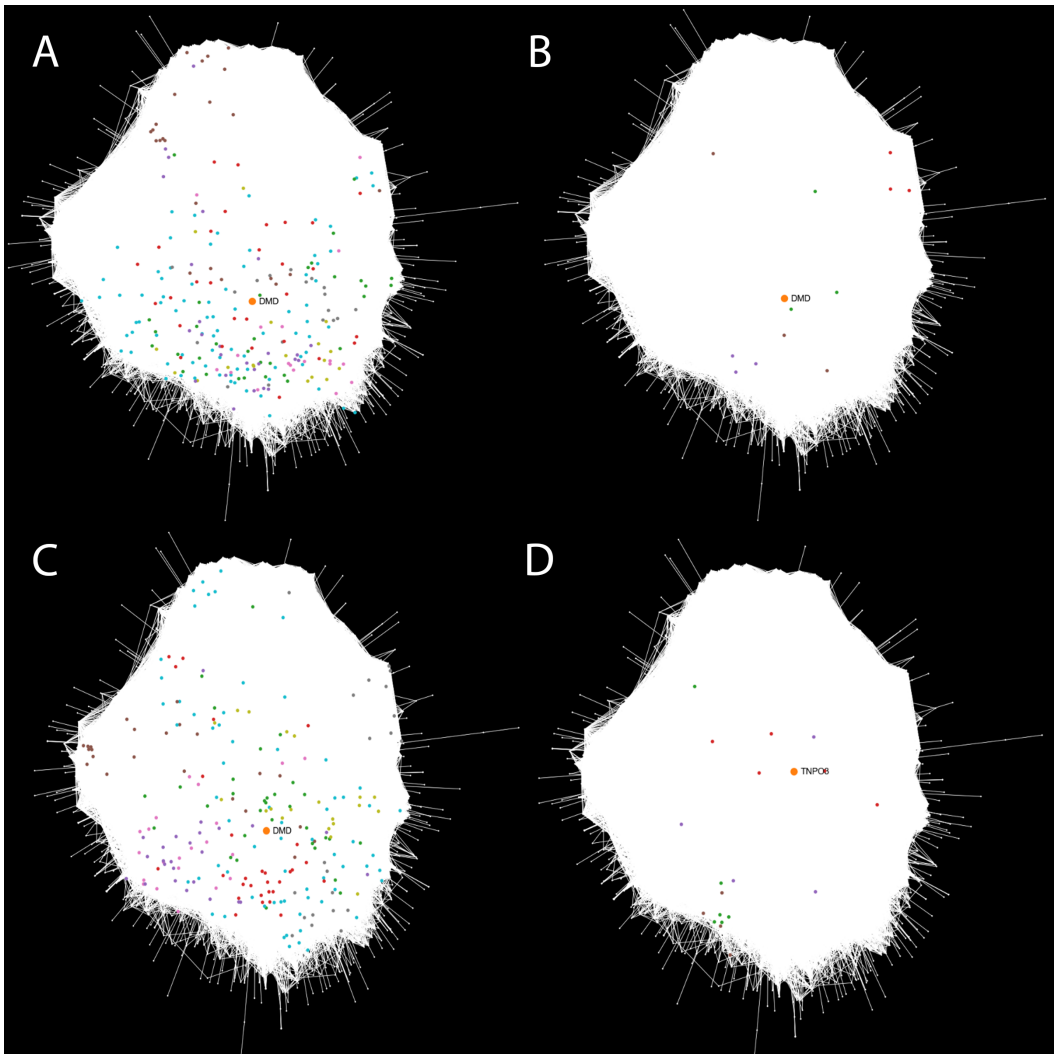

Figure S8. Protein-protein interaction networks based on the union of all expressed genes across datasets. For each dataset, the DEG clusters are shown in different colors, revealing their clustering patterns. The orange dot indicates the disease-related gene used as a reference (*DMD* and *TNPO3*). A) DMD_pCard, B) DMD_cfib, C) DMD_myob and D) LGMD_pbmc datasets.

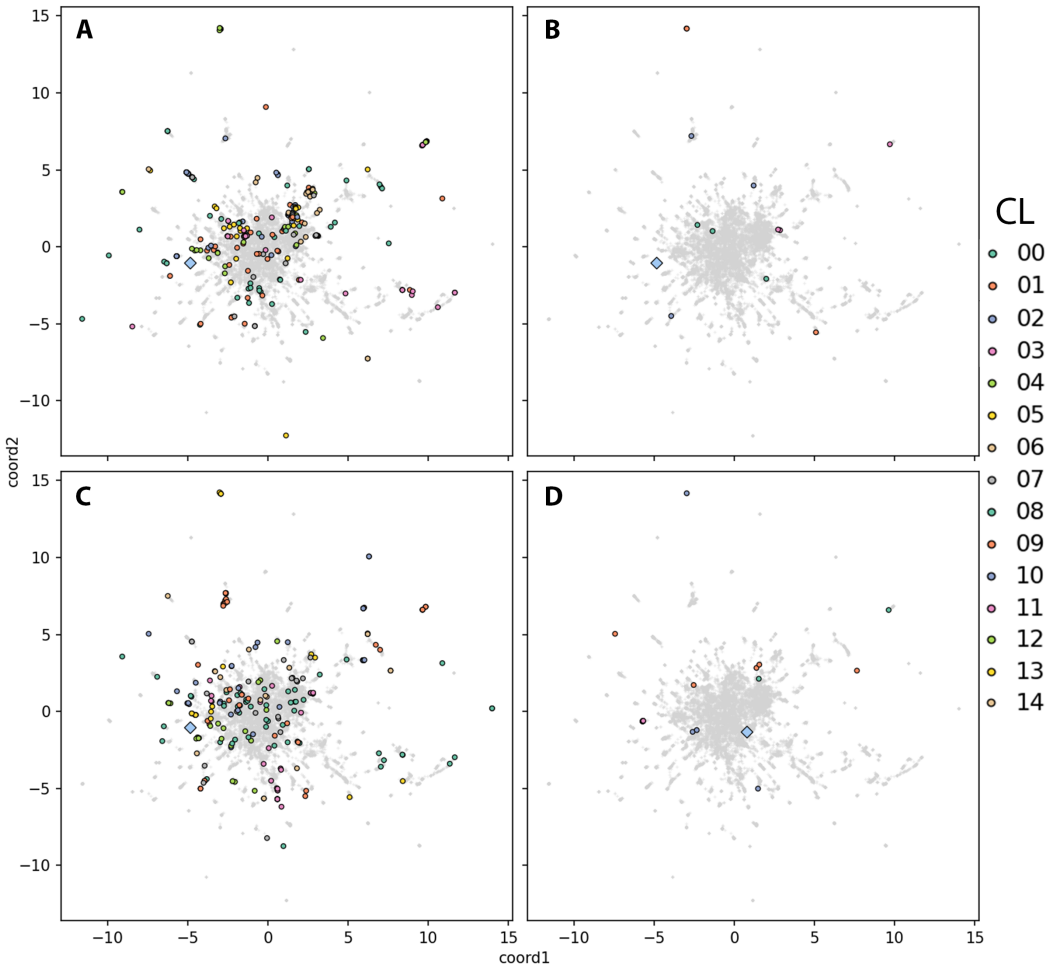

Figure S9. Two-dimensional embedding of gene interaction networks. Each panel represents the reduced vector space (coord1 vs. coord2) of the full gene interaction network based on STRING "Experimental" evidence. Background grey dots correspond to all expressed genes included in the network. The disease-causal gene (*seed*) for each condition (*DMD*, *TNPO3*, *C9ORF72* and *FUS*) is represented as a blue rhombus. Panels correspond to the DMD_pCard (A), DMD_cfib (B), DMD_myob (C) and LGMD_pbmc (D) datasets. Colored dots represent differentially expressed genes (DEGs), grouped by cluster (CL).

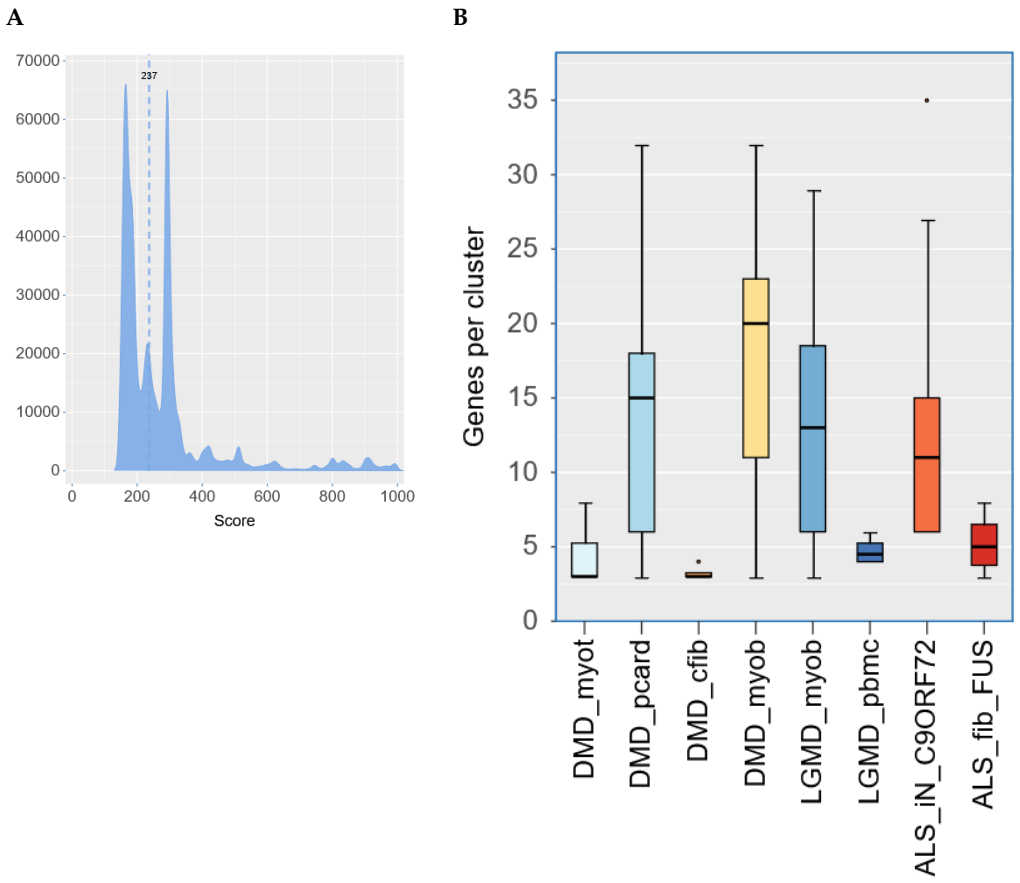

Figure S10. A) STRING "Experimental" channel scores distribution. B) Distribution of DEGs per cluster in all datasets (DMD_myot, DMD_pcard, DMD_cfib, DMD_myob, LGMD_myob, LGMD_pbmc, ALS_iN_C9ORF72 and ALS_fib_FUS).

Table S2. Top ten differentially expressed genes (DEGs) ranked by their log_2FC (from highest to lowest) for each LGMD dataset (LGMD_myob and LGMD_pbmc) and for each ALS dataset (ALS_iN_C9ORF72 and ALS_fib_FUS datasets.)

ID	log_2FC	Gene Symbol	Dataset
ENSG00000183549	9.76967333	*ACSM5*	LGMD_myob
ENSG00000275895	9.549062963	*LOC102724594*	LGMD_myob
ENSG00000228340	9.474992105	*MIR646HG*	LGMD_myob
ENSG00000183801	9.268052923	*OLFML1*	LGMD_myob
ENSG00000134184	9.166709346	*GSTM1*	LGMD_myob
ENSG00000225630	-9.348707076	*MTND2P28*	LGMD_myob
ENSG00000187688	-9.107161129	*TRPV2*	LGMD_myob
ENSG00000179455	-9.061608271	*MKRN3*	LGMD_myob
ENSG00000241945	-8.990805562	*PWP2*	LGMD_myob
ENSG00000275620	-8.720037735	*FLJ16779*	LGMD_myob
ENSG00000196611	6.246250316	*MMP1*	LGMD_PBMC
ENSG00000197632	6.072554488	*SERPINB2*	LGMD_PBMC
ENSG00000163735	5.765536582	*CXCL5*	LGMD_PBMC
ENSG00000074410	5.268491204	*CA12*	LGMD_PBMC
ENSG00000163734	5.247146242	*CXCL3*	LGMD_PBMC
ENSG00000228078	-5.692259109	*ENSG00000228078*	LGMD_PBMC
ENSG00000152672	-3.462160924	*CLEC4F*	LGMD_PBMC
ENSG00000249917	-2.336057043	*LINC00536*	LGMD_PBMC
ENSG00000278416	-2.152524027	*ENSG00000278416*	LGMD_PBMC
ENSG00000262402	-2.097058208	*ENSG00000262402*	LGMD_PBMC
ENSG00000250366	9.581682298	*TUNAR*	ALS_iN_C9ORF72
ENSG00000181291	9.575731861	*TMEM132E*	ALS_iN_C9ORF72
ENSG00000269994	9.540754174	*LINC02893*	ALS_iN_C9ORF72
ENSG00000204287	9.448847287	*HLA-DRA*	ALS_iN_C9ORF72
ENSG00000162998	9.398092467	*FRZB*	ALS_iN_C9ORF72
ENSG00000175793	-6.11840715	*SFN*	ALS_iN_C9ORF72
ENSG00000148671	-5.095503897	*ADIRF*	ALS_iN_C9ORF72
ENSG00000101210	-5.07538562	*EEF1A2*	ALS_iN_C9ORF72
ENSG00000011590	-4.611922474	*ZBTB32*	ALS_iN_C9ORF72
ENSG00000188042	-4.598501103	*ARL4C*	ALS_iN_C9ORF72
ENSG00000171724	2.199740968	*VAT1L*	ALS_fib_FUS
ENSG00000130720	1.917163957	*FIBCD1*	ALS_fib_FUS
ENSG00000204055	1.737743134	*ENSG00000204055*	ALS_fib_FUS
ENSG00000175084	1.718878774	*DES*	ALS_fib_FUS
ENSG00000117152	1.683603117	*RGS4*	ALS_fib_FUS
ENSG00000115457	-3.606169709	*IGFBP2*	ALS_fib_FUS
ENSG00000164530	-3.479066099	*PI16*	ALS_fib_FUS
ENSG00000106483	-2.811743963	*SFRP4*	ALS_fib_FUS
ENSG00000086289	-2.38666159	*EPDR1*	ALS_fib_FUS
ENSG00000168672	-2.139848876	*LRATD2*	ALS_fib_FUS

Table S3. Summary of the datasets used in this study, including the dataset name, the causative mutation, the time of cell culture differentiation (Time of CC dif.), the platform (Illumina sequencer) and layout selected, the average number of reads per sample (Avg. NRPS), the description of samples and cell type used to perform the differential expression analysis. DMD: Duchenne muscular distrophy. LGMD: limb-girdle muscular distrophy. ALS: amyotrophic lateral scoliosis. PBMC: peripheral blood ononuclear cells. iNPC: induced neural progenitor cells.

Dataset	Causative mutation	Time of CC dif.	Platform	Layout	Avg. NRPS	Samples description	Cell type
DMD_myot	DMD c.10141C>T	120 hours	NextSeq 500	Paired	1×10^7	9 isogenic controls and 9 DMD samples (three biological replicates of two genotypes at three time points)	Myotubes derived from reprogrammed skin fibroblasts
DMD_pcard	*DMD* ΔEx44	Day 50 post differentiation	NextSeq 500	Single	4×10^7	3 controls and 6 CRISPR-corrected samples (3 by exon-skipping and 3 by reframing)	Cardiomyocytes differentiated from iPSCs reprogrammed from PBMCs
DMD_cfib	DMD ΔEx49,50; ΔEx45,50; c.6913–1G>A	—	NextSeq 500	Paired	3×10^7	4 controls and 4 DMD patients	Human iPSC lines differentiated into cardiac fibroblasts
DMD_myob	Dup2, Dup2-9, Dup8-9	Day 8 of differentiation	NovaSeq 6000	Paired	4×10^7	3 controls and 9 patients (3 with Dup2, 3 with Dup2-9, 3 with Dup8-9)	Myoblasts
LGMD_myob	*TNPO3* c.2771delA	—	NovaSeq 6000	Paired	1.5×10^8	3 controls, 3 isogenic controls and 3 patients	Primary myoblasts
LGMD_pbmc	*TNPO3* c.2771delA	—	NextSeq 550	Single	2×10^7	10 controls and 10 patients from the same family	PBMCs
ALS_iN_C9ORF72	CS28iALS-C9nxx; RRID:CVCL_W558 & CS29iALS-C9nxx; RRID:CVCL_W559	Day 40	HiSeq 4000	Paired	4.5×10^7	6 controls (3 whole-cell transcriptomes, 3 cytoplasmic) and 6 patients	iNPC-derived iNeurons
ALS_fib_FUS	*FUS*(c.1545_1547dup, p.Glu515_Glu516insAsp), g.11194C>G (p.H517Q) - H517Q, FUS-R521G, *FUS*-P525L, FUS-R524W, c.1566G>A (p.R522R) - G522A, 1561C>G (p.R521G), *FUS* R495X	—	HiSeq 4000	Paired	3×10^7	13 controls, 12 FUS-ALS patients, 11 sporadic ALS patients	Fibroblasts

Table S4. Common differentially expressed non-coding RNAs not reported in the STRING interactome network.

Gene Symbol	log_2FC	Datasets
ENSG00000210082	6.66271761, 2.350697846	DMD_myot, DMD_myob
ENSG00000211459	6.856519466, 1.769644484	DMD_myot, DMD_myob
ENSG00000225746	8.46109789, -1.61975765, 7.744089875	DMD_myot, DMD_pcard, DMD_myob
ENSG00000228509	-1.500228158, -4.619953197	DMD_pcard, LGMD_myob
ENSG00000234665	-5.303053232, 6.612476245	LGMD_myob, ALS_iN_C9ORF72
ENSG00000236333	2.583009658, 6.398757626	DMD_myot, ALS_iN_C9ORF72
ENSG00000247134	7.082599117, -3.918320817, 1.052531096	DMD_myob, LGMD_myob, LGMD_pbmc
ENSG00000260604	-1.633556296, 2.770659015	DMD_myob, LGMD_myob
ENSG00000264490	2.620604786, -3.239567081	DMD_myob, LGMD_myob
ENSG00000267328	4.314826556, 0.939302737	DMD_myob, ALS_fib_FUS
ENSG00000268518	1.620199672, -3.410659159	DMD_myob, LGMD_myob
ENSG00000275266	-1.611720439, -3.11208589	DMD_pcard, LGMD_myob
ENSG00000281344	2.259690595, 3.131760569, 3.327599889	DMD_myob, LGMD_myob, ALS_iN_C9ORF72
H19	-1.671813204, -2.741273729	DMD_pcard, LGMD_myob
LINC01615	1.585841416, 3.337558246	DMD_myob, LGMD_myob
MEG3	9.329897687, -2.168895462, 8.564267813	DMD_myot, DMD_pcard, DMD_myob
MIR1915HG	-2.303961547, 1.167736333	DMD_cfib, ALS_fib_FUS
MIR206	1.777347446, -4.167276676	DMD_myob, LGMD_myob
MIR381HG	6.155384396, -2.648835477	DMD_myot, DMD_pcard
NEAT1	2.156031047, -1.464471	DMD_myob, ALS_fib_FUS
PAX8-AS1	4.242991409, 5.88822191, 6.566983729, -4.979454426, 4.695141887	DMD_myot, DMD_pcard, DMD_cfib, DMD_myob, LGMD_myob
PCOLCE-AS1	4.098649958, 2.550229585	DMD_myob, LGMD_myob
SBF2-AS1	3.71442084, -2.264803297	LGMD_myob, ALS_iN_C9ORF72
SCAT8	-1.550985117, -4.568172847	DMD_myob, LGMD_myob
SNHG18	5.533150502, -7.97648756	DMD_myob, LGMD_myob

Table S5. Clustered DEGs priorization results for the datasets DMD_myot, DMD_pcard, DMD_cfib and DMD_myot. Here we show the disease-causal gene identifiers (ENSG00000198947: *DMD*), their corresponding clustering scores (score), normalized ranks (Norm. rank), and raw ranks (considering the total expressed genes in the network), alongside their assigned causal gene group and correspond-ing dataset.

Causal gene	Score	Norm. rank	Rank	Group	Dataset
ENSG00000198947	0.77	0.02	400	3	DMD_myot
ENSG00000198947	0.04	0.35	5244	7	DMD_myot
ENSG00000198947	0.03	0.34	4986	6	DMD_myot
ENSG00000198947	0.02	0.26	3791	2	DMD_myot
ENSG00000198947	9.87e-04	0.78	11476	5	DMD_myot
ENSG00000198947	6.96e-04	0.81	11860	4	DMD_myot
ENSG00000198947	4.97e-04	0.59	8747	1	DMD_myot
ENSG00000198947	7.55e-05	0.54	7983	0	DMD_myot
ENSG00000198947	1.00	0.00	37	8	DMD_pcard
ENSG00000198947	0.05	0.32	4758	15	DMD_pcard
ENSG00000198947	0.03	0.37	5430	16	DMD_pcard
ENSG00000198947	0.01	0.38	5613	12	DMD_pcard
ENSG00000198947	0.00	0.27	4058	10	DMD_pcard
ENSG00000198947	0.00	0.64	9357	14	DMD_pcard
ENSG00000198947	0.00	0.46	6770	13	DMD_pcard
ENSG00000198947	4.87e-04	0.59	8705	11	DMD_pcard
ENSG00000198947	9.35e-06	0.34	4984	5	DMD_pcard
ENSG00000198947	9.15e-06	0.32	4696	6	DMD_pcard
ENSG00000198947	7.06e-06	0.36	5255	4	DMD_pcard
ENSG00000198947	8.29e-07	0.31	4611	3	DMD_pcard
ENSG00000198947	4.74e-08	0.44	6518	2	DMD_pcard
ENSG00000198947	2.32e-08	0.69	10176	7	DMD_pcard
ENSG00000198947	3.98e-10	0.85	12401	9	DMD_pcard
ENSG00000198947	4.14e-12	0.56	8210	1	DMD_pcard
ENSG00000198947	1.24e-15	0.73	10740	0	DMD_pcard
ENSG00000198947	1.00	0.00	3	0	DMD_cfib
ENSG00000198947	0.03	0.34	5220	1	DMD_cfib
ENSG00000198947	1.41e-03	0.81	12689	2	DMD_cfib
ENSG00000198947	0.02	0.42	6480	3	DMD_cfib
ENSG00000198947	4.65e-07	0.21	3245	0	DMD_myob
ENSG00000198947	2.43e-22	0.97	15047	1	DMD_myob
ENSG00000198947	1.07e-08	0.41	6386	2	DMD_myob
ENSG00000198947	1.34e-11	0.61	9445	3	DMD_myob
ENSG00000198947	1.03e-13	0.86	13395	4	DMD_myob
ENSG00000198947	4.05e-05	0.22	3410	5	DMD_myob
ENSG00000198947	1.30e-08	0.51	7919	6	DMD_myob
ENSG00000198947	0.94	0.00	41	7	DMD_myob
ENSG00000198947	7.75e-16	0.97	15066	8	DMD_myob
ENSG00000198947	1.09e-04	0.47	7334	9	DMD_myob
ENSG00000198947	0.11	0.10	1592	10	DMD_myob
ENSG00000198947	2.78e-03	0.32	4947	11	DMD_myob
ENSG00000198947	0.02	0.42	6468	12	DMD_myob

Table S6. Clustered DEGs priorization results for the datasets LGMD_myob, LGMD_pbmc, ALS_iN_C9ORF72 and ALS_fib_FUS. Here we show the disease-causal gene identifiers (ENSG00000064419: *TNPO3*, ENSG00000147894: *C9ORF72* and ENSG00000089280: *FUS*), their corresponding clustering scores (score), normalized ranks (Norm. rank), and raw ranks (considering the total expressed genes in the network), alongside their assigned causal gene group and correspond-ing dataset.

Causal gene	Score	Norm. rank	Rank	Group	Dataset
ENSG00000064419	5.42E-16	0.75	11605	0	LGMD_myob
ENSG00000064419	1.15E-10	0.51	7974	1	LGMD_myob
ENSG00000064419	2.07E-10	0.43	6714	2	LGMD_myob
ENSG00000064419	1.03E-09	0.68	10657	3	LGMD_myob
ENSG00000064419	8.48E-10	0.74	11528	4	LGMD_myob
ENSG00000064419	1.89E-06	0.51	7897	5	LGMD_myob
ENSG00000064419	9.25E-14	0.94	14600	6	LGMD_myob
ENSG00000064419	1.59E-05	0.36	5627	7	LGMD_myob
ENSG00000064419	8.72E-03	0.17	2644	8	LGMD_myob
ENSG00000064419	2.85E-06	0.82	12801	9	LGMD_myob
ENSG00000064419	1.09E-09	1.00	15528	10	LGMD_myob
ENSG00000064419	9.90E-04	0.59	9141	11	LGMD_myob
ENSG00000064419	1.53E-02	0.36	5546	12	LGMD_myob
ENSG00000064419	1.96E-04	0.75	11756	13	LGMD_myob
ENSG00000064419	9.73E-03	0.48	7452	14	LGMD_myob
ENSG00000064419	5.21E-07	0.89	13904	0	LGMD_pbmc
ENSG00000064419	2.78E-04	0.73	11417	1	LGMD_pbmc
ENSG00000064419	5.99E-02	0.21	3212	2	LGMD_pbmc
ENSG00000064419	1.05E-03	0.65	10148	3	LGMD_pbmc
ENSG00000147894	0.31	0.08	1231	9	ALS_iN_C9ORF72
ENSG00000147894	0.11	0.06	928	3	ALS_iN_C9ORF72
ENSG00000147894	4.18e-04	0.58	8573	7	ALS_iN_C9ORF72
ENSG00000147894	5.21e-05	0.73	10767	8	ALS_iN_C9ORF72
ENSG00000147894	1.83e-05	0.72	10614	4	ALS_iN_C9ORF72
ENSG00000147894	3.68e-06	0.83	12243	5	ALS_iN_C9ORF72
ENSG00000147894	1.43e-06	0.96	14067	6	ALS_iN_C9ORF72
ENSG00000147894	2.23e-07	0.43	6279	2	ALS_iN_C9ORF72
ENSG00000147894	1.58e-13	0.88	12956	1	ALS_iN_C9ORF72
ENSG00000147894	9.59e-23	0.91	13329	0	ALS_iN_C9ORF72
ENSG00000089280	0.02	0.35	5118	2	ALS_fib_FUS
ENSG00000089280	0.01	0.54	7901	3	ALS_fib_FUS
ENSG00000089280	4.07e-04	0.64	9366	1	ALS_fib_FUS
ENSG00000089280	2.35e-04	0.45	6648	0	ALS_fib_FUS

Table S7. Gene Ontology (GO) biological process enrichment results for cluster (CL) 3 in the DMD_myot dataset. Metrics include GeneRatio (GR), background ratio (BgRatio), RichFactor, Fold Enrichment, z-score, and adjusted p-value.

CL	ID	Description	GR	BgRatio	RichFactor	FoldEnrichment	zScore	p.adjust	geneID	Count
3	GO:0098911	regulation of ventricular cardiac muscle cell action potential	0.67	12/18888	0.17	1049.33	45.78	0.00007	*PKP2/DSC2*	2
3	GO:0086069	bundle of His cell to Purkinje myocyte communication	0.67	15/18888	0.13	839.47	40.94	0.00007	*PKP2/DSC2*	2
3	GO:0098901	regulation of cardiac muscle cell action potential	0.67	29/18888	0.07	434.21	29.43	0.00016	*PKP2/DSC2*	2
3	GO:0086004	regulation of cardiac muscle cell contraction	0.67	34/18888	0.06	370.35	27.17	0.00016	*PKP2/DSC2*	2
3	GO:0086005	ventricular cardiac muscle cell action potential	0.67	35/18888	0.06	359.77	26.78	0.00016	*PKP2/DSC2*	2
3	GO:0070252	actin-mediated cell contraction	0.67	113/18888	0.02	111.43	14.84	0.00047	*PKP2/DSC2*	2
3	GO:0030048	actin filament-based movement	0.67	138/18888	0.01	91.25	13.41	0.00066	*PKP2/DSC2*	2
3	GO:0006936	muscle contraction	0.67	355/18888	0.01	35.47	8.26	0.00283	*PKP2/DSC2*	2

Table S8. Gene Ontology (GO) biological process enrichment results for cluster (CL) 0 in the DMD_cfib dataset. Metrics such as GeneRatio (GR), background ratio (BgRatio), RichFactor, Fold Enrichment, z-score, and adjusted p-value are included.

CL	ID	Description	GR	BgRatio	RichFactor	FoldEnrichment	zScore	p.adjust	geneID	Count
0	GO:0002162	dystroglycan binding	0.25	5.94e-4	0.09	4.21e+2	20.48	7.96e-3	*DMD*	1
0	GO:0050998	nitric-oxide synthase binding	0.25	5.94e-4	0.09	4.21e+2	20.48	7.96e-3	*DMD*	1
0	GO:0017166	vinculin binding	0.25	6.48e-4	0.08	3.86e+2	19.60	7.96e-3	*DMD*	1
0	GO:0003779	actin binding	0.50	2.34e-2	0.00	2.13e+1	6.30	7.96e-3	*ANLN/DMD*	2
0	GO:0008307	structural constituent of muscle	0.25	2.27e-3	0.02	1.10e+2	10.42	1.81e-2	*DMD*	1

Table S9. Gene Ontology (GO) biological process enrichment results for cluster (CL) 8 in the DMD_pCard dataset. Displayed metrics include GeneRatio (GR), background ratio (BgRatio), RichFactor, Fold Enrichment, z-score, and adjusted p-value.

CL	ID	Description	GR	BgRatio	RichFactor	FoldEnrichment	zScore	p.adjust	geneID	Count
8	GO:0030195	negative regulation of blood coagulation	0.20	49/18888	0.06	77.09	15.04	0.00157	*PLAT/TFPI/SERPINE2*	3
8	GO:1900047	negative regulation of hemostasis	0.20	50/18888	0.06	75.55	14.88	0.00157	*PLAT/TFPI/SERPINE2*	3
8	GO:0050819	negative regulation of coagulation	0.20	53/18888	0.06	71.28	14.44	0.00157	*PLAT/TFPI/SERPINE2*	3
8	GO:0030193	regulation of blood coagulation	0.20	69/18888	0.04	54.75	12.61	0.00208	*PLAT/TFPI/SERPINE2*	3
8	GO:1900046	regulation of hemostasis	0.20	71/18888	0.04	53.21	12.42	0.00208	*PLAT/TFPI/SERPINE2*	3
8	GO:0060384	innervation	0.07	27/18888	0.04	46.64	6.69	0.09417	*SERPINE2*	1

Table S10. Gene Ontology (GO) biological process enrichment results for cluster (CL) 7 in the DMD_myob dataset. Metrics such as GeneRatio (GR), background ratio (BgRatio), RichFactor, Fold Enrichment, z-score, and adjusted p-value are included.

7	GO:0002162	dystroglycan binding	0.10	5.94e-4	0.18	1.68e+2	18.26	5.88e-3	*AGRN/DMD*	2
7	GO:0034185	apolipoprotein binding	0.10	9.18e-4	0.12	1.09e+2	14.64	7.24e-3	*LRP4/PCSK9*	2
7	GO:0005201	extracellular matrix structural constituent	0.15	8.96e-3	0.02	1.67e+1	6.70	1.86e-2	*AGRN/THBS2/PCOLCE*	3
7	GO:0004252	serine-type endopeptidase activity	0.15	9.40e-3	0.02	1.60e+1	6.52	1.86e-2	*MMP2/PCSK9/FAP*	3
7	GO:0008236	serine-type peptidase activity	0.15	1.03e-2	0.02	1.45e+1	6.19	1.86e-2	*MMP2/PCSK9/FAP*	3

Table S11. Gene Ontology (GO) biological process enrichment results for cluster (CL) 9 in the ALS_iN_C9ORF72 dataset. Metrics such as GeneRatio (GR), background ratio (BgRatio), RichFactor, Fold Enrichment, z-score, and adjusted p-value are included.

CL	ID	Description	GR	BgRatio	RichFactor	FoldEnrichment	zScore	p.adjust	geneID	Count
9	GO:0004252	serine-type endopeptidase activity	0.50	174/18522	0.01	53.22	7.19	0.02793	*RHBDL3*	1
9	GO:0008236	serine-type peptidase activity	0.50	191/18522	0.01	48.49	6.86	0.02793	*RHBDL3*	1
9	GO:0017171	serine hydrolase activity	0.50	195/18522	0.01	47.49	6.78	0.02793	*RHBDL3*	1
9	GO:0004175	endopeptidase activity	0.50	388/18522	0.00	23.87	4.73	0.04146	*RHBDL3*	1

Table S12. Gene Ontology (GO) biological process enrichment results for cluster (CL) 2 in the ALS_fib_FUS dataset. Metrics such as GeneRatio (GR), background ratio (BgRatio), RichFactor, Fold Enrichment, z-score, and adjusted p-value are included.

CL	ID	Description	GR	BgRatio	RichFactor	FoldEnrichment	zScore	p.adjust	geneID	Count
2	GO:0061448	connective tissue development	0.75	288/18888	0.01	49.19	11.99	0.00414	*SLC25A25/AMER1/NOG*	3
2	GO:0060612	adipose tissue development	0.50	59/18888	0.03	160.07	17.81	0.00854	*SLC25A25/AMER1*	2
2	GO:0090090	negative regulation of canonical Wnt signaling pathway	0.50	145/18888	0.01	65.13	11.28	0.03453	*AMER1/NOG*	2
2	GO:0030178	negative regulation of Wnt signaling pathway	0.50	177/18888	0.01	53.36	10.19	0.03817	*AMER1/NOG*	2
2	GO:0060828	regulation of canonical Wnt signaling pathway	0.50	268/18888	0.01	35.24	8.22	0.03817	*AMER1/NOG*	2
2	GO:0048704	embryonic skeletal system morphogenesis	0.25	94/18888	0.01	50.23	6.96	0.04312	*NOG*	1
2	GO:0048706	embryonic skeletal system development	0.25	128/18888	0.01	36.89	5.93	0.04676	*NOG*	1
2	GO:0009952	anterior/posterior pattern specification	0.25	219/18888	0.00	21.56	4.45	0.06119	*NOG*	1
2	GO:0003002	regionalization	0.25	429/18888	0.00	11.01	3.05	0.09321	*NOG*	1
2	GO:0007389	pattern specification process	0.25	474/18888	0.00	9.96	2.88	0.09866	*NOG*	1

Table S13. Ranking of isolated DEGs that were filtered out in the network clustering process. The disease-causing (causal) gene is indicated in the first column, followed by the score calculated for each DEG based on its distance from the causal gene. Norm. Rank: normalized rank.

Causal Gene	Score	Norm. Rank	Rank	Gene Symbol	Dataset
ENSG00000198947	1.00	6.42e-05	1	*DMD*	DMD_myot
ENSG00000198947	0.74	0.03	497	*ACKR3*	DMD_myot
ENSG00000198947	0.63	0.06	865	*ARHGAP20*	DMD_myot
ENSG00000198947	0.44	0.07	1158	*TEX14*	DMD_myot
ENSG00000198947	0.50	0.08	1208	*SIM2*	DMD_myot
ENSG00000198947	0.57	0.09	1386	*SLITRK5*	DMD_myot
ENSG00000198947	0.48	0.10	1491	*PDE3A*	DMD_myot
ENSG00000198947	0.44	0.10	1503	*LDB2*	DMD_myot
ENSG00000198947	0.49	0.10	1547	*TMEM182*	DMD_myot
ENSG00000198947	0.39	0.11	1674	*DPYSL5*	DMD_myot
ENSG00000198947	0.88	0.01	164	*ABCB11*	DMD_pcard
ENSG00000198947	0.84	0.01	220	*ZNF469*	DMD_pcard
ENSG00000198947	0.66	0.02	379	*CDR2L*	DMD_pcard
ENSG00000198947	0.66	0.03	470	*CDH12*	DMD_pcard
ENSG00000198947	0.63	0.03	510	*TBX18*	DMD_pcard
ENSG00000198947	0.73	0.04	550	*SHFL*	DMD_pcard
ENSG00000198947	0.66	0.04	593	*ABCA12*	DMD_pcard
ENSG00000198947	0.67	0.05	795	*JCAD*	DMD_pcard
ENSG00000198947	0.54	0.06	929	*FER1L6*	DMD_pcard
ENSG00000198947	0.63	0.06	954	*ARHGAP45*	DMD_pcard
ENSG00000198947	0.83	0.02	263	*PTPRN*	DMD_cfib
ENSG00000198947	0.66	0.03	470	*CDH12*	DMD_cfib
ENSG00000198947	0.54	0.06	891	*NEK9*	DMD_cfib
ENSG00000198947	0.55	0.08	1297	*UNC5C*	DMD_cfib
ENSG00000198947	0.44	0.11	1641	*LAMB1*	DMD_cfib
ENSG00000198947	0.41	0.12	1843	*SIM1*	DMD_cfib
ENSG00000198947	0.55	0.12	1871	*SLC46A3*	DMD_cfib
ENSG00000198947	0.39	0.15	2321	*STC1*	DMD_cfib
ENSG00000198947	0.37	0.15	2322	*USP2*	DMD_cfib
ENSG00000198947	0.29	0.19	2990	*BMP4*	DMD_cfib
ENSG00000198947	0.99	6.42e-04	10	*EPSTI1*	DMD_myob
ENSG00000198947	0.97	6.04e-03	94	*SCN5A*	DMD_myob
ENSG00000198947	0.85	8.09e-03	126	*DDIT4L*	DMD_myob
ENSG00000198947	0.86	8.35e-03	130	*EVI2A*	DMD_myob
ENSG00000198947	0.92	0.02	237	*ADRA1B*	DMD_myob
ENSG00000198947	0.63	0.04	647	*IL16*	DMD_myob
ENSG00000198947	0.57	0.06	863	*ACSS3*	DMD_myob
ENSG00000198947	0.63	0.06	865	*ARHGEF26*	DMD_myob
ENSG00000198947	0.60	0.06	903	*KIF1A*	DMD_myob
ENSG00000198947	0.56	0.06	908	*MACROD2*	DMD_myob

Table S14. Ranking of isolated DEGs that were filtered out in the network clustering process. The disease-causing (causal) gene is indicated in the first column, followed by the score calculated for each DEG based on its distance from the causal gene. Norm. Rank: normalized rank.

Causal Gene	Score	Norm. Rank	Rank	Gene Symbol	Dataset
ENSG00000064419	0.99	5.14e-04	8	*ST6GAL1*	LGMD_myob
ENSG00000064419	0.89	0.01	173	*P2RX5*	LGMD_myob
ENSG00000064419	0.66	0.03	394	*LPIN3*	LGMD_myob
ENSG00000064419	0.67	0.03	394	*TRPV2*	LGMD_myob
ENSG00000064419	0.75	0.03	511	*FOLR1*	LGMD_myob
ENSG00000064419	0.66	0.04	560	*EDNRB*	LGMD_myob
ENSG00000064419	0.83	0.04	566	*LOC102724594*	LGMD_myob
ENSG00000064419	0.69	0.04	611	*HS3ST3A1*	LGMD_myob
ENSG00000064419	0.70	0.04	635	*BCAM*	LGMD_myob
ENSG00000064419	0.96	0.04	645	*SRPK3*	LGMD_myob
ENSG00000064419	0.82	0.02	234	*EPHA1*	LGMD_pbmc
ENSG00000064419	0.84	0.05	764	*CHERP*	LGMD_pbmc
ENSG00000064419	0.61	0.05	854	*LGMN*	LGMD_pbmc
ENSG00000064419	0.54	0.06	858	*CHST15*	LGMD_pbmc
ENSG00000064419	0.51	0.08	1273	*CA12*	LGMD_pbmc
ENSG00000064419	0.51	0.09	1419	*NUDT12*	LGMD_pbmc
ENSG00000064419	0.54	0.10	1630	*MPZL2*	LGMD_pbmc
ENSG00000064419	0.43	0.12	1810	*TMEM178B*	LGMD_pbmc
ENSG00000064419	0.59	0.12	1836	*ADTRP*	LGMD_pbmc
ENSG00000064419	0.46	0.12	1903	*FAM13A*	LGMD_pbmc
ENSG00000147894	0.93	8.03e-03	125	*NREP*	ALS_iN_C9ORF72
ENSG00000147894	0.87	8.09e-03	126	*SPATA6*	ALS_iN_C9ORF72
ENSG00000147894	0.91	0.01	157	*MAP1LC3C*	ALS_iN_C9ORF72
ENSG00000147894	0.83	0.02	335	*TGFBRAP1*	ALS_iN_C9ORF72
ENSG00000147894	0.64	0.03	445	*ST8SIA1*	ALS_iN_C9ORF72
ENSG00000147894	0.65	0.04	690	*ZC4H2*	ALS_iN_C9ORF72
ENSG00000147894	0.52	0.07	1044	*DENND3*	ALS_iN_C9ORF72
ENSG00000147894	0.70	0.07	1062	*TMEM150C*	ALS_iN_C9ORF72
ENSG00000147894	0.53	0.08	1219	*CCND2*	ALS_iN_C9ORF72
ENSG00000147894	0.47	0.08	1318	*SYTL1*	ALS_iN_C9ORF72
ENSG00000089280	0.93	2.76e-03	43	*ZFP36L2*	ALS_fib_FUS
ENSG00000089280	0.94	3.72e-03	58	*HS3ST3A1*	ALS_fib_FUS
ENSG00000089280	1.00	3.98e-03	62	*JMJD6*	ALS_fib_FUS
ENSG00000089280	0.96	0.02	235	*VRK1*	ALS_fib_FUS
ENSG00000089280	0.80	0.02	247	*ZBTB1*	ALS_fib_FUS
ENSG00000089280	0.85	0.02	260	*SERTAD3*	ALS_fib_FUS
ENSG00000089280	0.71	0.02	343	*ADAMTS6*	ALS_fib_FUS
ENSG00000089280	0.69	0.05	702	*MN1*	ALS_fib_FUS
ENSG00000089280	0.71	0.05	738	*FNDC10*	ALS_fib_FUS
ENSG00000089280	0.58	0.06	875	*RSRP1*	ALS_fib_FUS

Table S15. List of differential expressed genes (DEGs) clustered for all DMD datasets (DMD_myot, DMD_pCard, DMD_cfib and DMD_myob). It includes the cluster (CL) identifier, the number of DEGs per cluster (#D), the name of the dataset and the Gene Symbol of each DEG.

CL	#D	Dataset	Gene Symbol
0	8	DMD_myot	*ACVRL1, SLIT3, BMP4, DCC, SOSTDC1, BMP2, ROBO2, MGP*
1	6	DMD_myot	*LRP1B, CLU, PLXNA4, WSCD1, ELN, STAB2*
2	5	DMD_myot	*H3C3, H2BC3, POLE2, CGAS, CTSK*
3	3	DMD_myot	*LRRC4C, PKP2, DSC2*
4	3	DMD_myot	*WNT2B, WNT11, SFRP5*
5	3	DMD_myot	*MYH2, MYH8, EFS*
6	3	DMD_myot	*DOK6, FLT1, FGF13*
7	3	DMD_myot	*TRHDE, EMILIN2, PLAC9*
0	32	DMD_pcard	*RNASEL, CDKN2B, TAGLN2, GRM1, CALB2, CALN1, HPCAL4, SLC34A2, FILIP1L, LOX, CD93, LTBP2, FBN1, FBLN5, FBLN2, EFEMP1, THBD, PRELP, COTL1, PCDH10, SLPI, RYR1, ALDH1A2, SEZ6L2, FBXO2, CHST15, BOC, ELN, CYBRD1, GPM6A, NTM, ANGPT2*
1	29	DMD_pcard	*LDHA, EGFR, DCN, CDH1, MYCN, ENO1, ICAM5, DDIT4, NDRG1, RGS4, CD248, PRKCB, SMAD7, MAPK13, BHLHE40, MBP, ANXA1, MAGEL2, S100A11, ADAMTS2, OSMR, PTPRT, TRPC4, BHLHE41, SPON2, SPARCL1, CDCP1, SLC7A7, PLEKHH2*
2	26	DMD_pcard	*PLAT, HGF, COL1A1, FN1, LOXL3, THBS2, MMP2, COL1A2, SPARC, MRC2, TFPI, BCAT1, COL11A1, COL3A1, TNC, ITGA2, VWF, TNFRSF21, ITGA4, ITGA1, SMOC2, TGFBI, PLA2R1, HHIP, CNTN6, SERPINE2*
3	23	DMD_pcard	*RPL3L, S100A10, CRIM1, AHNAK2, AMIGO2, IL31RA, PTGS1, PTGIS, KCNA5, KCNB1, CBLN2, EMILIN1, COL26A1, COL9A3, COL6A3, COL6A6, MUC16, F2RL2, PROCR, CCDC80, SDR42E1, GCNT1, PDYN*
4	18	DMD_pcard	*PDGFRB, KDR, CD9, CAPN6, PODXL, PTPRQ, LUM, MMP14, SCARA3, ATP2B2, NHERF1, SERPINF1, IL1R1, DOK5, TENM2, FNDC1, DHRS9, OLFML1*
5	18	DMD_pcard	*RHOU, DSC3, SPTBN2, PDLIM3, EHBP1L1, MYH11, BDNF, DOCK11, SLC2A1, KCNMA1, ACTG2, NEB, TRIM38, MYH8, DAAM2, CGN, ITGA8, BST1*
6	18	DMD_pcard	*EPPK1, NPR3, MYLK, PRKG2, SLC35F2, ITGB4, GNAL, HLA-B, DRD1, LPAR1, VIPR1, GNG2, NNMT, CCL2, HMCN1, KCTD16, NPY, INMT*
7	16	DMD_pcard	*NELL2, NR2F1, COLEC11, ENG, NOTCH3, EGFL7, RXFP1, BMP4, GPC3, GLIPR1, POU2F2, BMP2, GDF2, TMEM108, GDF7, C1QTNF6*
8	15	DMD_pcard	*FOS, TBX3, FOXC1, TLE1, SP100, CDC42EP1, MAF, RUNX1, SEL1L3, PLTP, DACH1, NFE2L3, FLI1, ELK3, ELF4*
9	13	DMD_pcard	*LRP2, SHTN1, ADAM12, MEGF10, IGF2, KRT7, PRSS23, DKK2, DKK1, IGFBP3, PEAR1, IL33, IGF1*
10	10	DMD_pcard	*VCAN, HAPLN1, SLITRK2, PTPRF, DPP4, THY1, PTPRZ1, CXCL14, CXCL12, FAP*
11	9	DMD_pcard	*NUPR1, PFKFB3, KRT19, SNTB1, DMD, LIMA1, KRT80, GPRC5A, PFKFB4*
12	6	DMD_pcard	*APOB, MAP3K21, BGN, TLR4, TRAF5, CD36*
13	5	DMD_pcard	*C1RL, CFI, C3, SERPING1, CFH*
14	3	DMD_pcard	*SEMA3F, NRP2, PKHD1L1*
15	3	DMD_pcard	*LMO3, MN1, LHX9*
16	3	DMD_pcard	*SULF2, SULF1, MFAP5*
0	4	DMD_cfib	*ANLN, LIACA, DMD, CD55*
1	3	DMD_cfib	*DCN, NTNG1, LUM*
2	3	DMD_cfib	*CDKN2B, CYTB, ANKRD6*
3	3	DMD_cfib	*VWA1, PCDHA10, PCDHA6*
0	32	DMD_myob	*SORCS2, TFRC, SPTB, KIF14, DPYSL5, MAP7, OAS3, BST2, CD9, MKI67, CEP55, CENPF, PKP2, GPAT3, ESCO2, APBA2, ASPM, ACTN3, ATP2A1, ASB4, BCHE, HELZ2, KNL1, SERPINB7, SP110, ANO2, NQO2, FHAD1, EFHD1, IFI44, EFR3B, CTHRC1*
1	29	DMD_myob	*ND4L, ND1, CYTB, COX2, RCCD1, IFI27, ND4, ROS1, MAP3K9, SHROOM2, SAMD9L, SCARA3, ITGB2, COX1, ND5, ND2, TTC24, SAMD9, SEZ6, NPIPB11, IFI6, NEU4, COL24A1, ND6, COX3, ADGRD1, SOD3, POSTN, TGFBI*
2	25	DMD_myob	*PLK1, FBN2, RRM2, PPP2R2B, IFIH1, KY, RIGI, PRKG2, LDOC1, RASGRF1, IFIT3, IFIT1, IFIT2, PNMA8A, ERC2, FBXL16, CCNO, RNF125, ARC, GUCY1A2, BCAS1, PALM, SKA1, RTL3, RIMS2*
3	23	DMD_myob	*IFITM3, TERT, COL22A1, H4C6, KIF11, SIK1, MX1, KIF18B, H1-5, HELLS, PODXL, H3C3, H3C2, H3C1, H3C8, H3C7, H1-3, H2BC17, POU5F1B, H2BC7, NAP1L3, GPX7, H2BC6*
4	21	DMD_myob	*PRDX2, ESR1, DTX3L, STAT1, XAF1, ASS1, PARP14, PBX1, LMO2, ARNT2, BIRC3, FOXM1, HOXA11, DEPDC1, HOXB9, IRX5, PARP9, HOXB13, LCNL1, MAB21L1, HOXA13*
5	21	DMD_myob	*PTPRD, BAALC, STK26, ADGRL3, LUM, LRRN1, FLRT2, ADGRA1, SLIT2, SLITRK5, PODNL1, PTPN13, FLT4, TEK, GLI4, DOK6, TENM3, GHR, FGF7, TENM4, ADGRF5*
6	20	DMD_myob	*LRRK2, AP1G2, GAREM1, CIB2, MYO5C, MAMSTR, STK32B, MYH1, ACTN2, ACTA2, GSTM2, IQGAP3, CHGB, ACTG2, EPS8L2, MYH2, ATP6, ATP8, CCDC144A, GSTM1*
7	20	DMD_myob	*LRP4, DCX, AGRN, PYGM, THBS2, MMP2, DMD, TTYH2, AGTR1, S100A1, PCSK9, SIRPA, PCOLCE, CKMT2, S1PR1, LIN7A, KCNJ12, CYTIP, FAP, C11orf87*
8	19	DMD_myob	*SLC15A3, FREM2, OASL, ISG15, NFATC4, MAPK15, KCNQ5, IFI35, SELENOP, HERC5, SBSPON, DDX60, NDP, FBXO16, DUSP26, DUSP15, BATF2, MCOLN3, ZDHHC15*
9	11	DMD_myob	*DYNC1I1, GRM4, CCDC8, TOP2A, AMOT, PLEKHG3, EMILIN1, CPNE8, FAM131B, SEPTIN1, MFAP2*
10	8	DMD_myob	*SHANK1, IL20RA, DIRAS2, DIRAS1, KALRN, DLGAP1, PLCH1, PLCL1*
11	7	DMD_myob	*PCDH10, PCDHGA12, PCDHGA10, PCDHGB2, PCDHGB4, PCDHB2, PCDHGB6*
12	3	DMD_myob	*VANGL2, SFRP1, FZD3*

Table S16. List of differential expressed genes (DEGs) clustered for all LGMD (LGMD_myob and LGMD_pbmc) and ALS (ALS_iN_C9ORF72, ALS_fib_FUS) datasets. It includes the cluster (CL) identifier, the number of DEGs per cluster (#D), the name of the dataset and the Gene Symbol of each DEG.

CL	#D	Dataset	Gene Symbol
0	29	LGMD_myob	*SEMA3G, SEMA6A, SEMA3A, SEMA5A, PLXNB3, SYT12, ATP1A2, TMC6, PCDHGB3, PKHD1, ATP2A3, BCL2, TMTC2, ATP2A1, FAIM2, CST6, PCDHB5, PCDHGA5, PCDHB3, PCDHGB4, VNN2, ATP1B4, SLN, EMILIN2, PCDHGB5, SEPTIN4, PLAC9, ST8SIA1, PCDHB4*
1	27	LGMD_myob	*DCN, C3orf52, CLEC11A, FBN2, GATD3, SMOC1, FBLN2, NTNG1, MXRA5, EPYC, LRRN1, FMOD, TPBG, LRRC15, CHRD, LAMA1, IGF2, F2R, F2RL2, TNC, PCOLCE, DDIT4L, FOXP2, ELN, CD36, ACAN, PROB1*
2	25	LGMD_myob	*ERBB3, SHD, MYL1, MYL11, MYL4, MYL3, MYL6B, PWP2, GRIA1, MYH6, EEF1A2, MYH1, MYH3, ACTC1, ACTN2, ACTG2, MYH2, MYH7, MYH8, FBXL22, UNC45B, MYOZ2, KCNN3, SGCG, SYNPO2L*
3	19	LGMD_myob	*ID1, FOS, HSD17B14, LDB3, MYOG, NFIA, KRT31, GYG2, MEF2C, SNAI1, NES, PIEZO2, PAX8, FOSB, MAFB, P2RX7, LHX4, ZDHHC15, CHRNG*
4	18	LGMD_myob	*KLHL40, AQP3, AQP1, TTN, OBSCN, CKB, PPM1E, PPM1L, XIRP2, NEB, LMOD3, TRIM54, PHOSPHO1, CKM, XIRP1, MYOM1, TSPAN33, ADAMTSL4*
5	16	LGMD_myob	*CORIN, VANGL2, AIF1L, TES, THRB, HRC, CASQ2, TRDN, RYR1, CSRP2, CAMK2B, JCHAIN, ANGPTL7, ANXA3, HMCN2, FZD9*
6	16	LGMD_myob	*COL22A1, MAPRE3, TUBB2B, SIK1, TAS1R1, DCLK1, VASH2, LOC102724428, AIRE, FBLN7, H3C8, H3C7, SERTAD4, POU4F1, SMYD1, L3MBTL1*
7	13	LGMD_myob	*CAPG, SCIN, TNNC1, TNNC2, ACTA1, TMOD1, TNNI1, TNNT1, TNNT3, TNNT2, NOX5, TNNI2, KISS1*
8	13	LGMD_myob	*FLT4, STYK1, CACNA1A, RAI2, SLCO4C1, CACNA1H, FGF13, DOK5, CACNA1S, CACNA1E, SCN4A, SCN7A, CACNG1*
9	7	LGMD_myob	*SHANK1, RERG, DIRAS2, SPTBN4, SPTB, EPB41, RAP1GAP2*
10	7	LGMD_myob	*LRP4, LDOC1, KREMEN2, APOL4, BCAS1, ST8SIA5, STAB2*
11	5	LGMD_myob	*ADRA1D, DPP4, FNDC5, ACKR3, CXCL12*
12	4	LGMD_myob	*PKP1, RNLS, FLG, ALDH3A1*
13	4	LGMD_myob	*COL21A1, COL4A6, COL4A5, C1QTNF1*
14	3	LGMD_myob	*RNF122, RSPO3, RSPO4*
0	6	LGMD_pbmc	*MMP9, CXCL2, CCL20, CXCL5, CXCL1, CXCL3*
1	5	LGMD_pbmc	*MYO5C, PALLD, CALD1, HLA-DQA1, HLA-DQB1*
2	4	LGMD_pbmc	*NLRP3, IL1R2, IL1B, IL1A*
3	4	LGMD_pbmc	*CXCL8, CCL2, CXCL9, CCL8*
0	35	ALS_iN_C9ORF72	*MAGI3, NELL2, TAGLN3, CALB1, RHBDL3, MYO5B, AK5, KIF5C, ENO2, FBLN1, EZR, NHS, NIBAN2, SYNE2, VCAN, EFEMP2, CRB2, ELAPOR2, DLG3, PODXL, LTBP1, PARM1, ACTC1, LCP1, SDC2, MRAP2, ABCA1, ACTL8, EFS, CCDC33, NEDD9, FILIP1, TIE1, BIK, MYCL*
1	27	ALS_iN_C9ORF72	*SOX2, KIF21B, IMPA2, EBF1, MEIS2, MYEF2, NAV2, MACROH2A2, MFGE8, EIF2S3B, SCN3B, HAL, PKP3, H2BC14, ASH2L, SERPINF1, SPINK5, ZNF462, ZNF608, SCD5, GLI3, FIGNL2, ENPP4, ST6GALNAC3, RBP1, SERPINB8, UCP2*
2	15	ALS_iN_C9ORF72	*SLC7A2, TTN, HLA-DRA, EEF1A2, MARCKS, MRC2, COL6A1, COL2A1, MFAP4, TMEM131L, COLGALT2, IGFN1, TSPAN5, CILP, CLEC1A*
3	13	ALS_iN_C9ORF72	*NANOG, PCDH15, BCAT1, TEX15, F2RL1, ARMCX4, ALPL, ARID3B, DCHS1, CDH7, CECR2, LRATD2, SALL4*
4	11	ALS_iN_C9ORF72	*SFN, MLLT11, RAB3IP, ARK2C, RAB15, P2RX5, CYP26C1, LEF1, DACT1, SMTNL2, ZNF154*
5	8	ALS_iN_C9ORF72	*CDH2, SRC, EPHB2, EPHA3, FMO5, MGAM, DDR2, TRIM7*
6	6	ALS_iN_C9ORF72	*MAGED2, ADGRG6, GUCY1B1, GNAL, FNDC5, NPR1*
7	6	ALS_iN_C9ORF72	*MPP2, TFPI, MEOX2, CNTNAP2, CNTN1, HHIP*
8	6	ALS_iN_C9ORF72	*RBPMS, RXRA, NCOA1, ZBTB16, SIM1, ZBTB32*
0	8	ALS_fib_FUS	*RIPK4, RHOBTB1, PANK1, TUBE1, KIF26B, RGS2, RGS4, IFIT2*
1	6	ALS_fib_FUS	*RND3, NOTCH1, DES, JUP, SLC12A7, CBARP*
2	4	ALS_fib_FUS	*IQCN, SLC25A25, AMER1, NOG*
3	3	ALS_fib_FUS	*HOXC6, HOXC4, HOXB5*

Parte VI

Análisis de vías compensatorias para enfermedades raras y cáncer

Capítulo 6

Desarrollo de sistema de priorización de genes compensadores. Aplicación en cáncer y enfermedades raras.

Sobre la base de los sistemas desarrollados y descritos en las secciones anteriores, conseguí una base sólida para el uso del análisis de redes en el estudio de diversas enfermedades raras. Con estas herramientas fue posible implementar técnicas de detección de comunidades, proyecciones sobre redes multipartitas, generación de embebido de nodos y su correspondiente priorización. El siguiente paso consiste en aplicar este conjunto de herramientas al problema de la priorización de genes compensadores. De este modo, partiendo de un gen afectado en una enfermedad, el objetivo es construir un sistema capaz de predecir qué otro gen podría desempeñar un papel compensador.

Para ello, se emplean diferentes fuentes de información biológica, se ajusta el sistema de priorización específicamente al problema de los genes compensadores y se evalúa su rendimiento tanto en genes asociados a enfermedad como en los candidatos a compensadores. Entre los resultados obtenidos, cabe destacar la identificación de genes implicados en enfermedades musculares y rasopatías, así como en otras enfermedades raras (por ejemplo, talasemias o laminopatías) y cáncer.

6.1. Desarrollo de sistema de priorización de genes compensadores usando redes de similitud en base a conocimiento biomédico actual.

Mi contribución:

En este trabajo me encargué de expandir la herramienta *NetAnalyzer* para la gestión e integración de distintas redes biológicas. Para ello, se recopiló información de diversas bases de datos, abarcando desde el nivel molecular hasta el fenotípico. Entre las bases de datos utilizadas, tenemos: STRING, HIPPIE, HPO, MONDO, GO, DepMap, Reactome, HGNC. Esta información fue procesada para la construcción de múltiples redes de similitud (18 en total), que posteriormente fueron embebidas mediante distintas estrategias, incluyendo enfoques clásicos basados en kernels y métodos de *random walks*, ya utilizados previamente. A partir de los espacios de embebido se calcularon medidas de similitud por coseno, que fueron integradas según distintos promedios.

Con el fin de ampliar las funcionalidades de *NetAnalyzer*, desarrollé dos nuevos módulos: (i) un módulo de integración (*integration*) de información basado en pares de similitud, y (ii) un módulo de análisis topológico, orientado a caracterizar propiedades estructurales de las redes biológicas tanto a nivel global como de nodo.

Adicionalmente, implementé el sistema de priorización génica, adaptándolo específicamente al problema de los genes compensadores. Para ello, construí un *diseasoma* a partir de datos previos sobre enfermedades y validé la metodología comparando los resultados frente a casos conocidos. Paralelamente, llevé a cabo una búsqueda manual para la construcción de un *gold standard* de genes compensadores, combinando palabras clave y la revisión de artículos relevantes. Dada la escasez inicial de información sobre mecanismos de compensación, planteé una generalización del problema que permitió realizar una primera evaluación de las estrategias de priorización de genes causantes de enfermedad. Este enfoque me facilitó la validación del sistema a nivel general para luego aplicarlo de manera específica al conjunto de pares de gen causal-gen compensador recopilado y demostrar que es posible identificar de manera sistemática genes compensadores de enfermedad.

Highlights

Predicting Gene Compensation in Disease with Graph Embedding Techniques

Federico García-Criado, Jesús Pérez-García, Elena Rojano, Pedro Seoane-Zonjic, Juan AG Ranea

- Genetic compensation can both alleviate genetic disease severity and drive cancer treatment resistance.
- We propose a network-based computational method to predict functional compensatory genes.
- Integration of multiplexed similarity networks enhances prediction performance across datasets.
- Node2vec embeddings improve compensatory gene prioritization, especially for single-gene queries.
- Our approach identifies validated compensatory genes and supports the development of taRNA-based therapies.

Predicting Gene Compensation in Disease with Graph Embedding Techniques

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Abstract

Genetic compensation plays a critical role in mitigating the effects of deleterious mutations in genetic diseases. Identifying functionally compensatory gene relationships represents a promising strategy for discovering therapeutic targets in inherited genetic disorders and cancer. We present a novel approach that combines multiplexed network analysis with graph embedding techniques to predict compensatory genes. Our method constructs a multi-source gene similarity network, embedding each source with kernels and node2vec methods, to finally integrate all sources in a comprehensive functional similarity gene network. Our approach demonstrates high predictive performance, successfully prioritizing known compensatory genes in disorders such as Duchenne muscular dystrophy, spinal muscular atrophy, and β-thalassemia. Furthermore, we extend its application to cancer, where genetic compensation mechanisms contribute to treatment resistance. Notable examples include androgen receptor (*AR*) in prostate cancer and *RBL1* suppression compensation by *RBL2* in breast cancer. These results demonstrate that embedding network representations is useful for prioritising compensatory genes.

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Keywords:
Gene function compensation, genetic functional networks, embedding, prioritization, genetic diseases

Introduction

Genetic variation is the primary driver of human genetic diseases including cancer. However, in genetic diseases, such variations do not always result in observable phenotypic effects. This phenomenon is particularly evident in germline mutations, where the human genome is estimated to carry an average of 100 true loss-of-function variants, with approximately 20 genes completely inactivated [1]. Similarly, cancer cell lines often remain fully functional despite harboring a high mutation burden [2]. This suggests that functional genetic compensation plays a key role, whereby a functionally similar gene compensates for the loss of another.

In recent years, a specific natural mechanism of genetic compensation has been identified [3]. When a premature termination codon appears in mRNA, nonsense-mediated decay is triggered, leading to the upregulation of paralogous genes to compensate for the dysfunctional gene [4, 3, 5]. Examples of paralogous compensation are observed in genetic diseases including human ciliopathies [6] and β-hemoglobinopathies [7], as well as in different types of cancer, including colorrectal [8], breast [9] and in MAPK pathway-driven cancers [10]. However, functional genetic compensation is not necessarily mediated by paralogous genes, as suggested by numerous studies [11, 12, 13, 14, 15, 16]. Therefore, much remains to be understood about this complex mechanism.

Despite the lack of clarity surrounding its mechanisms, functional compensation could offer significant benefits for the treatment and understanding of the 300 million individuals worldwide affected by rare genetic diseases [17], most of them caused by gene mutations [18]. In this context, genetic compensation could serve as an alternative to gene therapy, which remains technically challenging and limited by current medical advancements [19]. An increasingly viable strategy involves using small molecules to modulate the expression of compensatory genes, thereby restoring functional balance [20, 21, 22]. Such therapies are now being applied to rare diseases, including Duchenne muscular dystrophy (Ezutromid) [23, 24], spinal muscular atrophy (Nusinersen) [25, 26], β-thalassemia (Sirthalaclin) [27] and X-linked adrenoleukodystrophy (metformin) [28].

Another important consideration from a clinical perspective is the role of genetic robustness in cancer disorders. In this context, cancer cells mitigate gene disruption by harnessing other genes to compensate for the lost function. This explains cases of treatment resistance, as demonstrated in studies on cells resistant to MAPK pathway inhibitors [10], nilotinib in chronic myeloid leukemia [29], and lung adenocarcinoma [14], where resistance to *ARHGAP35* suppression is compensated through of *SRC* and *STAT3* intervention. Additionally, genetic compensation is potentially essential for cancer cell survival against mutations caused by dysregulated genetic controls. For example, a recent study demonstrated the crucial role of genetic compensation in colorectal cancer [8], allowing cells to overcome the loss of function in *SRSF6* gene and promote metastasis. These cases, including cancer and genetic disease treatments, highlight the urgent need for systematic methodologies to identify potential gene compensatory relationships across a range of diseases. Such approaches could not only deepen our understanding of disease mechanisms but also inform the development of more targeted and effective clinical interventions by addressing these compensatory pathways.

Given the complexity of studying the role of compensatory genes, network analysis offers a powerful framework for understanding the intricate interactions between genes [30], which is of particularly interest in the context of gene compensation mechanisms. Traditional gene-by-gene approaches may overlook the complexity of these interactions [31], but network-based methodologies can provide a more holistic view by mapping the relationships between genes, their products, and associated biological processes [32].

In this work, we propose an *in silico* network analysis to study gene function compensation, building on advancements in multiplexed network analysis [33, 34] and embedding techniques, such as node2vec [35, 36], and kernels [33, 37]. These methods have made significant advances in prioritizing causal genes [33, 34]. We extend these methodologies to explore gene function compensation by constructing a multiplexed gene-to-gene similarity network from various data sources. To overcome the scarcity of compensatory data while maintaining integrity and relevance, we employed a transfer learning strategy, leveraging knowledge from a broader dataset to enhance predictions for a specific target dataset [38, 39], a method used in drug discovery [40] and gene network reconstruction [41, 42]. In this case, we use disease-causing genes as the broader dataset (as described in Zampieri [33] and Buphamalai [34] studies) to optimize our prediction system. Finally, we use a specific gene subset to perform a case study of gene compensation predictions based

on curated compensatory genes from literature [15, 43, 12, 6] and the cell line screening data collected for this study [10, 44]. The results demonstrate a superior performance compared to similar systems for prioritizing disease-causing genes [33, 34], as well as its reliable application in compensatory gene prediction.

The software developed for this study is available at `https://github.com/federedef/GraphPrioritizer.git` and `https://github.com/federedef/BackupGenes.git`.

1. Materials and methodology

The framework presented in this study has been designed to prioritize compensatory genes. It consists on a multi-stage workflow which it is depicted in Figure 1A. The first stage consists on the data acquisition, in which gene-related data spanning molecular to phenotype levels are compiled into gene-feature tables. These gene-feature tables are transformed to Gene Similarity Matrix (GSM) capturing the gene similarity in each data source. This step is followed by the generation of embedded GSMs (eGSMs), in which the GSMs are processed and embedded using kernels and deep learning techniques. This transformation weights gene similarities taking into account the gene context in the network. GSMs and eGSMs are computed using our NetAnalyzer tool [45], an expansion of the python library NetworkX [46].

Once eGSMs are generated, the following stage consists on the selection and integration of the best eGSMs on the basis of their gene prediction performance. For this reason, each eGSM is evaluated using methodology based on disease-causing studies. Finally, the integrated eGSMs undergo validation through two primary assessments: 1) A comparison with original disease-causing studies to measure the eGSM integration improvement (referred Figure 1B as “Selection-Integration"), and 2) using a curated list of documented gene compensatory cases we assesed the prioritization of compensatory genes that is the ultimate goal of this study (1C, referred to as the “Evaluation").

1.1. Building GSM, eGSM and gene prioritizing with NetAnalyzer

Our group have developed a tool devoted to process networks in the biomedicine context, called NetAnalyzer, initially implemented in Ruby programming language [45]. This tool was designed originally to perform network projections and was used for calculating associations between pairs

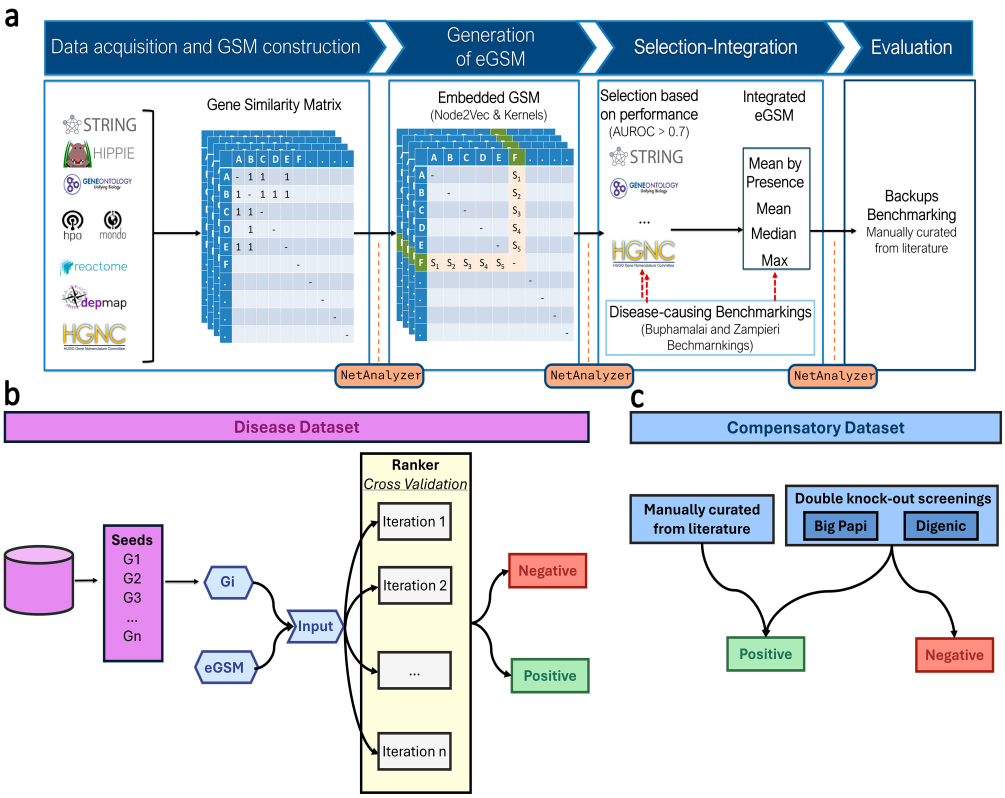

Figure 1: Workflow and evaluation description. **a** Main steps of the workflow for prioritizing compensatory genes. The process includes the following stages: 1) Data acquisition and GSM construction, 2) Individual eGSM generation, 3) eGSM selection and integration and 4) Evaluation on compensatory gene dataset. Each GSM transformation and gene prioritization are performed with the NetAnalyzer library. **b** Workflow of the disease evaluation process. Seeds are obtained from Orphanet (Buphamalai) and OMIM (Zampieri) disease groupings containing at least 20 or 30 annotated genes, respectively. Subsequently, cross-validation is conducted to derive positive and negative instances for performance assessment. **c** Workflow of the benchmarking process illustrating the selection and evaluation of compensatory gene cases. Compensatory gene cases were sourced from scientific literature or double CRISPR knock-out screenings, selected based on their significance across a majority of tested cellular lines.

of comorbid phenotypes [47], phenotype-genotype associations [48], and between protein domains and their functions [49]. Furthermore, we added basic graph embedding capabilities with kernel techniques [37]. For this study, we migrated NetAnalyzer to python (`https://github.com/seoanezonjic/`

`NetAnalyzer`) to take advantage of the NetworkX [46] library widely used in the field and to expand our tool with the recent graph embedding techniques based in deep learning.

In this way, NetAnalyzer was reestructured with the aim of allowing to process a network from raw data until generate the processed results in a supercomputing environment with command line interface. For this reason, we developed a DSL (Domain Specific Language) to describe the desired operations that would be applied onto the network. Each operation could be applied directly onto the graph or could create a new network representation in memory. For all this reasons, we designed the architecture of NetAnalyzer as shown in figure 2. We use NetworkX as main backend to storage the network itself and perform basic operations such as get node neighbors, calculate shortest paths, get network metrics, etc. More specialized calculations, such as network projections or kernel transformations are handle by NetAnalyzer on top of NetworkX. For this work, we also build an interface with our py_semtools library (under review) that is focused in perform ontology operations and calculate semantic similarity in ontology term sets as we described in Rojano *et al* [50]. To perform network embedding based on deep learning, we use the PecanPy library [36], a fast and parallelized implementation of the Node2vec algorithm [35]. All these functionalities have been incorporated into the NetAnalyzer module of the library.

In addition, we developed two new modules in NetAnalyzer to deal with this study needs: 1) ranker, a module that takes a adjacency matrix or embedding space from a network and prioritize the graph nodes using a initial node set (seed) as reference, and 2) embedding_integrator, responsible for integrating similarity values derived from different embedding spaces.

The GSMs and eGSMs of this study are computed using NetAnalyzer.

1.1.1. Generating Gene Similarity Matrix from features

The first step is build Gene Similarity Matrix (GSM) for the different features compiled in this study. There are two ways of perform this processing depending on the processed data source: by semantic similarity or by network projection.

In the case of semantic similarity, the features are comprised by gene ontology relations such as Gene Ontology (GO) or Human Phenotype Ontology (HPO). This makes that each gene has associated a list of terms that describes its properties, which are hierarchically related by the selected ontology structure. We then calculate the GSM using Best-Match Average

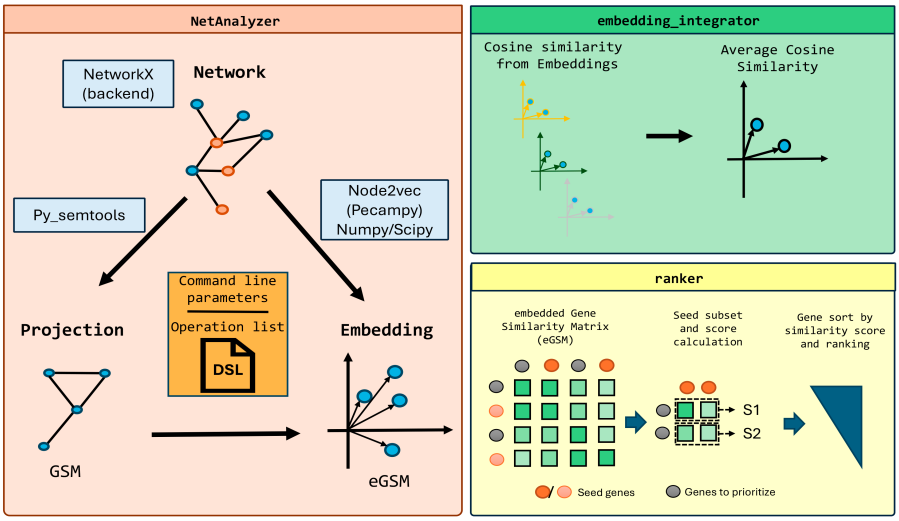

Figure 2: NetAnalyzer python library design. **NetAnalyzer**: Takes a network and process it as specifies the command line parameters o a DSL (Somain Specific Language) script for complex operations. These operations could be a network projection (that uses our py_semtools ontology library when are semantic similarity based) or an embedding using Node2vec or Kernels. **embedding_integrator**: Used to combine the information of several embeddings (represented as coloured axis) with different data sources in a single integrated embedding. **ranker**: Prioritize the genes presented in the input network embedding using a set of genes (seed) as reference computing their average distance the seed.

(BMA) with the Lin similarity metric [51]. Prior to similarity calculation, all parental terms present in each term set associated to each gene are removed. This methodology allowed for the quantification of similarity between gene term profiles and its performed by our py_semtools library.

Regarding network projection, the features form a gene-attribute network such the case of pathways in which one gene is associated to one or several Reactome pathway identifiers. In this way, we perform a network projection, transforming the bipartite network in a weighted gene network. In this projection, we use the Jaccard index or correlation measures to calculate the edges of this weighted gene network whose adjacency matrix is the GSM needed for the following steps.

The Jaccard index evaluates the proportion of shared attributes relative

to the total number of attributes observed in both genes. For two genes i and j, let n_i and n_j denote the sets of pathways associated with each gene. The weight of the edge connecting them is defined as equation 1

$$w_{ij}^{\text{Jaccard}} = \frac{|n_i \cap n_j|}{|n_i \cup n_j|} \tag{1}$$

In this expression, w_{ij}^{Jaccard} represents the degree of similarity between genes i and j. The sets n_i and n_j correspond to the neighborhood associated with gene i and j, respectively.

For the case of fully dense bipartite networks, such as gene essentiality data from the DepMap database, we also considered projections based on correlation measures, specifically Spearman and Pearson correlations. In this approach, each gene is represented by a vector that corresponds to its weighted connections across all attributes, and the similarity between two genes is defined as the correlation between their corresponding vectors. Formally, for two genes i and j, the edge weight is given by the equation 2, where A_i and A_j denote the attribute vectors of genes i and j, respectively.

$$w_{ij} = \text{Corr}(A_i, A_j) \tag{2}$$

In the case of Pearson correlation, the weight is defined in equation 3, where $\bar{A}_i$ and $\bar{A}_j$ represent the mean values of the vectors M_i and M_j, respectively.

$$w_{ij}^{\text{Pearson}} = \frac{\sum_{k=1}^{m} \left(A_{ik} - \bar{A}_i\right)\left(A_{jk} - \bar{A}_j\right)}{\sqrt{\sum_{k=1}^{m} \left(A_{ik} - \bar{A}_i\right)^2} \sqrt{\sum_{k=1}^{m} \left(A_{jk} - \bar{A}_j\right)^2}} \tag{3}$$

For Spearman correlation, the similarity is defined as the Pearson correlation applied to the rank-transformed vectors as equation 4, where R_i and R_j denote the rank transformations of M_i and M_j.

$$w_{ij}^{\text{Spearman}} = \text{Pearson}(R_i, R_j) \tag{4}$$

1.1.2. Transformation of GSM in embedded Gene Similarity Matrix

GSMs derived from the previous step underwent various graph embedding techniques to generate eGSMs. These techniques encompassed classical graph kernels (referred to as *kernel embedding*), and the deep learning method node2vec (*learning embedding*). Prior to conducting the embedding processes, the GSM was normalized by degree according to equation 5, where W denotes the original GSM, D is the diagonal degree matrix with entries $D_{ii} = \sum_j W_{ij}$, and $\hat{W}$ represents the normalized matrix.

$$\hat{W} = D^{-\frac{1}{2}} W D^{-\frac{1}{2}} \tag{5}$$

In the case of the kernel embedding we apply two popular kernels: the Random Forest kernel (RF) [52] and the Exponential Laplacian kernel (EL) [53]. For the learning embedding, Node2vec [35] is based on the simulation of random walks over the network, which are then used to train a Word2Vec model to learn low-dimensional embeddings. NetAnalyzer generates the random walks with PecanPy and process with Word2Vec the resulting sequences produce the embeddings. We employed the following parameters: *dimensions = 64*, *walk_length = 30*, *num_walks = 200*, *p = 1*, *q = 1*, and *window = 10*. These settings correspond to unbiased random walks, and generate embeddings that capture both local and global structural properties of the graph. The same parameters were applied to each network in order to compare the structural information without introducing bias from hyperparameter fine tuning. The resulting embedding assigns to each gene a vector in $\mathbb{R}^{64}$. The eGSM was then obtained by computing the cosine similarity between genes in the embedding space. Formally, if $z_i \in \mathbb{R}^{64}$ denotes the embedding vector of gene i, the similarity between genes i and j is defined in equation 6.

$$\text{eGSM}_{ij} = \frac{z_i \cdot z_j}{\|z_i\| \, \|z_j\|} = \frac{\sum_{k=1}^{64} z_{ik} \, z_{jk}}{\sqrt{\sum_{k=1}^{64} z_{ik}^2} \sqrt{\sum_{k=1}^{64} z_{jk}^2}} \tag{6}$$

1.1.3. Combining eGSM with the embedding integrator module

In addition, NetAnalyzer includes the module *embedding integrator* to process several eGSM and combine them in an integrated eGSM, summarizing the information for the selected data sources previously processed. This integration is performed using several point-wise strategies. For each gene

pair (i, j), let $\{s_{ij}^{(1)}, s_{ij}^{(2)}, \ldots, s_{ij}^{(L)}\}$ denote the set of eGSM similarity scores obtained from L different networks. The first strategy computes the mean score considering only the networks in which (i, j) has a valid score, referred to as integration mean by presence (IMP), and is defined as equation 7 where $\mathcal{L}_{ij} \subseteq \{1, \ldots, L\}$ is the set of layers containing a valid score for (i, j).

$$s_{ij}^{\text{IMP}} = \frac{1}{|\mathcal{L}_{ij}|} \sum_{\ell \in \mathcal{L}_{ij}} s_{ij}^{(\ell)} \tag{7}$$

Other alternative approach is the unrestricted mean across all layers defined by equation 8.

$$s_{ij}^{\text{Mean}} = \frac{1}{L} \sum_{\ell=1}^{L} s_{ij}^{(\ell)}. \tag{8}$$

Other strategies include taking the maximum score, as defined by equation 9

$$s_{ij}^{\text{Max}} = \max_{\ell \in \{1,\ldots,L\}} s_{ij}^{(\ell)}, \tag{9}$$

or the median of the scores, as equation 10.

$$s_{ij}^{\text{Median}} = \text{median}\left\{s_{ij}^{(1)}, s_{ij}^{(2)}, \ldots, s_{ij}^{(L)}\right\}. \tag{10}$$

All these strategies yield an integrated eGSM, which is subsequently used for prioritization analysis.

1.1.4. Prioritization algorithm with ranker

In this study, a “seed" is defined as a group of genes associated with a specific characteristic, disease, or disease-causing scenario, and as a target gene in compensatory gene identification analysis. The *ranker* module was developed to prioritize candidate genes. It accepts as input either a single eGSM or an integrated eGSM, together with a set of seed genes, and computes for each candidate gene a score that reflects its similarity to the seed set. The score is computed using the scalar product of two vectors as described in equation 11. The $eGSM$ term is the embedding matrix representation and

v_{seed} and v_{gene} are indicator vectors with ones at the positions of the seed genes and the candidate gene, respectively, and zeros elsewhere. Then, the genes are sorted in order of their average similarity value with respect to each gene on the seed. All candidate genes are then ranked in descending order of this score, so that those showing the highest average similarity to the seed set are prioritized as the most relevant and proposed as predicted candidates.

$$Score_{eGSM}(gene, Seed) =< v_{gene}, eGSM \cdot v_{seed} > \tag{11}$$

1.2. Gene feature retrieving and GSM construction

We selected a total of eight databases with different functional gene information to establish measurable relations between genes. Details are provided in Table 1. From these databases, we derived 19 gene functional networks with direct (i.e. PPI weighted networks) or gene attribute mediated (i.e. gene - disease) relations. These networks are used for constructing GSMs.

Each dataset was processed as follows.

For STRINGdb, both combined and specific channel scores were considered. Eight different GSMs were directly generated without further processing using the database raw score. One GSM comprises the combined score and seven derived from specific STRING channels: textmining, coexpression, cooccurrence, database, experimental, fusion, and neighborhood.

In the case of HIPPIE, we used the specific genes dataset described in Buphamalai and collaborators [34], keeping gene interactions supported by PubMed articles to create a binary GSM without further processing.

In our analysis, we employed three ontologies: Gene Ontology (GO) [54], Human Phenotype Ontology (HPO) [55], and Mondo Disease Ontology (MONDO, named as Disease in this study) [56]. In the case of GO, the procedure was applied to all three sub-ontologies: Molecular function, Biological process, and Cellular component.

In the DepMap dataset [57], the survival profile matrix (a dataset that contains information about the sensitivity of cancer cell lines to various treatments or perturbations) served as the basis for constructing the GSMs. Spearman and Pearson correlation analyses were conducted (referred to as DepMap Spearman and DepMap Pearson, respectively), with significant relationships retained using a p-value cutoff of 0.05. Furthermore, processed

data from Kim and collaborators (2019) [2] (referred to as DepMap Kim) were incorporated into our study without further processing.

With the Reactome dataset [58], we assessed gene similarity by calculating the Jaccard index between shared pathways. Gene pairs with at least three common associated pathways were selected.

And for the HGNC Gene group dataset [59], we used the defined gene groups as gene families in order to capture its structural information, function, homology, and encoded protein structure. To construct the GSM we computed the Jaccard index for common HGNC Gene Groups as one gene can be related to several groups.

1.3. Evaluation of eGSMs using disease datasets

Given the scarcity of data for prioritizing compensatory genes, we initially assessed the performance of our system in a standard disease-causing scenario. These results informed the optimization of eGSM selection and integration. We evaluated the performance of our system using two distinct disease-causing evaluation strategies, referred to as Zampieri and Buphamalai [33, 34]. Each eGSM was utilized in both strategies to predict disease-associated genes and to assess their performance. Additionally, the raw GSM was also utilized as a negative control eGSM. A visual representation of this process is illustrated in Figure 1B.

The first method, described by Zampieri and collaborators [33], uses a list of disease-associated genes obtained from the Online Mendelian Inheritance in Man (OMIM). Zampieri extracted the dataset from Goh and collaborators [64]. We regenerated this dataset by downloading the Morbid Map (MM) from OMIM on May 7, 2024. The MM file was filtered by the tag “(3)" to identify mutations found in the gene. Then, each disease was categorized in accordance with the original classification system proposed in [64], using the MONDO ontology with cross-references. Finally, following the procedure described in [33], we selected the same 20 OMIM disorder groups comprising at least 30 genes to perform a leave-one-out validation. For each seed, half of its size was randomly selected as negative examples from genes found in other seeds but not in the seed being studied.

The second strategy, described by Buphamalai and collaborators [34] extracts genes linked to rare diseases from the Orphanet database. We used the original Buphamalai dataset [34] of ORPHA rare genetic diseases, each containing a minimum of 20 associated genes. With this data, we assessed performance using a 10-fold cross-validation, which was executed for every

Table 1: Summary of the main characteristics of the databases used to create the GSMs.

Database & GSM	Description	Release date & URL
STRINGdb *STRING [60] (channel): combined, textmining, coexpression, neighborhood, experiments, cooccurrence, databases and fusion*	Protein-protein interactions (PPI) functional associations. We selected information from eight different channels: text mining, co-expression, co-occurrence, database, experimental, fusion, neighborhood and combined score.	2019-01-19 `https://string-db.org`
HIPPIE [61] *Hippie*	Reliable PPI in *Homo sapiens*. We used the data selected by Buphamalai et al (2021) (interactions with PubMed references) [34]	2019-02-14 `https://github.com/menchelab/MultiOme` `https://cbdm-01.zdv.uni-mainz.de/~mschaefer/hippie/`
Gene Ontology [54] *Biological Process, Celullar Component, Molecular Function*	Standardized terms to describe molecular functions, biological processes and cellular components of genes. We selected genes with annotations in the three sub-ontologies.	2018-11-15 `https://release.geneontology.org`
Human Phenotype Ontology [62, 34] *HPO*	Standardized terms to describe human disease clinical phenotypes. We selected genes from the ontology file and those described in Buphamalai and collaborators (2021) [34]	2018-10-09 `https://hpo.jax.org/`, `https://github.com/menchelab/MultiOme`
MONDO [56] *Disease*	Ontology of standardized terms to describe human diseases and their relationships. Both ontology and annotation files were used.	2018-12-02 `https://github.com/monarch-initiative/mondo`, `https://monarchinitiative.org`
DepMap [57] *DepMap Pearson, DepMap Spearman, DepMap Kim*	Includes information about the dependency of cancer cells on specific genes for their survival and proliferation. We use both general information and selected by Kim et al. (2019) [2].	2018-02 `https://depmap.org` (18Q1, gene_effect.csv)
Reactome [63] *Pathway*	Detailed signaling pathways and biological processes in humans. We selected the data used by Buphamalai et al. (2021) [34].	2019-01-30 `https://github.com/menchelab/MultiOme` `https://reactome.org/`
HGNC gene group [59] *HGNC group*	Grouping of genes within the HUGO Gene Nomenclature Committee (HGNC) related to similar biological functions or characteristics in the same set.	2020-07-01 `https://www.genenames.org/data/genegroup`

seed. In this process, the remaining genes which were not in the original disease group in the eGSM were designated as negatives for each respective seed. We intentionally use the data from Buphamalai and collaborators and we not updated them to the current database data. This is due to the need to preserve the same conditions and being able to compare the results of our

work with this study to see if our data selection and algorithm implementation outperforms the Buphamalai results. The cited study builds a multiplex network with several gene features that uses an Informed multiplex network propagation algorithm to prioritize genes based on their functional similarity.

1.4. eGSM selection and integration

For results obtained in both strategies to predict disease-associated genes, the predictive performance was assessed using Receiver Operating Characteristic (ROC) curves computed with the R package ROCR. This approach uses methodologies established in prior research [33, 34]. The eGSM selection to integrate is based on their performance, requiring an area under the ROC (AUROC) value exceeding 0.7 in both strategies. Once the best eGSM are selected, the individual eGSM scores were integrated pairwise for each type of GSM embedding. Finally, the improvement in integration was verified following the sames metrics and datasets used on the evaluation of individual eGSM.

In our study, we incorporated coverage metrics, referred to as gold standard coverage, for both individual and integrated eGSMs. Specifically, for each gene belonging to one or more disease groups, we assessed how frequently it forms connections within the eGSM to other genes in the same group during cross-validation. This process was performed for all genes in the reference dataset, and the final ratio was calculated relative to the total number of gene-disease connections in the dataset (Equation 12).

$$\text{GS-coverage} = \frac{\sum_{g \in D} c(g)}{\sum_{g \in D} s(g)} \tag{12}$$

Here, D represents the set of genes in the dataset, c(g) denotes the number of times the gene g have connections respect to a seed, and s(g) indicates the number of seeds to which g is related in the dataset.

1.5. Case study for compensatory genes prioritization

We applied our methodology to prioritize compensatory genes in a real-world scenario. Initially, we conducted a literature search to identify reliable compensatory genes. Bibliographic data were obtained from two types of sources: 1) low-throughput (LT) articles with different experimental validations, and 2) high-throughput (HT) articles employing double knock-out CRISPR screening.

On LT, articles were searched by a keyword list, including: “genetic/gene/transcriptional compensation", “backup/compensatory gene" and “compensatory response". Then, we selected those on which experimental validation by PCR or western blot validation in human cell cultures or patient biopsies was performed. For the HT articles, two double knockout screens studies were selected based on the number of cell lines used (number of cell lines > 5): i) the Ito screen (Digenic) [10], which screened 5,065 paralog pairs and 815 paralog families in eleven cell lines, and ii) the Najm screen (Big Papi) [44], performed in six cell lines. Here, to ensure robust gene compensation in both positive and negative cases, we selected candidates based on significant compensation in the majority of cell lines in each screening study.

Finally, for evaluation we used three metrics to asses the prioritization reliability of the proposed methodology. For coverage, is measure as described in disease dataset. For performance, top rankings (5, 10, 20 and 100) and ROC curves were added. The evaluation process was conducted as described in the disease-causing evaluations. A diagram outlining this process is provided in Figure 1C.

2. Results

2.1. Gene coverage by source and embedding impact in gene relationships

We evaluate the informativeness of GSMs and eGSMs based on two key metrics: gene coverage, defined as the number of genes represented in the similarity matrix, and edge density, which quantifies the proportion of observed edges relative to the total possible connections. Regarding gene coverage (Figure 3), DepMap Kim, HPO and Disease exhibit the lowest values (<8000) compared to the other GSMs, which exceed 17000 genes. Notably, the STRING combined GSM (incorporating protein-protein interactions from STRING using the combined score channel) and the GO Cellular Component GSM (based on semantic similarity of Gene Ontology annotations) show the highest coverage. Regarding edge density, it is important to consider that the embedding process plays a crucial role in gene relationships, as it can infer hidden connections and assign relative importance to them. Consequently, a detailed analysis of the GSM edge density ratio (Figure 3) reveals that GSMs derived from ontology databases (GO subcategories, Disease, and HPO) approach the highest possible value (1), whereas STRING combined, HIPPIE, HGNC Group, and DepMap Kim exhibit the lowest ratios.

However, the edge density ratio increases after embedding the GSMs, as observed in the cases of node2vec, RF, and EL (Figure 3, *Edge Density* column). Particularly, STRING Combined, HIPPIE, HGNC Group, and DepMap Kim nearly reach the highest edge density ratio following the embedding process, particularly with node2vec.

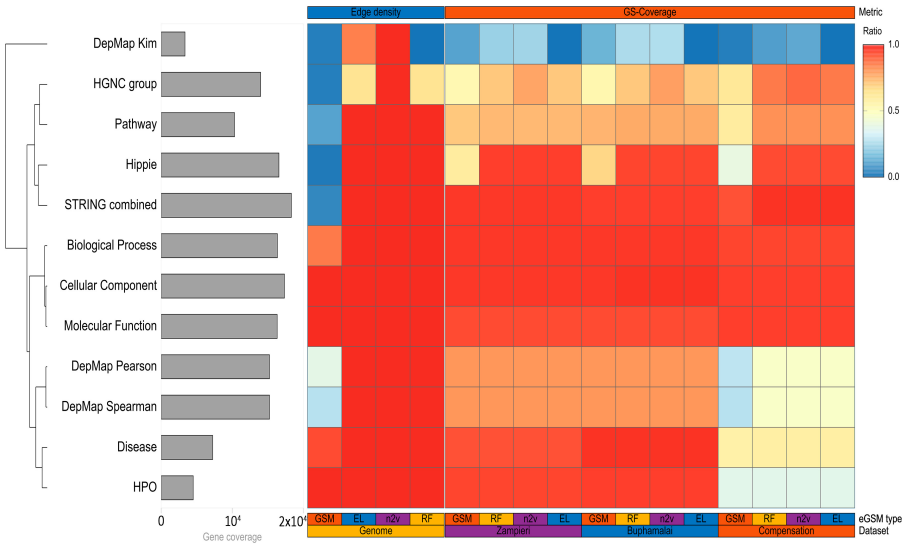

Figure 3: Representation of GSM and eGSM analysis results. Data sources are clustered based on edge density (ratio of actual to maximum potential edges in the matrix) and the gold standard (GS) coverage (defined in equation 12). The values are distributed in a ratio from 0 (blue) to 1 (red). The coverage value (total number of genes in the GSM) for each data source is represented by gray bars. The ratio calculated for the different types of embedding (GSM: GSM without embedding; EL: exponential laplacian; RF: random forest; n2v: node2vec) and the dataset used for the evaluation (Genome, Zampieri, Buphamalai and Compensation) are also shown.

2.2. Prioritization capacity assessment on individual layers

Focusing on the Zampieri evaluation, this approach leverages disease-associated gene groups inferred from OMIM as a reference dataset and generates a single ROC curve based on a leave-one-out strategy [33]. Each eGSM was used to generate a gene ranking, which was then evaluated using this methodology. A key factor influencing performance is the gold standard (GS) coverage of each eGSM (Figure 3, GS-Coverage), proportional to the number of curated interactions from a given dataset covered in th eGSM

(more details in Methodology section). All eGSMs have good GS-coverage ($> 80\%$), except for DepMap (especially Kim), HGNC group and Pathway, which have lower values. Regarding ROC curves (Figure 4A), the Disease, STRING combined, HPO and Pathway eGSMs exhibit robust performance, with AUROC values exceeding 0.7 (Supplementary Figure 1). The HGNC group, Hippie, and Biological process eGSMs show moderate performance, with AUROC values ranging from 0.65 to 0.7. And the DepMap, Cellular component and Molecular function eGSMs exhibit lower performance.

To further characterize performance, we employed the evaluation methods described by Buphamalai and collaborators [34]. In this approach, disease groups are derived from Orphanet, and the AUROC is calculated for each group individually using a 10-fold cross-validation. This method leverages AUROC distributions from individual disease groups to mitigate potential bias caused by outliers (e.g., complex or systemic diseases, as shown in Supplementary Figure 2), thereby providing a more robust performance assessment.

Regarding GS coverage (Figure 3), we observed similar values to those in the Zampieri dataset, demonstrating consistent performance across all individual eGSMs. The performance for each eGSM was calculated as the median AUROC distribution by seed and is presented in Figure 5A.

If we compare these results with those obtained using the Zampieri dataset, the Disease, STRING combined, HPO, Pathway and Biological process demonstrated the highest performance with median AUROC values reaching up to 0.7. We also observe that the median AUROCs of the HGNC group and Hippie decreases to similar values obtained for DepMap Kim, Cellular component and Molecular function eGSM, ranging from 0.6 to 0.7. And regarding the median AUROCs of both DepMap Spearman and Pearson, their performance were the lowest (median AUROC values ~ 0.5).

Two eGSMs consistently exhibit the highest performance across both evaluation methods: Disease and STRING combined. Notably, the Disease eGSM integrates datasets from OMIM and Orphanet [33, 34], leveraging the MONDO ontology to merge and expand these sources, which enhances its predictive capabilities. Regarding STRING combined, the eGSM aggregates scores from all data sources within the STRING database, constituting an integrated information source. Therefore, a comprehensive performance analysis was undertaken for each individual STRING channel (Supplementary Figures 4 and 5). Among these channels, textmining, coexpression, experiments, and databases exhibited the highest efficacy, demonstrating AUROC

values exceeding 0.7 in both Zampieri and Buphamalai datasets.

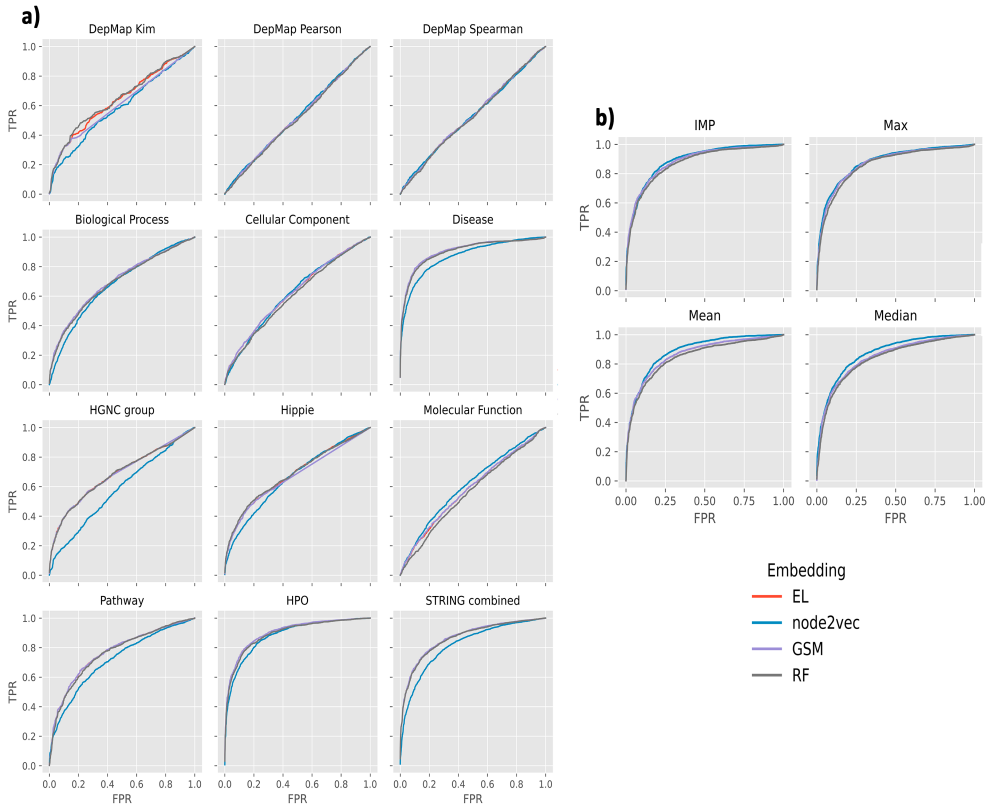

Figure 4: ROC curve analysis of eGSM performance on the Zampieri dataset. **a)** ROC curves for each individual eGSM, categorized by data source and color-coded by embedding method. **b)** ROC curve for the integrated model combining selected eGSMs, showing the improvement in classification performance through integration.

2.3. Selecting the best layers for integration

After evaluating each eGSM using both disease-related benchmarks, we proceed to integrate the best-performing eGSMs to enhance the gene predictions. Specifically, we select those with AUROC values exceeding 0.7 in both evaluations (Buphamalai, Figure 5A, and Zampieri, Supplementary Figure 1). Our integrated eGSM includes HPO, Pathway, and the STRING channels: textmining, coexpression, experiments, and databases. The Disease eGSM was excluded due to redundancy in information shared between MONDO and OMIM/Orphanet.

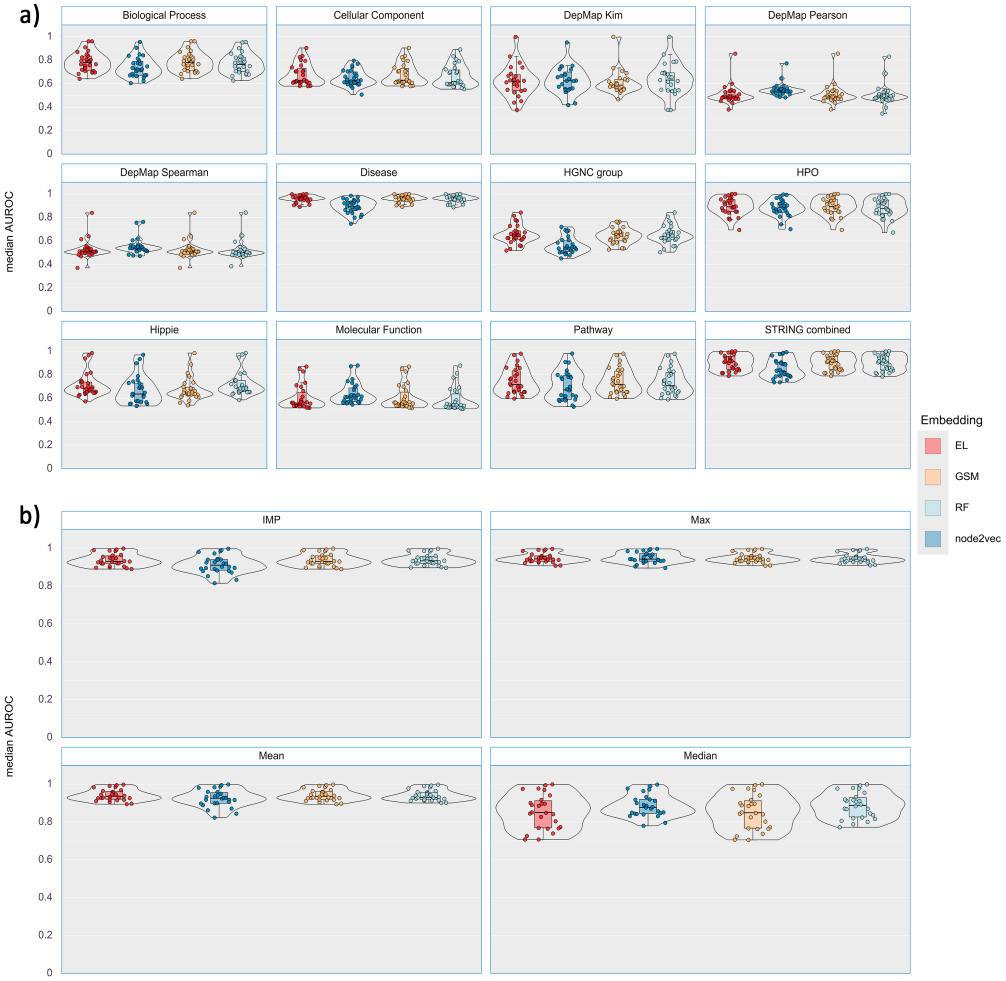

Figure 5: Median AUROC distributions for **a)** each individual eGSM, and **b)** each integrated eGSM on the Buphamalai dataset, categorized by source and embedding. The Y-axis represents the median of the 10-fold cross-validation AUROCs for each seed. IMP: integration mean by presence, EL: exponential Laplace, RF: random forest.

In the context of the Zampieri datasets, the integrated eGSMs did not demonstrate superior performance compared to the best individual one (Figure 4B). Nevertheless, our methodology achieves higher AUROC values (Supplementary Figure 1) compared to the original Zampieri system (AUROC = 0.80) [33]. Regarding integration behavior, different embedding processes

yielded similar performance, with node2vec showing a slightly superior AUROC (Figure 4B and Supplementary Figure 1).

As for the Buphamalai dataset, the distribution of median AUROCs by seed (Figure 5B) shows that the integrated eGSM outperforms the best individual layers. However, node2vec demonstrates the lowest performance among all integration methods studied, except for the median integration.

2.4. Case study: Application on manually curated compensatory genes dataset

Our primary objective is to identify potential compensatory genes for disease-associated genes. Using our integrated eGSM, which consolidates pertinent functional information linking gene pairs, we assessed its performance on a small dataset containing 28 genes with known compensatory relationships extracted from low-throughput (LT) articles. These genes served as positive cases in our evaluation (Table 2). Furthermore, we sourced 58 genes with known compensatory relationships from high-throughput (HT) articles, bringing the total number of positive cases to 86. Conversely, we identified 452 experimentally confirmed genes from these articles that lack compensatory relationships, serving as negative cases in our system.

All integrated eGSMs achieved nearly full coverage, except for HPO, which reached approximately 40% coverage (Figure 3 and Supplementary Figure 3). The integrated eGSMs achieved 100% coverage (Supplementary Figure 6). In terms of performance, our system accurately predicted compensatory genes within the top 5 and 10 integrated eGSMs (Figure 6). Node2vec achieved the highest performance (51.2% in the top 5) among all embedding processes across different integration methods. Moreover, regardless of the embedding technique used, the mean integration method proved to be the most effective in terms of top rankings.

The AUROCs ranged from 0.82 to 0.88, showing high values across most integration methods, except for the median integration method, which exhibited more heterogeneous AUROCs, generally lower than those observed for other integration methods. Particularly, the Max integration combined with RF embedding proved to be the most effective (Figure 7A and Figure 7B). However, no significant differences were observed among the various integration and embedding methods in terms of the AUROC metric (Figure 7B).

Table 2: Compensatory genes identified from LT articles.

Affected gene	Compensatory gene(s)	Disease or biological process involved	Source
RBL2	*RBL1*	Breast cancer	[9]
CD9	*CD63*	Melanoma	[65]
TGM1	*ABCA12*, *SLC27A4*, *ELOVL4*, *CYP4F22*, *CERS3*, *SDR9C7*, *ALOX12B*, *LIPN*	Congenital ichthyosis	[13]
HBB	*HBG1*, *HBG2*	β-hemoglobinopathies (Sickle cell disease and β-thalassemia)	[7, 66]
AR	*GR*	Prostate cancer	[67, 68]
BRD4	*POLR2A*	Myc-driven tumors	[69]
ARHGAP35	*SRC*, *STAT3*	Lung adenocarcinoma	[14]
ACTA1	*ACTC1*	Nemaline myopathy	[70]
KCNH2	*NFE2L2*	Breast cancer	[12]
HBA2	*HBA1*	α-thalassemia-2	[71]
PKLR	*M2PK*	Nonspherocytic hemolytic anemia	[16]
CEP290	*UNC119b*	Human ciliopathies	[6]
DMD	*UTRN*	Duchenne Muscular Dystrophy	[72]
HNF1B	*PDX1*	Pancreatic hypoplasia	[15]
HNRNPA1	*HNRNPA2*	mRNA metabolism	[73]
HNRNPA2	*HNRNPA1*	mRNA metabolism	[73]
GLT25D1	*GLT25D2*	Osteosarcoma	[43]
SMN1	*SMN2*	Spinal muscular atrophy	[26, 74]
ABCD1	*ABCD2*	X-Linked adrenoleukodystrophy	[75]

3. Discussion

Functional compensation masks the potentially deleterious effects of genetic variants, thereby preserving gene function. However, this mechanism is a double-edged sword. In genetic disorders, compensation can alleviate the severity of the pathology in affected individuals [6, 7, 70]. Unfortunately,

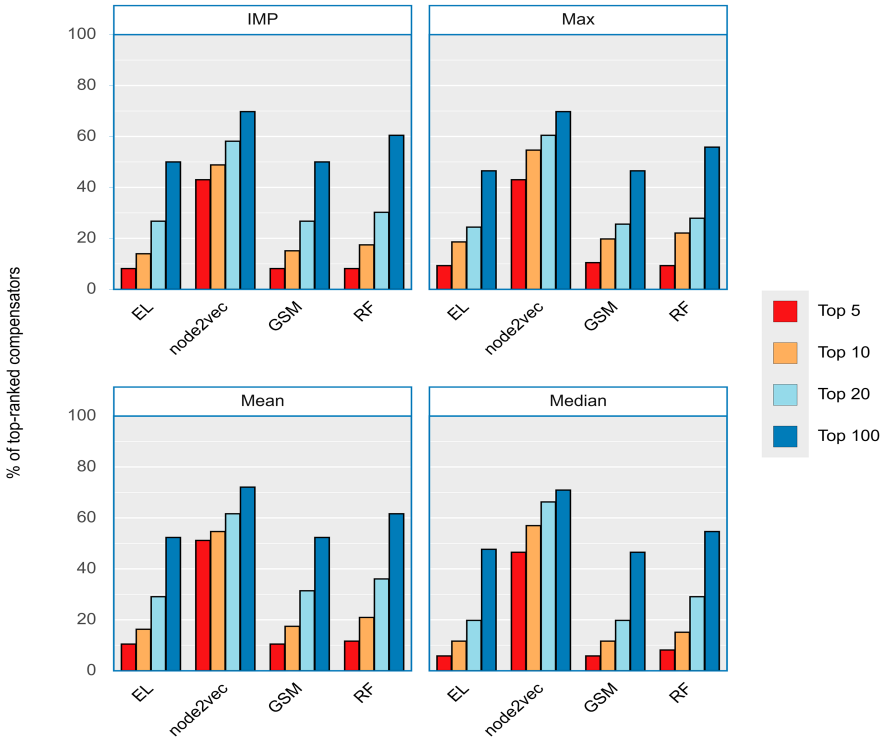

Figure 6: Top 5, 10, 20, and 100 results across various integrated eGSMs, categorized by integration method and color-coded according to the top number. IMP: integration mean by presence, EL: exponential Laplace, GSM: gene similarity matrix (control), RF: random forest.

cancer cells also exploit this process, leveraging genetic compensation to resist treatment [10, 29, 14] and protect themselves from the detrimental effects of their high mutation burden [8].

In the context of genetic disease treatment, functional genetic compensation offers a viable alternative to gene therapy, which remains technically challenging and limited by current medical advancements [19]. A more feasible strategy involves identifying genes with similar functions that can compensate for the loss of function, and then using small molecules to upregulate these compensatory genes. This approach offers a more feasible and effective treatment strategy, with successful applications in various genetic diseases, such as Duchenne muscular dystrophy [23, 24], spinal muscular at-

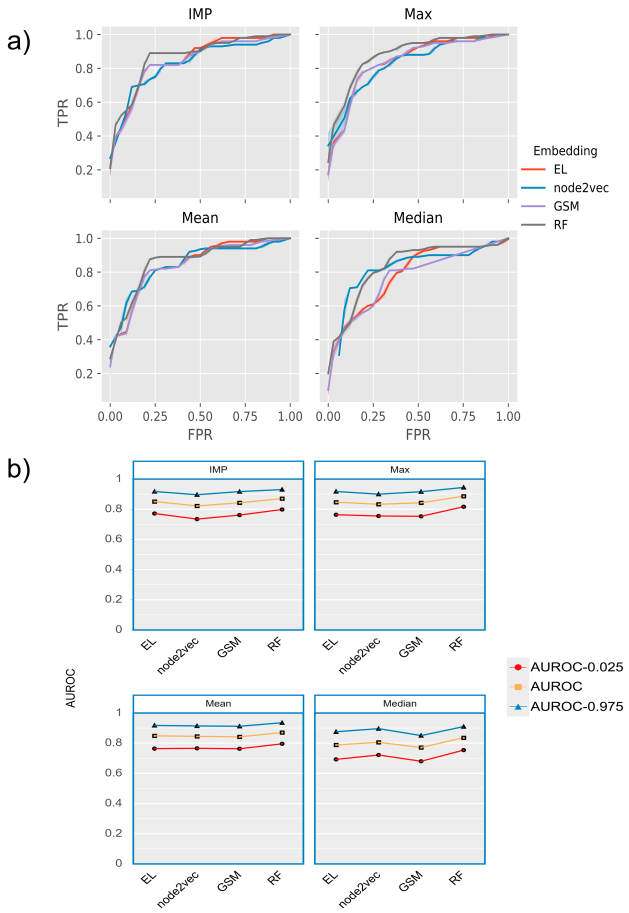

Figure 7: ROC curves and AUROC for each integrated eGSM using the curated dataset of compensatory genes. a) Curves categorized by integration method and color-coded according to the embedding process. IMP: integration mean by presence, EL: exponential Laplace, GSM: gene similarity matrix (control), RF: random forest, TPR: true positive rate, FPR: false positive rate. b) AUROC Confidence Interval (CI) in each integrated eGSM for Backups dataset. CI was obtained by a 1000 iteration bootstrap.

rophy [25, 26], and other genetic diseases [27, 28]. However, experimentally testing the function of all genes to identify those with equivalent functions is impractical. Consequently, computational strategies are crucial for addressing this challenge.

In this study, we demonstrated that a network-based approach, combined with multiple evaluation datasets, achieves high performance in the identi-

fication of functional compensatory genes. First, as a consequence of the limited availability of compensatory data, we selected and integrated the layers that has the best performance to predict gene related diseases. This integration outperformed previous algorithms in the field, as evidenced by our results with the Zampieri [33] and Buphamalai [34] benchmark frameworks. Additionally, our optimized system was applied to the prediction of functional compensation over a reference dataset curated from literature.

Our integration approach shows enhanced performance in the Buphamalai dataset compared to the best-performing individual layer (Figure 5). We have predictive performance (reaching a median AUROC = 0.94) that exceeds that observed in the Buphamalai study (median AUROC = 0.90) [34], using the same dataset and performance metric. For this reason, we used the source data at the dates used in Buphamalai study with the purpose of obtaining a fair comparison. In this way, our approach has higher accuracy and this is due to the data source selection and the integration data approach combined with the embedding transformation instead of use the Informed multiplex network propagation algorithm used in Buphamalai work [34].

Interestingly, this improvement is not observed in the Zampieri dataset (Figure 4) likely due to the differing methodologies used to measure performance. Zampieri dataset employs a general AUROC measure, while Buphamalai dataset utilizes a median AUROC calculated across each seed used, minimizing the impact of specific seed biases. This latter approach is robust in cases where seed sizes are not uniformly distributed and prioritization tends to yield poor results with large seed groups, as is the case for both datasets (Supplementary Figure 2). This occurs because larger gene groups often combine unrelated genes with highly heterogeneous functions, leading to poor gene predictions.

With respect to the different databases used in this study, STRING demonstrates the best performance for the Buphamalai dataset (Figure 5). This is particularly surprising when compared to the HIPPIE interactions, which do not show such high performance. The decomposition of the different channels allowed us to identify the textmining and coexpression channels as the primary contributors to this improvement (Supplementary Figure 4). This suggests a special enhancement when the interactions represent more of a functional or contextual relationship rather than a physical one. In fact, when we analyze the experimental channel alone (Supplementary Figure 4) the performance closely resembles that of HIPPIE (Figures 5 and 4). This demonstrates that splitting the STRING data across its various channels

and analyzing their performance separately allows for the identification of relevant information to establish functional similarity between genes. This is especially relevant when applying this approach to identify compensating gene pairs. It is worth noting that the coexpression channel captures both transcriptomic and proteomic contexts, which have a profound impact on functional similarity, even more so than the experimental protein interaction channel. Other databases that demonstrated strong performance include HPO (as observed in [34]) and MONDO. However, this is likely due to the strong relationship between the disease evaluation dataset and the way diseases are defined in MONDO, as well as their descriptions in HPO.

Regarding the compensatory dataset, node2vec shows a significant improvement in the final results, achieving top ranks with a substantial advantage over other embedding techniques and GSM controls (Figure 6). This is due to the problem with large seed sizes described before that has a great impact in node2vec, which enhances the non-linear nature of the algorithm when applied to disease datasets (Supplementary Figure 2). This limitation is overcome in the identification of compensatory genes, as the algorithm focuses on querying individual genes, showcasing its true potential in the top predictions. Moreover, all methods consistently achieved satisfactory results at the AUROC level, with values exceeding 0.75 (Figure 7B), and no significant differences were observed between embedding methods.

On the other hand, it is noteworthy that the gold standard (GS) coverage significantly decreases in the compensation dataset for databases related to pathological processes such as MONDO, Phenotype, and DepMap (Figure 3). This finding is particularly surprising considering the pathological context from which the compensatory genes were identified. Two potential explanations for this decline could be the difficulty in characterizing these genes in patients or cell lines due to their essential gene status, or the practical challenges associated with studying such interactions. Notably, compensation does not always involve direct interaction or coexpression between the compensatory and affected genes. This is reflected in the reduced coverage of the GS for the coexpression and experimental channels (Supplementary Figure 3).

From a clinical perspective, our prioritization system holds significant promise for treating genetic diseases by identifying potential targets for experimental validation. For example, in genetic diseases like spinal muscular atrophy, Duchenne muscular dystrophy, and β-thalassemia, our system successfully prioritized each known compensatory gene within the Top 5.

This success underscores the predictive capability of our approach in human genetic disorders, highlighting its potential to uncover novel gene targets for therapeutic interventions. In the case of Duchenne muscular dystrophy (DMD), the utrophin compensation mechanism has been mechanistically characterized in patient-derived cell lines [76]. Furthermore, this pathway has already been targeted in clinical trials with Ezutromid [23, 24], providing a clear example of a compensatory mechanism prioritized by our framework, mechanistically defined, and translated into clinical application.

Furthermore, given the role of genetic compensation in cancer, our methodology could also be extended to oncology, enabling the identification and prioritization of compensatory genes that contribute to treatment resistance and tumor progression. Among notable examples from Table 2, prostate cancer stands out, where cancer cells bypass androgen receptor blockade through the induction of the glucocorticoid receptor. Similarly, in breast cancer cells, the suppression of *RBL1* is compensated by *RBL2* [67, 68]. In addition, compensatory relationships extracted from high-throughput experiments arise from genetic blockade studies in various cancer cell lines. Based on this, our system has the potential to predict or help overcome treatment resistance in cancer.

Once compensatory genes are identified for a disease-associated gene, the next step involves addressing their activation using small molecules. Initially, potential drugs can be explored to induce the activation of these compensatory genes [24, 26, 27, 28]. Another alternative is the application of small RNAs, which are a versatile biotechnological tool of significant interest [21]. These small RNAs are already implicated in natural compensation mechanisms for paralogous genes [3, 5]. However, in cases where these mechanisms are not naturally induced (e.g., when no premature termination codon or paralogous gene is present), activation may require the use of exogenous small RNAs. In this context, recent studies have highlighted translation-activating RNAs (taRNAs) as promising tools [20], specifically designed to enhance the translation of target genes. These RNAs offer a straightforward delivery system, which could be particularly applicable in the treatment of RASopathies [20]. Other strategies include the use of siRNAs targeting inhibitors of the compensatory gene, as well as splice-switching antisense oligonucleotides (ASOs) to induce reading-frame alterations that introduce premature termination codons (PTCs) into the mRNA, thereby activating compensatory mechanisms mediated by paralogous genes [76]. Integrating our approach with these activation systems thus represents a promising strat-

egy for the development of novel therapeutic interventions.

4. Conclusion

This study clears the way for predicting robust compensatory relationships through network embeddings, with potential for experimental validation using screening techniques [77]. Future improvements could be achieved by incorporating additional databases, such as specific coexpression data from particular tissues, and integrating results through machine learning processes with transfer learning, as described in [78]. Moreover, applying our system in combination with guided activation RNA strategies could accelerate the development of treatments based on gene compensation.

5. CRediT authorship contribution statement

Conceptualization, J.A.G., F.G.C., and P.S.Z.; Methodology, J.A.G., F.G.C., and P.S.Z.; Software, F.G.C. and P.S.Z.; Formal analysis, J.A.G., F.G.C., J.P.G., E.R., and P.S.Z.; Writing – original draft, J.A.G., F.G.C., E.R., and P.S.Z.; Writing – review and editing, all authors; Funding acquisition, J.A.G. and P.S.Z.;Supervision, J.A.G. and P.S.Z.

6. Competing interests

The authors declare no competing interests.

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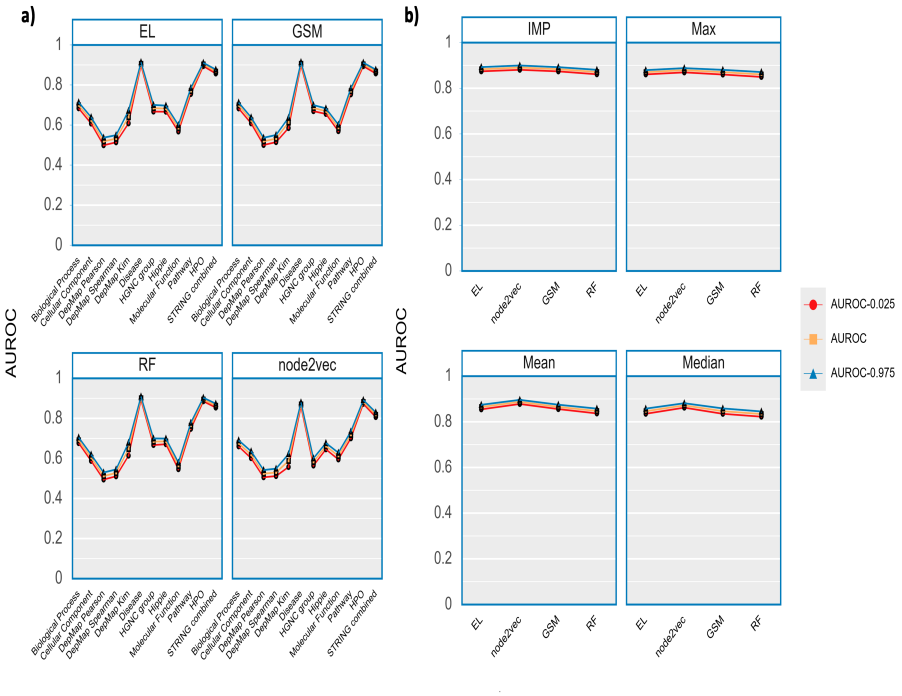
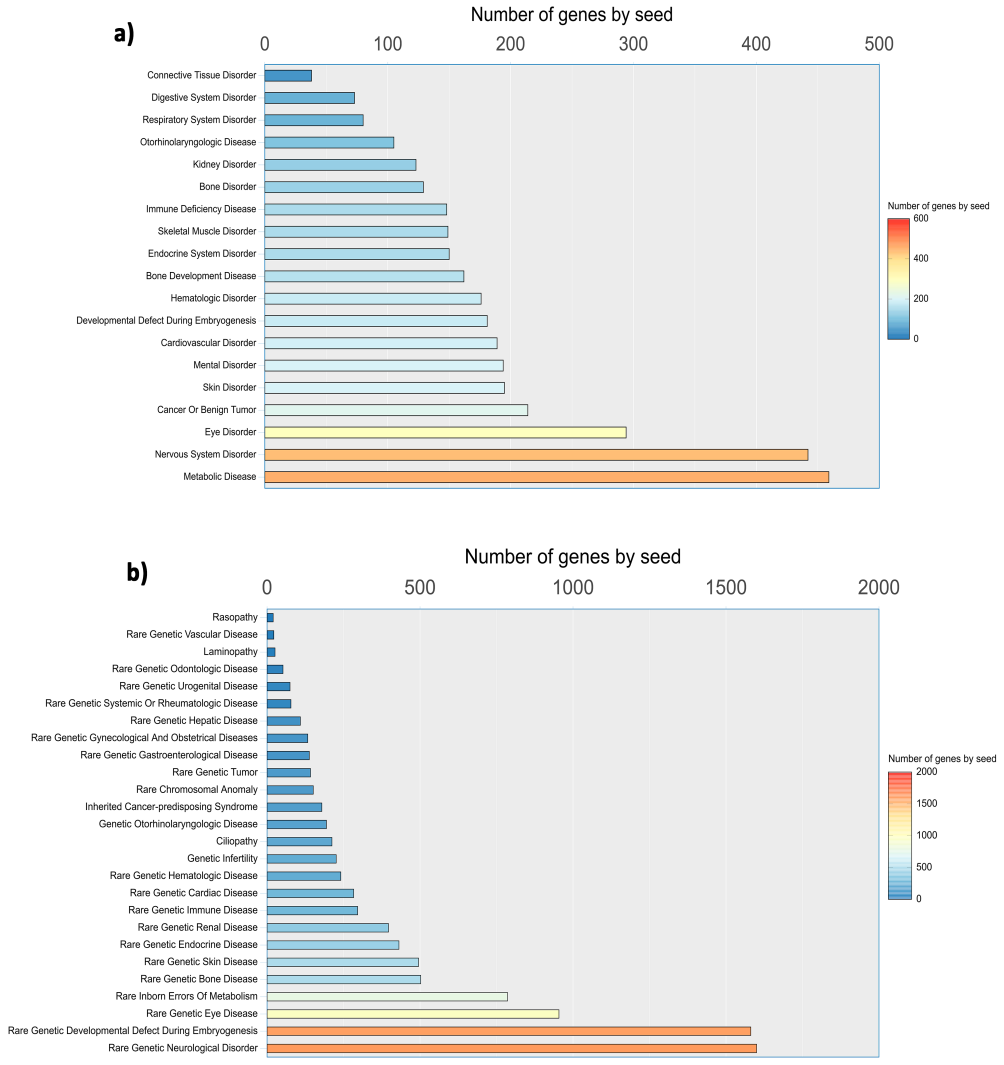
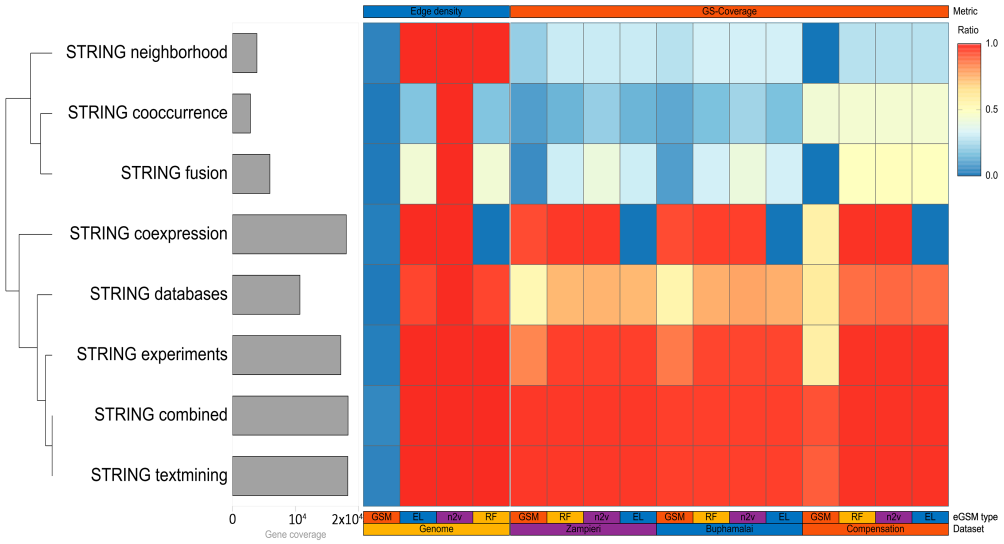
Supplementary figures



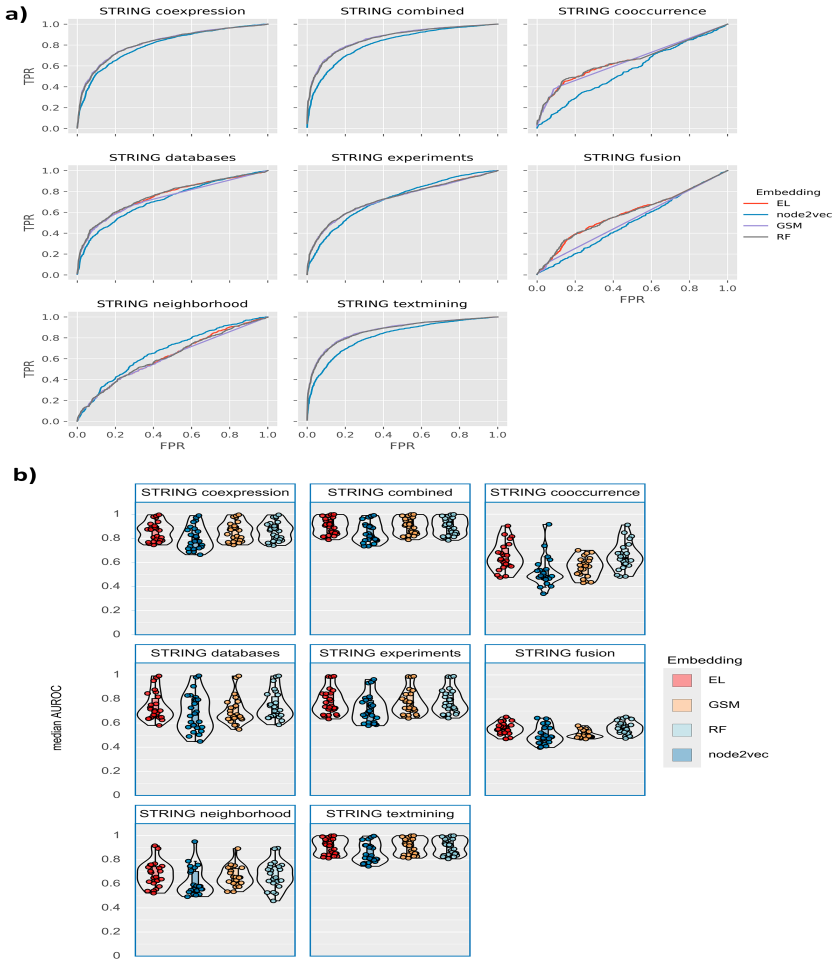
Supplementary Figure 1: AUROC Confidence Interval (CI) for individual and integrated eGSMs on the Zampieri dataset. **a)** AUROC CIs for each individual eGSM. **b)** AUROC CIs for each integrated eGSM. Confidence intervals were computed using a 1000-iteration bootstrap procedure [79].



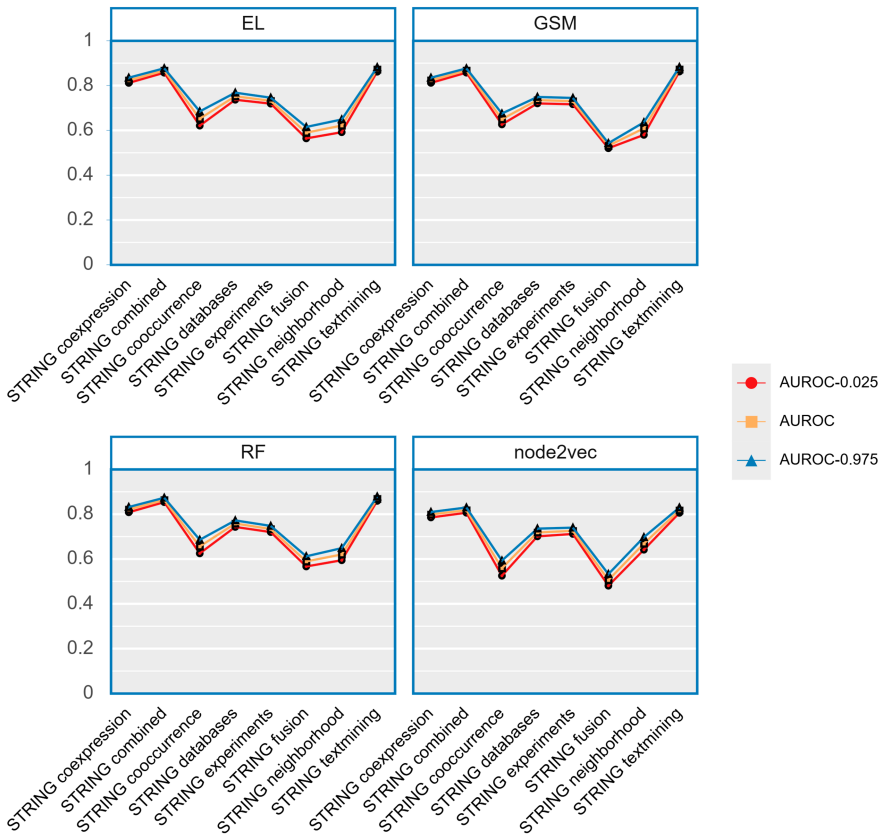
Supplementary Figure 2: Disease group size distribution across benchmarks. **a)** Number of diseases per group in the Zampieri benchmark. **b)** Number of diseases per group in the Buphamalai benchmark.



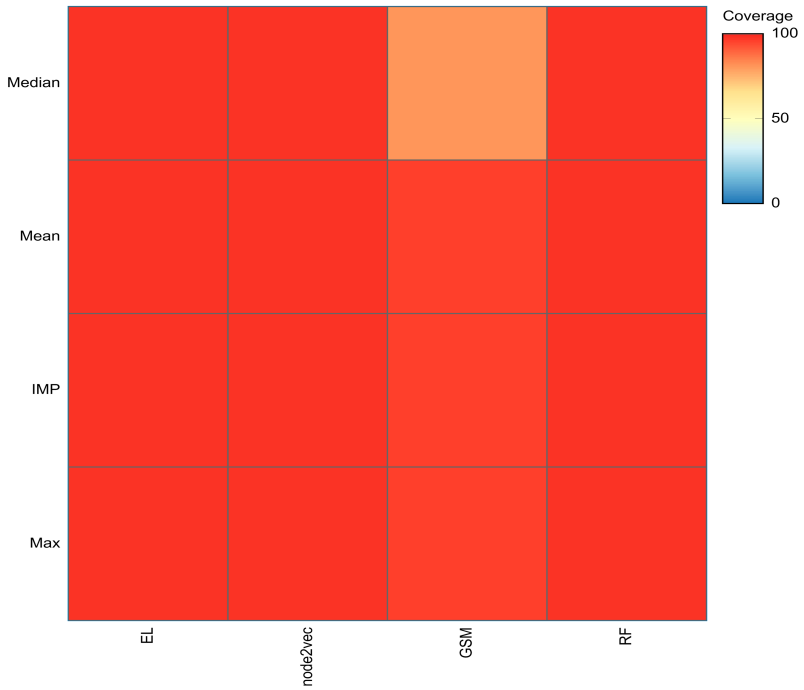
Supplementary Figure 3: Summary of gene coverage (grey bars), edge density (on left heatmap panel) and gold standard coverage (right heatmap panel) categorized by embedding technique and data source clustered by edge density and coverage similarity values. Gene coverage is defined as the total number of genes in the GSM. Edge density of the eGSM is calculated as the percentage ratio between the actual number of edges and the maximum potential number of edges represented in the matrix. Gold Standard coverage (GS-coverage) is defined in methods section (equation 12), indicated by bottom bar as Zampieri (violet), Buphamalai (blue) and compensation (red).



Supplementary Figure 4: ROC curve and median AUROC performance of individual eGSMs derived from STRING channels. **a)** ROC curves and median AUROC values for each individual eGSM in the Zampieri benchmark. **b)** ROC curves and median AUROC values for each individual eGSM in the Buphamalai benchmark.



Supplementary Figure 5: AUROC Confidence Interval (CI) in each string subchannel eGSM for Zampieri dataset. CI was obtained by a 1000 iteration bootstrap [79].



Supplementary Figure 6: Gold standard coverage (%) obtained in each integrated eGSM for the compensatory dataset categorized by embedding and integration technique.

Parte VII

Discusión y conclusiones

Capítulo 7

Discusión

A lo largo de esta tesis doctoral se ha abordado el problema de la compensación genética desde una perspectiva clínica. Partimos de la hipótesis de que la compensación entre genes requiere un cierto grado de similitud funcional, y que dicha similitud puede definirse y cuantificarse desde diferentes niveles biológicos. En este marco, uno de los ejes principales del trabajo fue la expansión de la herramienta *NetAnalyzer*, con el objetivo de explorar el problema mediante la construcción y el análisis de redes de similitud gen a gen.

El desarrollo se estructuró en varias etapas. En primer lugar, se trabajó con interacciones funcionales proteína–proteína derivadas de la base de datos STRING [31]. En los estudios de enfermedades neuromusculares (Capítulos 5.1 y 5.2), *NetAnalyzer* fue adaptado para generar representaciones de red basadas en embebidos de nodos, incorporando el algoritmo *node2vec* y módulos de análisis de clústeres en los espacios embebidos. Esto permitió obtener agrupaciones coherentes de enfermedades y genes en redes de tipo *diseasoma* e interactoma, facilitando análisis posteriores.

En una segunda etapa, se desarrolló un nuevo sistema de embebido de nodos orientado a mejorar la representación de complejos moleculares en la red de STRING. Este avance no solo resultó útil para la visualización de complejos proteicos, sino también para el estudio de mecanismos moleculares comunes en diferentes enfermedades, como en el caso de las RASopatías. Además, se incorporaron múltiples algoritmos de embebido de nodos, empleados en las comparaciones presentadas en el Capítulo 4, junto con nuevos métodos de detección y caracterización de comunidades, ampliando las capacidades analíticas de *NetAnalyzer*.

En los estudios de enfermedades neuromusculares y RASopatías se implementaron sistemas de priorización de genes basados en la similitud observada en los espacios embebidos. Estos sistemas se apoyaron en dos enfoques estadísticos complementarios: uno bayesiano y otro basado en prueba de combinación de probabilidades de Fisher. Ambos permitieron manejar la no linealidad inherente de los

embebidos y ofrecieron resultados consistentes y prometedores, tanto en el ámbito clínico como en el análisis exploratorio.

Una vez consolidada la capacidad de embeber redes y priorizar genes según su proximidad en dichos espacios, el trabajo se centró en el problema específico de los genes compensadores. Para ello, se extendió *NetAnalyzer* con módulos que integran similitudes por coseno derivadas de embebidos obtenidos de diferentes fuentes biológicas, métricas de red, nuevos métodos de proyección y herramientas específicas para el manejo de redes de ontologías.

En conjunto, estos desarrollos dieron lugar a una versión ampliada de *NetAnalyzer*, modular y ejecutable desde línea de comandos, capaz de integrar distintas bases de datos, procesarlas, embeberlas y priorizarlas con fines diversos. La herramienta permite tanto el análisis exploratorio y la priorización de genes asociados a enfermedad como la identificación de genes con potencial papel compensador. En otras palabras, *NetAnalyzer* se configura como un sistema versátil que posibilita la detección de procesos patológicos y de mecanismos compensatorios con relevancia clínica.

A continuación, se discuten los objetivos específicos de la tesis de manera individual. En la Figura 7.1 se presenta una síntesis de los principales resultados y aportaciones, en la que se señalan las relaciones con los distintos objetivos planteados.

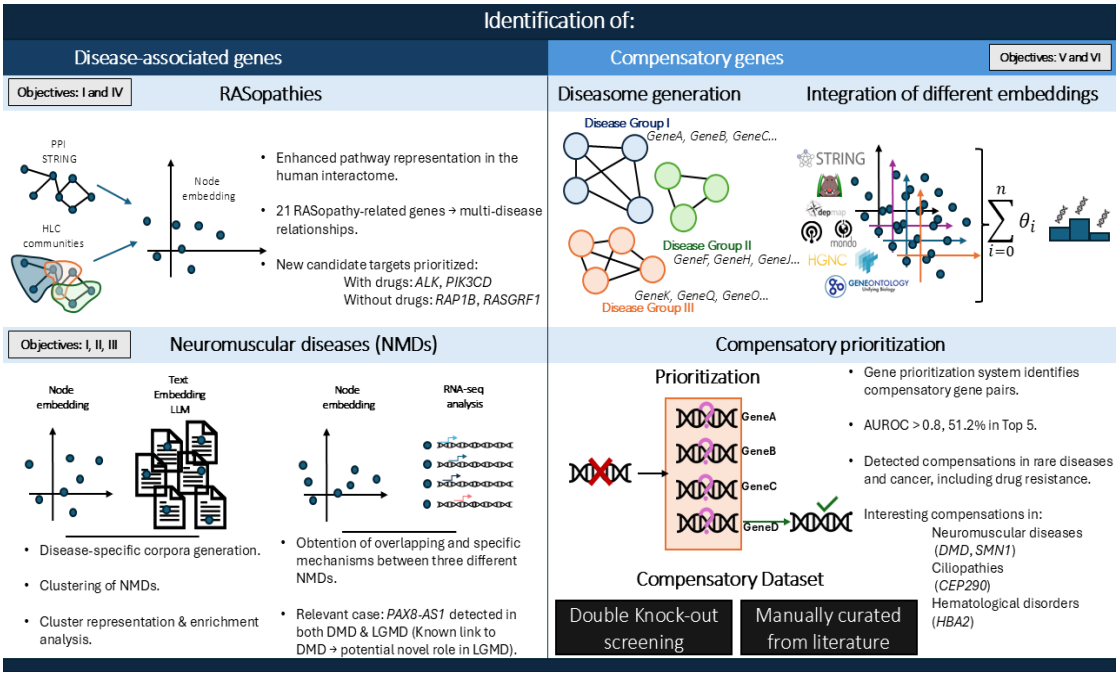


Figura 7.1: Síntesis de los principales resultados y aportaciones de la tesis, indicando la relación con los objetivos específicos.

7.1. Objetivo I. Diseñar y desarrollar un sistema de embebido de nodos que integre la información de comunidades solapantes e implementar un algoritmo de priorización basado en estos sistemas desarrollados.

En el Capítulo 4 presentamos un nuevo sistema de embebido de nodos que integra información de comunidades solapantes mediante el algoritmo *Hierarchical Link Clustering* (HLC) [88]. Aunque HLC es ampliamente utilizado, su aplicación a redes biológicas de gran tamaño presenta importantes limitaciones computacionales. Para superarlas, se desarrollaron dos implementaciones: HLCnp en Python y LKCMsp en R. La primera, integrada en la librería CDLIB, optimiza el procesamiento de redes densas, incorpora mayor flexibilidad en el agrupamiento y permite un filtrado eficiente de la matriz de distancias de Tanimoto. La segunda sigue más fielmente la formulación original de Ahn [88] respecto a la implementación del paquete *linkcomm* [89], mejorando tiempos de ejecución y eficiencia en memoria, como puede verse en la Figura 1. Las pruebas comparativas en redes biológicas y grafos densos (como Wikipedia) confirmaron no sólo la equivalencia entre las implementaciones sino también su superiodidad en términos de escalabilidad y eficacia (Figura 1C y 1D).

Por otro lado, la integración de comunidades en el proceso de embebido se evaluó tanto con métodos solapantes (HLC) como no solapantes (Louvain [90]). Los resultados mostraron la superioridad del enfoque solapante: HLC representó las rutas biológicas de manera más coherente en el espacio embebido, reflejando que muchas proteínas participan en procesos múltiples de forma simultánea (Figuras Suplementarias 6 y 7). Por el contrario, Louvain tiende a formar pocos clústeres grandes que fragmentan rutas relacionadas, dispersando genes funcionalmente próximos en regiones aisladas y sin revelar estructura interna significativa (Figuras Suplementarias 6 y 7). Como ejemplo, en los análisis UMAP (Figuras 5 y 6) de las principales rutas biológicas, HLC distinguió procesos como "Formation of a pool of free 40S subunits", "Mitochondrial translation elongation" y "L13a-mediated translational silencing of Ceruloplasmin expression", que en Louvain aparecieron fusionados en un único clúster. Este patrón se mantuvo al considerar las diez rutas más grandes, donde HLC preservó conexiones funcionales que Louvain no consiguió reflejar. Una vez demostrada la superioridad de los métodos solapantes frente a los no solapantes bajo su contexto biológico, comparamos nuestro sistema con otros diez algoritmos de embebido de nodos del estado del arte (Figura 4), entre los que se encontraban node2vec [47] y deepwalk [46]. Los resultados confirmaron que nuestra estrategia superó de manera consistente la representación de las rutas

biológicas en la red de interacciones proteína–proteína.

En conjunto, estos resultados muestran que HLC permite organizar las rutas de forma biológicamente significativa, resaltando el valor de las estructuras de comunidades solapantes en redes biológicas complejas. Este tipo de embebidos mejorados puede potenciar aplicaciones de aprendizaje profundo en genómica y proteómica, donde el embebido de nodos constituye un paso esencial de preprocesamiento [91, 92, 93, 94]. Además, la propia representación ofrece un marco útil para mapear genes de interés, explorar su conectividad con rutas biológicas e interpretar su papel funcional en un contexto de red.

Finalmente, todos los procesos de detección de comunidades solapantes y generación de embebidos de nodos fueron implementados en la herramienta *NetAnalyzer*. Sobre estos espacios se incorporaron sistemas de priorización basados en enfoques estadísticos, con el objetivo de identificar genes cercanos a un conjunto de referencia y con alta probabilidad de pertenecer al mismo. En particular, se desarrolló un modelo bayesiano que supera las limitaciones asociadas a la no linealidad de los espacios de embebido, sustituyendo la aproximación clásica mediante centroides por un esquema de ponderación probabilística que refina la selección de candidatos.

De forma complementaria, y como se muestra en la sección 5.1 del Capítulo V, se integró un segundo método de priorización basado en el test de Fisher. De este modo, *NetAnalyzer* cuenta actualmente con dos estrategias estadísticas alternativas y complementarias para la priorización en el espacio de embebido, lo que incrementa la flexibilidad y robustez en el análisis y la identificación de genes relevantes.

7.2. Objetivo II. Generar un corpus de conocimiento especializado en enfermedades neuromusculares mediante la combinación de redes de similitud textual y representaciones basadas en embebido de nodos.

Relacionado con este objetivo se encuentra el trabajo del Capítulo 5.1, donde se construyó una red de similitud textual mediante la herramienta *Sentence Transformer*, a partir de la asociación entre artículos de PubMed, enfermedades neuromusculares (NMDs) y sus fenotipos. Para ello fue esencial la codificación de las enfermedades con la ontología MONDO [30] y de los fenotipos con la HPO [29]. Como primer paso, evaluamos la coherencia y cobertura de las anotaciones, analizando la distribución de términos en ambas ontologías, lo que permitió com-

probar el grado de alineamiento entre las anotaciones derivadas de literatura y las ontologías biomédicas establecidas.

A nivel de enfermedad,según se muestra en la Figura 2 observamos una baja frecuencia de anotación, explicada por la dispersión de publicaciones que suelen centrarse en una única enfermedad, lo que limita la descripción conjunta de múltiples patologías. Algunos dominios presentan escasa representación, mientras que otros, como "hereditary neuromuscular disease" o "peripheral neuropathy", concentran la mayoría de las anotaciones. En contraste, el análisis de fenotipos (Figura 2B) mostró una representación más equilibrada, coherente con la naturaleza sistémica de las NMDs, que afectan a múltiples sistemas biológicos y generan fenotipos diversos [7]. Este patrón sugiere una cierta convergencia fenotípica, reflejada en la frecuencia de alteraciones musculoesqueléticas, neurológicas y metabólicas, categorías estrechamente vinculadas con este grupo de enfermedades [7]. En conjunto, este doble fenómeno, focalización en NMDs individuales a nivel de artículo y convergencia de fenotipos entre estas enfermedades, refleja un sesgo habitual en la investigación biomédica: el estudio aislado de enfermedades, pese a compartir rasgos fenotípicos relevantes. Esto es algo a tener en cuenta pues indica potenciales sesgos a la hora de estudiar las distintas enfermedades neuromusculares, como pueda ser el caso de la atrofia muscular espinal [8, 9] o la distrofia muscular de Duchenne (DMD) [10, 11].

Tras ello, siguiendo el esquema mostrado en la Figura 1, se construyeron distintas redes enfermedad–enfermedad en base a artículos compartidos. Sobre cada una de estas redes se llevaron a cabo distintos embebidos de nodos y se analizaron cuáles de ellos eran más coherentes respecto a la ontología. Para ello, se realizó una correlación por *Spearman* de la similitud por coseno entre las enfermedades dentro del espacio de embebido, respecto a la similitud semántica de estas enfermedades en el contexto de la ontología MONDO. El análisis mostró que la red construida a partir de texto, basada puramente en conexión por artículos compartidos, correlacionaba mejor con las relaciones semánticas definidas en la ontología MONDO, superando a las redes basadas en fenotipos (HPO) asociados comunes (Figura 2C). Cabe mencionar que parte de las limitaciones en la red basada en fenotipos se debe a falta de precisión en la asociación entre enfermedades y fenotipos. Esto se debe a que el enfoque empleado vincula todos los fenotipos identificados en un documento con todas las enfermedades mencionadas en el mismo. En este sentido, una mejora futura podría consistir en filtrar los fenotipos con bajo contenido de información o ponderar las asociaciones según la relevancia de la intersección entre fenotipos y enfermedades. Entre los métodos de embebido evaluados, *node2vec* fue el que mostró un rendimiento más consistente y robusto, por lo que se seleccionó para los análisis posteriores.

Una vez seleccionado el mejor embebido de nodos para la red enfermedad–

enfermedad, se pasó a la identificación de comunidades presentes dentro del espacio generado. Así, el estudio de las comunidades derivadas del embebido permitió observar cómo las enfermedades neuromusculares se agrupan con coherencia tanto a nivel ontológico (Tabla 1), en función de los parentales comunes con alto valor informativo dentro de MONDO, como fenotípico (Tabla 2), mediante la compartición de términos de la HPO. Estos resultados reflejan una consistencia de las agrupaciones a nivel semántico como clínico.

La aparición de subdivisiones claras dentro de dominios heterogéneos como la neuropatía periférica (Figura 3) confirma que la estrategia de representación mediante embebido de nodos puede capturar especializaciones funcionales y relaciones metabólicas que no son evidentes en enfoques tradicionales. Asimismo, la identificación de fenotipos enriquecidos en comunidades concretas, como la esclerosis lateral amiotrófica (ALS) familiar o la distrofia muscular (Tabla 1), constituye un recurso valioso para el diagnóstico diferencial y muestra la fortaleza del enfoque en la caracterización de las relaciones entre enfermedades y sus manifestaciones clínicas. Además, los fenotipos comunes entre la distrofia muscular y la esclerosis lateral amiotrófica cobra relevancia en el análisis desarrollado en el Capítulo 5.2, donde se estudian en mayor profundidad ambas enfermedades, comparándolas a nivel transcripcional e integrando dicha información con datos de interactómica.

Finalmente, el análisis de enriquecimiento funcional a nivel génico (Figura 4) confirmó la coherencia interna de las comunidades obtenidas. Las especializaciones funcionales y los procesos biológicos compartidos entre enfermedades respaldan la validez biológica de las agrupaciones y sugieren que estas representaciones pueden ser empleadas tanto para el estudio exploratorio de las NMDs como para la priorización de genes implicados en sus mecanismos patológicos. En conjunto, estos resultados subrayan el potencial del embebido de redes como herramienta para integrar conocimiento biomédico disperso y generar hipótesis sobre la organización funcional de las enfermedades neuromusculares.

7.3. Objetivo III. Integrar y analizar información transcriptómica de distintas enfermedades neuromusculares junto con datos de interacción proteína–proteína, con el fin de identificar mecanismos reguladores comunes.

Como se ha mencionado en el objetivo anterior, ya en el Capítulo 5.1 se observó que las enfermedades neuromusculares como la DMD, la distrofia muscular de cinturas (LGMD) y la esclerosis lateral amiotrófica (ALS), poseían semejanzas a

nivel fenotípico y tendían a agruparse al identificar artículos científicos que los mencionan en común significativamente. A partir de este hallazgo, el siguiente objetivo fue integrar el estudio de enfermedades que habitualmente se analizan de manera aislada, como la DMD y la ALS, y abordarlas de forma conjunta. A este análisis se añadió la LGMD, una patología con elevada similitud fenotípica respecto a la DMD, pero que ha recibido una atención investigadora notablemente menor. Con ello, se buscó integrar la información disponible de distintas enfermedades neuromusculares y, al mismo tiempo, mejorar la caracterización de una patología menos estudiada como la LGMD. Este planteamiento requiere una exploración más profunda de las bases genéticas y moleculares compartidas, la cuál se desarrolla en el Capítulo 5.2.

Estas enfermedades presentan orígenes genéticos diversos. Entre estas, nos encontramos con mutaciones en el gen *DMD* en el caso de la distrofia muscular de Duchenne, en *TNPO3* para LGMD [95], o en más de 40 genes asociados a ALS [96], entre los que destacan *C9ORF72* [97, 98], *FUS* [99, 100] y *TARDBP* [101]. Sin embargo, convergen en la alteración de procesos fundamentales como la homeostasis celular, la inflamación y la función sináptica [102, 103]. Aquí, el análisis transcriptómico (RNA-seq) constituye una herramienta clave para desentrañar estas alteraciones, identificar rutas desreguladas y proponer biomarcadores, especialmente cuando se combina información de diferentes tejidos (mioblastos, miotubos, fibroblastos cardiacos, iNeurons o PBMCs), como es el caso del estudio llevado a cabo en esta tesis. Una estrategia que permitió capturar tanto señales comunes como específicas de tejido, a pesar de la limitada disponibilidad de datos en enfermedades raras.

En términos globales, los resultados de expresión diferencial coincidieron con los descritos en los estudios originales, reforzando la robustez del *pipeline* empleado. La mayoría de los grupos de genes identificados mostró categorías funcionales coherentes relacionadas con morfogénesis, regulación neuronal, procesos sinápticos, integridad muscular, inflamación y organización de la matriz extracelular (Figura 2). Algunas discrepancias puntuales, como suceden en los conjuntos de datos DMD_cfib y ALS_fib_FUS, parecen deberse a diferencias metodológicas. A pesar de los contextos patológicos específicos, los resultados muestran una superposición en las categorías funcionales, lo que indica la existencia de mecanismos moleculares compartidos. Entre las funciones más recurrentemente enriquecidas se encontraron la unión a matriz extracelular y la unión a glicosaminoglicanos, presentes en la mayoría de los conjuntos de datos. Estos términos sugieren una desregulación conservada de las interacciones con la matriz extracelular a lo largo de las NMDs, lo que podría contribuir a la alteración de la integridad tisular y a una señalización celular aberrante en contextos tanto miogénicos como neurogénicos [104, 105].

La integración de redes de interacción proteína–proteína con expresión diferen-

cial permitió detectar agrupaciones de genes diferencialmente expresados próximos a los genes causales, revelando funciones altamente plausibles. En DMD, se asociaron con contracción muscular, electrofisiología cardiaca y organización del complejo distrofina (Tablas Suplementarias 7-10); en ALS con *C9ORF72* como gen causal (ALS-C9ORF72), con homeostasis neuronal, autofagia y mitocondrias (Tabla Suplementaria 11); y en ALS con *FUS* como gen causal (ALS-FUS, Tabla Suplementaria 12), con procesos de desarrollo del tejido conectivo y señalización Wnt, reflejando un impacto más allá de las neuronas. También identificamos genes aislados, no integrados en comunidades pero cercanos a genes causales en la red, que podrían tener roles indirectos o adaptativos. Entre ellos destacan el gen *HS3ST3A1*, común en LGMD y ALS; el gen *CDH12*, asociado a fibrosis cardiaca en DMD; y el gen *MAP1LC3C*, vinculado a la autofagia alterada en la enfermedad de ALS-C9ORF72 [98]. Asimismo, la proximidad en interactoma entre los genes *JMJD6* (implicado en la remodelación de la cromatina), *VRK1* (asociado con la degeneración de motoneuronas) respecto del gen *FUS* (Tabla Suplementaria 14), sugiere la convergencia de mecanismos epigenéticos y transcripcionales en ALS-FUS.

Un subconjunto de genes diferencialmente expresados no se pudo mapear en el interactoma humano (Tabla 2 y Tabla Suplementaria 4), entre los que destacan varios RNAs no codificantes de cadena larga (lncRNAs). Por ejemplo, la sobreexpresión del gen *PAX8-AS1* detectada en LGMD junto con su conocida asociación a modulacion inflamatoria y cardiaca en DMD [106], indica un potencial mecanismo patológico común y que puede caracterizarse para LGMD. Por otro lado, el gen *MEG3*, relacionado con la regulación de la masa muscular [107], mostró patrones opuestos de sobreexpresión en músculo esquelético y represión en corazón que sugieren un potencial papel compensador en músculo esquelético y otro regulador en el cardíaco. Finalmente, *NEAT1*, un gen crítico para la organización celular y regulación del RNA [108], presentó sobreexpresión en DMD y dismuyó su expresión en ALS-FUS, coherente con la disfunción nuclear mediada por el gen *FUS* [109]. Estos hallazgos apuntan a los lncRNAs como moduladores clave a tener en cuenta para potenciales biomarcadores y dianas terapeúticas.

7.4. Objetivo IV. Identificar potenciales genes implicados en modular la vía Ras/MAPK y caracterizar su impacto en la fisiopatología de las RASopatías

Durante el trabajo presentado en el Capítulo 4, se estudiaron potenciales genes que pudiesen estar modulando el desarrollo de las distintas RASopatías. Así, a partir de las listas de priorización obtenidas mediante embebido de nodos, seleccionamos genes y evaluamos la evidencia disponible en literatura, rutas de señalización y variantes patogénicas. Esto permitió distinguir genes con evidencia funcional o genética sólida, asociaciones indirectas y genes sin evidencia clara.

Se recopilaron 21 genes vinculados a 9 grupos de RASopatías a partir de MONDO y Monarch (Tabla 3), observándose en el embebido del interactoma humano (propuesto en el Capítulo 4) la agrupación en comunidades coherentes (Figura 7). Partiendo de este conjunto de genes ya caracterizados, se obtuvieron nuevos genes asociados mediante su priorización en base a su cercanía en el espacio de embebido respecto a los genes causales de las RASopatías (Tabla Suplementaria 2). Los resultados identificaron múltiples genes candidatos novedosos que podrían estar implicados en estas patologías. Varios de ellos aparecieron en más de una RASopatía, lo que pone de manifiesto una superposición significativa en los procesos biológicos subyacentes que comparten estos trastornos. La priorización conjunta destacó genes con funciones conocidas en la vía Ras/MAPK como *SHC1* [12], *MAPK7* y *MAP3K2* [13], así como *RAP1A/B* [110], previamente asociados a síndromes como Kabuki [111, 112]. Otros candidatos relevantes fueron *ANKS1A* [113], *SKAP1* [114] y *RAPGEF5* [115], que sugieren intracción cruzada entre las vías PI3K y Wnt/PCP. También se priorizaron miembros del eje PI3K/AKT como *PIK3R1*, cuya disfunción afecta la señalización por Ras [116], lo que refuerza la intersección funcional con síndromes relacionados [117]. Entre los genes con interés terapéutico destacan *ALK*, diana de inhibidores como crizotinib [118], y *PRKCA*, modulador potencial en contextos de señalización aberrante [13].

Al aplicar la metodología a genes asociados con RASopatías específicas, identificamos genes adicionales de interés. En el caso del síndrome de Noonan, destacó *RASA2*, que codifica una proteína activadora de GTPasas Ras y cuyas mutaciones de pérdida de función se han vinculado a fenotipos tipo Noonan [119, 120]. También se recuperaron genes relevantes como *HRAS* y *NF1*, asociados a otros síndromes (Costello y neurofibromatosis tipo 1), así como genes de la vía Ras, incluidos *BRAF* y diversos factores de intercambio de nucleótidos de guanina. Resultados similares se obtuvieron al analizar los genes asociados a Costello y síndrome cardio-facio-cutáneo (CFC), lo que resalta la superposición molecular

entre distintas RASopatías y apoya la hipótesis de una desregulación compartida de la vía Ras/MAPK como base de sus características fenotípicas comunes.

Además de analizar los genes asociados a RASopatías, investigamos la posición relativa de las rutas en las que participan (Figura 8). En este contexto, los genes implicados en la activación de MAPK1 (gen que codifica la proteína ERK2, una quinasa central en la vía de señalización RAS/MAPK) aparecen próximos a los de la ruta de señalización de *CSF3* (gen que codifica el factor estimulante de colonias de granulocitos, también denominado GM-CSF) y a los asociados a mutantes fosforilados de *KIT* (gen que codifica al receptor tirosina quinasa c-Kit). Tanto GM-CSF como c-Kit participan en la proliferación y supervivencia celular, y c-Kit es además un activador directo de la cascada RAS/MAPK [121, 122]. Esta proximidad espacial sugiere posibles interacciones o mecanismos de co-regulación que podrían contribuir a la señalización anómala observada en RASopatías. Cabe señalar, no obstante, que los genes directamente vinculados a RASopatías en esta ruta (los genes *MAPK1* y *MAP2K2*) aparecen relativamente alejados del resto en la representación.

Asimismo, la representación UMAP (técnica de reducción no lineal de la dimensionalidad) reveló que genes relacionados con la señalización de *FGFR4* (gen que codifica el receptor tirosina quinasa FGFR4), *PDGFRA* (gen que codifica el receptor α del factor de crecimiento derivado de plaquetas) e insulina (hormona que activa su propio receptor, INSR) se sitúan próximos a *MAPK1* y a la ruta de señalización de *CSF3* (ruta activada por el ligando CSF3/GM-CSF a través de su receptor CSF3R). Esto resulta relevante, ya que FGFR4 actúa como activador *upstream* de la cascada RAS/MAPK [123], PDGFRA se ha asociado a proliferación dependiente de Ras, y la señalización de insulina activa la vía Ras-MAPK a través de las proteínas adaptadoras Grb2 y SOS [124]. Esta proximidad funcional sugiere una posible convergencia de señales capaz de amplificar o modular la actividad de Ras, lo que abre la puerta a explorar genes intermedios localizados entre estas rutas en la representación UMAP.

En conjunto, estos resultados muestran que la metodología propuesta no sólo identifica genes ya conocidos en la etiología de las RASopatías, sino que también encuentra nuevos candidatos funcionales y rutas convergentes con potencial relevancia clínica. La capacidad de detectar tanto moduladores directos de la vía de señalización Ras/MAPK como elementos de interacción cruzada con otras cascadas (PI3K/AKT, Wnt, insulina, KIT) muestra la complejidad reguladora de estos síndromes y abre la posibilidad de explorar dianas terapéuticas comunes. En este contexto, el enfoque basado en embebido de nodos, capaz de visualizar y priorizar de genes próximos a nivel de interactoma, se presenta como una herramienta robusta para desentrañar mecanismos compartidos y guiar futuros estudios experimentales en el ámbito de las RASopatías.

7.5. Objetivo V. Generar un *diseasoma*, entendido como una red de enfermedades, actualizado para la validación del sistema de priorización de genes causales.

En esta tesis se replicó el sistema de evaluación por genes asociados a enfermedad de dos sistemas de referencia: el de Buphamalai y colaboradores [73] y el de Zampieri y colaboradores [35]. En ambos casos,desarrolados en el trabajo presentado en el Capítulo 6, la estrategia consistió en recopilar conjuntos de genes asociados a cada enfermedad y evaluar la capacidad del sistema para predecir dichas agrupaciones. En este último caso fue necesario actualizar el conjunto de datos, basado en el clásico *diseasoma* de Goh y colaboradores [55]. Esto puso de relieve una limitación habitual en este tipo de marcos de validación: la dependencia de estos respecto a bases de datos que evolucionan de manera constante, como OMIM [125]. A partir de la descarga del archivo Morbid Map (MM) con fecha 7 de mayo de 2024, y tras aplicar filtros y reclasificar las enfermedades con ayuda de la ontología MONDO, se reprodujo fielmente el procedimiento original de Zampieri. Esta actualización permitió no sólo validar los resultados de manera consistente con trabajos previos, sino también asegurar que las conclusiones extraídas se apoyan en información vigente y completa.

Más allá del interés metodológico, este ejercicio evidenció la importancia de disponer de sistemas capaces de actualizar de forma automática los conjuntos de validación. En particular, la actualización del *diseasoma* propuesto por Goh y colaboradores [55], ampliamente utilizado en estudios de análisis de redes, mostró el valor de mantener al día un recurso de referencia en constante uso. Con este objetivo, todo el proceso fue incorporado en la herramienta *PETS* [126], lo que garantiza la reproducibilidad y facilita la generación de nuevas versiones del *diseasoma* conforme avanza y evoluciona la información biomédica. Esto representa una aportación significativa, ya que en muchos estudios de priorización los marcos de validación permanecen estáticos y acaban quedando obsoletos frente al ritmo de nuevos descubrimientos en genética humana.

Finalmente, el propio concepto de *diseasoma* se presenta como una oportunidad de investigación en sí misma. Más allá de su uso como marco de validación, este *diseasoma* actualizado se puede explorar con fines analíticos, por ejemplo, aplicando metodologías de enriquecimiento de fenotipos y genes comunes, o analizando sus estructuras ontológicas y agrupaciones, en la línea de lo desarrollado en el Capítulo 5.1. De este modo, la actualización del *diseasoma* no sólo cumple un papel instrumental en la validación, sino que se perfila como un recurso valioso para profundizar en las relaciones entre enfermedades y sus bases moleculares

compartidas.

7.6. Objetivo VI. Identificar y caracterizar un conjunto de pares gen causal de la enfermedad – gen compensador, en el contexto de enfermedades raras y cáncer.

Uno de los primeros pasos de este trabajo, desarrollado en el Capítulo 6, fue reunir un conjunto de parejas conocidas de genes causal–compensador. A través de búsqueda manual caso por caso y extracción sobre cribados experimentales se recopilaron 86 parejas con relevancia clínica o validadas en distintas líneas celulares. Aunque el tamaño no permite su uso como base para entrenamiento supervisado, este recurso resulta valioso para la evaluación de los sistemas desarrollados y como colección de ejemplos representativos tanto en enfermedades raras como en cáncer, incluyendo casos de enfermedades neuromusculares y RASopatías.

En el conjunto de datos de compensación, *node2vec* mostró una mejora significativa en los resultados finales, alcanzando las primeras posiciones con una ventaja sustancial frente a las otras técnicas basadas en *Kernels*(Figuras 6 y 7). Todos los métodos alcanzaron resultados satisfactorios de validación a nivel de AUROC (área bajo la curva ROC), con valores superiores a 0.75, sin observarse diferencias significativas entre las distintas estrategias (Figura 7B). Todo ello demuestra, aunque con cierto margen de mejora, la capacidad del sistema de priorizar la pareja correcta de compensadores.

Por otro lado, resulta llamativo que la cobertura de los genes compensadores disminuya de manera significativa para bases de datos relacionadas con procesos patológicos, como MONDO, Phenotype y DepMap (Figura 3). Este hallazgo es especialmente sorprendente si se considera el contexto patológico a partir del cual se identificaron los genes compensadores. Dos posibles explicaciones para este descenso son, por un lado, la dificultad de caracterizar estos genes en pacientes o en líneas celulares debido a su condición de genes esenciales, y por otro, las limitaciones prácticas asociadas al estudio de este tipo de interacciones. Es importante destacar que la compensación no siempre implica interacción directa o coexpresión entre el gen compensador y el gen afectado, lo que se refleja en la menor cobertura de los compensadores en los canales de coexpresión y experimentales.

Desde una perspectiva clínica, nuestro sistema de priorización ofrece un gran potencial para el tratamiento de enfermedades genéticas, al identificar posibles dianas para validación experimental. Por ejemplo, en enfermedades como la atrofia muscular espinal, la distrofia muscular de Duchenne y la β-talasemia, casos modelo

sobre el uso clínico de la compensación, el sistema logró priorizar correctamente cada gen compensador conocido dentro del Top 5. Este éxito refuerza la capacidad predictiva de nuestro enfoque en trastornos genéticos humanos y subraya su potencial para descubrir nuevas dianas génicas con aplicaciones terapéuticas. Además, dado el papel de la compensación genética en cáncer, nuestra metodología podría extenderse al ámbito oncológico, permitiendo identificar y priorizar genes compensadores que contribuyen a la resistencia a tratamientos y a la progresión tumoral. Entre los ejemplos más destacados, en cáncer de próstata las células tumorales evaden el bloqueo del receptor de andrógenos mediante la inducción del receptor de glucocorticoides. De manera similar, en células de cáncer de mama, la supresión de *RBL1* es compensada por *RBL2* [127, 128].En este último caso, resulta de especial interés explorar la priorización del sistema en relación con PARP, considerando el éxito clínico de los inhibidores de PARP como estrategia terapéutica basada en la letalidad sintética en este tipo de tumores [80, 81, 87].

Una vez identificados los genes compensadores de un gen asociado a enfermedad, el siguiente paso consiste en explorar estrategias para su activación mediante pequeñas moléculas. Inicialmente, es posible considerar fármacos con capacidad de sobreexpresar estos genes compensadores [11, 129]. Otra alternativa es la aplicación de pequeños RNAs, herramientas biotecnológicas versátiles de gran interés [130], que ya están implicadas en mecanismos naturales de compensación de genes parálogos [60, 61]. En los casos donde dichos mecanismos no se inducen de manera natural (por ejemplo, en ausencia de un codón de terminación prematuro o de un gen parálogo), la activación podría requerir el uso de RNAs pequeños exógenos. En este contexto, estudios recientes han puesto de relieve el potencial de los *translation-activating RNAs* (taRNAs), diseñados específicamente para potenciar la traducción de genes diana [131]. Estos RNAs ofrecen un sistema de administración sencillo y podrían resultar especialmente útiles en el tratamiento de las RASopatías [131], grupo de enfermedades abordado en el Capítulo 4. Otras estrategias incluyen el empleo de siRNAs dirigidos a inhibidores del gen compensador, así como el uso de oligonucleótidos antisentido de cambio de splicing (ASOs) para inducir alteraciones en el marco de lectura que introduzcan codones de terminación prematuros (PTCs) en el ARNm, activando de este modo mecanismos de compensación mediados por genes parálogos [132]. La integración de nuestro enfoque junto al uso de estos sistemas activación representa, por tanto, una estrategia prometedora para el desarrollo de nuevas terapias.

En conjunto, los resultados obtenidos demuestran que la priorización de genes compensadores es factible y clínicamente relevante, incluso a partir de conjuntos de datos limitados. El sistema desarrollado basado en redes de similitud y la integración de sus embebidos, no sólo recupera compensadores conocidos en distintos contextos patológicos, sino que también abre la posibilidad de identificar nuevas

dianas terapéuticas y estrategias de intervención. La combinación de métodos de embebido de nodos, validación estadística y enfoques de activación génica (ya sea mediante fármacos o biotecnologías emergentes como los taRNAs) ofrece un marco sólido para avanzar en la comprensión y el aprovechamiento clínico de la compensación genética. Este enfoque sienta las bases para estudios experimentales futuros y para el diseño de terapias innovadoras en enfermedades raras y cáncer.

Capítulo 8

Conclusions

From the present thesis, the following conclusions can be drawn:

Conclusion I The embedding space generated in various studies proved valuable for exploring the relationships between genes for multiple diseases.

Conclusion II By integrating overlapping clustering with node embedding, our approach enhanced the representation of pathways in the human interactome.

Conclusion III The use of language models and network embeddings to explore associations between genetic diseases allows clusterize neuromuscular disorders and generate specific document corpora for each subgroup.

Conclusion IV The lncRNA gene *PAX8-AS1* was detected in both Duchenne muscular dystrophy (DMD) and limb-girdle muscular dystrophy (LGMD). Its previously reported association with DMD highlights it as a potential point of interest for further characterization in LGMD.

Conclusion V In the case of RASopathies we identified new putative treatment gene targets such as *ALK* and *PIK3CD* with known associated drugs and *RAP1B*, *RASGRF1* genes as promising targets but without associated drug information.

Conclusion VI The causal gene prioritization system proved effective in identifying compensatory gene pairs. Several are associated with neuromuscular diseases, such as *DMD* in Duchenne Muscular Dystrophy and *SMN1* in Spinal Muscular Atrophy (SMA), while others are linked to ciliopathies, including *CEP290*, or to hematological disorders such as α-thalassemia-2 caused by mutations in *HBA2*.

Conclusion VII The prioritization system applied to compensatory genes was able to detect cases of compensation in cancer. In fact, some of these compensations occurred in the context of drug resistance.

Parte VIII

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